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(12) **United States Patent**
Albertini et al.

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(54) **MUTATED GLYCOPROTEIN OF VESICULAR STOMATITIS VIRUS**

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(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 0 days.

This patent is subject to a terminal disclaimer.

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(65) **Prior Publication Data**

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Related U.S. Application Data

(63) Continuation of application No. 18/629,097, filed on Apr. 8, 2024, now Pat. No. 12,269,848, which is a continuation of application No. 18/473,720, filed on Sep. 25, 2023, now Pat. No. 12,030,915, which is a continuation of application No. 16/649,271, filed as application No. PCT/EP2018/075824 on Sep. 24, 2018, now Pat. No. 12,091,434.

(30) **Foreign Application Priority Data**

Sep. 22, 2017 (EP) 17306255

(51) **Int. Cl.**
C07K 14/005 (2006.01)
C12N 7/00 (2006.01)

(52) **U.S. Cl.**
CPC **C07K 14/005** (2013.01); **C12N 7/00** (2013.01); **C12N 2760/20222** (2013.01); **C12N 2760/20232** (2013.01); **C12N 2760/20245** (2013.01); **C12N 2760/20262** (2013.01)

(58) **Field of Classification Search**
None
See application file for complete search history.

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Assistant Examiner — Myron G Hill

(74) *Attorney, Agent, or Firm* — Banner & Witcoff, Ltd.

(57) **ABSTRACT**

The invention relates to an isolated non-naturally occurring protein comprising the amino acid sequence as set forth in SEQ ID NO: 1, and wherein the amino acid in position 8, 47, 209 and/or 354 is substituted by any amino acid different from the amino acid indicated at that position in said sequence SEQ ID NO: 1.

30 Claims, 11 Drawing Sheets

Specification includes a Sequence Listing.

(56)

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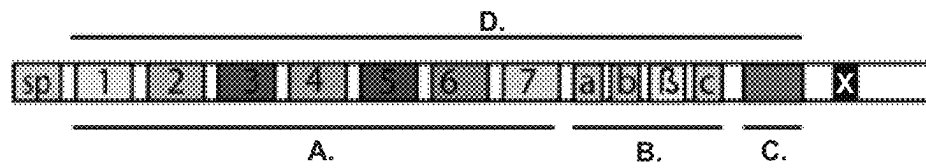


Fig. 1

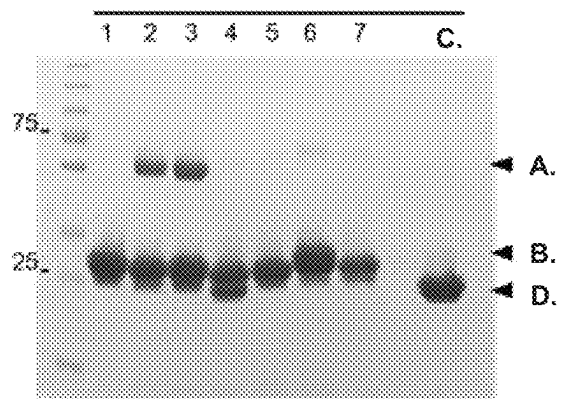


Fig. 2

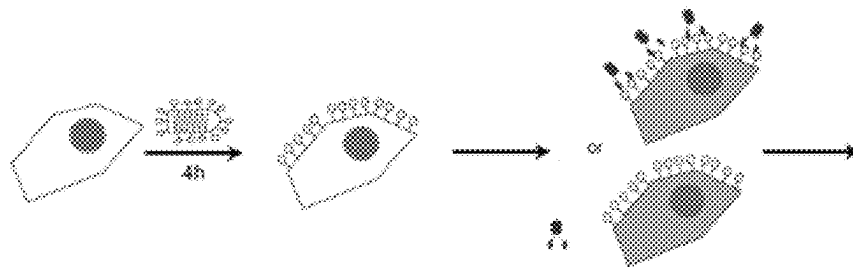


Fig. 3

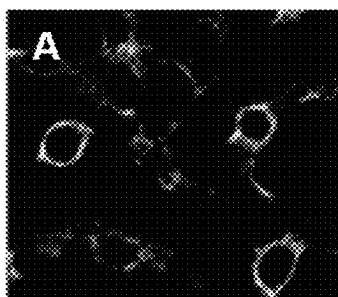


Fig. 4A

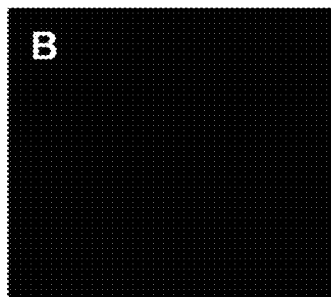


Fig. 4B

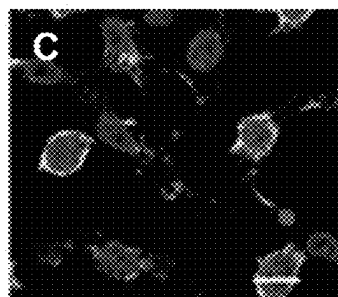


Fig. 4C

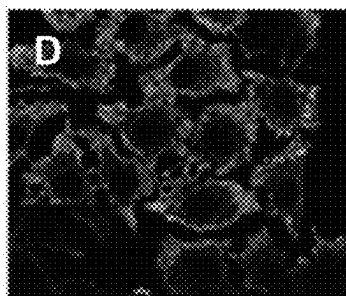


Fig. 4D



Fig. 4E

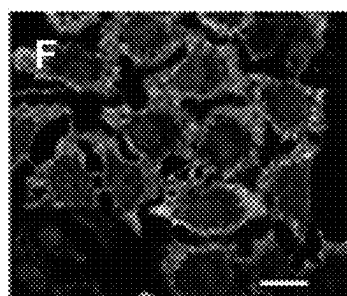


Fig. 4F

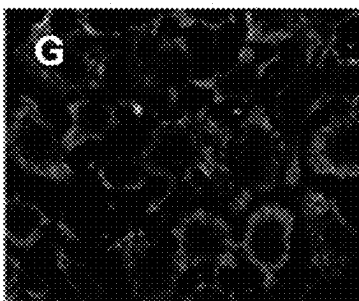


Fig. 4G



Fig. 4H

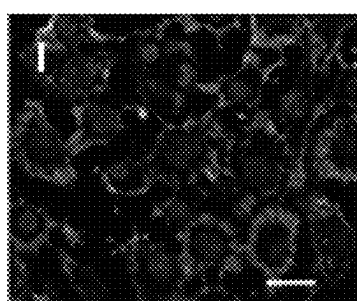
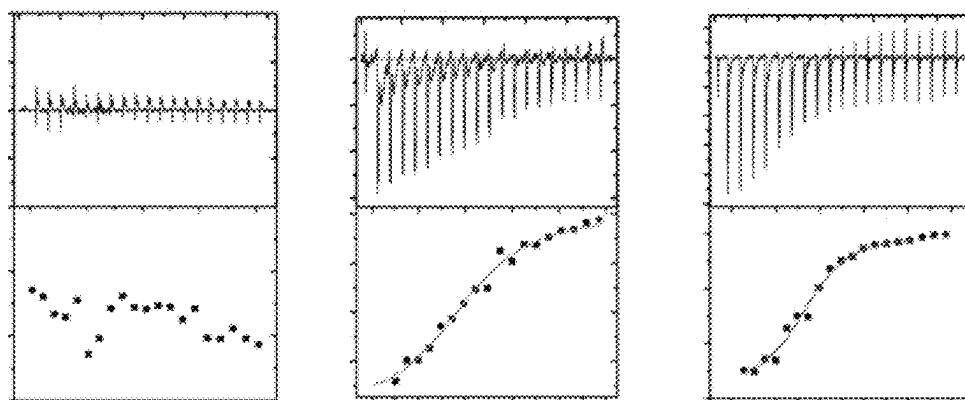


Fig. 4I



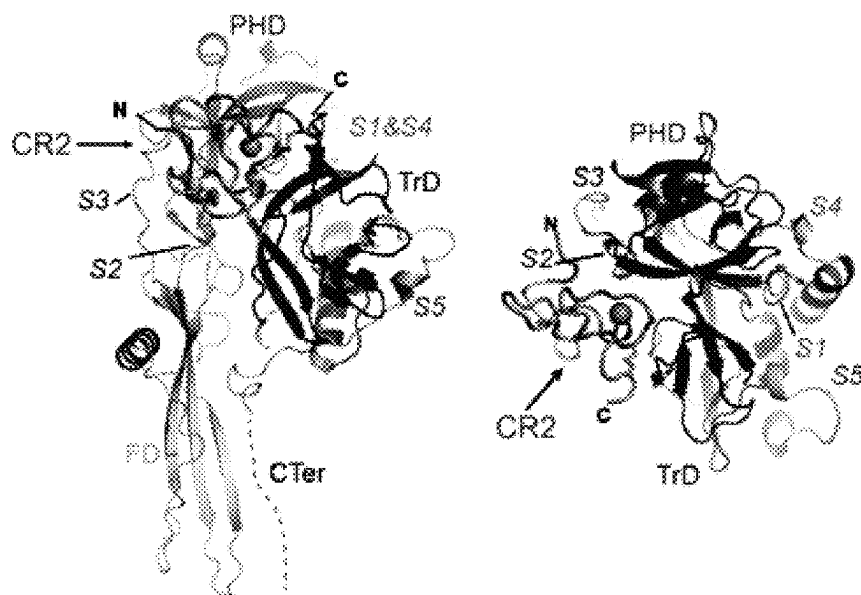


Fig. 7

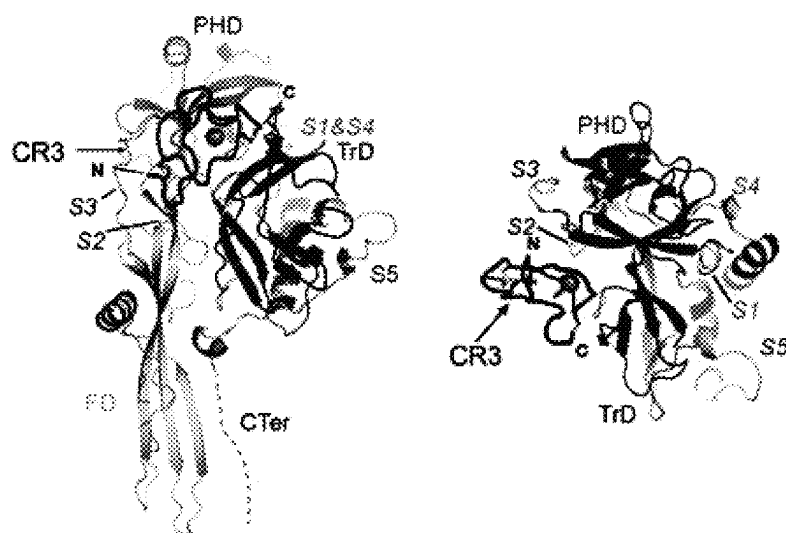


Fig. 8

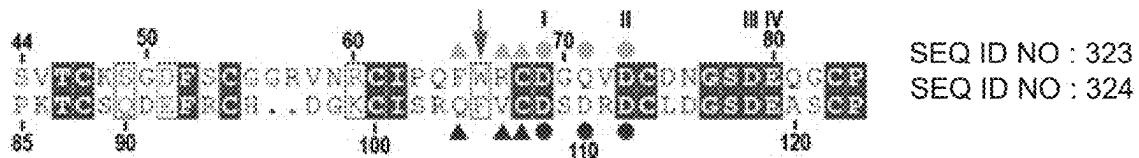


Fig. 9

A

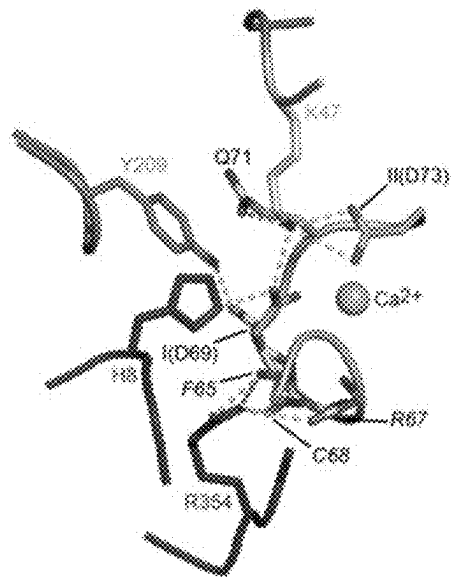


Fig. 10A

B

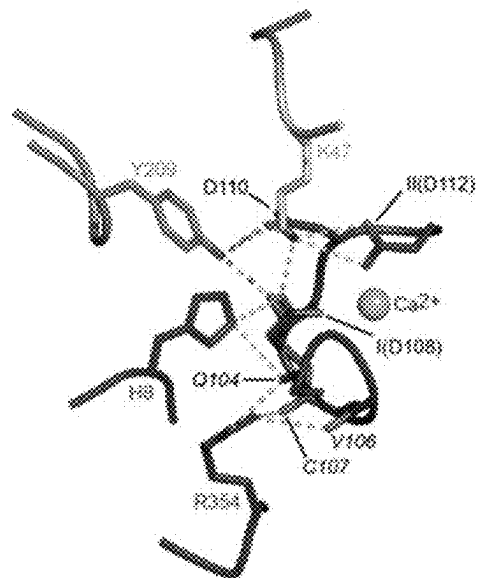


Fig. 10B

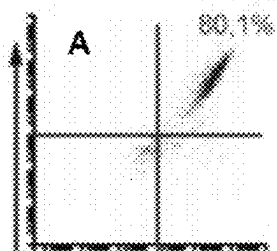


Fig. 11A

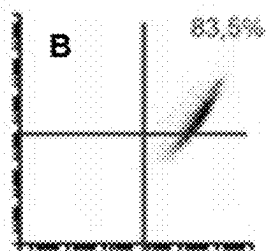


Fig. 11B

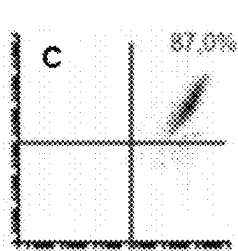


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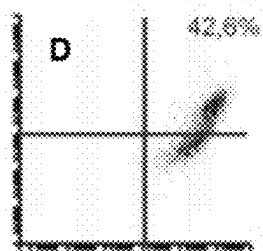


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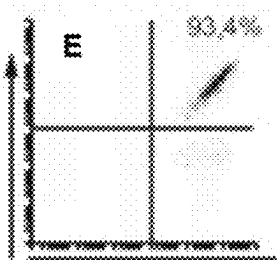


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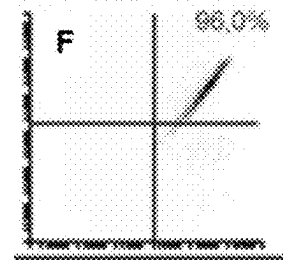


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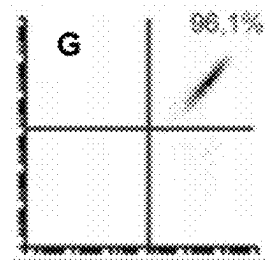


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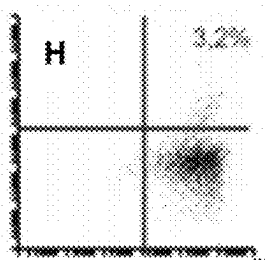


Fig. 11H

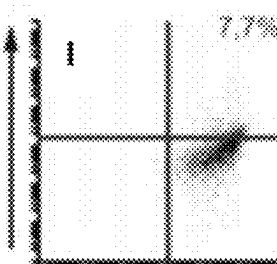


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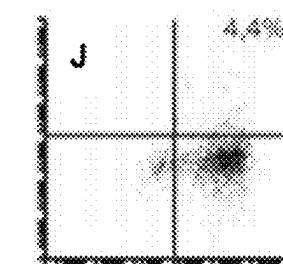


Fig. 11J

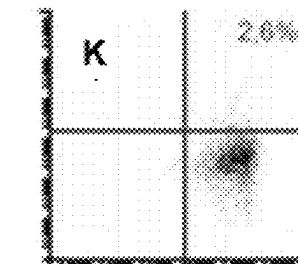


Fig. 11K

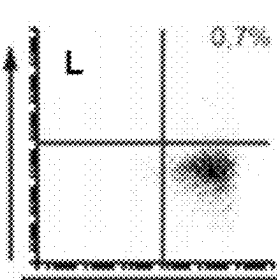


Fig. 11L

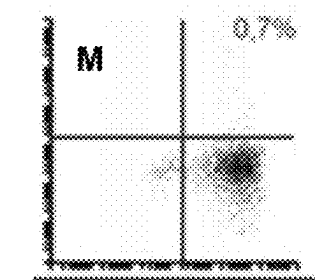


Fig. 11M

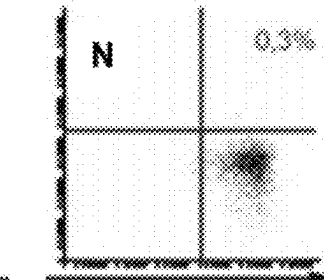


Fig. 11N

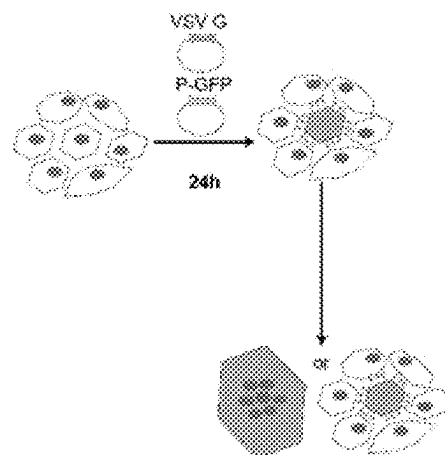


Fig. 12

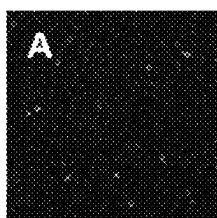


Fig. 13A

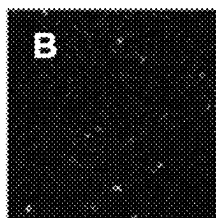


Fig. 13B

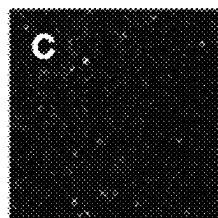


Fig. 13C

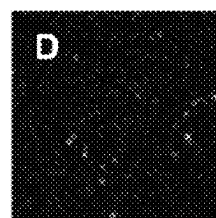


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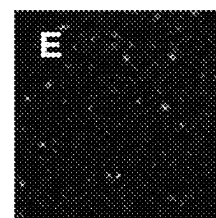


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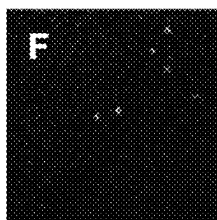


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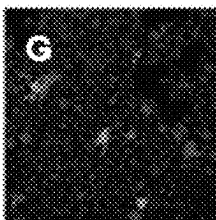


Fig. 13G

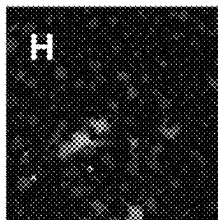


Fig. 13H

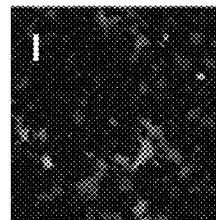


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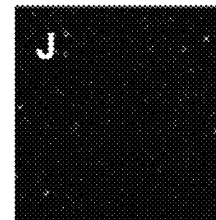


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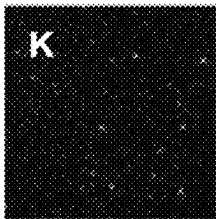


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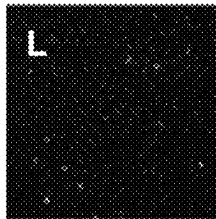


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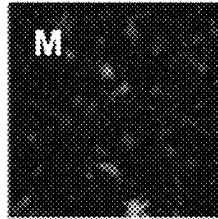


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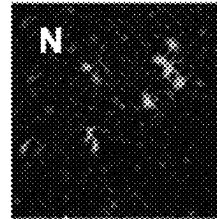


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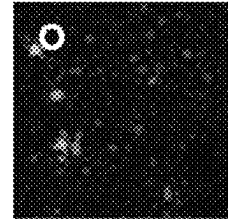


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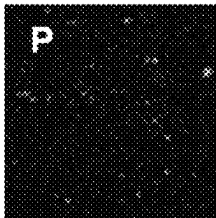


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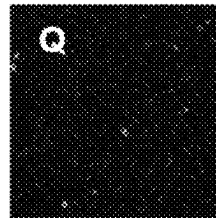


Fig. 13Q

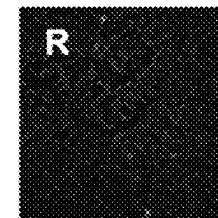


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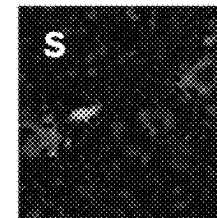


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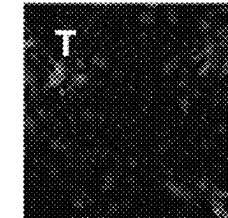


Fig. 13T

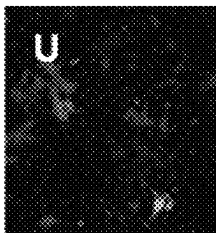


Fig. 13U

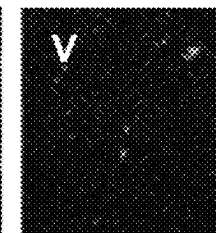


Fig. 13V

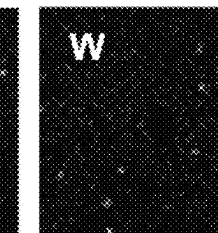


Fig. 13W

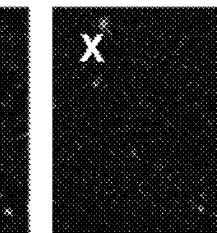


Fig. 13X

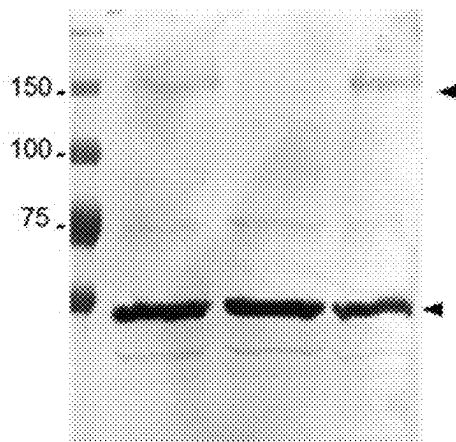


Fig. 14

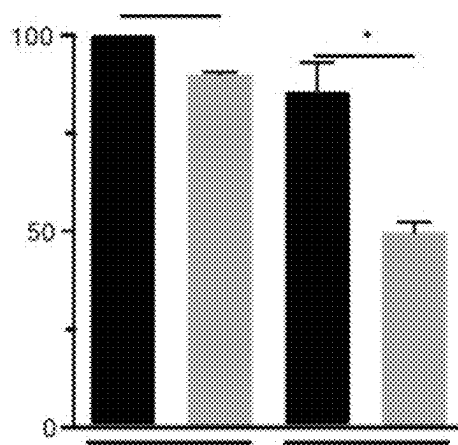


Fig. 15

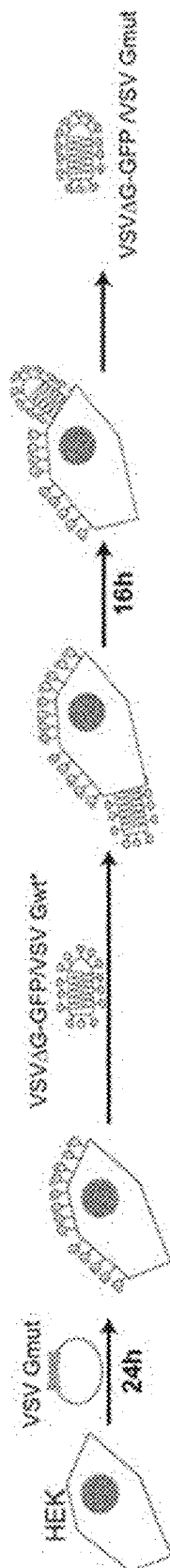


Fig. 16

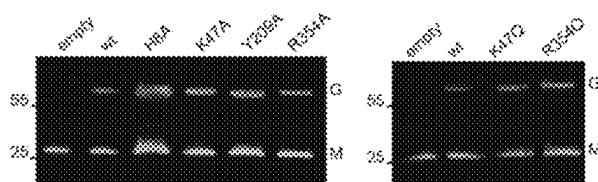


Fig. 17

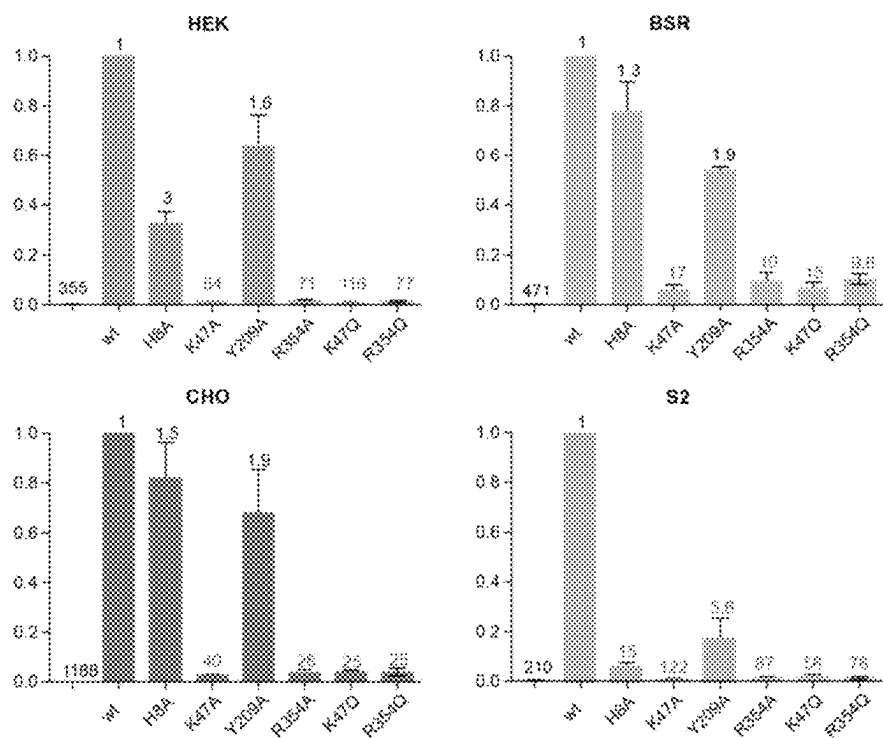


Fig. 18

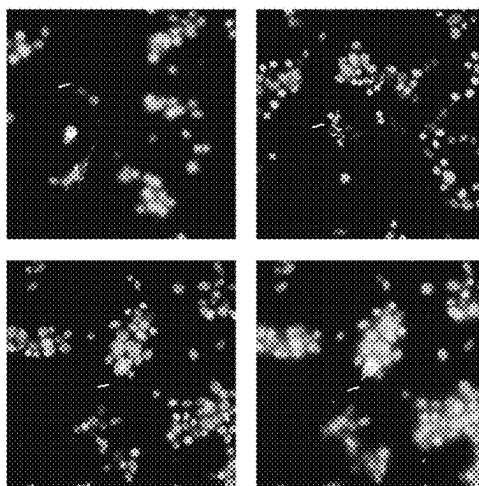


Fig. 19

1

MUTATED GLYCOPROTEIN OF VESICULAR STOMATITIS VIRUS**RELATED APPLICATIONS**

This application is a continuation application which claims priority to U.S. patent application Ser. No. 18/629,097, filed on Apr. 8, 2024 which is a continuation application which claims priority to U.S. patent application Ser. No. 18/473,720, filed on Sep. 25, 2023 which is a continuation application which claims priority to U.S. patent application Ser. No. 16/649,271, filed on Mar. 20, 2020, which is a National Stage Application under 35 U.S.C. 371 of expired PCT application number PCT/EP2018/075824 designating the United States and filed Sep. 24, 2018; which claims the benefit of EP application Ser. No. 17/306,255.5 and filed Sep. 22, 2017 each of which are hereby incorporated by reference in their entireties.

SEQUENCE LISTING

The instant application contains a Sequence Listing which has been submitted electronically in XML format and is hereby incorporated by reference in its entirety. Said XML copy, created on Sep. 25, 2023, is named "Sequence_Listing_009782.00003_ST26" and is 584,000 bytes in size.

FIELD OF THE INVENTION

The invention relates to a mutated viral protein, in particular a muted protein originating from an oncolytic virus.

BACKGROUND OF THE INVENTION

Vesicular stomatitis virus (VSV) is an enveloped, negative-strand RNA virus that belongs to the Vesiculovirus genus of the Rhabdovirus family. It is an arbovirus which can infect insects, cattle, horses and pigs. In mammals, its ability to infect and kill tumor cells while sparing normal cells makes it a promising oncolytic virus for the treatment of cancer (Barber, 2005; Fernandez et al., 2002; Hastie et al., 2013).

VSV genome encodes five structural proteins among which a single transmembrane glycoprotein (G). The glycoprotein is a classic type I membrane glycoprotein with an amino-terminal signal peptide, an ectodomain of about 450 amino acids, a single alpha helical transmembrane segment and a small intraviral carboxy-terminal domain. The signal peptide is cleaved in the lumen of the endoplasmic reticulum and the native glycoprotein consists in the ectodomain, the transmembrane domain and the intraviral domain.

G plays a critical role during the initial steps of virus infection (Albertini et al., 2012b). First, it is responsible for virus attachment to specific receptors. After binding, virions enter the cell by a clathrin-mediated endocytic pathway. In the acidic environment of the endocytic vesicle, G triggers the fusion between the viral and endosomal membranes, which releases the genome in the cytosol for the subsequent steps of infection. Fusion is catalyzed by a low-pH-induced large structural transition from a pre-toward a post-fusion conformation which are both trimeric (Roche et al., 2006; Roche et al., 2007).

The polypeptide chain of G ectodomain folds into three distinct domains which are the fusion domain (FD), the pleckstrin homology domain (PHD), and the trimerization domain (TrD). During the structural transition, the FD, the

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PHD and the TrD retain their tertiary structure. Nevertheless, they undergo large rearrangements in their relative orientation due to secondary changes in hinge segments (S1 to S5) which refold during the low-pH induced conformational change (Roche et al., 2006; Roche et al., 2007).

Recently, it has been shown that low-density lipoprotein receptor (LDL-R) and other members of this receptor family serve as VSV receptors (Finkelshtein et al., 2013).

The LDL-R is a type I transmembrane protein which regulates cholesterol homeostasis in mammalian cells (Brown and Goldstein, 1986). LDL-R removes cholesterol carrying lipoproteins from plasma circulation. Ligands bound extracellularly by LDL-R at neutral pH are internalized and then released in the acidic environment of the endosomes leading to their subsequent lysosomal degradation. The receptor then recycles back to the cell surface. LDL-R ectodomain is composed of a ligand-binding domain, an epidermal growth factor (EGF) precursor homology domain and a C-terminal domain enriched in O-linked oligosaccharides. The ligand binding domain is made of 7 cysteine-rich repeats (CR1 to CR7, FIG. 1). Each repeat is made of approximately 40 amino acids and contains 6 cysteine residues, engaged in 3 disulfide bridges, and an acidic residues cluster that coordinates a Ca^{2+} ion. The intracellular release of the cargo is driven by a low-pH-induced conformational change of LDL-R from an open to a closed conformation.

The LDL-R gene family consists of trans-membrane receptors that reside on the cell-surface, are involved in endocytic uptake of lipoproteins, and require Ca^{2+} for ligand binding. All these receptors have in common several CR repeats (up to several tens), EGF precursor-like repeats, a membrane-spanning region and an intracellular domain containing at least one internalization signal sequence. They are found ubiquitously in all animals including insects.

VSV-G has been widely used for pseudotyping other viruses and VSV-G-pseudotyped lentiviruses (VSV-G-LVs) exhibit the same broad tropism as VSV.

On the other hand, VSV-G-LVs do not allow efficient gene transfer into unstimulated T cells, B cells, and hematopoietic stem cells, as they have a very low expression level of LDL-R (Amirache et al., 2014).

The broad tropism of VSV and VSV-G LVs, due to the ubiquitous distribution of the LDL-R receptor family members, is a limitation of their therapeutic use. This is particularly the case in oncotherapy when one wants to target specifically tumor cells.

OBJECT OF THE INVENTION

One aim of the invention is to obviate this drawback.

One aim of the invention is to provide a new mutated VSV-G protein deficient in one of its properties in order to specifically target this protein.

Another aim of the invention is to provide a new VSV expressing such a protein and its use in oncotherapy.

SUMMARY OF THE INVENTION

The invention relates to an isolated non-naturally occurring protein comprising, consisting essentially of or consisting of the amino acid sequence as set forth in SEQ ID NO: 1, which corresponds to (which is) the amino acid sequence of the ectodomain of VSV Indiana strain,

wherein at least one of the amino acids at positions 8, 47, 209 and 354, said numbering being made from the position of the first amino acid in the sequence SEQ ID

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NO:1, is substituted by an amino acid different from the amino acid indicated at that position in said sequence SEQ ID NO: 1,

or any homologous protein derived from said protein as set forth in SEQ ID NO:1 by substitution, addition or deletion of at least one amino acid, provided that the derived protein retains at least 70% of identity with the amino acid sequence as set forth in SEQ ID NO: 1, and said derived protein retaining the ability to induce membrane fusion and retaining the ability to interact with LDL membrane receptor,

wherein at least one amino acid, of said homologous protein located at a position equivalent to the positions 8, 47, 209 and 354 of said sequence SEQ ID NO: 1, is substituted by an amino acid distinct from the amino acid indicated at that position in the sequence SEQ ID NO:1,

said isolated non-naturally occurring protein retaining the ability to induce membrane fusion and being unable to interact with LDL membrane receptor.

In a preferred embodiment, the amino acid at position 8 in SEQ ID NO: 1 of the isolated non-naturally occurring protein of invention (said numbering being made from the position of the first amino acid in the sequence SEQ ID NO: 1), or at the position equivalent in the homologous protein derived from said protein as set forth in SEQ ID NO:1 of the invention, cannot be a Y residue.

DETAILED DESCRIPTION

In a preferred embodiment, the amino acid at position 209 in SEQ ID NO: 1 of the isolated non-naturally occurring protein of invention (said numbering being made from the position of the first amino acid in the sequence SEQ ID NO: 1), or at the position equivalent in the homologous protein derived from said protein as set forth in SEQ ID NO:1 of the invention, cannot be a H residue.

Thus, in other words, the invention relates to an isolated non-naturally occurring protein comprising the amino acid sequence as set forth in SEQ ID NO: 1, which corresponds to (which is) the amino acid sequence of the ectodomain of VSV Indiana strain,

wherein at least one of the amino acids at positions 8, 47, 209 and 354, said numbering being made from the position of the first amino acid in the sequence SEQ ID NO: 1, is substituted by an amino acid different from the amino acid indicated at that position in said sequence SEQ ID NO: 1,

wherein the substitution at position 8 is by any amino acid different from the amino acid indicated at that position in said sequence SEQ ID NO: 1, except Y, and

wherein the substitution at position 209 is by any amino acid different from the amino acid indicated at that position in said sequence SEQ ID NO: 1, except H,

or any homologous protein derived from said protein as set forth in SEQ ID NO: 1 by substitution, addition or deletion of at least one amino acid, provided that the derived protein retains at least 70% of identity with the amino acid sequence as set forth in SEQ ID NO: 1, and said derived protein retaining the ability to induce membrane fusion and retaining the ability to interact with LDL membrane receptor,

wherein at least one amino acid, of said homologous protein located at a position equivalent to the positions 8, 47, 209 and 354 of said sequence SEQ ID NO: 1, is

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substituted by an amino acid distinct from the amino acid indicated at that position in the sequence SEQ ID No 1,

wherein the substitution of the amino acid located at a position equivalent to the position 8 is by any amino acid distinct from the amino acid indicated at that position in the sequence SEQ ID No 1, except Y, and wherein the substitution of the amino acid located at a position equivalent to the position 209 is by any amino acid distinct from the amino acid indicated at that position in the sequence SEQ ID No 1, except H,

said isolated non-naturally occurring protein retaining the ability to induce membrane fusion and being unable to interact with LDL membrane receptor.

Advantageously, the invention relates to an isolated non-naturally occurring protein comprising, consisting essentially of or consisting of the amino acid sequence as set forth in SEQ ID NO: 1, which corresponds to (which is) the amino acid sequence of the ectodomain of VSV Indiana strain,

wherein the amino acid at position 8, or at position 47, or at position 209, or at position 354, or at both positions 8 and 47, or at both positions 8 and 209, or at both positions 8 and 354, or at both positions 47 and 209, or at both positions 47 and 354, or at both positions 209 and 354, or at the positions 8 and 47 and 209, or at the positions 8 and 47 and 354, or at the positions 8 and 209 and 354, or at the positions 47 and 209 and 354, or at the position 8 and 47 and 209 and 354, said numbering being made from the position of the first amino acid in the sequence SEQ ID NO:1, are substituted by any amino acid different from the amino acid found in SEQ ID NO: 1,

or any homologous protein derived from said protein as set forth in SEQ ID NO:1 by substitution, addition or deletion of at least one amino acid, provided that the derived protein retains at least 70% of identity with the amino acid sequence as set forth in SEQ ID NO: 1, and said derived protein retaining the ability to induce membrane fusion and retaining the ability to interact with LDL membrane receptor,

wherein the amino acid, of said homologous protein, located at a position equivalent to position 8, or to position 47, or to position 209, or to position 354, or to both positions 8 and 47, or to both positions 8 and 209, or to both positions 8 and 354, or to both positions 47 and 209, or to both positions 47 and 354, or to both positions 209 and 354, or to the positions 8 and 47 and 209, or to the positions 8 and 47 and 354, or at the positions 8 and 209 and 354, or to the positions 47 and 209 and 354, or to the position 8 and 47 and 209 and 354, are substituted by any amino acid different from the amino acid found in SEQ ID NO: 1,

in particular the amino acid at position 8 is substituted by any amino acid except H, and preferably except Y, the amino acid at position 47 is substituted by any amino acid except K,

the amino acid at position 209 is substituted by any amino acid except Y and preferably except H,

the amino acid at position 354 is substituted by any amino acid except R,

the numbering being made from the position of the first amino acid in the sequence SEQ ID NO: 1,

said isolated non-naturally occurring protein retaining the ability to induce membrane fusion and being unable to interact with LDL membrane receptor.

Advantageously, the invention relates to an isolated non-naturally occurring protein comprising or consisting essentially of or consisting of the amino acid sequence as set forth in SEQ ID NO: 1,

RFTTVEFHNQKSNMKNVSNYHYCBSSDLMNDLIGTAIQVKMPKSHKAIQADGMCHASK
 WVTTCDEFWYGFKNITDSRSFTFSVEQCKESIEQTKQSTWLNPGFPFQSCGYATITDAEAVIMQVTFHH
 VLVDEYTCGMDSQFINCKSNYICPTMHNSTTHSDYKVKGLDSNLISMDITFFSEDGELSSLSKEGTGF
 RSNMFAVETGGKAKMAYQVCKHMCVRIPSGWFEEMADKDLFAAARFPECEPESSISAPSQTSVIVSITIQ
 VERHLDYSLCCPTWSKIRAGLPISVDLSYLAPEKNPGIGPAFTIINGITKYPETRIIRVDIAAPILSRMVMIS
 GTTTEHLELDMAFYEDVRLGPNGLDSSSMPLVMICGMLDLDLHLSKAQVFEHHIQDPAASQL
 PDDLSLFFGDTGLSKNPIELVEGWFSNRSIASFFFTISLIIGLFLVLRVGIHLCKIKLHHTKKRQIYTDIEMNRLG

K

or any protein derived from said protein as set forth in SEQ ID NO: 1 by substitution, addition or deletion of at least one amino acid, provided that said protein derived protein derived from said protein as set forth in SEQ ID NO: 1 retains the boxed amino acids as shown above,
 wherein the amino acid at position 8, or at position 47, or at position 209, or at position 354, or at both positions 8 and 47, or at both positions 8 and 209, or at both positions 8 and 354, or at both positions 47 and 209, or at both positions 47 and 354, or at both positions 209 and 354, or at the positions 8 and 47 and 209, or at the positions 8 and 47 and 354, or at the positions 8 and 209 and 354, or at the positions 47 and 209 and 354, or at the position 8 and 47 and 209 and 354, or the corresponding positions in the protein derived from said protein as set forth in SEQ ID NO: 1, are substituted by any amino acid different from the amino acid found in SEQ ID NO: 1,
 in particular the amino acid at position 8 is substituted by any amino acid except H, and preferably except Y, the amino acid at position 47 is substituted by any amino acid except K,
 the amino acid at position 209 is substituted by any amino acid except Y and preferably except H,
 the amino acid at position 354 is substituted by any amino acid except R,
 the numbering being made from the position of the first amino acid in the sequence SEQ ID NO: 1,
 said isolated non-naturally occurring protein retaining the ability to induce membrane fusion and is unable to interact with LDL membrane receptor.
 Advantageously, the invention relates to an isolated non-naturally occurring protein comprising, consisting essentially of or consisting of the amino acid sequence as set forth in SEQ ID NO: 1, which corresponds to (which is) the amino acid sequence of the ectodomain of VSV Indiana strain,

wherein the amino acid at position 47 or at position 354, or at both positions 47 and 354, said numbering being made from the position of the first amino acid in the sequence SEQ ID NO:1, are substituted by any amino acid, in particular by any amino acid except K or R,
 or any homologous protein derived from said protein as set forth in SEQ ID NO:1 by substitution, addition or deletion of at least one amino acid, provided that the derived protein retains at least 70% of identity with the amino acid sequence as set forth in SEQ ID NO: 1, and said derived protein retaining the ability to induce membrane fusion and retaining the ability to interact with LDL membrane receptor,
 wherein the amino acid, of said homologous protein, located at a position equivalent to position 47 or to position 354, or to both positions 47 and 354, are substituted by any amino acid, in particular any amino acid except K or R,
 the numbering being made from the position of the first amino acid in the sequence SEQ ID NO: 1,
 said isolated non-naturally occurring protein retaining the ability to induce membrane fusion and being unable to interact with LDL membrane receptor.

In one embodiment, said isolated non-naturally occurring protein further comprises a substitution of the amino acid at position 8, or at position 209, or at both positions 8 and 209, by any amino acid, said numbering being made from the position of the first amino acid in the sequence SEQ ID NO:1,

preferably wherein the amino acid at position 8 is substituted by any amino acid except H or Y, and preferably wherein the amino acid at position 209 is substituted by any amino acid except H or Y.

Advantageously, the invention relates to an isolated non-naturally occurring protein as defined above comprising, consisting essentially of or consisting of the amino acid sequence as set forth in SEQ ID NO: 1,

RFTTVEFHNQKSNMKNVSNYHYCBSSDLMNDLIGTAIQVKMPKSHKAIQADGMCHASK
 WVTTCDEFWYGFKNITDSRSFTFSVEQCKESIEQTKQSTWLNPGFPFQSCGYATITDAEAVIMQVTFHH
 VLVDEYTCGMDSQFINCKSNYICPTMHNSTTHSDYKVKGLDSNLISMDITFFSEDGELSSLSKEGTGF

-continued

RSN**E**AVETGGK**A**CH**Q**Y**Q**KH**M**VR**Q**PSG**V**W**E**MA**D**K**L**FAA**R**F**E**Q**P**EGSS**T**SA**S**QTS**V**Q**V**S**L**Q**Q**

VER**I**LD**Y**S**L**CC**P**W**S**K**I**RAG**L**P**I**S**E**V**L**S**L**AP**K**NP**C**T**G**P**E**FT**I**NG**T**L**K**Y**E**TR**R**I**R**VD**I**AA**P**ILSR**M**V**E**M**I**S

GT**T**TE**R**EL**D**DR**A**Y**E**D**V**EL**G**P**N**G**L**Q**T**SS**M**FP**L**Y**M**I**C**H**G**ML**D**S**D**LHLSS**K**AQ**V**F**E**HH**I**Q**D**AS**L**

PDD**E**SL**F**F**G**DT**C**L**S**KN**P**H**L**VE**G**W**F**S**S**Q**H**SIAS**F**FF**I**Q**L**I**I**GL**F**LV**L**Q**V**GIHL**C**IK**L**K**H**T**K**R**Q**I**Y**T**I**EM**N**RL**G**

K

or any protein derived from said protein as set forth in SEQ ID NO: 1 by substitution, addition or deletion of at least one amino acid, provided that said protein derived protein derived from said protein as set forth in SEQ ID NO: 1 retains the boxed amino acids as shown above,

wherein the amino acid at position 47 or at position 354, or at both positions 47 and 354, or the corresponding positions in the protein derived from said protein as set forth in SEQ ID NO: 1, are substituted by any amino acid, in particular by any amino acid except K or R, the numbering being made from the position of the first amino acid in the sequence SEQ ID NO: 1, said isolated non-naturally occurring protein retaining the ability to induce membrane fusion and is unable to interact with LDL membrane receptor.

In one embodiment, said isolated non-naturally occurring protein further comprises a substitution of the amino acid at position 8, or at position 209, or at both positions 8 and 209, by any amino acid distinct from the amino acid indicated at that position in the sequence SEQ ID No 1 or at the position equivalent in said homologous protein, said numbering being made from the position of the first amino acid in the sequence SEQ ID NO:1,

preferably wherein the amino acid at position 8 is substituted by any amino acid except H or Y, and

preferably wherein the amino acid at position 209 is substituted by any amino acid except H or Y.

The invention is based on the unexpected observation made by the inventors that a substitution of at least one amino acid residues at positions 8, 47, 209 or 354, or the combination of two or three or the four amino acids, affects the ability of VSV G protein to interact with its receptor (LDL membrane receptor) but retain its property to induce membrane fusion, in particular at low pH.

The invention encompasses proteins containing the amino acid sequence SEQ ID NO: 1, which corresponds to the native form of the Indiana strain of VSV, and which lacks the signal peptide. The invention also encompasses any G protein from VSV strains provided that said protein retains the amino acids that are represented with a box in SEQ ID NO: 1.

The G proteins form VSV strains may differ by addition, substitution or insertion of at least one amino acid which are not the amino acid represented with a bow in SEQ ID NO: 1.

Regarding the numbering of the amino acid, this numbering is in the invention conventionally based on the numbering of the amino acids of the native form of the G protein of VSV G Indiana as set forth in SEQ ID NO: 1. The skilled person knows the sequence alignment algorithms and programs (ClustalW for instance) and could easily compare the sequences of different G proteins and recalculate the exact position for a determined G protein compared to the

numbering obtain in SEQ ID NO: 1. For sake of clarity, the amino acid at positions 8, 47, 209 and 354 are indicated in bold in the above SEQ ID NO: 1.

The invention encompasses proteins containing the amino acid sequence SEQ ID NO: 1. The invention also encompasses any homologous G protein from VSV strains provided that said protein retains at least 70% of identity with the amino acid sequence SEQ ID NO: 1.

By "at least 70% of identity", it is meant in the invention 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99% and 100% of identity with the sequence SEQ ID NO: 1.

Regarding the percentage of identity, it is defined by the percentage of amino acid residues of SEQ ID NO: 1 which align with the same amino acid in the sequence of the homologous protein. The sequence alignment is performed using dedicated algorithms and programs (such as ClustalW, for instance).

Therefore, the protein according to the invention may derive from the following amino acid sequences:

SEQ ID NO: 2, the full length VSV G protein from Indiana strain, by substitution of the amino acids at position 63, or at position 370, or both by any amino acid except K or R,

SEQ ID NO: 3, the ectodomain of the VSV G protein from Marraba strain, by substitution of the amino acids at position 47, or at position 354, or both by any amino acid except K or R,

SEQ ID NO: 4, the full length VSV G protein from Marraba strain, by substitution of the amino acids at position 63, or at position 370, or both by any amino acid except K or R,

SEQ ID NO: 5, the ectodomain of the VSV G protein from New Jersey strain, by substitution of the amino acids at position 47, or at position 358, or both by any amino acid except K or R,

SEQ ID NO: 6, the full length VSV G protein from New Jersey strain, by substitution of the amino acids at position 63, or at position 374, or both by any amino acid except K or R,

SEQ ID NO: 7, the ectodomain of the VSV G protein from Carajas strain, by substitution of the amino acids at position 47, or at position 358, or both by any amino acid except K or R,

SEQ ID NO: 8, the full length VSV G protein from Carajas strain, by substitution of the amino acids at position 63, or at position 374, or both by any amino acid except K or R,

SEQ ID NO: 9, the ectodomain of the VSV G protein from Alagoa strain, by substitution of the amino acids at position 47, or at position 354, or both by any amino acid except K or R,

SEQ ID NO: 10, the full length VSV G protein from Alagoa strain, by substitution of the amino acids at position 64, or at position 371, or both by any amino acid except K or R,

SEQ ID NO: 11, the ectodomain of the VSV G protein from Cocal strain, by substitution of the amino acids at position 47, or at position 354, or both by any amino acid except K or R, and

SEQ ID NO: 12, the full length VSV G protein from Cocal strain, by substitution of the amino acids at position 64, or at position 371, or both by any amino acid except K or R.

SEQ ID NO: 13, the ectodomain of the VSV G protein from Morreton strain, by substitution of the amino acids at position 47, or at position 354, or both by any amino acid except K or R, and

SEQ ID NO: 14, the full length VSV G protein from Morreton strain, by substitution of the amino acids at position 64, or at position 371, or both by any amino acid except K or R.

Via the crystallographic characterization of the G protein, the inventors showed that residues K47 and R354 are highly critical for the interaction with the LDL derived receptor. When one or both residues are substituted by another amino acid residue have physical and chemical different properties, the resulting G protein loses its capacity to interact with cellular receptor. By contrast, the same resulting protein, in appropriate condition of pH retains its fusogenic property.

In the invention, the protein is isolated, which means that the protein has been isolated from its natural context. The protein is non-naturally occurring, which means that the only way to obtain this protein is to carry out a substitution, in a laboratory, by using technological methods man-made, well known in the art.

More advantageously, the invention relates to the isolated non-naturally occurring protein previously disclosed, wherein said protein comprises, or consists essentially of or consists of one of the following amino acid sequence:

SEQ ID NO: 15-20;

SEQ ID NO: 21-26;

SEQ ID NO: 27-32;

SEQ ID NO: 33-38;

SEQ ID NO: 39-44;

SEQ ID NO: 45-50; and

SEQ ID NO: 51-56,

wherein the amino acid at position 47 or at position 354, or at both positions 47 and 354, or the corresponding positions in the protein derived from said protein as set forth in SEQ ID NO: 1, are any amino acid except K or R. In other words, in the invention the amino acids Xaa corresponds to any amino acid except R or K.

SEQ ID NO: 15 corresponds to the ectodomain of the VSV G protein from Indiana strain having a substitution at position 47, by any amino acid except K or R.

SEQ ID NO: 16 corresponds to the ectodomain of the VSV G protein from Indiana strain having a substitution at position 354, by any amino acid except K or R.

SEQ ID NO: 17 corresponds to the ectodomain of the VSV G protein from Indiana strain having a substitution at positions 47 and 354, by any amino acid except K or R.

SEQ ID NO: 18 corresponds to the full length VSV G protein from Indiana strain having a substitution at position 63, by any amino acid except K or R.

SEQ ID NO: 19 corresponds to the full length VSV G protein from Indiana strain having a substitution at position 370, by any amino acid except K or R.

SEQ ID NO: 20 corresponds to the full length VSV G protein from Indiana strain having a substitution at positions 63 and 370, by any amino acid except K or R.

SEQ ID NO: 21 corresponds to the ectodomain of the VSV G protein from Marraba strain having a substitution at position 47, by any amino acid except K or R.

SEQ ID NO: 22 corresponds to the ectodomain of the VSV G protein from Marraba strain having a substitution at position 354, by any amino acid except K or R.

SEQ ID NO: 23 corresponds to the ectodomain of the VSV G protein from Marraba strain having a substitution at positions 47 and 354, by any amino acid except K or R.

SEQ ID NO: 24 corresponds to the full length VSV G protein from Marraba strain having a substitution at position 63, by any amino acid except K or R.

SEQ ID NO: 25 corresponds to the full length VSV G protein from Marraba strain having a substitution at position 370, by any amino acid except K or R.

SEQ ID NO: 26 corresponds to the full length VSV G protein from Marraba strain having a substitution at positions 63 and 370, by any amino acid except K or R.

SEQ ID NO: 27 corresponds to the ectodomain of the VSV G protein from New Jersey strain having a substitution at position 47, by any amino acid except K or R.

SEQ ID NO: 28 corresponds to the ectodomain of the VSV G protein from New Jersey strain having a substitution at position 358, by any amino acid except K or R.

SEQ ID NO: 29 corresponds to the ectodomain of the VSV G protein from New Jersey strain having a substitution at positions 47 and 358, by any amino acid except K or R.

SEQ ID NO: 30 corresponds to the full length VSV G protein from New Jersey strain having a substitution at position 63, by any amino acid except K or R.

SEQ ID NO: 31 corresponds to the full length VSV G protein from New Jersey strain having a substitution at position 374, by any amino acid except K or R.

SEQ ID NO: 32 corresponds to the full length VSV G protein from New Jersey strain having a substitution at positions 63 and 374, by any amino acid except K or R.

SEQ ID NO: 33 corresponds to the ectodomain of the VSV G protein from Carajas strain having a substitution at position 47, by any amino acid except K or R.

SEQ ID NO: 34 corresponds to the ectodomain of the VSV G protein from Carajas strain having a substitution at position 358, by any amino acid except K or R.

SEQ ID NO: 35 corresponds to the ectodomain of the VSV G protein from Carajas strain having a substitution at positions 47 and 358, by any amino acid except K or R.

SEQ ID NO: 36 corresponds to the full length VSV G protein from Carajas strain having a substitution at position 68, by any amino acid except K or R.

SEQ ID NO: 37 corresponds to the full length VSV G protein from Carajas strain having a substitution at position 379, by any amino acid except K or R.

SEQ ID NO: 38 corresponds to the full length VSV G protein from Carajas strain having a substitution at positions 68 and 379, by any amino acid except K or R.

SEQ ID NO: 39 corresponds to the ectodomain of the VSV G protein from Alagoa strain having a substitution at position 47, by any amino acid except K or R.

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SEQ ID NO: 40 corresponds to the ectodomain of the VSV G protein from Alagoa strain having a substitution at position 354, by any amino acid except K or R.

SEQ ID NO: 41 corresponds to the ectodomain of the VSV G protein from Alagoa strain having a substitution at positions 47 and 354, by any amino acid except K or R.

SEQ ID NO: 42 corresponds to the full length VSV G protein from Alagoa strain having a substitution at position 64, by any amino acid except K or R.

SEQ ID NO: 43 corresponds to the full length VSV G protein from Alagoa strain having a substitution at position 371, by any amino acid except K or R.

SEQ ID NO: 44 corresponds to the full length VSV G protein from Alagoa strain having a substitution at positions 64 and 371, by any amino acid except K or R.

SEQ ID NO: 45 corresponds to the ectodomain of the VSV G protein from Cocal strain having a substitution at position 47, by any amino acid except K or R.

SEQ ID NO: 46 corresponds to the ectodomain of the VSV G protein from Cocal strain having a substitution at position 354, by any amino acid except K or R.

SEQ ID NO: 47 corresponds to the ectodomain of the VSV G protein from Cocal strain having a substitution at positions 47 and 354, by any amino acid except K or R.

SEQ ID NO: 48 corresponds to the full length VSV G protein from Cocal strain having a substitution at position 64, by any amino acid except K or R.

SEQ ID NO: 49 corresponds to the full length VSV G protein from Cocal strain having a substitution at position 371, by any amino acid except K or R.

SEQ ID NO: 50 corresponds to the full length VSV G protein from Cocal strain having a substitution at positions 64 and 371, by any amino acid except K or R.

SEQ ID NO: 51 corresponds to the ectodomain of the VSV G protein from Morreton strain having a substitution at position 47, by any amino acid except K or R.

SEQ ID NO: 52 corresponds to the ectodomain of the VSV G protein from Morreton strain having a substitution at position 354, by any amino acid except K or R.

SEQ ID NO: 53 corresponds to the ectodomain of the VSV G protein from Morreton strain having a substitution at positions 47 and 354, by any amino acid except K or R.

SEQ ID NO: 54 corresponds to the full length VSV G protein from Morreton strain having a substitution at position 64, by any amino acid except K or R.

SEQ ID NO: 55 corresponds to the full length VSV G protein from Morreton strain having a substitution at position 371, by any amino acid except K or R.

SEQ ID NO: 56 corresponds to the full length VSV G protein from Morreton strain having a substitution at positions 64 and 371, by any amino acid except K or R.

In other word, the invention relates advantageously to an isolated non-naturally occurring protein comprising or consisting of one the following sequences SEQ ID NO: 15-56, wherein Xaa corresponds to any amino acid except R or K.

Advantageously, the invention relates to the isolated non-naturally occurring protein as defined above, wherein the amino acid at position 47 or at position 354, or at both positions 47 and 354 are substituted by A, G, F or Q, preferably A or Q.

In other word, the invention relates advantageously to an isolated non-naturally occurring protein comprising or con-

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sisting of one the following sequences: SEQ ID NO: 15-56, wherein Xaa corresponds to any amino acid except R or K.

Advantageously, the invention relates to an isolated protein comprising, consisting essentially of, or consisting of one of the following sequences SEQ ID NO: 155-322. In other words, the invention relates advantageously to an isolated protein as defined above comprising, consisting essentially of, or consisting of one of the following sequences SEQ ID NO: 155, SEQ ID NO: 156, SEQ ID NO: 157, SEQ ID NO: 158, SEQ ID NO: 159, SEQ ID NO: 160, SEQ ID NO: 161, SEQ ID NO: 162, SEQ ID NO: 163, SEQ ID NO: 164, SEQ ID NO: 165, SEQ ID NO: 166, SEQ ID NO: 167, SEQ ID NO: 168, SEQ ID NO: 169, SEQ ID NO: 170, SEQ ID NO: 171, SEQ ID NO: 172, SEQ ID NO: 173, SEQ ID NO: 174, SEQ ID NO: 175, SEQ ID NO: 176, SEQ ID NO: 177, SEQ ID NO: 178, SEQ ID NO: 179, SEQ ID NO: 180, SEQ ID NO: 181, SEQ ID NO: 182, SEQ ID NO: 183, SEQ ID NO: 184, SEQ ID NO: 185, SEQ ID NO: 186, SEQ ID NO: 187, SEQ ID NO: 188, SEQ ID NO: 189, SEQ ID NO: 190, SEQ ID NO: 191, SEQ ID NO: 192, SEQ ID NO: 193, SEQ ID NO: 194, SEQ ID NO: 195, SEQ ID NO: 196, SEQ ID NO: 197, SEQ ID NO: 198, SEQ ID NO: 199, SEQ ID NO: 200, SEQ ID NO: 201, SEQ ID NO: 202, SEQ ID NO: 203, SEQ ID NO: 204, SEQ ID NO: 205, SEQ ID NO: 206, SEQ ID NO: 207, SEQ ID NO: 208, SEQ ID NO: 209, SEQ ID NO: 210, SEQ ID NO: 211, SEQ ID NO: 212, SEQ ID NO: 213, SEQ ID NO: 214, SEQ ID NO: 215, SEQ ID NO: 216, SEQ ID NO: 217, SEQ ID NO: 218, SEQ ID NO: 219, SEQ ID NO: 220, SEQ ID NO: 221, SEQ ID NO: 222, SEQ ID NO: 223, SEQ ID NO: 224, SEQ ID NO: 225, SEQ ID NO: 226, SEQ ID NO: 227, SEQ ID NO: 228, SEQ ID NO: 229, SEQ ID NO: 230, SEQ ID NO: 231, SEQ ID NO: 232, SEQ ID NO: 233, SEQ ID NO: 234, SEQ ID NO: 235, SEQ ID NO: 236, SEQ ID NO: 237, SEQ ID NO: 238, SEQ ID NO: 239, SEQ ID NO: 240, SEQ ID NO: 241, SEQ ID NO: 242, SEQ ID NO: 243, SEQ ID NO: 244, SEQ ID NO: 245, SEQ ID NO: 246, SEQ ID NO: 247, SEQ ID NO: 248, SEQ ID NO: 249, SEQ ID NO: 250, SEQ ID NO: 251, SEQ ID NO: 252, SEQ ID NO: 253, SEQ ID NO: 254, SEQ ID NO: 255, SEQ ID NO: 256, SEQ ID NO: 257, SEQ ID NO: 258, SEQ ID NO: 259, SEQ ID NO: 260, SEQ ID NO: 261, SEQ ID NO: 262, SEQ ID NO: 263, SEQ ID NO: 264, SEQ ID NO: 265, SEQ ID NO: 266, SEQ ID NO: 267, SEQ ID NO: 268, SEQ ID NO: 269, SEQ ID NO: 270, SEQ ID NO: 271, SEQ ID NO: 272, SEQ ID NO: 273, SEQ ID NO: 274, SEQ ID NO: 275, SEQ ID NO: 276, SEQ ID NO: 277, SEQ ID NO: 278, SEQ ID NO: 279, SEQ ID NO: 280, SEQ ID NO: 281, SEQ ID NO: 282, SEQ ID NO: 283, SEQ ID NO: 284, SEQ ID NO: 285, SEQ ID NO: 286, SEQ ID NO: 287, SEQ ID NO: 288, SEQ ID NO: 289, SEQ ID NO: 290, SEQ ID NO: 291, SEQ ID NO: 292, SEQ ID NO: 293, SEQ ID NO: 294, SEQ ID NO: 295, SEQ ID NO: 296, SEQ ID NO: 297, SEQ ID NO: 298, SEQ ID NO: 299, SEQ ID NO: 300, SEQ ID NO: 301, SEQ ID NO: 302, SEQ ID NO: 303, SEQ ID NO: 304, SEQ ID NO: 305, SEQ ID NO: 306, SEQ ID NO: 307, SEQ ID NO: 308, SEQ ID NO: 309, SEQ ID NO: 310, SEQ ID NO: 311, SEQ ID NO: 312, SEQ ID NO: 313, SEQ ID NO: 314, SEQ ID NO:

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315, SEQ ID NO: 316, SEQ ID NO: 317, SEQ ID NO: 318, SEQ ID NO: 319, SEQ ID NO: 320, SEQ ID NO: 321 and SEQ ID NO: 322.

More advantageously, the invention relates to the isolated non-naturally occurring protein as defined above, wherein the amino acid at position 8 of SEQ ID NO: 1 or the corresponding positions in the protein derived from said protein as set forth in SEQ ID NO: 1, is substituted by any amino acid except H or Q or Y.

More advantageously, the invention relates to the isolated non-naturally occurring protein as defined above, wherein the amino acid at position 209 of SEQ ID NO: 1 or the corresponding positions in the protein derived from said protein as set forth in SEQ ID NO: 1, is substituted by any amino acid except Y or H.

More advantageously, the invention relates to the isolated non-naturally occurring protein as defined above, wherein said protein comprises or consists essentially of one of the following amino acid sequence: SEQ ID NO 57-154.

SEQ ID NO 57-154 represent proteins wherein the amino acid at position 47 or at position 354, or at both positions 47 and 354, or the corresponding positions in the protein derived from said protein as set forth in SEQ ID NO: 1, are substituted by any amino acid except K or R, and wherein the amino acid at position 8 of SEQ ID NO: 1 or the corresponding positions in the protein derived from said protein as set forth in SEQ ID NO: 1, is substituted by any amino acid except H or Q or preferably except Y

the amino acid at position 209 of SEQ ID NO: 1 or the corresponding positions in the protein derived from said protein as set forth in SEQ ID NO: 1, is substituted by any amino acid except Y or H, or

the amino acid at position 8 of SEQ ID NO: 1 or the corresponding positions in the protein derived from said protein as set forth in SEQ ID NO: 1, is substituted by any amino acid except H or Q or preferably except Y and the amino acid at position 209 of SEQ ID NO: 1 or the corresponding positions in the protein derived from said protein as set forth in SEQ ID NO: 1, is substituted by any amino acid except Y or H.

More advantageously, the invention relates to the isolated non-naturally occurring protein as defined above, further comprising an insertion of a peptide between the amino acids at positions

192 to 202, or
240 to 257, or
347 to 353, or
364 to 366, or

376 to 379, said peptide originating from a protein different from the protein as set forth in SEQ ID NO: 1.

The inventors identified that an insertion of a peptide, said peptide originating from a protein different from the G protein of VSV, within the above mentioned region does not alter the fusion property of the protein according to the invention.

In the invention, an insertion "between the amino acids positions 192 to 202, 240 to 257, 347 to 353, 364 to 366, or 376 to 379" means that the peptide is inserted

either between two consecutive amino acids, for instance between the amino acids at positions 192 and 193, 193 and 194, 194 and 195, 195 and 196, 196 and 197, 197 and 198, 198 and 199, 199 and 200, 200 and 201, 201 and 202, 240 and 241, 241 and 242, 242 and 243, 243 and 244, 244 and 245, 245 and 246, 246 and 247, 247 and 248, 248 and 249, 249 and 250, 250 and 251, 251 and 252, 252 and 253, 253 and 254, 254 and 255, 255

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and 256, 256 and 257, 347 and 348, 348 and 349, 349 and 350, 350 and 351, 351 and 352, 352 and 353, 364 and 365, 365 and 366, 376 and 377, 377 and 378, and 378 and 379,

or between two non-consecutive numbered amino acids due to a deletion of one or more amino acid; for instance a peptide can be inserted between the amino acids at positions 192 and 194, because the amino acid at position 193 was deleted or replaced by the inserted peptide (etc.).

With the above explanations, the skilled person will be able to determine the positions of the insertion encompassed by the invention.

More advantageously, the invention relates to the isolated non-naturally occurring protein as defined above, further comprising an insertion of a peptide between the amino acids at position 351 and 352, said peptide originating from a protein different from the protein as set forth in SEQ ID NO: 1. Advantageously, the invention relates to the isolated non-naturally occurring protein as defined above, further comprising an insertion of a peptide in position 1 (in other words, at the N-terminal extremity, i.e. upstream of the amino acid at position 1), said peptide originating from a protein different from the protein as set forth in SEQ ID NO: 1. In the invention, an insertion of a peptide in position 1 means an insertion of said peptide at the N-terminal extremity of the non-naturally occurring protein as defined above. Thus, in this embodiment, the first amino acid residue of the sequence of said non-naturally occurring protein as defined above is preserved (maintained). In other words, said first amino acid residue at position 1 of the sequence of said non-naturally occurring protein as defined above is not deleted. In an alternative embodiment, the invention relates to the isolated non-naturally occurring protein as defined above, further comprising an insertion of a peptide in replacement of the amino acid residue at position 1, said peptide originating from a protein different from the protein as set forth in SEQ ID NO: 1. Thus, in this alternative embodiment, the first amino acid residue of the sequence of said non-naturally occurring protein as defined above is deleted and replaced (i.e. substituted) with the sequence of said peptide.

Advantageously, the inventors identified that an insertion in position 1 (in other words, at the N-terminal extremity) or between the amino acids 351 and 352 of SEQ ID NO: 1, or the corresponding position in SEQ ID NO: 2-14, and having a mutation at the position 47, or 354 or both, or the corresponding position in SEQ ID NO: 2-14 does not modify the fusion properties of the mutated protein.

Therefore, the inventors propose to insert in the VSV G protein enable to interact with its receptor, between these two amino acids, a tag peptide, a luminescent, a nanobody or any peptide that recognize specifically a membrane protein.

In other words, and with a specific advantage, the inventors therefore propose to provide a mutated VSV G protein, in which is inserted, between the two above mentioned amino acids, any peptide that would allow a specific targeting of cells of interest.

More advantageously, the invention relates to the isolated non-naturally occurring protein as defined above, wherein said peptide is at least a part of a ligand of a cellular receptor.

With the mutated protein according to the invention, in which it is inserted a peptide or a nanobody, it become possible to produce a VSV that specifically target a cell of

interest, in particular a tumoral cell, and therefore specifically kill this determined cell by using the oncolytic properties of the virus.

Indeed, in this case, the G protein according to the invention would not interact with its natural receptor (LDL-R) but will recognize a receptor which is a target to the peptide inserted between the amino acids at position 351 and 352 of SEQ ID NO: 1, or the corresponding positions in SEQ ID NO: 2-14. As the protein according to the invention retains its fusogenic properties, the protein would allow the virus entry, and the virus could therefore kill the targeted cell.

For instance, the inserted peptide could be an anti-HER2 nanobody, an anti-MUC18 nanobody or an anti-PD-1 nanobody.

The invention also relates to a nucleic acid molecule coding for an isolated non-naturally occurring protein as defined above.

In other words, the invention relates to a nucleic acid molecule coding for any protein as set forth in SEQ ID NO: 15-322, as defined above.

In another aspect, the invention relates to a recombinant virus expressing an isolated non-naturally occurring protein as defined above. An advantageous virus is a VSV expressing all the viral protein in their wild type form except the G protein which corresponds to the mutated protein according to the invention.

Advantageously, the invention relates to a recombinant virus comprising a nucleic acid molecule as defined above.

The invention also relates to a eukaryotic cell containing or expressing a non-naturally occurring protein as defined above, or containing a nucleic acid molecule as defined above. Advantageously, the invention relates to a eukaryotic cell infected by a virus as defined above.

The invention also relates to a composition comprising one at least of the followings:

- a protein according as defined above; or
- a nucleic acid molecule as defined above;
- a virus as defined above; or,
- a eukaryotic cell as defined above.

In particular, the invention relates to a composition comprising a virus coding for a G protein comprising or consisting of one of the following sequences SEQ ID NO: 15-322.

The invention also relates to a composition comprising one at least of the followings:

- a protein according as defined above; or
 - a nucleic acid molecule as defined above;
 - a virus as defined above; or,
 - a eukaryotic cell as defined above,
- for its use as drug.

In particular, the invention relates to a composition comprising a virus coding for a G protein comprising or consisting of one of the following sequences SEQ ID NO: 15-322, for its use as drug.

Advantageously, the invention relates to the composition as defined above, for its use for treating cancer.

As mentioned above, a virus expressing a mutated protein according to the invention can be used to specifically target a determined cell and therefore to specifically kill them.

The invention relates to the in vitro use of a protein as defined above, anchored in surface, advantageously, a lipid membrane for targeting said surface, advantageously a lipid membrane, to a specific target, for instance a cell, in particular a cell to be killed, such as a cancer cell.

In the invention, it is proposed to use in vitro a protein according to the invention which is anchored to a membrane

to specifically address such membrane to a target destination. For instance, the mutated protein according to the invention can be anchored in the membrane of a liposome, of a vesicle, of an exosome, on a capsule, on a nanoparticle . . . such that the protein according to the invention can specifically target this liposome, vesicle, capsule or nanoparticle to a specific target, for instance a cell, in particular a cell to be killed, such as a cancer cell.

This use is particularly advantageous in a drug-delivery targeting purpose, wherein the protein according to the invention allow a specific target of a drug.

The drug can be a therapeutic molecule, a protein, and a nucleic acid.

The invention relates to a protein as defined above, anchored in a surface, advantageously, a lipid membrane, for use for targeting said surface, advantageously a lipid membrane, to a specific target, for instance a cell, in particular a cell to be killed, such as a cancer cell.

In the invention, it is proposed a protein according to the invention which is anchored to a membrane for use to specifically address such membrane to a target destination. For instance, the mutated protein according to the invention can be anchored in the membrane of a liposome, of a vesicle, of an exosome, on a capsule, on a nanoparticle . . . such that the protein according to the invention can specifically target this liposome, vesicle, capsule or nanoparticle to a specific target, for instance a cell, in particular a cell to be killed, such as a cancer cell.

This use is particularly advantageous in a drug-delivery targeting purpose, wherein the protein according to the invention allows to specifically target a drug to a specific target, for instance a cell, in particular a cell to be killed, such as a cancer cell.

Thus, the invention relates to a mutated protein according to the invention for use as a drug delivery-system, wherein said mutated protein is anchored in the membrane of a liposome, or of a vesicle, or of an exosome, or on a capsule, or on a nanoparticle.

The drug can be a therapeutic molecule, a protein, and a nucleic acid.

BRIEF DESCRIPTION OF THE FIGURES

The invention will be better understood from the following figures and examples.

LEGEND TO THE FIGURES

FIG. 1 is a schematic representation of the modular organization of the LDL-R indicating the 7 CR modules (1 to 7), the 3 EGF repeats (a b and c), the seven-bladed β -propeller domain (β) of the epidermal growth factor precursor like domain (B.), and the C-terminal domain containing O-linked oligosaccharides (C.). SP=signal peptide; X=transmembrane domain. A.: CR domains and D.: ectodomain.

FIG. 2 represents a result of a SDS PAGE analysis of interaction experiments between the 7 GST-CR domains (1-7), bounded to GSH magnetic beads, and Gth at pH 8. C. represents control. The migration position of the proteins are indicated by an arrow: A.: Gth, B.: CRx-GST and D.: GST.

FIG. 3 illustrates the experiments presented in FIG. 6 and FIG. 7. After 4 h of infection, BSR cells were labelled with an antibody directed against VSV nucleoprotein (Anti VSV N) to visualize the infection (green fluorescence) and a GST-CRATTO550 to probe CR domain recognition by the surface displayed glycoprotein (red fluorescence).

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FIGS. 4A-4I are photographs of labelling of G at the surface of BSR cells infected with VSV using fluorescent GST-CR1ATTO550, GST-CR2ATTO550 and GST-CR3ATTO550. At 4 h post infection, cells were incubated with the appropriate GST-CRATTO550 at 4° C. during 30 minutes prior fixation and permeabilization and then immuno-labelled using an anti-VSV N antibody to visualize the infection. DAPI was used to stain the nuclei. Scale bars 20 μ m.

FIG. 4A represents the labelling of cells using anti-VSV N antibody.

FIG. 4B represents the labelling of cells using the fluorescent GST-CR1ATTO550.

FIG. 4C represent the superposition of the fluorescence in FIG. 6A and FIG. 6B.

FIG. 4D represents the labelling of cells using anti-VSV N antibody.

FIG. 4E represents the labelling of cells using the fluorescent GST-CR2ATTO550.

FIG. 4F represent the superposition of the fluorescence in FIG. 6D and FIG. 6E.

FIG. 4G represents the labelling of cells using anti-VSV N antibody.

FIG. 4H represents the labelling of cells using the fluorescent GST-CR3ATTO550.

FIG. 4I represent the superposition of the fluorescence in FIG. 6G and FIG. 6H.

FIG. 5 represents representative plots of each Isothermal titration calorimetry (ITC) analyses between Gth and CR1, Gth and CR2, Gth and CR3 at 20° C. Binding parameters were determined by curve fitting analysis with a single-site binding model. The values indicated in the panel are those corresponding to the curves that are presented. Kd values given in the text are means of 3 independent experiments $\pm$ standard errors. B-C Inhibition of VSV infection by soluble forms of CR domains. Upper x-axis: time (min); upper Y-axis: μ cal/s; lower x-axis: molar ratio and lower y-axis: kcal·mol⁻¹ of injectant. Left panel: CR1, middle panel: CR2 and right panel: CR3.

FIGS. 6A-6F represents photographs of BSR cells infected with VSV-eGFP preincubated with GST-CR1, GST-CR2, GST-CR3 (A-C), CR1, CR2, or CR3 monovalent domains (D-F) at the indicated concentrations. Cells were fixed 4 h post infection. Only infected cells are expressing eGFP. Neither CR1 nor GST-CR1 construction protect cells from infection. DAPI was used to stain the nuclei. Scale bars 100 μ m.

FIG. 6A represents photograph of BRS cells infected with VSV-eGFP preincubated with GST-CR1 at the indicated concentrations.

FIG. 6B represents photograph of BRS cells infected with VSV-eGFP preincubated with GST-CR2 at the indicated concentrations.

FIG. 6C represents photograph of BRS cells infected with VSV-eGFP preincubated with GST-CR3 at the indicated concentrations.

FIG. 6D represents photograph of BRS cells infected with VSV-eGFP preincubated with CR1 monovalent domain at the indicated concentrations.

FIG. 6E represents photograph of BRS cells infected with VSV-eGFP preincubated with CR2 monovalent domain at the indicated concentrations.

FIG. 6F represents photograph of BRS cells infected with VSV-eGFP preincubated with CR3 monovalent domain at the indicated concentrations.

FIG. 7 is a tridimensional representation of GthCR2 crystalline structures in ribbon representation.

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FIG. 8 is a tridimensional representation of GthCR3 crystalline structures in ribbon representation.

In both complexes the CR domain is nested in the same cavity of G. N- and C-terminal extremities of each CR are indicated. The trimerization domain (TrD) the pleckstrin homology domain (PHD) and the fusion domain (FD) of Gth are represented.

FIG. 9 is a sequence alignment of LDL-R CR2 and CR3. Conserved residues are in a grey box and similar residues are boxed. Acidic residues involved in the binding of the Ca²⁺ ion are indicated by I, II, III, and IV. CR residues involved in polar contacts with G are labelled with grey symbols (light grey for CR2 and black for CR3; dots when the contact is established via the lateral chain and triangles when the contact is established via the main chain) on each CR sequence. The aromatic residue which protrudes from the CR modules and establishes hydrophobic interactions with G is indicated by the arrow.

FIG. 10A corresponds to a close-up view on the Gth-CR interface showing the docking of G basic residues on the acidic patch of CR2. The G residues H8, K47, Y209 and R354 are involved in the interaction. Residues labels on each CR domain are in italic letters when the contact is established via the main chain; putative bonds are shown as light grey dashed lines.

FIG. 10B corresponds to a close-up view on the Gth-CR interface showing the docking of G basic residues on the acidic patch of CR3. The G residues H8, K47, Y209 and R354 are involved in the interaction. Residues labels on each CR domain are in italic letters when the contact is established via the main chain; putative bonds are shown as light grey dashed lines.

FIGS. 11A-11N represent flow cytometry analysis of the expression of WT and mutant glycoproteins at the surface of HEK293T cells and of the binding of fluorescent GST-CR2 (A-D and I-K) and GST-CR3 (E-H and L-M). After 24 h of transfection, cell surface expression of WT and mutant G was assessed using monoclonal anti-G antibody 8G5F11 directly on living cells at 4° C. during 1 h. Cells were then incubated simultaneously with anti-mouse Alexa fluor 488 and the indicated GST-CRATTO550 dye. Cells transfected with a G construct that was still able to bind GST-CR domains exhibited red fluorescence due the ATTO550 dye. In each plot, the percentage of ATTO550 positive cells is indicated.

FIG. 11A represents the experiment of binding of fluorescent GST-CR2 with WT glycoprotein.

FIG. 11B represents the experiment of binding of fluorescent GST-CR2 with the glycoprotein H8A mutant.

FIG. 11C represents the experiment of binding of fluorescent GST-CR2 with the glycoprotein Y209A mutant.

FIG. 11D represents the experiment of binding of fluorescent GST-CR2 with the glycoprotein K47A mutant.

FIG. 11E represents the experiment of binding of fluorescent GST-CR3 with WT glycoprotein.

FIG. 11F represents the experiment of binding of fluorescent GST-CR3 with the glycoprotein H8A mutant.

FIG. 11G represents the experiment of binding of fluorescent GST-CR3 with the glycoprotein Y209A mutant.

FIG. 11H represents the experiment of binding of fluorescent GST-CR3 with the glycoprotein K47A mutant.

FIG. 11I represents the experiment of binding of fluorescent GST-CR2 with the glycoprotein K47Q mutant.

FIG. 11J represents the experiment of binding of fluorescent GST-CR2 with the glycoprotein R354A mutant.

FIG. 11K represents the experiment of binding of fluorescent GST-CR2 with the glycoprotein R354Q mutant.

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FIG. 11L represents the experiment of binding of fluorescent GST-CR3 with the glycoprotein K47Q mutant.

FIG. 11M represents the experiment of binding of fluorescent GST-CR3 with the glycoprotein R354A mutant.

FIG. 11N represents the experiment of binding of fluorescent GST-CR3 with the glycoprotein R354Q mutant.

FIG. 12 is a schematic representation of the cell-cell fusion assay. BSR cells are co-transfected with plasmids expressing VSV G (either WT or mutant G) and P-GFP (a cytoplasmic marker). 24 h post-transfection cells are exposed for 10 min to media adjusted to the indicated pH which is then replaced by DMEM at pH 7.4. The cells are then kept at 37° C. for 1 h before fixation. Upon fusion, the P-GFP diffuses in the syncytia.

FIGS. 13A-13X are photographs corresponding to the results of the experiment described in FIG. 12.

FIG. 13A represents the result of experiments with empty vector at pH 5.0.

FIG. 13B represents the result of experiments with empty vector at pH 5.5.

FIG. 13C represents the result of experiments with empty vector at pH 6.0.

FIG. 13D represents the result of experiments with empty vector at pH 6.5.

FIG. 13E represents the result of experiments with empty vector at pH 7.0.

FIG. 13F represents the result of experiments with empty vector at pH 7.5.

FIG. 13G represents the result of experiments with vector expressing K47A mutant at pH 5.0.

FIG. 13H represents the result of experiments with vector expressing GK47A mutant at pH 5.5.

FIG. 13I represents the result of experiments with vector expressing K47A mutant at pH 6.0.

FIG. 13J represents the result of experiments with vector expressing K47A mutant at pH 6.5.

FIG. 13K represents the result of experiments with vector expressing K47A mutant at pH 7.0.

FIG. 13L represents the result of experiments with vector expressing K47A mutant at pH 7.5.

FIG. 13M represents the result of experiments with vector expressing R354A mutant at pH 5.0.

FIG. 13N represents the result of experiments with vector expressing R354A mutant at pH 5.5.

FIG. 13O represents the result of experiments with vector expressing R354A mutant at pH 6.0.

FIG. 13P represents the result of experiments with vector expressing R354A mutant at pH 6.5.

FIG. 13Q represents the result of experiments with vector expressing R354A mutant at pH 7.0.

FIG. 13R represents the result of experiments with vector expressing R354A mutant at pH 7.5.

FIG. 13S represents the result of experiments with vector expressing WT G protein at pH 5.0.

FIG. 13T represents the result of experiments with vector expressing WT G protein at pH 5.5.

FIG. 13U represents the result of experiments with vector expressing WT G protein at pH 6.0.

FIG. 13V represents the result of experiments with vector expressing WT G protein at pH 6.5.

FIG. 13W represents the result of experiments with vector expressing WT G protein at pH 7.0.

FIG. 13X represents the result of experiments with vector expressing WT G protein at pH 7.5.

FIG. 14 represent an analysis of LDL-R expression in wild-type HAP-1 cells (A), LDL-RKO HAP-1 (B) cells and HEK293T (C). The immunoblot was performed on crude

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cell extracts and revealed with anti LDL-R (EP1553Y-1.). Tubulin (tub) was also immunoblotted as a loading control (2.).

FIG. 15 is an histogram showing the effect of the RAP protein on the susceptibility of LDL-R deficient HAP-1 cells to VSV-eGFP infection. VSV-eGFP was used to infect HAP-1 (A) and HAP-1 LDL-RKO (B) cells in presence of RAP (grey column) or not (black column). Infectivity was determined by counting the number of cells expressing eGFP using a flow cytometer. Data depict the mean with standard error for experiments performed in triplicate. p values were determined using an unpaired Student t test (* p <0.01; *** non-significant).

FIG. 16 is a schematic representation of the generation of VSVAG-GFP virus pseudotyped with VSV G mutants. Transfected HEK-293T cells expressing mutant G at their surface were infected with VSVAG-GFP pseudotyped with VSV G wild type. After 16 h of infection, VSVAG-GFP virions pseudotyped with mutant VSV G were harvested from the supernatant.

FIG. 17 represents the incorporation of wild type and mutant G in VSVAG-GFP viral particles. VSVAG-GFP pseudotyped with the wild type VSV G was used to infect HEK-293T cells transfected with the indicated mutant (MOI 1). At 16 h post infection, viral supernatants were collected, concentrated and analyzed by Western blot (using an anti-VSV G and an anti-VSV M antibody).

FIG. 18 represents histograms showing the infectivity of VSVAG-GFP pseudotyped with WT and mutant glycoproteins. VSVAG-GFP pseudotyped with WT VSV G was used to infect HEK-293T cells previously transfected with the indicated mutated glycoprotein (MOI 1). VSVAG-GFP viruses pseudotyped by WT or mutant glycoproteins were used to infect HEK-293T, BSR, CHO and S2 cells during 6 h; the percentage of infected cells was determined by counting GFP expressing cells by flow cytometry. Data depict the mean with standard error for three independent experiments. Above each bar, the reduction factor of the titer (compared to VSVAG-GFP, pseudotyped by WT G which was normalized to 1) is indicated.

FIG. 19 are photographs of HEK293T cells transfected with a pCAGGS plasmid encoding for VSV glycoprotein modified by the insertion of the mCherry protein in N-terminal extremity (position 1 of the mature protein) and by the insertion of the mCherry protein in between AA 351 and 352. Red fluorescence is present at the cell surface in both case indicating that the protein was correctly refolded and transported through the Golgi apparatus. This suggests that those two positions on G are potentially interesting to insert any peptide.

EXAMPLES

Example 1: Structural Basis of Low-Density Lipoprotein Receptor Recognition by VSV Glycoprotein

The inventors identified that VSV G is able to independently bind two distinct CR (cysteine-rich) domains (CR2 and CR3) of LDL-R and they report crystal structures of VSV G in complex with those domains. The structures reveal that the binding sites of CR2 and CR3 on G are identical. We show that HAP-1 cells in which the LDL-R gene has been knocked out are still susceptible to VSV infection confirming that VSV G can use receptors other than LDL-R for entry. However, mutations of basic residues, which are key for interaction with LDL-R CR domains,

abolish VSV infectivity in mammalian as well as insect cells. This indicates that the only receptors of VSV in mammalian and in insect cells are members of the LDL-R family and that VSV G has specifically evolved to interact with their CR domains.

LDL-R CR2 and CR3 Domains Bind VSV G and Neutralize Viral Infectivity

The inventors have expressed individually each LDL-R CR domain in fusion with the glutathione S-transferase (GST) in *E. coli*. Each fusion protein was incubated at pH 8 with magnetic beads coated with glutathione before addition of a soluble form of the ectodomain of G (VSV Gth, amino acid (AA) residues 1-422, generated by thermolysine limited proteolysis of viral particles (FIG. 2). After 20 minutes of incubation at 4° C., the beads were washed and the associated proteins were analyzed by SDS/PAGE followed by Coomassie blue staining. This revealed that only CR2 and CR3 domains are able to directly bind VSV G (FIG. 2) at pH 8. Additionally, GST-CR2 and GST-CR3 (but not GST-CR1) fluorescently labeled with ATTO550 (FIG. 3 and FIG. 4) specifically recognized VSV G expressed at the surface of infected cells. The inventors also used isothermal titration calorimetry (ITC) to investigate the binding parameters of CR1, CR2 and CR3 to Gth in solution (FIG. 5). Here again, no interaction between G and CR1 was detected. On the other hand, for both CR2 and CR3, the binding reactions appear to be exothermic, show a 1:1 stoichiometry and exhibit similar Kds (4.3+/-1 µM for CR3 and 7.3+/-1.5 µM for CR2).

Furthermore, recombinant soluble CR2 and CR3 domains, either alone or in fusion with GST, are also able to neutralize viral infectivity when incubated with the viral inoculum prior infection (FIG. 6).

Crystal Structures of VSV G Ectodomain in Complex with LDL-R CR Domains

The inventors crystallized Gth in complex with either CR2 or CR3. The binding site of CR domains on G is the same in both crystal forms (FIG. 7 and FIG. 8).

Two basic residues of G (H8 from the TrD and K47 from PHD) are pointing toward two acidic residues which belong to the octahedral calcium cage of the CR domains (D69 and D73 on CR2; D108 and D112 on CR3 labelled I and II—FIG. 9). Together with Y209 and R354, they seem to be key for the interaction (FIGS. 10A and B).

K47 and R354 are Key Residues of G Required for LDL-R CR Domains Binding

To investigate their contribution to LDL-R CR domains binding, the inventors mutated residues H8, K47, Y209 and R354 of G into an alanine or a glutamine. HEK293T cells were transfected with a plasmid encoding wild-type or mutant VSV G glycoproteins (WT, H8A, K47A, K47Q, Y209A, R354A and R354Q). Twenty-four hours post-transfection, the cells were incubated with a MAb against G ectodomain. Then, green fluorescent anti IgG secondary antibodies and GST-CR fusion proteins fluorescently labelled with ATTO550 were simultaneously added. Immunofluorescence labelling indicated that WT and all G mutants are efficiently transported to the cell surface (FIG. 11). Mutants H8A and Y209A bind GST-CR domains as WT G whereas the other mutants are affected in their binding ability (FIG. 11). Mutants K47Q, R354A and R354Q bind neither GST-CR2 nor GST-CR3. Finally, although no interaction is detected between mutant K47A and CR3, a residual binding activity is observed between this mutant and CR2 (FIG. 11).

The inventors also checked the fusion properties of mutants K47A and R354A. For this, BSR cells were trans-

ected with pCAGGS plasmids encoding wild-type or mutant VSV G glycoproteins (WT, K47A and R354A). The cells expressing mutant G protein have a fusion phenotype similar to that of WT G (FIG. 13). This confirms that the mutant glycoproteins are correctly folded and demonstrates that it is possible to decouple G fusion activity and receptor recognition.

Other LDL-R Family Members are Alternative Receptors of VSV

HAP-1 cells in which the LDL-R gene has been knocked out (HAP-1 LDL-RKO) (FIG. 14) are as susceptible to VSV infection as WT HAP-1 cells (FIG. 15). This demonstrates that VSV receptors other than the LDL-R are present at the surface of HAP-1 cells.

To evaluate the role of other LDL-R family members as VSV receptors, the inventors took advantage of the properties of the receptor-associated protein (RAP), a common ligand of all LDL-R family members which blocks ligand binding to all LDL-R family members with the exception of LDL-R itself (Finkelshtein et al., 2013). RAP significantly inhibits VSV infection in HAP-1 LDL-RKO but not in WT HAP-1 cells (FIG. 15). Those results are consistent with previous data suggesting that VSV can use other LDL-R family members as alternative receptors (Finkelshtein et al., 2013).

G Mutants Affected in their CR Domain Binding Site cannot Rescue a Recombinant VSV Lacking the G Gene

The inventors then examined whether the mutant glycoproteins described above are able to sustain viral infection. The inventors used a recombinant VSV (VSVAG-GFP) in which the G envelope gene was replaced by the green fluorescent protein (GFP) gene and which was pseudotyped with the VSV G glycoprotein. This pseudotyped recombinant was used to infect HEK cells either transfected or not transfected by a plasmid encoding WT or mutant glycoproteins (Ferlin et al., 2014). After 8 h, the infected cells supernatant was collected (FIG. 16). Mutant glycoproteins incorporation into the envelope of the particles present in the supernatant was verified by western blot (FIG. 17) and the infectivity of the pseudotyped particles was analyzed in different cell lines (mammalian HEK, BSR, CHO and Drosophila S2 cells) by counting the cells expressing GFP by flow cytometry 4 h post-infection (p.i.) (FIG. 18). Mutants K47A, K47Q, R354A and R354Q did not rescue the infectivity of VSVAG-GFP. Compared to WT G, the infectivity decreased by a factor of 10 up to 120 (FIG. 18). The decrease was more important in HEK and S2 cell lines than in the two hamster cell lines. In mammalian cell lines, mutants H8A and Y209A can rescue the infectivity of VSVAG-GFP, but at a lower level than that of WT. In S2 cell line, their infectivity is significantly decreased (by a factor of 15 for mutant H8A and ~6 for Y209A) (FIG. 18).

As the fusion activity of the mutants is unaffected, the loss of infectivity of pseudotypes bearing a mutant glycoprotein can be safely attributed to their disability to recognize a cellular receptor. These results indicate that mutants K47A, K47Q, R354A and R354Q which are unable to bind LDL-R CR domains are also severely impaired in their ability to bind other VSV receptors.

Discussion

LDL-R has been demonstrated to be the major entry port of VSV and lentivirus pseudotyped by VSV-G (Finkelshtein et al., 2013). Here, the inventors demonstrate that VSV-G is able to bind two CR domains of the LDL-R with similar affinities. The biological relevance of this interaction was

demonstrated by the ability of both CR2 and CR3 to inhibit VSV infection. The crystal structures of VSV G in complex with CR2 and CR3 reveal that they both occupy the same site at the surface of the glycoprotein in its pre-fusion conformation and that the same G residues ensure the correct anchoring of the CR domains. This binding site is split apart when G is in its post-fusion conformation, which explains why G is unable to bind CR domains at low pH. This may disrupt the interaction between G and LDL-R in the endosomal lumen and favour the transport of the virion to an appropriate fusion site.

CR domain recognition by VSV G involves basic residues K47 and R354 pointing toward the calcium-coordinating acidic residues. This mode of binding is very similar to what is observed for endogenous ligand recognition by CR domains of the LDL-R family members and, indeed, mutant glycoproteins in which either K47 or R354 is replaced by an alanine or a glutamine, are unable to bind CR domains. It is worth noting that those key residues are not conserved among vesiculoviruses. Therefore, the use of LDL-R as a viral receptor cannot be generalized to the other members of the genus. Indeed, the inventors have shown that CHAV G, which does not possess basic residues in positions corresponding to VSV residues 47 and 354, does not bind CR domains.

The inventor's functional analysis confirms that LDL-R is not the only receptor of VSV as HAP-1 LDL-RKO can be infected as efficiently as HAP-1 cells. However, the mutant glycoproteins which are unable to bind CR domains cannot restore VSVAG-GFP infectivity neither in mammalian nor in insect cells. The most parsimonious interpretation of this result is that the only receptors of VSV in HEK cells are members of the LDL-R family. The molecular basis of the interaction is the same for all those receptors and involves G ability to bind their CR domains. This is in agreement with the decrease of infectivity observed in presence of RAP protein which is an antagonist of other members of the LDL-R family. Overall this study demonstrates that VSV G has specifically evolved to interact with CR domains of the members of the LDL-R family. The ubiquitous nature of this receptor family (which is also widespread among invertebrates) explains the pantropism of VSV.

The demonstration that the receptors of VSV are all members of the LDL-R family together with the characterization of the molecular basis of CR domains recognition by G paves the way to develop recombinant VSVs with modified tropism. Indeed, a glycoprotein having (i) a point mutation which ablates the natural receptor tropism and (ii) an insertion of a protein domain or a peptide targeting specifically a tumor cell (Ammayappan et al., 2013) should allow the design of fully retargeted oncolytic VSVs. Such viruses should be able to eliminate cancerous cells while sparing normal ones.

Cells and Viruses

BSR, a clone of BHK-21 (Baby Hamster Kidney cells; ATCC CCL-10) and HEK-293T (human embryonic kidney cells expressing simian virus 40 T antigen; ATCC CRL-3216) cells were grown in Dulbecco's modified Eagle's medium (DMEM) supplemented with 10% fetal calf serum (FCS). HAP-1 wt and HAP-1 LDL-R deficient cells (HAP-1 LDL-RKO) purchased from Horizon Discovery were grown in Iscove's Modified Dulbecco's Medium (IMDM) supplemented with 10% FCS. CHO (cell line derived from Chinese hamster ovaries) cells were grown in Ham's F12 medium supplemented with 2 mM glutamine and 10% FCS. All mammalian cell lines were maintained at 37° C. in a

humidified incubator with 5% CO₂. *Drosophila* S2 cells were grown in Schneider's medium supplemented with 10% FCS at 28° C.

Wild-type VSV (Mudd-Summer strain, Indiana serotype), VSVAG-GCHAV (Rose et al., 2000) and VSV-eGFP were propagated in BSR cells.

VSVAG-GFP is a recombinant VSV which was derived from a full-length cDNA clone of the VSV genome (Indiana serotype) in which the coding region of the G protein was replaced by a modified version of the GFP gene and pseudotyped with the VSV G protein (Ferlin et al., 2014). VSVAG-GFP was propagated on HEK-293T cells that had been previously transfected with pCAGGS-VSVG.

Plasmids and Cloning

Point mutations were created starting from the cloned VSV G gene (Indiana Mudd-Summer strain) in the pCAGGS plasmid. Briefly, forward and reverse primers containing the desired mutation were combined separately with one of the primers flanking the G gene to generate two PCR products. These two G gene fragments overlap in the region containing the mutation and were assembled into the pCAGGS linearized vector using Gibson assembly reaction kit (New England Biolabs).

Protein Expression, Purification and Labelling

VSV Gth was obtained by limited proteolysis of viral particles and purified as previously described (Albertini et al., 2012a).

DNA sequences encoding the 7 CR domains of the human LDL-R (NM_000527, GenBank) were synthesized (MWG biotech) and subcloned in the pGEX-6P1 bacterial expression vector (Invitrogen). Each protein construct contains at its N-terminus a GST tag and a preScission protease cleavage site. Each CR domain was purified using the following protocol derived from (Harper and Speicher, 2011). C41 bacteria transformed with the CR construct were cultured at 37° in LB-ampicillin medium until OD reached 0.6 AU. Protein expression was then induced with 1 mM IPTG during 5 h at 37° C. Cells were sonicated in lysis buffer (500 mM NaCl, 20 mM Tris-HCl pH 8, 2 mM CaCl₂, 2% w/v sarkosyl and 1 mM DTT). The clarified supernatant was incubated with glutathione agarose beads (Thermo Fisher Scientific) in presence of 0.2% Triton X100 during 2 h. After incubation, beads were then extensively washed with equilibration buffer (200 mM NaCl, 50 mM Tris HCl pH 8, 2 mM CaCl₂, 1 mM PMSF). The GST-CR construct was then eluted with the same buffer supplemented with 20 mM GSH. Purification of each GST-CR was achieved with a gel filtration step using a Superdex 200 column (Ge Healthcare). To isolate CR domains, purified GST-CR was incubated with preScission protease and injected on a gel filtration column Superdex 75 (Ge Healthcare). Fractions containing pure CR domains were then pooled, concentrated at 1 mM and stored at -80° C. until use.

One milligram of purified GST-CR2 (or GST-CR3) was labelled with the fluorescent dye ATTO550 NHS ester (Sigma Aldrich) using the instruction of the manufacturer. The labelled proteins were then diluted at a concentration of 50 µM and stored at -80° C. until use. The labelling ratio was estimated to be around 2 dyes per molecule.

Characterization of the Binding Between G and CR Domains

Purified GST-CR domains were incubated with magnetic beads coated with GSH (Eurogentec) under agitation during 20 min at 4° C. Then, the slurry was washed with the equilibration buffer at the appropriate pH (200 mM NaCl, 2 mM CaCl₂, 50 mM Tris-HCl pH 8 or 50 mM MES-NaOH pH 6). Purified Gth or viral particles were preincubated in

this same buffer for 20 min and added to the magnetic beads bound to GST-CR construction or GST alone. After 20 min of incubation under soft agitation, the slurry was washed two times with the equilibration buffer at the appropriate pH (either 8 or 6). Beads were re-suspended in the gel loading buffer and directly analyzed on a SDS PAGE. Binding of CR Domains to Cells Expressing G (Either WT or Mutants)

For microscopy, BSR cells were infected for 4 h and were then incubated with GST-CR2ATTO550 or GST-CR3ATTO550 at 4° C. for 30 min. Cells were fixed with 4% paraformaldehyde and then permeabilized with 0.5% Triton X-100. Nucleoprotein was detected by using a mouse monoclonal anti-VSV N antibody. Goat anti-mouse Alexa fluor 488 (Invitrogen) was used as a secondary antibody. Images were captured using a Leica SP8 confocal microscope (63× oil-immersion objective).

For flow cytometry experiments, HEK-293T cells were transfected with pCAGGS plasmids encoding WT or mutant G using polyethylenimine (PEI, Sigma-Aldrich). 24 h after transfection, cells were collected and incubated with a mouse-monoclonal anti-G antibody that recognizes G ectodomain (8G5F11, KeraFast). Goat anti-mouse Alexa fluor 488 and GST-CR2ATTO550 (or GST-CR3ATTO550) were then simultaneously added to the cells. The fluorescence of cells was determined using a BD Accuri C6 flow cytometer.

Pseudotypes

HEK-293T cells at 80% confluence were transfected by pCAGGS encoding WT or mutant VSV G using PEI. At 24 h after transfection, cells were infected with VSVΔG-GFP at an MOI of 1. Two hours p.i., cells were washed to remove residual viruses from the inoculum. Cell supernatants containing the pseudotyped viral particles were collected at 16 h p.i. The infectious titers of the pseudotyped viruses were determined on non-transfected cells by counting cells expressing the GFP using a BD Accuri C6 flow cytometer at 4 h p.i. WT and mutant G incorporation in the pseudotyped particles was assessed after supernatant concentration by SDS PAGE and western blot analysis using an anti-VSV G and an anti-VSV M.

HAP-1 Cells Infection

HAP-1 cells were plated at 70% confluence and incubated, or not, with 50 nM of RAP during 15 min. Cells were then infected with VSV-eGFP at an MOI of 1. RAP was maintained during all the infection time. The percentage of infected cells (GFP-positive) was determined 4 h p.i. using a BD Accuri C6 flow cytometer.

ITC

ITC experiments were performed at 293 K using a Micro-Cal iTC200 apparatus (GE Healthcare) in a buffer composed of 150 mM NaCl, 20 mM Tris-HCl pH 8.0 and 2 mM CaCl₂. Gth, at a concentration of 50 μM, was titrated by successive injections of CR domains at a concentration of 600 μM. The titration sequence included a first 1 μL injection followed by 19 injections of 2 μL each with a spacing of 180 or 240 s between injections. OriginLab software (GE Healthcare) was used to analyze the raw data. Binding parameters were extracted from curve fitting analysis with a single-site binding model.

Cell-Cell Fusion Assay

Cell-cell fusion assay was performed as previously described (Ferlin et al., 2014). Briefly, BSR cells plated on glass coverslips at 70% confluence were co-transfected with pCAGGS plasmids encoding WT G or mutant G, and P-GFP plasmid encoding the phosphoprotein of Rabies virus fused to GFP. Twenty four hours after transfection, cells were

incubated with fusion buffer (DMEM-10 mM MES) at various pHs (from 5.0 to 7.5) for 10 minutes at 37°. Cells were then washed once and incubated with DMEM-10 mM HEPES-NaOH buffered at pH 7.4, 1% BSA at 37° C. for 1 hour. Cells were fixed with 4% paraformaldehyde in 1×PBS for 15 min. Cells nuclei were stained with DAPI and syncytia formation was analyzed with Zeiss Axiovert 200 fluorescence microscope with a 10× lens.

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Example 2: Preparation of Plasmid Encoding Modified G

Construction of pCAGGS plasmids containing the desired coding G sequence with the mCherry inserted at various position were generated using Gibson assembly reaction. The empty vector pCAGGS was linearized using EcoRI restriction enzyme. Then 3 PCR products with overlapping parts were generated. The product I is the fragment of G before the insertion site; it is generated running a PCR on the VSV G gene using primers Ia and Ib₁ to insert the mCherry in position 1 or Ia and Ib₃₅₁ to insert the mCherry in position 351. The product II is the mCherry gene (using primers IIa₁ and IIb₁ to insert the mCherry in position 1 or IIa₃₅₁ and IIb₃₅₁ to insert the mCherry in position 351). The product III is the fragment of G after the insertion site; it is generated using primers IIIa₁ and IIIb to insert the mCherry in position 1 or IIIa₃₅₁ and IIIb to insert the mCherry in position 351. Primer Sequences were Synthesized by Eurofins Genomics

Ia:

(SEQ ID NO: 332)

TCTCATCATTTTGGCAAAGATGAAGTGCCTTTTGTACTTAG

27

-continued

Ib₁:
 (SEQ ID NO: 333)
 TTGCTCACCATGCAATTCACCCCAATGAATAAAAG

Ib₃₅₁:
 (SEQ ID NO: 334)
 GCTCACCATAGTTCCTACTGATCATTCCGACC

IIa₁:
 (SEQ ID NO: 335)
 CATTGGGGTGAATTGCATGGTGAAGCAAGGGC

IIa₃₅₁:
 (SEQ ID NO: 336)
 AATGATCAGTGGAATATGGTGAAGCAAGGGC

IIB₁:
 (SEQ ID NO: 337)
 AAAIIB1CTATGGTGAAGTCTTGTACAGCTCGTCC

IIB₃₅₁:
 (SEQ ID NO: 338)
 GTTCCTTTCTGTGGTCTTGTACAGCTCGTCC

IIIA₁:
 (SEQ ID NO: 339)
 GAGCTGTACAAGAAGTTCACCATAGTTTTTCCACACA

IIIA₃₅₁:
 (SEQ ID NO: 340)
 CTGTACAAGACCACAGAAAGGGAAGTGT

28

-continued

IIIB:
 (SEQ ID NO: 341)
 CCGCCCGGAGCTCGTTACTTTCCAAGTCGGTTC

5 After purification of each fragment on agarose gel, the 3 fragments plus the purified digested pCAGGS vector are then combined in equimolar concentration and assembled by Gibson assembly reaction. The DNA is then transformed into bacteria, and a correct plasmid product amplified after
 10 been identified by restriction digest and/or sequencing.

Example 3: Transient Expression of Modified VSV Glycoproteins

15 The transfection protocol will depend of the kind of cells to transduce. For HEK cells the inventors use PolyEthylenimine (PEI) transfection protocol. For BHK the inventors use Ca-Phosphate transfection protocol or PEI.

Cells grown on coverslips were transfected with pCAGGS plasmid encoding for VSV modified glycoprotein. After 20 hour of transfection the cells were fixed with 4% paraformaldehyde in PBS. After washing (3 times with PBS) coverslips were mounted with immu-mount DAPI (thermo-fisher) and examined with a Zeiss microscope. Red fluorescence is present at the cell surface in both case indicating that the protein was correctly folded throw the Golgi apparatus (FIG. 19).

The invention is not limited to the above-mentioned embodiments.

SEQUENCE LISTING

Sequence total quantity: 341

SEQ ID NO: 1 moltype = AA length = 495
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 mol_type = protein
 organism = Vesicular stomatitis virus

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 ASKWVTTCDF RWYGPKYITQ SIRSFPTPSVE QCKESIEQTK QGTWLNPGFP PQSCGYATVT 120
 DAEAVIVQVT PHHVLVDEYT GEWVDSQFIN GKCSNYICPT VHNSTTWHSD YKVKGLCDN 180
 LISMDITFFS EDGELSSLGK EGTGFRSNYF AYETGGKACK MQYCKHWGVR LPSGVWFEMA 240
 DKDLFAAARF PECPEGSSIS APSQTSVDVS LIQDVERILD YSLCQETWSK IRAGLPISPV 300
 DLSYLAPKNP GTGPAFTIIN GTLKYPETRY IRVDIAAPIL SRMVGMIISGT TTERELWDDW 360
 APYEDVEIGP NGVLRITSSGY KFPLYMIGHG MLDSDLHLSS KAQVFEHPHI QDAASQLPDD 420
 ESLPFGDTGL SKNPIELVEG WFSSWKSSIA SFFFIIGLII GLFLVLVRGI HLCIKLKHTK 480
 KRQIYTDIEM NRLGK 495

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 mol_type = protein
 organism = Vesicular stomatitis virus

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 MPKSHKAIQA DGWMCHASKW VTTCDPRWYG PKYITQSIRS FTSPVEQCKE SIEQTKQGTW 120
 LNPGFPPQSC GYATVTDAAE VIVQVTPHHV LVDEYTGWEV DSQFINGKCS NYICPTVHNS 180
 TTWHSYKVK GLCDNSNLIM DITFFSEDEGE LSSLGKEGTG FRSNYFAYET GGKACKMQYC 240
 KHVGRLPSG VWFEMADKDL FAAARFPECP EGSSISAPSQ TSVDVSLIQD VERILDYSLC 300
 QETWSKIRAG LPISPVDLSY LAPKNPGTGP AFTIINGTLK YFETRYIRVD IAAPILSRMV 360
 GMISGTTTER ELWDDWAPYE DVEIGPNVGL RTSSGYKFPL YMIGHGMLDS DLHLSKKAQV 420
 FEHPHIQDAA SQLPDDDSL FGDGTLGSKNP IELVEGWFSK WKSSIASFFF IIGLIIGLFL 480
 VLRVGIHLCL KLAHTKKRQI YTDIEMNRLG K 511

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 organism = Vesicular stomatitis virus

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 AAKWVTTCDF RWYGPKYIITH SIHMSPTLE QCKTSIEQTK QGVWINPGFP PQSCGYATVT 120
 DAEVVVVQAT PHHVLVDEYT GEWIDSQLVG GKCSKEVCQT VHNSTVWHAD YKITGLCESN 180

-continued

LASVDITFFS	EDGQKTSLGK	PNTGFRSNHF	AYESGEKACR	MQYCTQWGIR	LPSGVWFELV	240
DKDLFQAQAKL	PECPRGSSIS	APSQTSVDVS	LIQDVERILD	YSLCQETWSK	IRAKLPVSPV	300
DLSYLAPKNP	GS GPAFTIIN	GTLKYFETRY	IRVDISNPII	PHMVGTMSGT	TTERELWNDW	360
YPYEDVEIGP	NGVLKPTPTGF	KFPLYMIGHG	MLDSDLHKSS	QAQVFEHPHA	KDAASQLPDD	420
ETLFFPGDTGL	SKNPVELVEG	WFSSWKSTLA	SFFLIIGLGV	ALIFIIRIIV	AIRYKYKGRK	480
TQKIYNDVEM	SRLGNK					496

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 mol_type = protein
 organism = Vesicular stomatitis virus

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IPKSHKAIQA	DGWMCHAACK	VTTCDPRWYG	PKYITHSIHS	MSPTLEQCKT	SIEQTKQGVW	120
INPGPPPQSC	GYATVTDAEV	VVVQATPHHV	LVDEYTGWEI	DSQLVGGKCS	KEVCQTVHNS	180
TVVHADYKIT	GLCESNLASV	DITFFSEDGQ	KTSLGKPNTG	FRSNHFAYES	GEKACRMQYC	240
TQWGI RLPSG	VWFELVDKDL	FQA AKLPECP	RGSSISAPSQ	TSVDVSLIQD	VERILDYSLC	300
QETWSKIRAK	LPVSPVDLSY	LAPKNPGSGP	AFTIINGTLK	YFETRYIRVD	ISNPIIPHMV	360
GTMSGTTTER	ELWNDWYPYE	DVEIGPNGVL	KTPGTGFKPPL	YMIGHGMLDS	DLHKSSQAQV	420
FEHPHAKDAA	SQLPDDETLF	FGDTGLSKNP	VELVEGWFS	WKSTLASFFL	IIGLGV ALIF	480
IIRIIVAIRY	KYKGRKTQKI	YNDVEMSRLG	NK			512

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 mol_type = protein
 organism = Vesicular stomatitis virus

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SALWMTTCDF	RWYGPKYIITH	SIHNEEPTDY	QCLEAIKAYK	DGVSFNPGFP	PQSCGYGTVT	120
DAEAHIVTVT	PHSVKVD EYT	GEWIDPHFIG	GRCKGQICET	VHNSTKWFTS	SDGESVCSQL	180
FTLVGGTFPS	DSEIITSMGL	PETGIRSNYF	PVSTEGICK	MPFCRKPGYK	LKNDLWFQIT	240
DPDLDKTVRD	LPHIKDCDLS	SSIVTPGEHA	TDISLISDVE	RILDYALCQN	TWSKIEAGEP	300
ITPVDLSYLG	PKNPGAGPVF	TIINGSLHYF	MSKYLRVELE	SPVIPRMEGK	VAGTRIVRQL	360
WDQWFPFGEV	EIGPNGVLKT	KQGYKPLPHI	IGTGEVDNDI	KMERIVKHWE	HPHIEAAQTF	420
LKKDDTEEV L	YYGDTGVSKN	PVELVEGWFS	GWRSSIMGVL	AVIIGFVILI	FLIRLIGVLS	480
SLFRQKRRPI	YKSDVEMAHF	R				501

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 organism = Vesicular stomatitis virus

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FNP GPPPQSC	GYGTVTDAEA	HIVTVTPHSV	KVDEYTGWEI	DPHFIGGRCK	GQICETVHNS	180
TKWFTSSDGE	SVCSQLFTLV	GGTFFSDSEE	ITSMGLPETG	IRSNYFPYVS	TEGICKMPFC	240
RKPGYKLN D	LWFQITDPDL	DKTVRDLPHI	KDCDLSSSIV	TPGEHATDIS	LISDVERILD	300
YALCQNTWSK	IEAGEPITPV	DLSYLGPKNP	GAGPVFTIIN	GSLHYFMSKY	LRVELESPVI	360
PRMEGKVAGT	RIVRQLWDQW	PFPGEVEIGP	NGVLKTKQGY	KFPLHIIGTG	EVDNDIKMER	420
IVKHWEHPHI	EAAQTF LKKD	DTEEVLYYGD	TGVSKNPVEL	VEGWFSGWRS	SIMGVLAVII	480
GFVILIFLIR	LIGLSSSLFR	QKRRIYKSD	VEMAHFR			517

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AARWITTCDF	RWYGPKYIITH	SIHSFRPSND	QCKEAIRLTN	EGNWINPGFP	PQSCGYASVT	120
DSESVVTVTT	KHQVLVDEYS	GSWIDSQFPG	GSCTSPICDT	VHNSTLWHAD	HTLDSICDQE	180
FVAMDAVLFT	ESGKFEEFGK	PNSGIRSNYF	PYESLKDVCO	MDFCRKRGFK	LPSGVWFEIE	240
DAEKSHKAQV	ELKIKRCPHG	AVISAPNQNA	ADINLIMDVE	RILDYSLCQA	TWSKIQNK EA	300
LTPIDISYLG	PKNPGPGPAF	TIINGTLHYF	NTRYIRVDIA	GPVTKEITGF	VSGTSTSRVL	360
WDQWFPYGEN	SIGPNGLLKT	ASGYKPLFM	VTGVLADADI	HKLGEATVIE	HPHAK EAQKV	420
VDDSEVIFFG	DTGVSKNPVE	VVEGWFSGWR	SSLMSIFGII	LLIVCLVLIV	RILIALKYCC	480
VRHKKRTIYK	EDLEMGRIPR	RA				502

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 organism = Vesicular stomatitis virus

SEQUENCE: 8

MKMKMVIAGL	ILCIGILPAI	GKITISFPQS	LKGDWRPV PK	GYNYCPTSAD	KNLHGLDIDI	60
GLRLRAPKSF	KGISADGWM C	HAARWITTC D	FRWYGPKYIT	HSIHSFRPSN	DQCKEAIRLT	120

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NEGNWINPGF	PPQSCGYASV	TDSESVVTV	TKHQVLVDEY	SGSWIDSQFP	GGSCTSPICD	180
TVHNSTLWHA	DHTLDSICDQ	EFVAMDAVLF	TESGKFEEFG	KPNSGIRSNY	FPYESLKDVC	240
QMDPCKRKGF	KLPSGVWFEI	EDAESHKAQ	VELKIKRCPH	GAVISAPNQ	AADINLIMDV	300
ERILDYSLCQ	ATWSKIQNKE	ALTPIDISYL	GPKNPGPGPA	FTIINGTLHY	FNTRYIRVDI	360
AGPVTKEITG	FVSGTSTRV	LWDQWFFPYGE	NSIGPNGLLK	TASGYKYPLF	MVGTGVLDAD	420
IHKLGATVI	EHPHAKAQK	VVDDSEVIF	GDTGVSKNPV	EVVEGWFSGW	RSSLMSIFGI	480
ILLIVCLVLI	VRILIALKYC	CVRHKKRTIY	KEDLEMGRIP	RAA		523

SEQ ID NO: 9 moltype = AA length = 494
 FEATURE Location/Qualifiers
 source 1..494
 mol_type = protein
 organism = Vesicular stomatitis virus

SEQUENCE: 9

KFTIVFPQSQ	KGDWKDVPPN	YRYCPSSADQ	NWHGDLGVN	IRAKMPKVHK	AIKADGWMCH	60
AAKWTTCDY	RWYGPQYITH	SIHSFIPTKA	QCEESIKQTK	EGVWINPGFP	PKNCGYASVS	120
DAESIIVQAT	AHSMIDEYS	GDWLDSQFPT	GRCTGSTCET	IHNSTLWYAD	YQVTGLCDSA	180
LVSTEVTFYS	EDGLMTSISR	QNTGYRSNYF	PYKGAACR	MKYCTHEGIR	LPSGVWFEMV	240
DKELLESVQM	PECPAGLTIS	APTQTSVDVS	LILDVERMLD	YSLCQETWSK	VHSGLPISPV	300
DLGYIAPKNP	GAGPAFTIVN	GTLKYFDTRY	LRIDIEGPVL	KKMTGKVSQT	PTKRELWTEW	360
FPYDDVEIGP	NGVLKTPREG	KFPLYMIGHG	LLDSDLQKTS	QAEVFHHPQI	AEAVQKLPPD	420
ETLFFGDTGI	SKNPVEVIEG	WFSNWRSSVM	AIVFAILLLV	ITVLMVRLCV	AFRHFCCQKR	480
HKIYNDLEMN	QLRR					494

SEQ ID NO: 10 moltype = AA length = 511
 FEATURE Location/Qualifiers
 source 1..511
 mol_type = protein
 organism = Vesicular stomatitis virus

SEQUENCE: 10

MTPAFILCML	LAGSSWAKFT	IVFPQSQKGD	WKDVPPNYRY	CPSSADQNW	GDLGVNIRA	60
KMPKVHKAIK	ADGWMCHAAK	WVTTCDYRWY	GPQYITHSIH	SFIPTKAQCE	ESIKQTKQEV	120
WINPGFPPKN	CGYASVSDAE	SIIVQATAHS	VMIDEYSGDW	LDSQFPTGRC	TGSTCETIHN	180
STLWYADYQV	TGLCDALVS	TEVTFYSEDG	LMTSIGRQNT	GYRSNYFPYE	KGAAACRMKY	240
CTHEGIRLPS	GVWFEMVDKE	LLESVQMPEC	PAGLTISAPT	QTSVDVSLIL	DVERMLDYSL	300
CQETWSKVHS	GLPISPVDLG	YIAPKNPGAG	PAFTIVNGTL	KYFDTRYLRI	DIEGPVLKMK	360
TGKVSQTPTK	RELWTEWFPY	DDVEIGPENG	LKTPEGYKFP	LYMIGHGLLD	SDLQKTSQAE	420
VFHHQPQIAEA	VQKLPPDET	FFGDTGISK	PVEVIEGWFS	NWRSSVMAIV	FAILLLVITV	480
LMVRLCVAFR	HFCQCKRHKI	YNDLEMNQLR	R			511

SEQ ID NO: 11 moltype = AA length = 495
 FEATURE Location/Qualifiers
 source 1..495
 mol_type = protein
 organism = Vesicular stomatitis virus

SEQUENCE: 11

KFSIVFPQSQ	KGNWKNVPSS	YHYCPSSSDQ	NWHNDLLGIT	MKVKMPKTHK	AIQADGWMCH	60
AAKWTTCDY	RWYGPQYITH	SIHSIQPTSE	QCKESIKQTK	QGTWMSPGFP	PQNCGYATVT	120
DSVAVVVQAT	PHHLVDEYT	GEWIDSQFPN	GKCEETECET	VHNSTVWYSD	YKVTGLCDAT	180
LVDTEITFFS	EDGKESIGK	PNTGYRSNYF	AYEKGDVKCK	MNYCKHAGVR	LPSGVWFEFV	240
DQDVYAAAKL	PECPVGATIS	APTQTSVDVS	LILDVERILD	YSLCQETWSK	IRSKQPVSPV	300
DLSYLAPKNP	GTGPAFTIIN	GTLKYFETRY	IRIDIDNPPI	SKMVGKISGS	QTERELWTEW	360
FPYEGVEIGP	NGILKTPTGY	KFPLFMIGHG	MLDSDLHKTS	QAEVFEHPLH	AEAPKQLPEE	420
ETLFFGDTGI	SKNPVELIEG	WFSWKSTTVV	TFFFAIGVFI	LLYVVARIVI	AVRYRYQGSN	480
NKRIYNDIEM	SRFRK					495

SEQ ID NO: 12 moltype = AA length = 512
 FEATURE Location/Qualifiers
 source 1..512
 mol_type = protein
 organism = Vesicular stomatitis virus

SEQUENCE: 12

MNFLLLTFIV	LPLCSHAKFS	IVFPQSQKGN	WKNVPSSYHY	CPSSSDQNW	NLLGITMKV	60
KMPKTHKAIQ	ADGWMCHAAK	WITTCDFRWY	GPKYITHSIH	SIQPTSEQCK	ESIKQTKQGT	120
WMSPGFPPQN	CGYATVTDVS	AVVVQATPHH	VLVDEYTGW	IDSQFPNGKC	ETEECETVHN	180
STVWYSDYKV	TGLCDATLVD	TEITFFSEDG	KKESIGKPNT	GYRSNYFAYE	KGDVKCKMNY	240
CKHAGVRLPS	GVWFEEVDQD	VYAAAKLPEC	PVGATISAPT	QTSVDVSLIL	DVERILDYSL	300
CQETWSKIRS	KQPVSVDLS	YLAPKNPGTG	PAFTIINGTL	KYFETRYIRI	DIDNPIISKM	360
VGKISGSQTE	RELWTEWFPY	EGVEIGPNGI	LKPTPTGYKFP	LFMIGHGLMD	SDLHKTSQAE	420
VFEHPLHAEA	PKQLPEEETL	FFGDTGISK	PVELIEGWFS	SWKSTVVTFF	FAIGVFILLY	480
VVARIVIAVR	YRYQGSNNKR	IYNDIEMSRF	RK			512

SEQ ID NO: 13 moltype = AA length = 496
 FEATURE Location/Qualifiers
 source 1..496
 mol_type = protein
 organism = Vesicular stomatitis virus

SEQUENCE: 13

KFTIVFPQSQ	KGNWKNVPAN	YQYCPSSSDL	NWHNGLIGTS	LQVKMPKSHK	AIQADGWMCH	60
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AAKWVTTCD F RWYGP KYVTH SIKSMIPTVD QCKESIAQTK QGTWLNPGFP PQSCGYASVT 120
DAEAVIVKAT PHQVLVDEYT GEWVDSQFPT GKCNKDICTP VHNSTTWHS D YKVTGLCDAN 180
LISM DITFFS EDGKLTSLGK EGTGFRSNYF AYENGDKACR MQYCKHWGVR LPSGVWFEMA 240
DKDIYNDAKF PDCPEGSSIA APSQTSVDVS LIQDVERILD YSLCQETWSK IRAHLPISPV 300
DLSYLSPKNP GTGPAFTIIN GTLKYFETRY IRVDIAGPII PQMRGVISGT TTERELWTDW 360
YPYEDVEIGP NGVLKTATGY KFPLYMIGHG MLDSDLHISS KAQVFEHPHI QDAASQLPDD 420
ETLFFGDTGL SKNPIELVEG WFSGWKSTIA SFFFIIGLVI GLYLVLIRIGI ALCKICRVQE 480
KRPKIYTDVE MNRLDR 496

SEQ ID NO: 14      moltype = AA length = 513
FEATURE           Location/Qualifiers
source            1..513
                  mol_type = protein
                  organism = Vesicular stomatitis virus

SEQUENCE: 14
MLVLVLLLSL LALGAQCKFT IVFPHNQKGN WKNVPANYQY CPSSSDLNWH NGLIGTSLQV 60
KMPKSHKAIQ ADGWMCHAAK WVTTCDFRWY GPKYVTHSIK SMIPTVDQCK ESIAQTKQGT 120
WLNPGFPPQS CGYASVTDAE AVIVKATPHQ VLVDEYTG EW VDSQFPTGKC NKDICTPVHN 180
STTWHS DYKV TGLCDANLIS MDITFFSEDG KLTSLGKEGT GFRSNYPAYE NGDKACRMQY 240
CKHWGVR LPS GGVWFEMADK IYNDAKFPDC PEGSSIAAPS QTSVDVSLIQ DVERILDYSL 300
CQETWSKIRA HLPISPV DLS YLSPKNPGTG PFTIINGTL KYFETRYIRV DIAGPIIQM 360
RGVISGTTTE RELWTDWYPY EDVEIGPNGV LKTATGYKFP LYMIGHGMLD SDLHISSKAQ 420
VFEHPHIQDA ASQLPDD ETL FFGDTGLSKN PIELVEGWFS GWKSTIASFF FIIGLVIGLY 480
LVLRI GIALC IKCRVQEKRP KIYTDVEMNR LDR 513

SEQ ID NO: 15      moltype = AA length = 495
FEATURE           Location/Qualifiers
VARIANT          47
                  note = Xaa can be any naturally occurring amino acid except
                  Lys or Arg
source            1..495
                  mol_type = protein
                  organism = Vesicular stomatitis virus

SEQUENCE: 15
KFTIVFPHNQ KGNWKNVPSN YHYCPSSSDL NWHNDLIGTA IQVKMPKSHK AIQADGWMCH 60
ASKWVTTCD F RWYGP KYITQ SIRSF TSPSVE QCKESIEQTK QGTWLNPGFP PQSCGYATVT 120
DAEAVIVQVT PHHVLVDEYT GEWVDSQFIN GKCSNYICPT VHNSTTWHS D YKVKGLCDSN 180
LISM DITFFS EDGELSSLGK EGTGFRSNYF AYETGGKACK MQYCKHWGVR LPSGVWFEMA 240
DKDLFAAARF PECPEGSSIS APSQTSVDVS LIQDVERILD YSLCQETWSK IRAGLPISPV 300
DLSYLAPKNP GTGPAFTIIN GTLKYFETRY IRVDIAAPIL SRMVGMISGT TTERELWDDW 360
APYEDVEIGP NGVLTSSGY KFPLYMIGHG MLDSDLHLSS KAQVFEHPHI QDAASQLPDD 420
ESLFFGDTGL SKNPIELVEG WFSWKSSIA SFFFIIGLII GLFLVLVRGI HLCIKLKHTK 480
KRQIYTDIEM NRLGK 495

SEQ ID NO: 16      moltype = AA length = 495
FEATURE           Location/Qualifiers
source            1..495
                  mol_type = protein
                  organism = Vesicular stomatitis virus
VARIANT          354
                  note = Xaa can be any naturally occurring amino acid except
                  Lys or Arg

SEQUENCE: 16
KFTIVFPHNQ KGNWKNVPSN YHYCPSSSDL NWHNDLIGTA IQVKMPKSHK AIQADGWMCH 60
ASKWVTTCD F RWYGP KYITQ SIRSF TSPSVE QCKESIEQTK QGTWLNPGFP PQSCGYATVT 120
DAEAVIVQVT PHHVLVDEYT GEWVDSQFIN GKCSNYICPT VHNSTTWHS D YKVKGLCDSN 180
LISM DITFFS EDGELSSLGK EGTGFRSNYF AYETGGKACK MQYCKHWGVR LPSGVWFEMA 240
DKDLFAAARF PECPEGSSIS APSQTSVDVS LIQDVERILD YSLCQETWSK IRAGLPISPV 300
DLSYLAPKNP GTGPAFTIIN GTLKYFETRY IRVDIAAPIL SRMVGMISGT TTEXELWDDW 360
APYEDVEIGP NGVLTSSGY KFPLYMIGHG MLDSDLHLSS KAQVFEHPHI QDAASQLPDD 420
ESLFFGDTGL SKNPIELVEG WFSWKSSIA SFFFIIGLII GLFLVLVRGI HLCIKLKHTK 480
KRQIYTDIEM NRLGK 495

SEQ ID NO: 17      moltype = AA length = 495
FEATURE           Location/Qualifiers
source            1..495
                  mol_type = protein
                  organism = Vesicular stomatitis virus
VARIANT          354
                  note = Xaa can be any naturally occurring amino acid except
                  Lys or Arg
VARIANT          47
                  note = Xaa can be any naturally occurring amino acid except
                  Lys or Arg

SEQUENCE: 17
KFTIVFPHNQ KGNWKNVPSN YHYCPSSSDL NWHNDLIGTA IQVKMPKSHK AIQADGWMCH 60
ASKWVTTCD F RWYGP KYITQ SIRSF TSPSVE QCKESIEQTK QGTWLNPGFP PQSCGYATVT 120
DAEAVIVQVT PHHVLVDEYT GEWVDSQFIN GKCSNYICPT VHNSTTWHS D YKVKGLCDSN 180
LISM DITFFS EDGELSSLGK EGTGFRSNYF AYETGGKACK MQYCKHWGVR LPSGVWFEMA 240

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DKDLFAAARF PECPEGSSIS APSQTSVDVS LIQDVERILD YSLCQETWSK IRAGLPISPV 300
 DLSYLAPKNP GTGPAFTIIN GTLKYPETRY IRVDIAAPIL SRMVGMIISGT TTEXELWDDW 360
 APYEDVEIGP NGVLTSSGY KPPLYMIGHG MLDSDLHLSS KAQVFEHPHI QDAASQLPDD 420
 ESLFFGDTGL SKNPIELVEG WFSSWKSSIA SFFFIIIGLII GLFLVLRVGI HLCIKLKHTK 480
 KRQIYTDIEM NRLGK 495

SEQ ID NO: 18 moltype = AA length = 511
 FEATURE Location/Qualifiers
 source 1..511
 mol_type = protein
 organism = Vesicular stomatitis virus

VARIANT 63
 note = Xaa can be any naturally occurring amino acid except
 Lys or Arg

SEQUENCE: 18
 MKCLLYLAFI FIGVNCKFTI VFPHNQKGNW KNVPSNYHYC PSSSDLNWHN DLIGTAIQVK 60
 MPXSHKAIQA DGWMCHASKW VTTCDFRWYG PKYITQSIRS FTPSVEQCKE SIEQTKQGTW 120
 LNPGFPPQSC GYATVTDAEA VIVQVTPHHV LVDEYTGWEV DSQFINGKCS NYICPTVHNS 180
 TTWHSYKVKV GLCDNLISM DITFFSEDGE LSSLGKEGTG FRSNYFAYET GGKACKMQYC 240
 KHWGVRPLSG VWFEMADKDL FAAARFPECP EGSSISAPSQ TSVDVSLIQD VERILDYSLC 300
 QETWSKIRAG LPISPVDLSY LAPKNPGTGP AFTIINGTLK YFETRYIRVD IAAPILSRMV 360
 GMISGTTTTER ELWDDWAPYE DVEIGPNGVL RTSSGYKFPL YMIGHGMLDS DLHLSSKAQV 420
 FEHPHIQDAA SQLPDDESIF FGDGTGLSKNP IELVEGWFSK WKSSIASFFF IIGLIIGLFL 480
 VLRVGIHLICI KLKHTKKRQI YTDIEMNRLG K 511

SEQ ID NO: 19 moltype = AA length = 511
 FEATURE Location/Qualifiers
 source 1..511
 mol_type = protein
 organism = Vesicular stomatitis virus

VARIANT 370
 note = Xaa can be any naturally occurring amino acid except
 Lys or Arg

SEQUENCE: 19
 MKCLLYLAFI FIGVNCKFTI VFPHNQKGNW KNVPSNYHYC PSSSDLNWHN DLIGTAIQVK 60
 MPXSHKAIQA DGWMCHASKW VTTCDFRWYG PKYITQSIRS FTPSVEQCKE SIEQTKQGTW 120
 LNPGFPPQSC GYATVTDAEA VIVQVTPHHV LVDEYTGWEV DSQFINGKCS NYICPTVHNS 180
 TTWHSYKVKV GLCDNLISM DITFFSEDGE LSSLGKEGTG FRSNYFAYET GGKACKMQYC 240
 KHWGVRPLSG VWFEMADKDL FAAARFPECP EGSSISAPSQ TSVDVSLIQD VERILDYSLC 300
 QETWSKIRAG LPISPVDLSY LAPKNPGTGP AFTIINGTLK YFETRYIRVD IAAPILSRMV 360
 GMISGTTTTER ELWDDWAPYE DVEIGPNGVL RTSSGYKFPL YMIGHGMLDS DLHLSSKAQV 420
 FEHPHIQDAA SQLPDDESIF FGDGTGLSKNP IELVEGWFSK WKSSIASFFF IIGLIIGLFL 480
 VLRVGIHLICI KLKHTKKRQI YTDIEMNRLG K 511

SEQ ID NO: 20 moltype = AA length = 511
 FEATURE Location/Qualifiers
 source 1..511
 mol_type = protein
 organism = Vesicular stomatitis virus

VARIANT 63
 note = Xaa can be any naturally occurring amino acid except
 Lys or Arg

VARIANT 370
 note = Xaa can be any naturally occurring amino acid except
 Lys or Arg

SEQUENCE: 20
 MKCLLYLAFI FIGVNCKFTI VFPHNQKGNW KNVPSNYHYC PSSSDLNWHN DLIGTAIQVK 60
 MPXSHKAIQA DGWMCHASKW VTTCDFRWYG PKYITQSIRS FTPSVEQCKE SIEQTKQGTW 120
 LNPGFPPQSC GYATVTDAEA VIVQVTPHHV LVDEYTGWEV DSQFINGKCS NYICPTVHNS 180
 TTWHSYKVKV GLCDNLISM DITFFSEDGE LSSLGKEGTG FRSNYFAYET GGKACKMQYC 240
 KHWGVRPLSG VWFEMADKDL FAAARFPECP EGSSISAPSQ TSVDVSLIQD VERILDYSLC 300
 QETWSKIRAG LPISPVDLSY LAPKNPGTGP AFTIINGTLK YFETRYIRVD IAAPILSRMV 360
 GMISGTTTTER ELWDDWAPYE DVEIGPNGVL RTSSGYKFPL YMIGHGMLDS DLHLSSKAQV 420
 FEHPHIQDAA SQLPDDESIF FGDGTGLSKNP IELVEGWFSK WKSSIASFFF IIGLIIGLFL 480
 VLRVGIHLICI KLKHTKKRQI YTDIEMNRLG K 511

SEQ ID NO: 21 moltype = AA length = 496
 FEATURE Location/Qualifiers
 source 1..496
 mol_type = protein
 organism = Vesicular stomatitis virus

VARIANT 47
 note = Xaa can be any naturally occurring amino acid except
 Lys or Arg

SEQUENCE: 21
 KFTIVFPHHQ KGNWKNVPST YHYCPSSSDQ NWHNDLTGVS LHVKIPXSHK AIQADGWMCH 60
 AAKWVTTCDF RWYGPYIITH SIHMSPTLE QCKTSIEQTK QGVWINPGFP PQSCGYATVT 120
 DAEVVVQAT PHHVLVDEYT GEWIDSQLVG GKCSKEVCQT VHNSTVWHAD YKITGLCESN 180
 LASVDITFFS EDGQKTSLGK PNTGFRSNHF AYESGEKACR MQYCTQWGIR LPSGVWFELV 240

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DKDLFQAACL PECPRGSSIS APSQTSVDVS LIQDVERILD YSLCQETWSK IRAKLPVSPV 300
 DLSYLAPKNP GSGPAFTIIN GTLKYPETRY IRVDISNP II PHMVGTMSTG TTERELWNWD 360
 YPYEDVEIGP NGVLKTPTGF KPPLYMIGHG MLDSDLHKSS QAQVFEHPHA KDAASQLPDD 420
 ETLFFGDTGL SKNPVELVEG WFSSWKSTLA SFFLIIGLGV ALIFIIRIIV AIRYKYKGRK 480
 TQKIYNDVEM SRLGNK 496

SEQ ID NO: 22 moltype = AA length = 496
 FEATURE Location/Qualifiers
 source 1..496
 mol_type = protein
 organism = Vesicular stomatitis virus

VARIANT 354
 note = Xaa can be any naturally occurring amino acid except
 Lys or Arg

SEQUENCE: 22
 KFTIVFPHHQ KGNWKNVPST YHYCPSSSDQ NWHNDLTGVS LHVKIPKSHK AIQADGWMCH 60
 AAKWVTTCDF RWYGP KYI TH SIHMSPTLE QCKTSIEQTK QGVWINPGFP PQSCGYATVT 120
 DAEVVVVQAT PHHVLVDEYT GEWIDSQLVG GKCSKEVCQT VHNSTVWHAD YKITGLCESN 180
 LASVDITFFS EDGQKTS LGK PNTGFRSNHF AYESGEKACR MQYCTQWGIR LPSGVWFELV 240
 DKDLFQAACL PECPRGSSIS APSQTSVDVS LIQDVERILD YSLCQETWSK IRAKLPVSPV 300
 DLSYLAPKNP GSGPAFTIIN GTLKYPETRY IRVDISNP II PHMVGTMSTG TTERELWNWD 360
 YPYEDVEIGP NGVLKTPTGF KPPLYMIGHG MLDSDLHKSS QAQVFEHPHA KDAASQLPDD 420
 ETLFFGDTGL SKNPVELVEG WFSSWKSTLA SFFLIIGLGV ALIFIIRIIV AIRYKYKGRK 480
 TQKIYNDVEM SRLGNK 496

SEQ ID NO: 23 moltype = AA length = 496
 FEATURE Location/Qualifiers
 source 1..496
 mol_type = protein
 organism = Vesicular stomatitis virus

VARIANT 354
 note = Xaa can be any naturally occurring amino acid except
 Lys or Arg

VARIANT 47
 note = Xaa can be any naturally occurring amino acid except
 Lys or Arg

SEQUENCE: 23
 KFTIVFPHHQ KGNWKNVPST YHYCPSSSDQ NWHNDLTGVS LHVKIPKSHK AIQADGWMCH 60
 AAKWVTTCDF RWYGP KYI TH SIHMSPTLE QCKTSIEQTK QGVWINPGFP PQSCGYATVT 120
 DAEVVVVQAT PHHVLVDEYT GEWIDSQLVG GKCSKEVCQT VHNSTVWHAD YKITGLCESN 180
 LASVDITFFS EDGQKTS LGK PNTGFRSNHF AYESGEKACR MQYCTQWGIR LPSGVWFELV 240
 DKDLFQAACL PECPRGSSIS APSQTSVDVS LIQDVERILD YSLCQETWSK IRAKLPVSPV 300
 DLSYLAPKNP GSGPAFTIIN GTLKYPETRY IRVDISNP II PHMVGTMSTG TTERELWNWD 360
 YPYEDVEIGP NGVLKTPTGF KPPLYMIGHG MLDSDLHKSS QAQVFEHPHA KDAASQLPDD 420
 ETLFFGDTGL SKNPVELVEG WFSSWKSTLA SFFLIIGLGV ALIFIIRIIV AIRYKYKGRK 480
 TQKIYNDVEM SRLGNK 496

SEQ ID NO: 24 moltype = AA length = 512
 FEATURE Location/Qualifiers
 source 1..512
 mol_type = protein
 organism = Vesicular stomatitis virus

VARIANT 63
 note = Xaa can be any naturally occurring amino acid except
 Lys or Arg

SEQUENCE: 24
 MLRLFLFCFL ALGAHSKFTI VFPHHQKGNW KNPSTYHYC PSSSDQNWHN DLTGVS LHVK 60
 IPXSHKAIQA DGWMCHAAKW VTTCDFRWYG PKYITHSIHS MSPTLEQCKT SIEQTKQGVW 120
 INPGFPPQSC GYATVTD A EV VVVQATPHHV LVDEYTG EWI DSQLVGGKCS KEVCQTVHNS 180
 TVWHADYKIT GLCESNLASV DITFFSEDGQ KTS LGKPNTG FRSNHFAYES GEKACRMQYC 240
 TQWGIRLPSG VWFELVDKDL FQA AKLPECP RGSSISAPSQ TSVDVSLIQD VERILDYSLC 300
 QETWSKIRAK LPVSPVDLSY LAPKNPGSGP AFTIINGTLK YFETRYIRVD ISNPIIPHMV 360
 GTMSGTTTER ELWNDWYPYE DVEIGPNGVL KPTTGFKFPL YMIGHGMLDS DLHKSSQAQV 420
 FEHPHAKDAA SQLPDDETLF FGDTGLSKNP VELVEGWFS WKSTLASFFL IIGLGV ALIF 480
 IIRIIVAIRY KYKGRKTQKI YNDVEMSRLG NK 512

SEQ ID NO: 25 moltype = AA length = 512
 FEATURE Location/Qualifiers
 source 1..512
 mol_type = protein
 organism = Vesicular stomatitis virus

VARIANT 370
 note = Xaa can be any naturally occurring amino acid except
 Lys or Arg

SEQUENCE: 25
 MLRLFLFCFL ALGAHSKFTI VFPHHQKGNW KNPSTYHYC PSSSDQNWHN DLTGVS LHVK 60
 IPXSHKAIQA DGWMCHAAKW VTTCDFRWYG PKYITHSIHS MSPTLEQCKT SIEQTKQGVW 120
 INPGFPPQSC GYATVTD A EV VVVQATPHHV LVDEYTG EWI DSQLVGGKCS KEVCQTVHNS 180
 TVWHADYKIT GLCESNLASV DITFFSEDGQ KTS LGKPNTG FRSNHFAYES GEKACRMQYC 240

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TQWGIRLPSG VWFELVDKDL FQAAKLPECP RGSSISAPSQ TSVDVSLIQD VERILDYSLC 300
 QETWSKIRAK LPVSPVDLSY LAPKNPGSGP AFTIINGTLK YFETRYIRVD ISNPIIPHMV 360
 GTMSGTTTEX ELWNDWYPYE DVEIGPNGVL KTPTGFKFPL YMIGHGMLDS DLHKSSQAQV 420
 FEHPHAKDAA SQLPDDETLF FGDGTGLSKNP VELVEGWFS WKSTLASFFL IIGLGVALIF 480
 IIRIIVAIRY KYKGRKTQKI YNDVEMSRLG NK 512

SEQ ID NO: 26 moltype = AA length = 512
 FEATURE Location/Qualifiers
 source 1..512
 mol_type = protein
 organism = Vesicular stomatitis virus
 VARIANT 63
 note = Xaa can be any naturally occurring amino acid except
 Lys or Arg
 VARIANT 370
 note = Xaa can be any naturally occurring amino acid except
 Lys or Arg

SEQUENCE: 26
 MLRLFLFCFL ALGAHSKFTI VFPFHQKGNW KNVPSTYHYC PSSSDQNWNN DLTGVS LHVK 60
 IPXSHKAIQA DGWMCHAAKW VTTCDFRWYG PKYITHSIHS MSPBLEQCKT SIEQTKQGVW 120
 INPGFPQSC GYATVTD AEV VVVQATPHHV LVDEYTGWEI DSQLVGGKCS KEVCQTVHNS 180
 TVWHADYKIT GLCESNLASV DITFFSEDGQ KTSLGKPN TG FRSNHFAYES GEKACRMQYC 240
 TQWGIRLPSG VWFELVDKDL FQAAKLPECP RGSSISAPSQ TSVDVSLIQD VERILDYSLC 300
 QETWSKIRAK LPVSPVDLSY LAPKNPGSGP AFTIINGTLK YFETRYIRVD ISNPIIPHMV 360
 GTMSGTTTEX ELWNDWYPYE DVEIGPNGVL KTPTGFKFPL YMIGHGMLDS DLHKSSQAQV 420
 FEHPHAKDAA SQLPDDETLF FGDGTGLSKNP VELVEGWFS WKSTLASFFL IIGLGVALIF 480
 IIRIIVAIRY KYKGRKTQKI YNDVEMSRLG NK 512

SEQ ID NO: 27 moltype = AA length = 501
 FEATURE Location/Qualifiers
 VARIANT 47
 note = Xaa can be any naturally occurring amino acid except
 Lys or Arg
 source 1..501
 mol_type = protein
 organism = Vesicular stomatitis virus

SEQUENCE: 27
 KIEIVFPQHT TGDWKRVPHE YNYCPTSADK NSHGTQTGIP VELTMPXGLT THQVDGFMCH 60
 SALWMTTCDF RWYGPKYI TH SIHNEEPTDY QCLEAIKAYK DGVSFNPGFP PQSCGYGTVT 120
 DAEAHIVTVT PHSVKVDEYT GEWIDPHFIG GRCKGQICET VHNSTKWFTS SDGESVCSQL 180
 FTLVGGTFFS DSEBITSMGL PETGIRSNYF PYVSTEGICK MPFCRKPGYK LKNDLWFQIT 240
 DPDLDKTVRD LPHIKDCDLS SSI VTPGEHA TDISLISDVE RILDYALCQN TWSKIEAGEP 300
 ITPVDLSYLG PKNPGAGPVF TIINGSLHYF MSKYL RVELE SPVIPRMEGK VAGTRIVRQL 360
 WDQWFPFGEV EIGPNGVLKT KQGYKFPLHI IGTGEVDNDI KMERIVKHWE HPHIEAAQTF 420
 LKDDTEEV L YYGDTGVSKN PVELVEGWFS GWRSSIMGVL AVIIGFVILI FLIRLIGVLS 480
 SLFRQKRRPI YKSDVEMAHF R 501

SEQ ID NO: 28 moltype = AA length = 501
 FEATURE Location/Qualifiers
 VARIANT 358
 note = Xaa can be any naturally occurring amino acid except
 Lys or Arg
 source 1..501
 mol_type = protein
 organism = Vesicular stomatitis virus

SEQUENCE: 28
 KIEIVFPQHT TGDWKRVPHE YNYCPTSADK NSHGTQTGIP VELTMPKGLT THQVDGFMCH 60
 SALWMTTCDF RWYGPKYI TH SIHNEEPTDY QCLEAIKAYK DGVSFNPGFP PQSCGYGTVT 120
 DAEAHIVTVT PHSVKVDEYT GEWIDPHFIG GRCKGQICET VHNSTKWFTS SDGESVCSQL 180
 FTLVGGTFFS DSEBITSMGL PETGIRSNYF PYVSTEGICK MPFCRKPGYK LKNDLWFQIT 240
 DPDLDKTVRD LPHIKDCDLS SSI VTPGEHA TDISLISDVE RILDYALCQN TWSKIEAGEP 300
 ITPVDLSYLG PKNPGAGPVF TIINGSLHYF MSKYL RVELE SPVIPRMEGK VAGTRIVXQL 360
 WDQWFPFGEV EIGPNGVLKT KQGYKFPLHI IGTGEVDNDI KMERIVKHWE HPHIEAAQTF 420
 LKDDTEEV L YYGDTGVSKN PVELVEGWFS GWRSSIMGVL AVIIGFVILI FLIRLIGVLS 480
 SLFRQKRRPI YKSDVEMAHF R 501

SEQ ID NO: 29 moltype = AA length = 501
 FEATURE Location/Qualifiers
 VARIANT 47
 note = Xaa can be any naturally occurring amino acid except
 Lys or Arg
 VARIANT 358
 note = Xaa can be any naturally occurring amino acid except
 Lys or Arg
 source 1..501
 mol_type = protein
 organism = Vesicular stomatitis virus

SEQUENCE: 29
 KIEIVFPQHT TGDWKRVPHE YNYCPTSADK NSHGTQTGIP VELTMPXGLT THQVDGFMCH 60

-continued

SALWMTTCDF RWYGPKYITH SIHNEETDY QCLEAIKAYK DGVSNPFGFP PQSCGYGTVT 120
 DAEAHIVTVT PHSVKVDEYT GEWIDPHFIG GRCKGQICET VHNSTKWFTS SDGESVCSQL 180
 FTLVGGTFPS DSEETSMGL PETGIRSNYF PYVSTEGICK MPFCRKPGYK LKNDLWFQIT 240
 DPDLDKTVRD LPHIKDCDLS SSIVTPGEHA TDISLISDVE RILDYALCQN TWSKIEAGEP 300
 ITPVDLSYLG PKNPGAGPVF TIINGSLHYF MSKYLRVELE SPVIPRMEGK VAGTRIVXQL 360
 WDQWFPFGVE EIGPNGVLKT KQGYKFPLHI IGTGEVDNDI KMERIVKHWE HPHIEAAQTF 420
 LKKDDTEEVLY YGDTGVSKN PVELVEGWFS GWRSSIMGVL AVIIGFVILI FLIRLIGVLS 480
 SLFRQKRRPI YKSDVEMAHF R 501

SEQ ID NO: 30 moltype = AA length = 517
 FEATURE Location/Qualifiers
 VARIANT 63
 note = Xaa can be any naturally occurring amino acid except
 Lys or Arg
 source 1..517
 mol_type = protein
 organism = Vesicular stomatitis virus

SEQUENCE: 30
 MLSYLIFALV VSPILGKIEI VFPQHTTGDW KRVPHYNYC PTSADKNSHG TQTGIPVELT 60
 MPXGLTTHQV DGFMCCHSALW MTTCDFRWYG PKYITHSIHN EEPTDYQCLE AIKAYKDGVN 120
 FNPFGPPQSC GYGTVTDAEA HIVTVTPHSV KVDEYTGWEI DPHFIGGRCK GQICETVHNS 180
 TKWFTSSDGE SVCSQLFTLV GGTFFSDSEE ITSMGLPETG IRSNYFPYVS TEGICKMPFC 240
 RKPQGYKLKND LWFQITDPDL DKTVRDLPHI KDCDLSSSIV TPGEHATDIS LISDVERILD 300
 YALCQNTWSK IEAGEPITPV DLSYLGPKNP GAGPVFTIIN GSLHYFMSKY LRVELESPVI 360
 PRMEGKVAGT RIVRQLWDQW FPFGEVEIGP NGVLKTKQGY KFPLHIIGTG EVDNDIKMER 420
 IVKHWEHPHI EAAQTFLKND DTEEVLYYGD TGVSKNPVEL VEGWFSGWRS SIMGVLAVII 480
 GFVILIFLIR LIGVLSLFR QKRRPIYKSD VEMAHFR 517

SEQ ID NO: 31 moltype = AA length = 517
 FEATURE Location/Qualifiers
 VARIANT 374
 note = Xaa can be any naturally occurring amino acid except
 Lys or Arg
 source 1..517
 mol_type = protein
 organism = Vesicular stomatitis virus

SEQUENCE: 31
 MLSYLIFALV VSPILGKIEI VFPQHTTGDW KRVPHYNYC PTSADKNSHG TQTGIPVELT 60
 MPXGLTTHQV DGFMCCHSALW MTTCDFRWYG PKYITHSIHN EEPTDYQCLE AIKAYKDGVN 120
 FNPFGPPQSC GYGTVTDAEA HIVTVTPHSV KVDEYTGWEI DPHFIGGRCK GQICETVHNS 180
 TKWFTSSDGE SVCSQLFTLV GGTFFSDSEE ITSMGLPETG IRSNYFPYVS TEGICKMPFC 240
 RKPQGYKLKND LWFQITDPDL DKTVRDLPHI KDCDLSSSIV TPGEHATDIS LISDVERILD 300
 YALCQNTWSK IEAGEPITPV DLSYLGPKNP GAGPVFTIIN GSLHYFMSKY LRVELESPVI 360
 PRMEGKVAGT RIVRQLWDQW FPFGEVEIGP NGVLKTKQGY KFPLHIIGTG EVDNDIKMER 420
 IVKHWEHPHI EAAQTFLKND DTEEVLYYGD TGVSKNPVEL VEGWFSGWRS SIMGVLAVII 480
 GFVILIFLIR LIGVLSLFR QKRRPIYKSD VEMAHFR 517

SEQ ID NO: 32 moltype = AA length = 517
 FEATURE Location/Qualifiers
 VARIANT 63
 note = Xaa can be any naturally occurring amino acid except
 Lys or Arg
 VARIANT 374
 note = Xaa can be any naturally occurring amino acid except
 Lys or Arg
 source 1..517
 mol_type = protein
 organism = Vesicular stomatitis virus

SEQUENCE: 32
 MLSYLIFALV VSPILGKIEI VFPQHTTGDW KRVPHYNYC PTSADKNSHG TQTGIPVELT 60
 MPXGLTTHQV DGFMCCHSALW MTTCDFRWYG PKYITHSIHN EEPTDYQCLE AIKAYKDGVN 120
 FNPFGPPQSC GYGTVTDAEA HIVTVTPHSV KVDEYTGWEI DPHFIGGRCK GQICETVHNS 180
 TKWFTSSDGE SVCSQLFTLV GGTFFSDSEE ITSMGLPETG IRSNYFPYVS TEGICKMPFC 240
 RKPQGYKLKND LWFQITDPDL DKTVRDLPHI KDCDLSSSIV TPGEHATDIS LISDVERILD 300
 YALCQNTWSK IEAGEPITPV DLSYLGPKNP GAGPVFTIIN GSLHYFMSKY LRVELESPVI 360
 PRMEGKVAGT RIVRQLWDQW FPFGEVEIGP NGVLKTKQGY KFPLHIIGTG EVDNDIKMER 420
 IVKHWEHPHI EAAQTFLKND DTEEVLYYGD TGVSKNPVEL VEGWFSGWRS SIMGVLAVII 480
 GFVILIFLIR LIGVLSLFR QKRRPIYKSD VEMAHFR 517

SEQ ID NO: 33 moltype = AA length = 502
 FEATURE Location/Qualifiers
 VARIANT 47
 note = Xaa can be any naturally occurring amino acid except
 Lys or Arg
 source 1..502
 mol_type = protein
 organism = Vesicular stomatitis virus

SEQUENCE: 33
 KITISFPQSL KGDWRPVPGK YNYCPTSADK NLHGDLIDIG LRLRAPXSPK GISADGWMCH 60

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AARWITTCDF RWYGPYIYTH SIHSFRPSND QCKEAIRLTN EGNWINPGFP PQSCGYASVT 120
DSESVVTVT KHQVLVDEYS GSWIDSQFPF GSCTSPICDT VHNSTLWHAD HTLDSICDQE 180
FVAMDAVLFT ESGKFEEFGK PNSGIRSNYF PYESLKDVCQ MDFCKRKGFK LPSGVWFEIE 240
DAEKSHKAQV ELKIKRCPHG AVISAPNQNA ADINLIMDVE RILDYSLCQA TWSKIQNKEA 300
LTPIDISYLG PKNPGPGPAF TIINGTLHYF NTRYIRVDIA GPVTKEITGF VSGTSTSRVL 360
WDQWFPYGEN SIGPNGLLKT ASGYKYPLFM VGTGVLDADI HKLGEATVIE HPHAKEAQKV 420
VDDSEVIFPG DTGVSKNPVE VVEGWFSGWR SSLMSIFGII LLIVCLVLIV RILIALKYCC 480
VRHKKRTIYK EDLEMGRIPR RA 502

SEQ ID NO: 34      moltype = AA length = 502
FEATURE           Location/Qualifiers
VARIANT           358
                  note = Xaa can be any naturally occurring amino acid except
                  Lys or Arg
source            1..502
                  mol_type = protein
                  organism = Vesicular stomatitis virus

SEQUENCE: 34
KITISFPQSL KGDWRPVPKG YNYCPTSADK NLHGLDIDIG LRLRAPKSPK GISADGWMCH 60
AARWITTCDF RWYGPYIYTH SIHSFRPSND QCKEAIRLTN EGNWINPGFP PQSCGYASVT 120
DSESVVTVT KHQVLVDEYS GSWIDSQFPF GSCTSPICDT VHNSTLWHAD HTLDSICDQE 180
FVAMDAVLFT ESGKFEEFGK PNSGIRSNYF PYESLKDVCQ MDFCKRKGFK LPSGVWFEIE 240
DAEKSHKAQV ELKIKRCPHG AVISAPNQNA ADINLIMDVE RILDYSLCQA TWSKIQNKEA 300
LTPIDISYLG PKNPGPGPAF TIINGTLHYF NTRYIRVDIA GPVTKEITGF VSGTSTSRVL 360
WDQWFPYGEN SIGPNGLLKT ASGYKYPLFM VGTGVLDADI HKLGEATVIE HPHAKEAQKV 420
VDDSEVIFPG DTGVSKNPVE VVEGWFSGWR SSLMSIFGII LLIVCLVLIV RILIALKYCC 480
VRHKKRTIYK EDLEMGRIPR RA 502

SEQ ID NO: 35      moltype = AA length = 502
FEATURE           Location/Qualifiers
VARIANT           47
                  note = Xaa can be any naturally occurring amino acid except
                  Lys or Arg
VARIANT           358
                  note = Xaa can be any naturally occurring amino acid except
                  Lys or Arg
source            1..502
                  mol_type = protein
                  organism = Vesicular stomatitis virus

SEQUENCE: 35
KITISFPQSL KGDWRPVPKG YNYCPTSADK NLHGLDIDIG LRLRAPKSPK GISADGWMCH 60
AARWITTCDF RWYGPYIYTH SIHSFRPSND QCKEAIRLTN EGNWINPGFP PQSCGYASVT 120
DSESVVTVT KHQVLVDEYS GSWIDSQFPF GSCTSPICDT VHNSTLWHAD HTLDSICDQE 180
FVAMDAVLFT ESGKFEEFGK PNSGIRSNYF PYESLKDVCQ MDFCKRKGFK LPSGVWFEIE 240
DAEKSHKAQV ELKIKRCPHG AVISAPNQNA ADINLIMDVE RILDYSLCQA TWSKIQNKEA 300
LTPIDISYLG PKNPGPGPAF TIINGTLHYF NTRYIRVDIA GPVTKEITGF VSGTSTSRVL 360
WDQWFPYGEN SIGPNGLLKT ASGYKYPLFM VGTGVLDADI HKLGEATVIE HPHAKEAQKV 420
VDDSEVIFPG DTGVSKNPVE VVEGWFSGWR SSLMSIFGII LLIVCLVLIV RILIALKYCC 480
VRHKKRTIYK EDLEMGRIPR RA 502

SEQ ID NO: 36      moltype = AA length = 523
FEATURE           Location/Qualifiers
VARIANT           68
                  note = Xaa can be any naturally occurring amino acid except
                  Lys or Arg
source            1..523
                  mol_type = protein
                  organism = Vesicular stomatitis virus

SEQUENCE: 36
MKMKMVIAGL ILCIGILPAI GKITISFPQS LKGDWRPVPK GYNYCPTSAD KNLHGLDIDI 60
GLRLRAPXSF KGISADGWMC HAARWITTCDF FRWYGPYIYTH HSIHSFRPSN DQCKEAIRLT 120
NEGNWINPGF PPQSCGYASV TDSESVVTVT TKHQVLVDEY SGWSIDSQFP GGCTSPICD 180
TVHNSTLWHA DHTLDSICDQ EFVAMDAVLFT ESGKFEEFGK PKNSGIRSNY FPYESLKDVC 240
QMDFCRKRGF KLPSGVWFEI EDAEKSHKAQ VELKIKRCPH GAVISAPNQN AADINLIMDV 300
ERILDYSLCQ ATWSKIQNKE ALTPIDISYL GPKNPGPGPA FTIINGTLHY FNTRYIRVDI 360
AGPVTKEITG FVSGTSTSRV LWDQWFPYGE NSIGPNGLLK TASGYKYPLF MVGTGVLDAD 420
IHKLGEATVI HPHAKEAQK VVDDSEVIFG GDTGVSKNPV EVVEGWFSGW RSSLMSIFGI 480
ILLIVCLVLI VRILIALKYC CVRHKKRTIY KEDLEMGRIP RRA 523

SEQ ID NO: 37      moltype = AA length = 523
FEATURE           Location/Qualifiers
VARIANT           379
                  note = Xaa can be any naturally occurring amino acid except
                  Lys or Arg
source            1..523
                  mol_type = protein
                  organism = Vesicular stomatitis virus

SEQUENCE: 37
MKMKMVIAGL ILCIGILPAI GKITISFPQS LKGDWRPVPK GYNYCPTSAD KNLHGLDIDI 60

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GLRLRAPKSF KGISADGWM C HAARWITTC D FRWYGPKYIT HSIHSFRPSN DQCKEAIRLT 120
NEGNWINPGF PPQSCGYASV TDES VVVTV TKHQVLVDEY SGSWIDSQFP GGSCTSPICD 180
TVHNSTLWHA DHTLDSICDQ EFMAMDAVL F TESGKFEEFG KPNSGIRSNY FPYESLKDVC 240
QMDFCRKGKGF KLPSGVWF E EDAEKSHKAQ VELKIKRCPH GAVISAPNQ N AADINLIMDV 300
ERILDYSLCQ ATWSKIQNKE ALTPIDISYL GPKNPGPGPA FTIINGTLHY FNTRYIRVDI 360
AGPVTKEITG FVSGTSTS X VLDQWFPYGE NSIGPNGLLK TASGYKYPLF MVGTGVLAD 420
IHKLGATVI EHPHAKAQK VVDDSEVIF GDTGVSKNPV EVVEGWFSGW RSSLMSIFGI 480
ILLIVCLVLI VRILIALKYC CVRHKKRTIY KEDLEMGRIP RRA 523

SEQ ID NO: 38      moltype = AA length = 523
FEATURE           Location/Qualifiers
VARIANT           68
                  note = Xaa can be any naturally occurring amino acid except
                  Lys or Arg
VARIANT           379
                  note = Xaa can be any naturally occurring amino acid except
                  Lys or Arg
source            1..523
                  mol_type = protein
                  organism = Vesicular stomatitis virus

SEQUENCE: 38
MKMKMVIAGL ILCIGILPAI GKITISFPQS LKGDWRPVPK GYNYCPTSAD KNLHGDLIDI 60
GLRLRAPKSF KGISADGWM C HAARWITTC D FRWYGPKYIT HSIHSFRPSN DQCKEAIRLT 120
NEGNWINPGF PPQSCGYASV TDES VVVTV TKHQVLVDEY SGSWIDSQFP GGSCTSPICD 180
TVHNSTLWHA DHTLDSICDQ EFMAMDAVL F TESGKFEEFG KPNSGIRSNY FPYESLKDVC 240
QMDFCRKGKGF KLPSGVWF E EDAEKSHKAQ VELKIKRCPH GAVISAPNQ N AADINLIMDV 300
ERILDYSLCQ ATWSKIQNKE ALTPIDISYL GPKNPGPGPA FTIINGTLHY FNTRYIRVDI 360
AGPVTKEITG FVSGTSTS X VLDQWFPYGE NSIGPNGLLK TASGYKYPLF MVGTGVLAD 420
IHKLGATVI EHPHAKAQK VVDDSEVIF GDTGVSKNPV EVVEGWFSGW RSSLMSIFGI 480
ILLIVCLVLI VRILIALKYC CVRHKKRTIY KEDLEMGRIP RRA 523

SEQ ID NO: 39      moltype = AA length = 494
FEATURE           Location/Qualifiers
VARIANT           47
                  note = Xaa can be any naturally occurring amino acid except
                  Lys or Arg
source            1..494
                  mol_type = protein
                  organism = Vesicular stomatitis virus

SEQUENCE: 39
KFTIVFPQSQ KGDWKDVPPN YRYCPSSADQ NWHGDL LGVN IRAKMPXVHK AIKADGWMCH 60
AAKWVTTCDY RWYGPQYITH SIHSFIPTKA QCEESIKQTK EGVWINPGFP PKNCGYASVS 120
DAESIIVQAT AHSVIMIDEYS GDWLD SQFPT GRCTGSTCET IHNSTLWYAD YQVTGLCDSA 180
LVSTEVTFYS EDGLMTS IGR QNTGYRSNYF PYEKGAAACR MKYCTHEGIR LPSGVWFEMV 240
DKELLESVQM PECPAGLTIS APTQTSVDVS LILDVERMLD YSLCQETWSK VHSGLPISPV 300
DLGYIAPKNP GAGPAFTIVN GTLKYFDTRY LRIDIEGPVL KKMTGKVS GT PTKRELWTEW 360
FPYDDVEIGP NGVLKTP EGY KPPLYMIGHG LLDSDLQKTS QAEVFHHPQI AEA VQKL PDD 420
ETLFFGDTGI SKNPVEVIEG WFSNWRSSVM AIVFAILL LV ITVLMVRLCV AFRHFCCQKR 480
HKIYNDLEMN QLRR 494

SEQ ID NO: 40      moltype = AA length = 494
FEATURE           Location/Qualifiers
VARIANT           354
                  note = Xaa can be any naturally occurring amino acid except
                  Lys or Arg
source            1..494
                  mol_type = protein
                  organism = Vesicular stomatitis virus

SEQUENCE: 40
KFTIVFPQSQ KGDWKDVPPN YRYCPSSADQ NWHGDL LGVN IRAKMPXVHK AIKADGWMCH 60
AAKWVTTCDY RWYGPQYITH SIHSFIPTKA QCEESIKQTK EGVWINPGFP PKNCGYASVS 120
DAESIIVQAT AHSVIMIDEYS GDWLD SQFPT GRCTGSTCET IHNSTLWYAD YQVTGLCDSA 180
LVSTEVTFYS EDGLMTS IGR QNTGYRSNYF PYEKGAAACR MKYCTHEGIR LPSGVWFEMV 240
DKELLESVQM PECPAGLTIS APTQTSVDVS LILDVERMLD YSLCQETWSK VHSGLPISPV 300
DLGYIAPKNP GAGPAFTIVN GTLKYFDTRY LRIDIEGPVL KKMTGKVS GT PTKRELWTEW 360
FPYDDVEIGP NGVLKTP EGY KPPLYMIGHG LLDSDLQKTS QAEVFHHPQI AEA VQKL PDD 420
ETLFFGDTGI SKNPVEVIEG WFSNWRSSVM AIVFAILL LV ITVLMVRLCV AFRHFCCQKR 480
HKIYNDLEMN QLRR 494

SEQ ID NO: 41      moltype = AA length = 494
FEATURE           Location/Qualifiers
VARIANT           47
                  note = Xaa can be any naturally occurring amino acid except
                  Lys or Arg
VARIANT           354
                  note = Xaa can be any naturally occurring amino acid except
                  Lys or Arg
source            1..494
                  mol_type = protein

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organism = Vesicular stomatitis virus

SEQUENCE: 41
KFTIVFPQSQ KGDWKDVPPN YRYCPSSADQ NWHGDL LGVN IRAKMPXVHK AIKADGWMCH 60
AAKWVTTCDY RWYGPQYITH SIHSFIPTKA QCEESIKQTK EGVWINPGFP PKNCGYASVS 120
DAESIIVQAT AHSVMIDEYS GDWLD SQFPT GRCTGSTCET IHNSTLWYAD YQVTGLCDSA 180
LVSTEVTFYS EDGLMTSISR QNTGYRSNYF PYEKGAAACR MKYCTHEGIR LPSGVWFEMV 240
DKELLESVQM PECPAGLTIS APTQTSVDVS LILDVERMLD YSLCQETWSK VHSGLPISPV 300
DLGYIAPKNP GAGPAFTIVN GTLKYFDTRY LRIDIEGPVL KKMTGKVSQT PTKXELWTEW 360
FPYDDVEIGP NGVLKTPEGY KFPLYMIGHG LLDSDLQKTS QAEVFHHPQI AEAQVQLPDD 420
ETLFFGDTGI SKNPVEVIEG WFSNWRSSVM AIVFAILLV ITVLMVRLCV AFRHFCCQKR 480
HKIYNDLEMN QLRR 494

SEQ ID NO: 42      moltype = AA length = 511
FEATURE           Location/Qualifiers
VARIANT           64
                  note = Xaa can be any naturally occurring amino acid except
                  Lys or Arg
source            1..511
                  mol_type = protein
                  organism = Vesicular stomatitis virus

SEQUENCE: 42
MTPAFILCML LAGSSWAKFT IVFPQSQKGD WKDVPPNYRY CPSSADQNW GDL LGVNIRA 60
KMPKVHKAIAK ADGWMCHAAK WVTTC DYRWY GPQYITHSIH SFIPTKAQCE ESIKQTKEGV 120
WINPGFPPKN CGYASVSDAE SIIVQATAHS VMIDEYSGDW LDSQFPTGRC TGSTCETIHN 180
STLWYADYQV TGLCDSALVS TEVTFYSEDG LMTSISRQNT GYRSNYFPYE KGAAACRMKY 240
CTHEGIRLPS GVVWFEMV DKE LLESVQMPEC PAGLTISAPT QTSVDVSLIL DVERMLDYSL 300
CQETWSKVHS GLPISPV DLG YIAPKNPGAG PFTIVNGTL KYFDTRYLRI DIEGPVLKKM 360
TGKVSQTPTK RELWTEWFPY DDVEIGPNGV LKTPEGYKFP LYMIGHGLLD SDLQKTSQAE 420
VFHHPQIAEA VQKLPPDET L FFGDTGISK N PVEVIEGWFS NWRSSVMAIV FAILLLVITV 480
LMVRLCVAFR HFCCQKRHKI YNDLEMNQLR R 511

SEQ ID NO: 43      moltype = AA length = 511
FEATURE           Location/Qualifiers
VARIANT           371
                  note = Xaa can be any naturally occurring amino acid except
                  Lys or Arg
source            1..511
                  mol_type = protein
                  organism = Vesicular stomatitis virus

SEQUENCE: 43
MTPAFILCML LAGSSWAKFT IVFPQSQKGD WKDVPPNYRY CPSSADQNW GDL LGVNIRA 60
KMPKVHKAIAK ADGWMCHAAK WVTTC DYRWY GPQYITHSIH SFIPTKAQCE ESIKQTKEGV 120
WINPGFPPKN CGYASVSDAE SIIVQATAHS VMIDEYSGDW LDSQFPTGRC TGSTCETIHN 180
STLWYADYQV TGLCDSALVS TEVTFYSEDG LMTSISRQNT GYRSNYFPYE KGAAACRMKY 240
CTHEGIRLPS GVVWFEMV DKE LLESVQMPEC PAGLTISAPT QTSVDVSLIL DVERMLDYSL 300
CQETWSKVHS GLPISPV DLG YIAPKNPGAG PFTIVNGTL KYFDTRYLRI DIEGPVLKKM 360
TGKVSQTPTK RELWTEWFPY DDVEIGPNGV LKTPEGYKFP LYMIGHGLLD SDLQKTSQAE 420
VFHHPQIAEA VQKLPPDET L FFGDTGISK N PVEVIEGWFS NWRSSVMAIV FAILLLVITV 480
LMVRLCVAFR HFCCQKRHKI YNDLEMNQLR R 511

SEQ ID NO: 44      moltype = AA length = 511
FEATURE           Location/Qualifiers
VARIANT           64
                  note = Xaa can be any naturally occurring amino acid except
                  Lys or Arg
VARIANT           371
                  note = Xaa can be any naturally occurring amino acid except
                  Lys or Arg
source            1..511
                  mol_type = protein
                  organism = Vesicular stomatitis virus

SEQUENCE: 44
MTPAFILCML LAGSSWAKFT IVFPQSQKGD WKDVPPNYRY CPSSADQNW GDL LGVNIRA 60
KMPKVHKAIAK ADGWMCHAAK WVTTC DYRWY GPQYITHSIH SFIPTKAQCE ESIKQTKEGV 120
WINPGFPPKN CGYASVSDAE SIIVQATAHS VMIDEYSGDW LDSQFPTGRC TGSTCETIHN 180
STLWYADYQV TGLCDSALVS TEVTFYSEDG LMTSISRQNT GYRSNYFPYE KGAAACRMKY 240
CTHEGIRLPS GVVWFEMV DKE LLESVQMPEC PAGLTISAPT QTSVDVSLIL DVERMLDYSL 300
CQETWSKVHS GLPISPV DLG YIAPKNPGAG PFTIVNGTL KYFDTRYLRI DIEGPVLKKM 360
TGKVSQTPTK RELWTEWFPY DDVEIGPNGV LKTPEGYKFP LYMIGHGLLD SDLQKTSQAE 420
VFHHPQIAEA VQKLPPDET L FFGDTGISK N PVEVIEGWFS NWRSSVMAIV FAILLLVITV 480
LMVRLCVAFR HFCCQKRHKI YNDLEMNQLR R 511

SEQ ID NO: 45      moltype = AA length = 495
FEATURE           Location/Qualifiers
VARIANT           47
                  note = Xaa can be any naturally occurring amino acid except
                  Lys or Arg
source            1..495
                  mol_type = protein

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organism = Vesicular stomatitis virus
SEQUENCE: 45
KFSIVFPQSQ KGNWKNVPSS YHYCPSSSDQ NWHNDLLGIT MKVKMPXTHK AIQADGWMCH 60
AAKWITTCDF RWYGPKYITH SIHSIQPTSE QCKESIKQTK QGTWMSPGFP PQNCGYATVT 120
DSVAVVVQAT PHHVLVDEYT GEWIDSQFPN GKCETEECET VHNSTVWYSD YKVTGLCDAT 180
LVDTEITFFS EDGKKEISIGK PNTGYRSNYF AYEKGDKVCK MNYCKHAGVR LPSGVWFEFV 240
DQDVYAAAKL PECPVGATIS APTQTSVDVS LILDVERILD YSLCQETWSK IRSKQPVSPV 300
DLSYLAPKNP GTGPAFTIIN GTLKYFETRY IRIDIDNP II SKMVGKISGS QTERELWTEW 360
FPYEGVEIGP NGILKTPTGY KFPLFMIGHG MLDSDLHKTS QAEVFEHPL AEAPKQLPEE 420
ETLFFGDTGI SKNPVELIEG WFSSWKSTVV TFFFAIGVFI LLYVVARIVI AVRYRYQGSN 480
NKRIYNDIEM SRFRK 495

SEQ ID NO: 46      moltype = AA length = 495
FEATURE           Location/Qualifiers
VARIANT           354
                  note = Xaa can be any naturally occurring amino acid except
                  Lys or Arg
source            1..495
                  mol_type = protein
                  organism = Vesicular stomatitis virus

SEQUENCE: 46
KFSIVFPQSQ KGNWKNVPSS YHYCPSSSDQ NWHNDLLGIT MKVKMPXTHK AIQADGWMCH 60
AAKWITTCDF RWYGPKYITH SIHSIQPTSE QCKESIKQTK QGTWMSPGFP PQNCGYATVT 120
DSVAVVVQAT PHHVLVDEYT GEWIDSQFPN GKCETEECET VHNSTVWYSD YKVTGLCDAT 180
LVDTEITFFS EDGKKEISIGK PNTGYRSNYF AYEKGDKVCK MNYCKHAGVR LPSGVWFEFV 240
DQDVYAAAKL PECPVGATIS APTQTSVDVS LILDVERILD YSLCQETWSK IRSKQPVSPV 300
DLSYLAPKNP GTGPAFTIIN GTLKYFETRY IRIDIDNP II SKMVGKISGS QTEXELWTEW 360
FPYEGVEIGP NGILKTPTGY KFPLFMIGHG MLDSDLHKTS QAEVFEHPL AEAPKQLPEE 420
ETLFFGDTGI SKNPVELIEG WFSSWKSTVV TFFFAIGVFI LLYVVARIVI AVRYRYQGSN 480
NKRIYNDIEM SRFRK 495

SEQ ID NO: 47      moltype = AA length = 495
FEATURE           Location/Qualifiers
VARIANT           47
                  note = Xaa can be any naturally occurring amino acid except
                  Lys or Arg
VARIANT           354
                  note = Xaa can be any naturally occurring amino acid except
                  Lys or Arg
source            1..495
                  mol_type = protein
                  organism = Vesicular stomatitis virus

SEQUENCE: 47
KFSIVFPQSQ KGNWKNVPSS YHYCPSSSDQ NWHNDLLGIT MKVKMPXTHK AIQADGWMCH 60
AAKWITTCDF RWYGPKYITH SIHSIQPTSE QCKESIKQTK QGTWMSPGFP PQNCGYATVT 120
DSVAVVVQAT PHHVLVDEYT GEWIDSQFPN GKCETEECET VHNSTVWYSD YKVTGLCDAT 180
LVDTEITFFS EDGKKEISIGK PNTGYRSNYF AYEKGDKVCK MNYCKHAGVR LPSGVWFEFV 240
DQDVYAAAKL PECPVGATIS APTQTSVDVS LILDVERILD YSLCQETWSK IRSKQPVSPV 300
DLSYLAPKNP GTGPAFTIIN GTLKYFETRY IRIDIDNP II SKMVGKISGS QTEXELWTEW 360
FPYEGVEIGP NGILKTPTGY KFPLFMIGHG MLDSDLHKTS QAEVFEHPL AEAPKQLPEE 420
ETLFFGDTGI SKNPVELIEG WFSSWKSTVV TFFFAIGVFI LLYVVARIVI AVRYRYQGSN 480
NKRIYNDIEM SRFRK 495

SEQ ID NO: 48      moltype = AA length = 512
FEATURE           Location/Qualifiers
VARIANT           64
                  note = Xaa can be any naturally occurring amino acid except
                  Lys or Arg
source            1..512
                  mol_type = protein
                  organism = Vesicular stomatitis virus

SEQUENCE: 48
MNFLLLTFIV LPLCSHAKFS IVFPQSQKGN WKNVPSSYHY CPSSSDQNW NHNDLLGITMKV 60
KMPXTHKAIQ ADGWMCHAAK WITTCDFRWY GPKYITHSIH SIQPTSEQCK ESIKQTKQGT 120
WMSPGFPPQN CGYATVTDVS AVVVQATPHH VLVDEYTG EW IDSQFPNGKC ETEECETVHN 180
STVWYSDYKV TGLCDATLVD TEITFFSEDG KKEISIGKPT GYRSNYFAYE KGDVKCKMNY 240
CKHAGVRLPS GVWFEFVDQD VYAAAKLPEC PVGATISAPT QTSVDVSLIL DVERILDYSL 300
CQETWSKIRS KQPVSPV DLS YLAPKNPGTG PAFTIINGTL KYFETRYIRI DIDNP IISKM 360
VGKISGSQTE RELWTEWFPY EGVEIGPNGI LKPTPTGYKFP LFMIGHGMLD SDLHKTSQAE 420
VFEHPLAEAE PKQLPEEETL FFGDTGISK N PVELIEGWFS SWKSTVVTFP FAIGVFILLY 480
VVARIVIAVR YRYQGSNNKR IYNDIEMSRF RK 512

SEQ ID NO: 49      moltype = AA length = 512
FEATURE           Location/Qualifiers
VARIANT           371
                  note = Xaa can be any naturally occurring amino acid except
                  Lys or Arg
source            1..512
                  mol_type = protein

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organism = Vesicular stomatitis virus

SEQUENCE: 49
MNFLLLTFIV LPLCSHAKFS IVFPQSQKGN WKNVPSSYHY CPSSSDQNW NDLGITMKV 60
KMPKTHKAIQ ADGWMCHAAC WITTCDFRWY GPKYITHSIH SIQPTSEQCK ESIKQTKQGT 120
WMSPGFPPQN CGYATVTDV AVVVQATPHH VLVDEYTG EW IDSQFPNGKC ETEECETVHN 180
STVWYS DYK TGLCDATLVD TEITFFSEDG KKE SIGKPT GYRSNYFAYE KGDVKCKMNY 240
CKHAGVRLPS GVVWFEFVDQD VYAAAKLPEC PVGATISAPT QTSVDVSLIL DVERILDYSL 300
CQETWSKIRS KQVSPVDLS YLAPKNPGTG PAFTIINGTL KYFETRYIRI DIDNPIISKM 360
VGKISGSQTE XELWTEWFPY EGVEIGPNGI LKTPGTGYKFP LFMIGHGMLD SDLHKTSQAE 420
VFEHPLHAEA PKQLPEEETL FFGDTGISK N PVELIEGWFS SWKSTVVTF FAIGVFILLY 480
VVARIVIAVR YRYQGSNNKR IYNDIEMSRF RK 512

SEQ ID NO: 50      moltype = AA length = 512
FEATURE           Location/Qualifiers
VARIANT           64
                  note = Xaa can be any naturally occurring amino acid except
                  Lys or Arg
VARIANT           371
                  note = Xaa can be any naturally occurring amino acid except
                  Lys or Arg
source            1..512
                  mol_type = protein
                  organism = Vesicular stomatitis virus

SEQUENCE: 50
MNFLLLTFIV LPLCSHAKFS IVFPQSQKGN WKNVPSSYHY CPSSSDQNW NDLGITMKV 60
KMPKTHKAIQ ADGWMCHAAC WITTCDFRWY GPKYITHSIH SIQPTSEQCK ESIKQTKQGT 120
WMSPGFPPQN CGYATVTDV AVVVQATPHH VLVDEYTG EW IDSQFPNGKC ETEECETVHN 180
STVWYS DYK TGLCDATLVD TEITFFSEDG KKE SIGKPT GYRSNYFAYE KGDVKCKMNY 240
CKHAGVRLPS GVVWFEFVDQD VYAAAKLPEC PVGATISAPT QTSVDVSLIL DVERILDYSL 300
CQETWSKIRS KQVSPVDLS YLAPKNPGTG PAFTIINGTL KYFETRYIRI DIDNPIISKM 360
VGKISGSQTE XELWTEWFPY EGVEIGPNGI LKTPGTGYKFP LFMIGHGMLD SDLHKTSQAE 420
VFEHPLHAEA PKQLPEEETL FFGDTGISK N PVELIEGWFS SWKSTVVTF FAIGVFILLY 480
VVARIVIAVR YRYQGSNNKR IYNDIEMSRF RK 512

SEQ ID NO: 51      moltype = AA length = 496
FEATURE           Location/Qualifiers
VARIANT           47
                  note = Xaa can be any naturally occurring amino acid except
                  Lys or Arg
source            1..496
                  mol_type = protein
                  organism = Vesicular stomatitis virus

SEQUENCE: 51
KFTIVFPHNQ KGNWKNVPAN YQYCPSSSDL NWHNGLIGTS LQVKMPKSHK AIQADGWMCH 60
AAKWVTTCD F RWYGPKYVTH SIKSMIPTVD QCKESIAQTK QGTWLNPGFP PQSCGYASVT 120
DAEAVIVKAT PHQVLVDEYT GEWVDSQFPT GKCNKDICT VHNSTTWHS YKVTGLCDAN 180
LISMDITFFS EDGKLTS LGK EGTGFRSNYF AYENGDKACR MQYCKHWGVR LPSGVWFEMA 240
DKDIYNDAK F PDCPEGSSIA APSQTSVDVS LIQDVERILD YSLCQETWSK IRAHLPISPV 300
DLSYLSPKNP GTGPAFTIIN GTLKYPETRY IRVDIAGPII PQMRGVISGT TTERELWTDW 360
YPYEDVEIGP NGVLKTATGY KFPLYMIGHG MLDSDLHISS KAQVFEHPhi QDAASQLPDD 420
ETLFFGDTGL SKNPIELVEG WFSGWKSTIA SFFFIIGLVI GLYLVLIRIGI ALCIKCRVQE 480
KRPKIYTDVE MNRLLDR 496

SEQ ID NO: 52      moltype = AA length = 496
FEATURE           Location/Qualifiers
VARIANT           354
                  note = Xaa can be any naturally occurring amino acid except
                  Lys or Arg
source            1..496
                  mol_type = protein
                  organism = Vesicular stomatitis virus

SEQUENCE: 52
KFTIVFPHNQ KGNWKNVPAN YQYCPSSSDL NWHNGLIGTS LQVKMPKSHK AIQADGWMCH 60
AAKWVTTCD F RWYGPKYVTH SIKSMIPTVD QCKESIAQTK QGTWLNPGFP PQSCGYASVT 120
DAEAVIVKAT PHQVLVDEYT GEWVDSQFPT GKCNKDICT VHNSTTWHS YKVTGLCDAN 180
LISMDITFFS EDGKLTS LGK EGTGFRSNYF AYENGDKACR MQYCKHWGVR LPSGVWFEMA 240
DKDIYNDAK F PDCPEGSSIA APSQTSVDVS LIQDVERILD YSLCQETWSK IRAHLPISPV 300
DLSYLSPKNP GTGPAFTIIN GTLKYPETRY IRVDIAGPII PQMRGVISGT TTEXELWTDW 360
YPYEDVEIGP NGVLKTATGY KFPLYMIGHG MLDSDLHISS KAQVFEHPhi QDAASQLPDD 420
ETLFFGDTGL SKNPIELVEG WFSGWKSTIA SFFFIIGLVI GLYLVLIRIGI ALCIKCRVQE 480
KRPKIYTDVE MNRLLDR 496

SEQ ID NO: 53      moltype = AA length = 496
FEATURE           Location/Qualifiers
VARIANT           47
                  note = Xaa can be any naturally occurring amino acid except
                  Lys or Arg
VARIANT           354
                  note = Xaa can be any naturally occurring amino acid except

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-continued

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source          Lys or Arg
                1..496
                mol_type = protein
                organism = Vesicular stomatitis virus

SEQUENCE: 53
KFTIVFPHNQ KGNWKNVPAN YQYCPSSSDL NWHNGLIGTS LQVKMPXSHK AIQADGWMCH 60
AAKWVTTCDF RWYGPKYVTH SIKSMIPTVD QCKESIAQTK QGTWLNPGFP PQSCGYASVT 120
DAEAVIVKAT PHQVLVDEYT GEWVDSQFPT GKCNDICPT VHNSTTWHSD YKVTGLCDAN 180
LISMDITFFS EDGKLTSLGK EGTGFRSNYF AYENGDKACR MQYCKHWGVR LPSGVWFEMA 240
DKDIYNDAKF PDCPEGSSIA APSQTSVDVS LIQDVERILD YSLCQETWSK IRAHLPISPV 300
DLSYLSPKNP GTGPAFTIIN GTLKYFETRY IRVDIAGPII PQMRGVISGT TTEXELWTDW 360
YPYEDVEIGP NGVLKTATGY KPPLYMIGHG MLDSDLHISS KAQVFEHPHI QDAASQLPDD 420
ETLFFGDTGL SKNPIELVEG WFSGWKSTIA SFFFIIGLVI GLYLVLIRIGI ALCIKCRVQE 480
KRPKIYTDVE MNRLDR                                     496

SEQ ID NO: 54      moltype = AA length = 513
FEATURE           Location/Qualifiers
VARIANT           64
                  note = Xaa can be any naturally occurring amino acid except
                  Lys or Arg
source            1..513
                  mol_type = protein
                  organism = Vesicular stomatitis virus

SEQUENCE: 54
MLVLYLLLSL LALGAQCKFT IVFPHNQKGN WKNVPANYQY CPSSSDLNWH NGLIGTSLQV 60
KMPXSHKAIQ ADGWMCHAAK WVTTCDFRWY GPKYVTHSIK SMIPTVDQCK ESIAQTKQGT 120
WLNPGFPPQS CGYASVTDAA AVIVKATPHQ VLVDEYTGGEW VDSQFPTGKC NKDICPTVHN 180
STTWHSDYKV TGLCDANLIS MDITFFSEDG KLTSLGKEGT GFRSNYFAYE NGDKACRMQY 240
CKHWGVR LPS GVVWFEMADKD IYNDAKFPDC PEGSSIAAPS QTSVDVSLIQ DVERILDYSL 300
CQETWSKIRA HLPISPV DLS YLSPKNPGTG PFTIINGTL KYFETRYIRV DIAGPIIPQM 360
RGVISGTTTE RELWTDWYPY EDVEIGPNGV LKTATGYKFP LYMIGHGMLD SDLHISSKAQ 420
VFEHPHIQDA ASQLPDD ETL FFGDTGLSKN PIELVEGWFS GWKSTIASFF FIIGLVIGLY 480
LVLIRIGIALC IKCRVQE KRP KIYTDVEMNR LDR                                     513

SEQ ID NO: 55      moltype = AA length = 513
FEATURE           Location/Qualifiers
VARIANT           371
                  note = Xaa can be any naturally occurring amino acid except
                  Lys or Arg
source            1..513
                  mol_type = protein
                  organism = Vesicular stomatitis virus

SEQUENCE: 55
MLVLYLLLSL LALGAQCKFT IVFPHNQKGN WKNVPANYQY CPSSSDLNWH NGLIGTSLQV 60
KMPXSHKAIQ ADGWMCHAAK WVTTCDFRWY GPKYVTHSIK SMIPTVDQCK ESIAQTKQGT 120
WLNPGFPPQS CGYASVTDAA AVIVKATPHQ VLVDEYTGGEW VDSQFPTGKC NKDICPTVHN 180
STTWHSDYKV TGLCDANLIS MDITFFSEDG KLTSLGKEGT GFRSNYFAYE NGDKACRMQY 240
CKHWGVR LPS GVVWFEMADKD IYNDAKFPDC PEGSSIAAPS QTSVDVSLIQ DVERILDYSL 300
CQETWSKIRA HLPISPV DLS YLSPKNPGTG PFTIINGTL KYFETRYIRV DIAGPIIPQM 360
RGVISGTTTE RELWTDWYPY EDVEIGPNGV LKTATGYKFP LYMIGHGMLD SDLHISSKAQ 420
VFEHPHIQDA ASQLPDD ETL FFGDTGLSKN PIELVEGWFS GWKSTIASFF FIIGLVIGLY 480
LVLIRIGIALC IKCRVQE KRP KIYTDVEMNR LDR                                     513

SEQ ID NO: 56      moltype = AA length = 513
FEATURE           Location/Qualifiers
VARIANT           64
                  note = Xaa can be any naturally occurring amino acid except
                  Lys or Arg
VARIANT           371
                  note = Xaa can be any naturally occurring amino acid except
                  Lys or Arg
source            1..513
                  mol_type = protein
                  organism = Vesicular stomatitis virus

SEQUENCE: 56
MLVLYLLLSL LALGAQCKFT IVFPHNQKGN WKNVPANYQY CPSSSDLNWH NGLIGTSLQV 60
KMPXSHKAIQ ADGWMCHAAK WVTTCDFRWY GPKYVTHSIK SMIPTVDQCK ESIAQTKQGT 120
WLNPGFPPQS CGYASVTDAA AVIVKATPHQ VLVDEYTGGEW VDSQFPTGKC NKDICPTVHN 180
STTWHSDYKV TGLCDANLIS MDITFFSEDG KLTSLGKEGT GFRSNYFAYE NGDKACRMQY 240
CKHWGVR LPS GVVWFEMADKD IYNDAKFPDC PEGSSIAAPS QTSVDVSLIQ DVERILDYSL 300
CQETWSKIRA HLPISPV DLS YLSPKNPGTG PFTIINGTL KYFETRYIRV DIAGPIIPQM 360
RGVISGTTTE RELWTDWYPY EDVEIGPNGV LKTATGYKFP LYMIGHGMLD SDLHISSKAQ 420
VFEHPHIQDA ASQLPDD ETL FFGDTGLSKN PIELVEGWFS GWKSTIASFF FIIGLVIGLY 480
LVLIRIGIALC IKCRVQE KRP KIYTDVEMNR LDR                                     513

SEQ ID NO: 57      moltype = AA length = 495
FEATURE           Location/Qualifiers
VARIANT           8
                  note = Xaa can be any naturally occurring amino acid except

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VARIANT Lys or Arg
47
note = Xaa can be any naturally occurring amino acid except
Lys or Arg

source 1..495
mol_type = protein
organism = Vesicular stomatitis virus

SEQUENCE: 57

KFTIVFPXNQ	KGNWKNVPSN	YHYCPSSSDL	NWHNDLIGTA	IQVKMPXSHK	AIQADGWMCH	60
ASKWVTTCDF	RWYGPKYITQ	SIRSFTPSVE	QCKESIEQTK	QGTWLNPGFP	PQSCGYATVT	120
DAEAVIVQVT	PHHVLVDEYT	GEWVDSQFIN	GKCSNYICPT	VHNSTTWHS	YKVKGLCDSN	180
LISMDITFFS	EDGELSSLGK	EGTGFRSNYF	AYETGGKACK	MQYCKHWGVR	LPSGVWFEMA	240
DKDLFAAARF	PECPEGSSIS	APSQTSVDVS	LIQDVERILD	YSLCQETWSK	IRAGLPISPV	300
DL SylAPKNP	GTGPAFTIIN	GTLKYFETRY	IRVDIAAPIL	SRMVGMISGT	TTERELWDDW	360
APYEDVEIGP	NGVLTSSGY	KFPLYMIGHG	MLDSDLHLSS	KAQVFEHPHI	QDAASQLPDD	420
ESLFFGDTGL	SKNPIELVEG	WFSSWKSSIA	SFFFIIGLII	GLFLVLRVGI	HLCIKLKHTK	480
KRQIYTDIEM	NRLGK					495

SEQ ID NO: 58 moltype = AA length = 495
FEATURE Location/Qualifiers
VARIANT 8
note = Xaa can be any naturally occurring amino acid except
Lys or Arg

VARIANT 354
note = Xaa can be any naturally occurring amino acid except
Lys or Arg

source 1..495
mol_type = protein
organism = Vesicular stomatitis virus

SEQUENCE: 58

KFTIVFPXNQ	KGNWKNVPSN	YHYCPSSSDL	NWHNDLIGTA	IQVKMPXSHK	AIQADGWMCH	60
ASKWVTTCDF	RWYGPKYITQ	SIRSFTPSVE	QCKESIEQTK	QGTWLNPGFP	PQSCGYATVT	120
DAEAVIVQVT	PHHVLVDEYT	GEWVDSQFIN	GKCSNYICPT	VHNSTTWHS	YKVKGLCDSN	180
LISMDITFFS	EDGELSSLGK	EGTGFRSNYF	AYETGGKACK	MQYCKHWGVR	LPSGVWFEMA	240
DKDLFAAARF	PECPEGSSIS	APSQTSVDVS	LIQDVERILD	YSLCQETWSK	IRAGLPISPV	300
DL SylAPKNP	GTGPAFTIIN	GTLKYFETRY	IRVDIAAPIL	SRMVGMISGT	TTEXELWDDW	360
APYEDVEIGP	NGVLTSSGY	KFPLYMIGHG	MLDSDLHLSS	KAQVFEHPHI	QDAASQLPDD	420
ESLFFGDTGL	SKNPIELVEG	WFSSWKSSIA	SFFFIIGLII	GLFLVLRVGI	HLCIKLKHTK	480
KRQIYTDIEM	NRLGK					495

SEQ ID NO: 59 moltype = AA length = 495
FEATURE Location/Qualifiers
VARIANT 8
note = Xaa can be any naturally occurring amino acid except
Lys or Arg

VARIANT 47
note = Xaa can be any naturally occurring amino acid except
Lys or Arg

VARIANT 354
note = Xaa can be any naturally occurring amino acid except
Lys or Arg

source 1..495
mol_type = protein
organism = Vesicular stomatitis virus

SEQUENCE: 59

KFTIVFPXNQ	KGNWKNVPSN	YHYCPSSSDL	NWHNDLIGTA	IQVKMPXSHK	AIQADGWMCH	60
ASKWVTTCDF	RWYGPKYITQ	SIRSFTPSVE	QCKESIEQTK	QGTWLNPGFP	PQSCGYATVT	120
DAEAVIVQVT	PHHVLVDEYT	GEWVDSQFIN	GKCSNYICPT	VHNSTTWHS	YKVKGLCDSN	180
LISMDITFFS	EDGELSSLGK	EGTGFRSNYF	AYETGGKACK	MQYCKHWGVR	LPSGVWFEMA	240
DKDLFAAARF	PECPEGSSIS	APSQTSVDVS	LIQDVERILD	YSLCQETWSK	IRAGLPISPV	300
DL SylAPKNP	GTGPAFTIIN	GTLKYFETRY	IRVDIAAPIL	SRMVGMISGT	TTEXELWDDW	360
APYEDVEIGP	NGVLTSSGY	KFPLYMIGHG	MLDSDLHLSS	KAQVFEHPHI	QDAASQLPDD	420
ESLFFGDTGL	SKNPIELVEG	WFSSWKSSIA	SFFFIIGLII	GLFLVLRVGI	HLCIKLKHTK	480
KRQIYTDIEM	NRLGK					495

SEQ ID NO: 60 moltype = AA length = 495
FEATURE Location/Qualifiers
VARIANT 47
note = Xaa can be any naturally occurring amino acid except
Lys or Arg

VARIANT 209
note = Xaa can be any naturally occurring amino acid except
Lys or Arg

source 1..495
mol_type = protein
organism = Vesicular stomatitis virus

SEQUENCE: 60

KFTIVFPXNQ	KGNWKNVPSN	YHYCPSSSDL	NWHNDLIGTA	IQVKMPXSHK	AIQADGWMCH	60
ASKWVTTCDF	RWYGPKYITQ	SIRSFTPSVE	QCKESIEQTK	QGTWLNPGFP	PQSCGYATVT	120

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DAEAVIVQVT PHHVLVDEYT GEWVDSQFIN GKCSNYICPT VHNSTTWHSD YKVKGLCDSN 180
 LISMDITFFS EDGELSSLGK EGTGFRSNXF AYETGGKACK MQYCKHWGVR LPSGVWFEMA 240
 DKDLFAAARF PECPEGSSIS APSQTSVDVS LIQDVERILD YSLCQETWSK IRAGLPISPV 300
 DLSYLAPKNP GTGPAFTIIN GTLKYFETRY IRVDIAAPIL SRMVGMIISGT TTERELWDDW 360
 APYEDVEIGP NGVLTSSGY KFPLYMIGHG MLDSDLHLSS KAQVFEHPHI QDAASQLPDD 420
 ESLFFGDTGL SKNPIELVEG WFSSWKSSIA SFFFIIGLII GLFLVLRVGI HLCIKLKHTK 480
 KRQIYTDIEM NRLGK 495

SEQ ID NO: 61 moltype = AA length = 495
 FEATURE Location/Qualifiers
 VARIANT 209
 note = Xaa can be any naturally occurring amino acid except
 Lys or Arg
 VARIANT 354
 note = Xaa can be any naturally occurring amino acid except
 Lys or Arg
 source 1..495
 mol_type = protein
 organism = Vesicular stomatitis virus

SEQUENCE: 61
 KFTIVFPHNQ KGNWKNVPSN YHYCPSSSDL NWHNDLIGTA IQVKMPKSHK AIQADGWMCH 60
 ASKWVTTCDF RWYGPKYITQ SIRSFPTPSVE QCKESIEQTK QGTWLNPGFP PQSCGYATVT 120
 DAEAVIVQVT PHHVLVDEYT GEWVDSQFIN GKCSNYICPT VHNSTTWHSD YKVKGLCDSN 180
 LISMDITFFS EDGELSSLGK EGTGFRSNXF AYETGGKACK MQYCKHWGVR LPSGVWFEMA 240
 DKDLFAAARF PECPEGSSIS APSQTSVDVS LIQDVERILD YSLCQETWSK IRAGLPISPV 300
 DLSYLAPKNP GTGPAFTIIN GTLKYFETRY IRVDIAAPIL SRMVGMIISGT TTEXELWDDW 360
 APYEDVEIGP NGVLTSSGY KFPLYMIGHG MLDSDLHLSS KAQVFEHPHI QDAASQLPDD 420
 ESLFFGDTGL SKNPIELVEG WFSSWKSSIA SFFFIIGLII GLFLVLRVGI HLCIKLKHTK 480
 KRQIYTDIEM NRLGK 495

SEQ ID NO: 62 moltype = AA length = 495
 FEATURE Location/Qualifiers
 VARIANT 47
 note = Xaa can be any naturally occurring amino acid except
 Lys or Arg
 VARIANT 209
 note = Xaa can be any naturally occurring amino acid except
 Lys or Arg
 VARIANT 354
 note = Xaa can be any naturally occurring amino acid except
 Lys or Arg
 source 1..495
 mol_type = protein
 organism = Vesicular stomatitis virus

SEQUENCE: 62
 KFTIVFPHNQ KGNWKNVPSN YHYCPSSSDL NWHNDLIGTA IQVKMPKSHK AIQADGWMCH 60
 ASKWVTTCDF RWYGPKYITQ SIRSFPTPSVE QCKESIEQTK QGTWLNPGFP PQSCGYATVT 120
 DAEAVIVQVT PHHVLVDEYT GEWVDSQFIN GKCSNYICPT VHNSTTWHSD YKVKGLCDSN 180
 LISMDITFFS EDGELSSLGK EGTGFRSNXF AYETGGKACK MQYCKHWGVR LPSGVWFEMA 240
 DKDLFAAARF PECPEGSSIS APSQTSVDVS LIQDVERILD YSLCQETWSK IRAGLPISPV 300
 DLSYLAPKNP GTGPAFTIIN GTLKYFETRY IRVDIAAPIL SRMVGMIISGT TTEXELWDDW 360
 APYEDVEIGP NGVLTSSGY KFPLYMIGHG MLDSDLHLSS KAQVFEHPHI QDAASQLPDD 420
 ESLFFGDTGL SKNPIELVEG WFSSWKSSIA SFFFIIGLII GLFLVLRVGI HLCIKLKHTK 480
 KRQIYTDIEM NRLGK 495

SEQ ID NO: 63 moltype = AA length = 495
 FEATURE Location/Qualifiers
 VARIANT 8
 note = Xaa can be any naturally occurring amino acid except
 Lys or Arg
 VARIANT 47
 note = Xaa can be any naturally occurring amino acid except
 Lys or Arg
 VARIANT 209
 note = Xaa can be any naturally occurring amino acid except
 Lys or Arg
 source 1..495
 mol_type = protein
 organism = Vesicular stomatitis virus

SEQUENCE: 63
 KFTIVFPXNQ KGNWKNVPSN YHYCPSSSDL NWHNDLIGTA IQVKMPKSHK AIQADGWMCH 60
 ASKWVTTCDF RWYGPKYITQ SIRSFPTPSVE QCKESIEQTK QGTWLNPGFP PQSCGYATVT 120
 DAEAVIVQVT PHHVLVDEYT GEWVDSQFIN GKCSNYICPT VHNSTTWHSD YKVKGLCDSN 180
 LISMDITFFS EDGELSSLGK EGTGFRSNXF AYETGGKACK MQYCKHWGVR LPSGVWFEMA 240
 DKDLFAAARF PECPEGSSIS APSQTSVDVS LIQDVERILD YSLCQETWSK IRAGLPISPV 300
 DLSYLAPKNP GTGPAFTIIN GTLKYFETRY IRVDIAAPIL SRMVGMIISGT TTERELWDDW 360
 APYEDVEIGP NGVLTSSGY KFPLYMIGHG MLDSDLHLSS KAQVFEHPHI QDAASQLPDD 420
 ESLFFGDTGL SKNPIELVEG WFSSWKSSIA SFFFIIGLII GLFLVLRVGI HLCIKLKHTK 480
 KRQIYTDIEM NRLGK 495

SEQ ID NO: 64 moltype = AA length = 511
FEATURE Location/Qualifiers
VARIANT 24
 note = Xaa can be any naturally occurring amino acid except
 Lys or Arg
VARIANT 63
 note = Xaa can be any naturally occurring amino acid except
 Lys or Arg
source 1..511
 mol_type = protein
 organism = Vesicular stomatitis virus

SEQUENCE: 64
MKCLLYLAFI FIGVNCFTI VFPXNQKGNW KNPVSNHYHC PSSSDLNWNH DLIGTAIQVK 60
MPXSHKAIQA DGMWCHASKW VTTCDFRWYG PKYITQSIRS FTSPVEQCKE SIEQTKQGTW 120
LNPGFPPQSC GYATVTDAAE VIVQVTPHHV LVDEYTGGEWV DSQFINGKCS NYICPTVHNS 180
TTWHS DYKVKV GLCDSNLISM DITFFSEDGE LSSLGKEGTG FRSNYFAYET GKGACKMQYC 240
KHWGVRLP SG VWFEMADKDL FAAARFPPECP EGSSISAPSQ TSDVDSL IQD VERILDYSLC 300
QETWSKIRAG LPISPDVLSY LAPKNPGTGP AFTIINGTLK YFETRYIRVD IAAPILSRMV 360
GMISGTTTER ELWDDWAPYE DVEIGPNGVL RTSSGYKFPL YMIGHGMLDS DLHLSSKAQV 420
FEHPHIQDAA SQLPDDDES LF FGDGTGLSKNP IELVEGW FSS WKSSIASFFF IIGLIIGLFL 480
VLRVGIHL CI KLKHTTKRQI YTDIEMNRLG K 511

SEQ ID NO: 65 moltype = AA length = 511
FEATURE Location/Qualifiers
VARIANT 24
 note = Xaa can be any naturally occurring amino acid except
 Lys or Arg
VARIANT 370
 note = Xaa can be any naturally occurring amino acid except
 Lys or Arg
source 1..511
 mol_type = protein
 organism = Vesicular stomatitis virus

SEQUENCE: 65
MKCLLYLAFI FIGVNCFTI VFPXNQKGNW KNPVSNHYHC PSSSDLNWNH DLIGTAIQVK 60
MPKSHKAIQA DGMWCHASKW VTTCDFRWYG PKYITQSIRS FTSPVEQCKE SIEQTKQGTW 120
LNPGFPPQSC GYATVTDAAE VIVQVTPHHV LVDEYTGGEWV DSQFINGKCS NYICPTVHNS 180
TTWHS DYKVKV GLCDSNLISM DITFFSEDGE LSSLGKEGTG FRSNYFAYET GKGACKMQYC 240
KHWGVRLP SG VWFEMADKDL FAAARFPPECP EGSSISAPSQ TSDVDSL IQD VERILDYSLC 300
QETWSKIRAG LPISPDVLSY LAPKNPGTGP AFTIINGTLK YFETRYIRVD IAAPILSRMV 360
GMISGTTTEX ELWDDWAPYE DVEIGPNGVL RTSSGYKFPL YMIGHGMLDS DLHLSSKAQV 420
FEHPHIQDAA SQLPDDDES LF FGDGTGLSKNP IELVEGW FSS WKSSIASFFF IIGLIIGLFL 480
VLRVGIHL CI KLKHTTKRQI YTDIEMNRLG K 511

SEQ ID NO: 66 moltype = AA length = 511
FEATURE Location/Qualifiers
VARIANT 24
 note = Xaa can be any naturally occurring amino acid except
 Lys or Arg
VARIANT 63
 note = Xaa can be any naturally occurring amino acid except
 Lys or Arg
VARIANT 370
 note = Xaa can be any naturally occurring amino acid except
 Lys or Arg
source 1..511
 mol_type = protein
 organism = Vesicular stomatitis virus

SEQUENCE: 66
MKCLLYLAFI FIGVNCFTI VFPXNQKGNW KNPVSNHYHC PSSSDLNWNH DLIGTAIQVK 60
MPXSHKAIQA DGMWCHASKW VTTCDFRWYG PKYITQSIRS FTSPVEQCKE SIEQTKQGTW 120
LNPGFPPQSC GYATVTDAAE VIVQVTPHHV LVDEYTGGEWV DSQFINGKCS NYICPTVHNS 180
TTWHS DYKVKV GLCDSNLISM DITFFSEDGE LSSLGKEGTG FRSNYFAYET GKGACKMQYC 240
KHWGVRLP SG VWFEMADKDL FAAARFPPECP EGSSISAPSQ TSDVDSL IQD VERILDYSLC 300
QETWSKIRAG LPISPDVLSY LAPKNPGTGP AFTIINGTLK YFETRYIRVD IAAPILSRMV 360
GMISGTTTEX ELWDDWAPYE DVEIGPNGVL RTSSGYKFPL YMIGHGMLDS DLHLSSKAQV 420
FEHPHIQDAA SQLPDDDES LF FGDGTGLSKNP IELVEGW FSS WKSSIASFFF IIGLIIGLFL 480
VLRVGIHL CI KLKHTTKRQI YTDIEMNRLG K 511

SEQ ID NO: 67 moltype = AA length = 511
FEATURE Location/Qualifiers
VARIANT 63
 note = Xaa can be any naturally occurring amino acid except
 Lys or Arg
VARIANT 225
 note = Xaa can be any naturally occurring amino acid except
 Lys or Arg
source 1..511

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mol_type = protein
organism = Vesicular stomatitis virus

SEQUENCE: 67
MKCLLYLAF L FIGVNCKFTI VFPHNQKGNW KNVPSNYHYC PSSSDLNWHN DLIGTAIQVK 60
MPXSHKAIQA DGWMCHASKW VTTCDPRWYG PKYITQSIRS FTPSVEQCKE SIEQTKQGTW 120
LNP GPPQSC GYATVTDAAE VIVQVTPHHV LVDEYTGGEWV DSQFINGKCS NYICPTVHNS 180
TTWHS DYKVK GLCDNLISM DITFFSEDGE LSSLGKEGTG FRSNXFAYET GGKACKMQYC 240
KHWGVRLPSG VWFEMADKDL FAAARFPECP EGSSISAPSQ TSVDVSLIQD VERILDYSLC 300
QETWSKIRAG LPISPVDLSY LAPKNPGTGP AFTIINGTLK YFETRYIRVD IAAPILSRMV 360
GMISGTTTER ELWDDWAPYE DVEIGPNGVL RTSSGYKFPL YMIGHGMLDS DLHLSSKAQV 420
FEHPHIQDAA SQLPDDESLE FGDGTGLSKNP IELVEGWFS WKSSIASFFF IIGLIIGLFL 480
VLRVGIHL CI KLKHTKKRQI YTDIEMNRLG K 511

SEQ ID NO: 68      moltype = AA length = 511
FEATURE           Location/Qualifiers
VARIANT           225
                  note = Xaa can be any naturally occurring amino acid except
                  Lys or Arg
VARIANT           370
                  note = Xaa can be any naturally occurring amino acid except
                  Lys or Arg
source            1..511
                  mol_type = protein
                  organism = Vesicular stomatitis virus

SEQUENCE: 68
MKCLLYLAF L FIGVNCKFTI VFPHNQKGNW KNVPSNYHYC PSSSDLNWHN DLIGTAIQVK 60
MPKSHKAIQA DGWMCHASKW VTTCDPRWYG PKYITQSIRS FTPSVEQCKE SIEQTKQGTW 120
LNP GPPQSC GYATVTDAAE VIVQVTPHHV LVDEYTGGEWV DSQFINGKCS NYICPTVHNS 180
TTWHS DYKVK GLCDNLISM DITFFSEDGE LSSLGKEGTG FRSNXFAYET GGKACKMQYC 240
KHWGVRLPSG VWFEMADKDL FAAARFPECP EGSSISAPSQ TSVDVSLIQD VERILDYSLC 300
QETWSKIRAG LPISPVDLSY LAPKNPGTGP AFTIINGTLK YFETRYIRVD IAAPILSRMV 360
GMISGTTTEX ELWDDWAPYE DVEIGPNGVL RTSSGYKFPL YMIGHGMLDS DLHLSSKAQV 420
FEHPHIQDAA SQLPDDESLE FGDGTGLSKNP IELVEGWFS WKSSIASFFF IIGLIIGLFL 480
VLRVGIHL CI KLKHTKKRQI YTDIEMNRLG K 511

SEQ ID NO: 69      moltype = AA length = 511
FEATURE           Location/Qualifiers
VARIANT           63
                  note = Xaa can be any naturally occurring amino acid except
                  Lys or Arg
VARIANT           225
                  note = Xaa can be any naturally occurring amino acid except
                  Lys or Arg
VARIANT           370
                  note = Xaa can be any naturally occurring amino acid except
                  Lys or Arg
source            1..511
                  mol_type = protein
                  organism = Vesicular stomatitis virus

SEQUENCE: 69
MKCLLYLAF L FIGVNCKFTI VFPHNQKGNW KNVPSNYHYC PSSSDLNWHN DLIGTAIQVK 60
MPXSHKAIQA DGWMCHASKW VTTCDPRWYG PKYITQSIRS FTPSVEQCKE SIEQTKQGTW 120
LNP GPPQSC GYATVTDAAE VIVQVTPHHV LVDEYTGGEWV DSQFINGKCS NYICPTVHNS 180
TTWHS DYKVK GLCDNLISM DITFFSEDGE LSSLGKEGTG FRSNXFAYET GGKACKMQYC 240
KHWGVRLPSG VWFEMADKDL FAAARFPECP EGSSISAPSQ TSVDVSLIQD VERILDYSLC 300
QETWSKIRAG LPISPVDLSY LAPKNPGTGP AFTIINGTLK YFETRYIRVD IAAPILSRMV 360
GMISGTTTEX ELWDDWAPYE DVEIGPNGVL RTSSGYKFPL YMIGHGMLDS DLHLSSKAQV 420
FEHPHIQDAA SQLPDDESLE FGDGTGLSKNP IELVEGWFS WKSSIASFFF IIGLIIGLFL 480
VLRVGIHL CI KLKHTKKRQI YTDIEMNRLG K 511

SEQ ID NO: 70      moltype = AA length = 511
FEATURE           Location/Qualifiers
VARIANT           24
                  note = Xaa can be any naturally occurring amino acid except
                  Lys or Arg
VARIANT           63
                  note = Xaa can be any naturally occurring amino acid except
                  Lys or Arg
VARIANT           225
                  note = Xaa can be any naturally occurring amino acid except
                  Lys or Arg
VARIANT           370
                  note = Xaa can be any naturally occurring amino acid except
                  Lys or Arg
source            1..511
                  mol_type = protein
                  organism = Vesicular stomatitis virus

SEQUENCE: 70
MKCLLYLAF L FIGVNCKFTI VFPXNQKGNW KNVPSNYHYC PSSSDLNWHN DLIGTAIQVK 60

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MPXSHKAIQA DGWMCHASKW VTTCDPRWYG PKYITQSIRS FTPSVEQCKE SIEQTKQGTW 120
LNPGFPPQSC GYATVTDAAE VIVQVTPHHV LVDEYTGEEV DSQFINGKCS NYICPTVHNS 180
TTWHS DYKVK GLCDSNLISM DITFFSEDGE LSSLGKEGTG FRSNXFAYET GKGACKMQYC 240
KHWGVRLP SG VWFEMADKDL FAAARFPECP EGSSISAPSQ TSDVVS LIQD VERILDYSLC 300
QETWSKIRAG LPISPVDSL Y LAPKNPGTGP AFTIINGTLK YFETRYIRVD IAAPILSRMV 360
GMISGTTTEX ELWDDWAPYE DVEIGPNGVL RTSSGYKFPL YMIGHGMLDS DLHLSSKAQV 420
FEHPHIQDAA SQLPDDES LF FGD TGLSKNP IELVEGWFS WKSSIASFFF IIGLIIGLFL 480
VLRVGIHL CI KKKHTKKRQI YTDIEMNRLG K 511

SEQ ID NO: 71      moltype = AA length = 496
FEATURE           Location/Qualifiers
VARIANT           8
                  note = Xaa can be any naturally occurring amino acid except
                  Lys or Arg
VARIANT           47
                  note = Xaa can be any naturally occurring amino acid except
                  Lys or Arg
source            1..496
                  mol_type = protein
                  organism = Vesicular stomatitis virus

SEQUENCE: 71
KFTIVFPXHQ KGNWKNVPST YHYCPSSSDQ NWHNDLTGVS LHVKIPXSHK AIQADGWMCH 60
AAKWVTTCDF RWYGPKYITH SIHMSPTLE QCKTSIEQTK QGVWINPGFP PQSCGYATVT 120
DAEVVVVQAT PHHVLVDEYT GEWIDSQLVG GKCSKEVCQT VHNSTVWHAD YKITGLCESN 180
LASVDITFFS EDGQKTS LGK PNTGFRSNHF AYESGEKACR MQYCTQWGIR LPSGVWFELV 240
DKDLFQA AKL PECPRGSSIS APSQTSVDVS LIQDVERILD YSLCQETWSK IRAKLPVSPV 300
DLSYLAPKNP GSGPAFTIIN GTLKYFETRY IRVDISNP II PHMVGTM SGT TTERELWNDW 360
YPYEDVEIGP NGVLKTP TGF KFLPYMIGHG MLDSDLHKSS QAQVFEHPHA KDAASQLPDD 420
ETLFFGDTGL SKNPVELVEG WFSWKSTLA SFFLIIGLGV ALIFIIRIIV AIRYKYKGRK 480
TQKIYNDVEM SRLGNK 496

SEQ ID NO: 72      moltype = AA length = 496
FEATURE           Location/Qualifiers
VARIANT           8
                  note = Xaa can be any naturally occurring amino acid except
                  Lys or Arg
VARIANT           354
                  note = Xaa can be any naturally occurring amino acid except
                  Lys or Arg
source            1..496
                  mol_type = protein
                  organism = Vesicular stomatitis virus

SEQUENCE: 72
KFTIVFPXHQ KGNWKNVPST YHYCPSSSDQ NWHNDLTGVS LHVKIPXSHK AIQADGWMCH 60
AAKWVTTCDF RWYGPKYITH SIHMSPTLE QCKTSIEQTK QGVWINPGFP PQSCGYATVT 120
DAEVVVVQAT PHHVLVDEYT GEWIDSQLVG GKCSKEVCQT VHNSTVWHAD YKITGLCESN 180
LASVDITFFS EDGQKTS LGK PNTGFRSNHF AYESGEKACR MQYCTQWGIR LPSGVWFELV 240
DKDLFQA AKL PECPRGSSIS APSQTSVDVS LIQDVERILD YSLCQETWSK IRAKLPVSPV 300
DLSYLAPKNP GSGPAFTIIN GTLKYFETRY IRVDISNP II PHMVGTM SGT TTEXELWNDW 360
YPYEDVEIGP NGVLKTP TGF KFLPYMIGHG MLDSDLHKSS QAQVFEHPHA KDAASQLPDD 420
ETLFFGDTGL SKNPVELVEG WFSWKSTLA SFFLIIGLGV ALIFIIRIIV AIRYKYKGRK 480
TQKIYNDVEM SRLGNK 496

SEQ ID NO: 73      moltype = AA length = 496
FEATURE           Location/Qualifiers
VARIANT           8
                  note = Xaa can be any naturally occurring amino acid except
                  Lys or Arg
VARIANT           47
                  note = Xaa can be any naturally occurring amino acid except
                  Lys or Arg
VARIANT           354
                  note = Xaa can be any naturally occurring amino acid except
                  Lys or Arg
source            1..496
                  mol_type = protein
                  organism = Vesicular stomatitis virus

SEQUENCE: 73
KFTIVFPXHQ KGNWKNVPST YHYCPSSSDQ NWHNDLTGVS LHVKIPXSHK AIQADGWMCH 60
AAKWVTTCDF RWYGPKYITH SIHMSPTLE QCKTSIEQTK QGVWINPGFP PQSCGYATVT 120
DAEVVVVQAT PHHVLVDEYT GEWIDSQLVG GKCSKEVCQT VHNSTVWHAD YKITGLCESN 180
LASVDITFFS EDGQKTS LGK PNTGFRSNHF AYESGEKACR MQYCTQWGIR LPSGVWFELV 240
DKDLFQA AKL PECPRGSSIS APSQTSVDVS LIQDVERILD YSLCQETWSK IRAKLPVSPV 300
DLSYLAPKNP GSGPAFTIIN GTLKYFETRY IRVDISNP II PHMVGTM SGT TTEXELWNDW 360
YPYEDVEIGP NGVLKTP TGF KFLPYMIGHG MLDSDLHKSS QAQVFEHPHA KDAASQLPDD 420
ETLFFGDTGL SKNPVELVEG WFSWKSTLA SFFLIIGLGV ALIFIIRIIV AIRYKYKGRK 480
TQKIYNDVEM SRLGNK 496

SEQ ID NO: 74      moltype = AA length = 496

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FEATURE Location/Qualifiers
 VARIANT 47
 note = Xaa can be any naturally occurring amino acid except
 Lys or Arg
 VARIANT 209
 note = Xaa can be any naturally occurring amino acid except
 Lys or Arg
 source 1..496
 mol_type = protein
 organism = Vesicular stomatitis virus

SEQUENCE: 74
 KFTIVFPHHQ KGNWKNVPST YHYCPSSSDQ NWHNDLTGVS LHVKIPXSHK AIQADGWMCH 60
 AAKWVTTCDF RWYGPKYITH SIHMSPTLE QCKTSIEQTK QGVWINPGFP PQSCGYATVT 120
 DAEVVVVQAT PHHVLVDEYT GEWIDSQLVG GKCSKEVCQT VHNSTVWHAD YKITGLCESN 180
 LASVDITFFS EDGQKTSLGK PNTGFRSNXF AYESGEKACR MQYCTQWGIR LPSGVWFELV 240
 DKDLFQAACL PECPRGSSIS APSQTSVDVS LIQDVERILD YSLCQETWSK IRAKLPVSPV 300
 DLSYLAPKNP GSGPAFTIIN GTLKYFETRY IRVDISNPPI PHMVGTMSTG TTERELWNDW 360
 YPYEDVEIGP NGVLKTPTFG KFPLYMIGHG MLDSDLHKSS QAQVFEHPHA KDAASQLPDD 420
 ETLFFGDTGL SKNPVELVEG WFSWKSTLA SFFLIIGLGV ALIFIIRIIV AIRYKYKGRK 480
 TQKIYNDVEM SRLGNK 496

SEQ ID NO: 75 moltype = AA length = 496
 FEATURE Location/Qualifiers
 VARIANT 209
 note = Xaa can be any naturally occurring amino acid except
 Lys or Arg
 VARIANT 354
 note = Xaa can be any naturally occurring amino acid except
 Lys or Arg
 source 1..496
 mol_type = protein
 organism = Vesicular stomatitis virus

SEQUENCE: 75
 KFTIVFPHHQ KGNWKNVPST YHYCPSSSDQ NWHNDLTGVS LHVKIPXSHK AIQADGWMCH 60
 AAKWVTTCDF RWYGPKYITH SIHMSPTLE QCKTSIEQTK QGVWINPGFP PQSCGYATVT 120
 DAEVVVVQAT PHHVLVDEYT GEWIDSQLVG GKCSKEVCQT VHNSTVWHAD YKITGLCESN 180
 LASVDITFFS EDGQKTSLGK PNTGFRSNXF AYESGEKACR MQYCTQWGIR LPSGVWFELV 240
 DKDLFQAACL PECPRGSSIS APSQTSVDVS LIQDVERILD YSLCQETWSK IRAKLPVSPV 300
 DLSYLAPKNP GSGPAFTIIN GTLKYFETRY IRVDISNPPI PHMVGTMSTG TTEXELWNDW 360
 YPYEDVEIGP NGVLKTPTFG KFPLYMIGHG MLDSDLHKSS QAQVFEHPHA KDAASQLPDD 420
 ETLFFGDTGL SKNPVELVEG WFSWKSTLA SFFLIIGLGV ALIFIIRIIV AIRYKYKGRK 480
 TQKIYNDVEM SRLGNK 496

SEQ ID NO: 76 moltype = AA length = 496
 FEATURE Location/Qualifiers
 VARIANT 47
 note = Xaa can be any naturally occurring amino acid except
 Lys or Arg
 VARIANT 209
 note = Xaa can be any naturally occurring amino acid except
 Lys or Arg
 VARIANT 354
 note = Xaa can be any naturally occurring amino acid except
 Lys or Arg
 source 1..496
 mol_type = protein
 organism = Vesicular stomatitis virus

SEQUENCE: 76
 KFTIVFPHHQ KGNWKNVPST YHYCPSSSDQ NWHNDLTGVS LHVKIPXSHK AIQADGWMCH 60
 AAKWVTTCDF RWYGPKYITH SIHMSPTLE QCKTSIEQTK QGVWINPGFP PQSCGYATVT 120
 DAEVVVVQAT PHHVLVDEYT GEWIDSQLVG GKCSKEVCQT VHNSTVWHAD YKITGLCESN 180
 LASVDITFFS EDGQKTSLGK PNTGFRSNXF AYESGEKACR MQYCTQWGIR LPSGVWFELV 240
 DKDLFQAACL PECPRGSSIS APSQTSVDVS LIQDVERILD YSLCQETWSK IRAKLPVSPV 300
 DLSYLAPKNP GSGPAFTIIN GTLKYFETRY IRVDISNPPI PHMVGTMSTG TTEXELWNDW 360
 YPYEDVEIGP NGVLKTPTFG KFPLYMIGHG MLDSDLHKSS QAQVFEHPHA KDAASQLPDD 420
 ETLFFGDTGL SKNPVELVEG WFSWKSTLA SFFLIIGLGV ALIFIIRIIV AIRYKYKGRK 480
 TQKIYNDVEM SRLGNK 496

SEQ ID NO: 77 moltype = AA length = 496
 FEATURE Location/Qualifiers
 VARIANT 8
 note = Xaa can be any naturally occurring amino acid except
 Lys or Arg
 VARIANT 47
 note = Xaa can be any naturally occurring amino acid except
 Lys or Arg
 VARIANT 209
 note = Xaa can be any naturally occurring amino acid except
 Lys or Arg

-continued

VARIANT 354
note = Xaa can be any naturally occurring amino acid except
Lys or Arg

source 1..496
mol_type = protein
organism = Vesicular stomatitis virus

SEQUENCE: 77

KFTIVFPXHQ	KGNWKNVPST	YHYCPSSSDQ	NWHNDLTGVS	LHVKIPXSHK	AIQADGWMCH	60
AAKWVTTCDF	RWYGPKYITH	SIHMSPTLE	QCKTSIEQTK	QGVWINPGFP	PQSCGYATVT	120
DAEVVVVQAT	PHHVLVDEYT	GEWIDSQLVG	GKCSKEVCQT	VHNSTVWHAD	YKITGLCESN	180
LASVDITFFS	EDGQKTSLGK	PNTGFRSNXF	AYESGEKACR	MQYCTQWGIR	LPSGVWFELV	240
DKDLFQAACL	PECPRGSSIS	APSQTSVDVS	LIQDVERILD	YSLCQETWSK	IRAKLPVSPV	300
DLSYLAPKNP	GSGPAFTIIN	GTKLYFETRY	IRVDISNPPI	PHMVGTMSTG	TTEXELWNDW	360
YPYEDVEIGP	NGVLKTPITG	KFPLYMIGHG	MLDSDLHKSS	QAQVFEHPHA	KDAASQLPDD	420
ETLFFGDTGL	SKNPVELVEG	WFSSWKSTLA	SFFLIIGLV	ALIFIIRIIV	AIRYKYKGRK	480
TQKIYNDVEM	SRLGNK					496

SEQ ID NO: 78 moltype = AA length = 512
FEATURE Location/Qualifiers
VARIANT 24
note = Xaa can be any naturally occurring amino acid except
Lys or Arg

VARIANT 63
note = Xaa can be any naturally occurring amino acid except
Lys or Arg

source 1..512
mol_type = protein
organism = Vesicular stomatitis virus

SEQUENCE: 78

MLRLFLFCFL	ALGAHSKFTI	VFPXHQKGNW	KNVPSTYHYC	PSSSDQNWHN	DLTGVSLSHVK	60
IPXSHKAIQA	DGWMCHAAKW	VTTCDFRWYG	PKYITHSIHS	MSPTLEQCKT	SIEQTKQGVW	120
INPGPPPQSC	GYATVTDAAV	VVVQATPHHV	LVDEYTGWEI	DSQLVGGKCS	KEVCQTVHNS	180
TVWHADYKIT	GLCESNLASV	DITFFSEDGQ	KTSLGKPNTE	FRSNHFAYES	GEKACRMQYC	240
TQWGIRLPSG	VWFELVDKDL	FQAAKLPECP	RGSSISAPSQ	TSVDVSLIQD	VERILDYSLC	300
QETWSKIRAK	LPVSPVDLSY	LAPKNPGSGP	AFTIINGTLK	YFETRYIRVD	ISNPIIPHMV	360
GTMSGTTTTER	ELWNDWYPYE	DVEIGPNGVL	KTPTGFKFPL	YMIGHGMLDS	DLHKSSQAQV	420
FEHPHAKDAA	SQLPDDDTLF	FGDTGLSKNP	VELVEGWFS	WKSTLASFFL	IIGLGVALIF	480
IIRIIVAIKY	KYKGRKTKI	YNDVEMSRLG	NK			512

SEQ ID NO: 79 moltype = AA length = 512
FEATURE Location/Qualifiers
VARIANT 24
note = Xaa can be any naturally occurring amino acid except
Lys or Arg

VARIANT 370
note = Xaa can be any naturally occurring amino acid except
Lys or Arg

source 1..512
mol_type = protein
organism = Vesicular stomatitis virus

SEQUENCE: 79

MLRLFLFCFL	ALGAHSKFTI	VFPXHQKGNW	KNVPSTYHYC	PSSSDQNWHN	DLTGVSLSHVK	60
IPKSHKAIQA	DGWMCHAAKW	VTTCDFRWYG	PKYITHSIHS	MSPTLEQCKT	SIEQTKQGVW	120
INPGPPPQSC	GYATVTDAAV	VVVQATPHHV	LVDEYTGWEI	DSQLVGGKCS	KEVCQTVHNS	180
TVWHADYKIT	GLCESNLASV	DITFFSEDGQ	KTSLGKPNTE	FRSNHFAYES	GEKACRMQYC	240
TQWGIRLPSG	VWFELVDKDL	FQAAKLPECP	RGSSISAPSQ	TSVDVSLIQD	VERILDYSLC	300
QETWSKIRAK	LPVSPVDLSY	LAPKNPGSGP	AFTIINGTLK	YFETRYIRVD	ISNPIIPHMV	360
GTMSGTTTTER	ELWNDWYPYE	DVEIGPNGVL	KTPTGFKFPL	YMIGHGMLDS	DLHKSSQAQV	420
FEHPHAKDAA	SQLPDDDTLF	FGDTGLSKNP	VELVEGWFS	WKSTLASFFL	IIGLGVALIF	480
IIRIIVAIKY	KYKGRKTKI	YNDVEMSRLG	NK			512

SEQ ID NO: 80 moltype = AA length = 512
FEATURE Location/Qualifiers
VARIANT 24
note = Xaa can be any naturally occurring amino acid except
Lys or Arg

VARIANT 63
note = Xaa can be any naturally occurring amino acid except
Lys or Arg

VARIANT 370
note = Xaa can be any naturally occurring amino acid except
Lys or Arg

source 1..512
mol_type = protein
organism = Vesicular stomatitis virus

SEQUENCE: 80

MLRLFLFCFL	ALGAHSKFTI	VFPXHQKGNW	KNVPSTYHYC	PSSSDQNWHN	DLTGVSLSHVK	60
IPXSHKAIQA	DGWMCHAAKW	VTTCDFRWYG	PKYITHSIHS	MSPTLEQCKT	SIEQTKQGVW	120
INPGPPPQSC	GYATVTDAAV	VVVQATPHHV	LVDEYTGWEI	DSQLVGGKCS	KEVCQTVHNS	180

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TVWHADYKIT	GLCESNLASV	DITFFSEDGQ	KTSLGKPNTG	FRSNHFAYES	GEKACRMQYC	240
TQWGIRLPSG	VWFELVDKDL	FQAAKLPECP	RGSSISAPSQ	TSVDVSLIQD	VERILDYSLC	300
QETWSKIRAK	LPVSPVDLSY	LAPKNPGSGP	AFTIINGTLK	YFETRYIRVD	ISNPIIPHMV	360
GTMSGTTTEX	ELWNDWYPYE	DVEIGPNGVL	KTPTGFKFPL	YMIGHGMLDS	DLHKSSQAQV	420
FEHPHAKDAA	SQLPDDETLF	FGDTGLSKNP	VELVEGWFS	WKSTLASFFL	IIGLGVALIF	480
IIRIIVAIRY	KYKGRKTQKI	YNDVEMSRLG	NK			512

SEQ ID NO: 81 moltype = AA length = 512
FEATURE Location/Qualifiers
VARIANT 63
 note = Xaa can be any naturally occurring amino acid except
 Lys or Arg
VARIANT 225
 note = Xaa can be any naturally occurring amino acid except
 Lys or Arg
source 1..512
 mol_type = protein
 organism = Vesicular stomatitis virus

SEQUENCE: 81

MLRLFLFCFL	ALGAHSKFTI	VFPHHQKGNW	KNVPSTYHYC	PSSSDQNWHN	DLTGVSLHVK	60
IPXSHKAIQA	DGWMCHAAKW	VTTCDPRWYG	PKYITHSIHS	MSPTLEQCKT	SIEQTKQGVW	120
INPGPPPQSC	GYATVTDAEV	VVVQATPHHV	LVDEYTGWEI	DSQLVGGKCS	KEVCQTVHNS	180
TVWHADYKIT	GLCESNLASV	DITFFSEDGQ	KTSLGKPNTG	FRSNXFAYES	GEKACRMQYC	240
TQWGIRLPSG	VWFELVDKDL	FQAAKLPECP	RGSSISAPSQ	TSVDVSLIQD	VERILDYSLC	300
QETWSKIRAK	LPVSPVDLSY	LAPKNPGSGP	AFTIINGTLK	YFETRYIRVD	ISNPIIPHMV	360
GTMSGTTTER	ELWNDWYPYE	DVEIGPNGVL	KTPTGFKFPL	YMIGHGMLDS	DLHKSSQAQV	420
FEHPHAKDAA	SQLPDDETLF	FGDTGLSKNP	VELVEGWFS	WKSTLASFFL	IIGLGVALIF	480
IIRIIVAIRY	KYKGRKTQKI	YNDVEMSRLG	NK			512

SEQ ID NO: 82 moltype = AA length = 512
FEATURE Location/Qualifiers
VARIANT 225
 note = Xaa can be any naturally occurring amino acid except
 Lys or Arg
VARIANT 370
 note = Xaa can be any naturally occurring amino acid except
 Lys or Arg
source 1..512
 mol_type = protein
 organism = Vesicular stomatitis virus

SEQUENCE: 82

MLRLFLFCFL	ALGAHSKFTI	VFPHHQKGNW	KNVPSTYHYC	PSSSDQNWHN	DLTGVSLHVK	60
IPKSHKAIQA	DGWMCHAAKW	VTTCDPRWYG	PKYITHSIHS	MSPTLEQCKT	SIEQTKQGVW	120
INPGPPPQSC	GYATVTDAEV	VVVQATPHHV	LVDEYTGWEI	DSQLVGGKCS	KEVCQTVHNS	180
TVWHADYKIT	GLCESNLASV	DITFFSEDGQ	KTSLGKPNTG	FRSNXFAYES	GEKACRMQYC	240
TQWGIRLPSG	VWFELVDKDL	FQAAKLPECP	RGSSISAPSQ	TSVDVSLIQD	VERILDYSLC	300
QETWSKIRAK	LPVSPVDLSY	LAPKNPGSGP	AFTIINGTLK	YFETRYIRVD	ISNPIIPHMV	360
GTMSGTTTEX	ELWNDWYPYE	DVEIGPNGVL	KTPTGFKFPL	YMIGHGMLDS	DLHKSSQAQV	420
FEHPHAKDAA	SQLPDDETLF	FGDTGLSKNP	VELVEGWFS	WKSTLASFFL	IIGLGVALIF	480
IIRIIVAIRY	KYKGRKTQKI	YNDVEMSRLG	NK			512

SEQ ID NO: 83 moltype = AA length = 512
FEATURE Location/Qualifiers
VARIANT 63
 note = Xaa can be any naturally occurring amino acid except
 Lys or Arg
VARIANT 225
 note = Xaa can be any naturally occurring amino acid except
 Lys or Arg
VARIANT 370
 note = Xaa can be any naturally occurring amino acid except
 Lys or Arg
source 1..512
 mol_type = protein
 organism = Vesicular stomatitis virus

SEQUENCE: 83

MLRLFLFCFL	ALGAHSKFTI	VFPHHQKGNW	KNVPSTYHYC	PSSSDQNWHN	DLTGVSLHVK	60
IPXSHKAIQA	DGWMCHAAKW	VTTCDPRWYG	PKYITHSIHS	MSPTLEQCKT	SIEQTKQGVW	120
INPGPPPQSC	GYATVTDAEV	VVVQATPHHV	LVDEYTGWEI	DSQLVGGKCS	KEVCQTVHNS	180
TVWHADYKIT	GLCESNLASV	DITFFSEDGQ	KTSLGKPNTG	FRSNXFAYES	GEKACRMQYC	240
TQWGIRLPSG	VWFELVDKDL	FQAAKLPECP	RGSSISAPSQ	TSVDVSLIQD	VERILDYSLC	300
QETWSKIRAK	LPVSPVDLSY	LAPKNPGSGP	AFTIINGTLK	YFETRYIRVD	ISNPIIPHMV	360
GTMSGTTTEX	ELWNDWYPYE	DVEIGPNGVL	KTPTGFKFPL	YMIGHGMLDS	DLHKSSQAQV	420
FEHPHAKDAA	SQLPDDETLF	FGDTGLSKNP	VELVEGWFS	WKSTLASFFL	IIGLGVALIF	480
IIRIIVAIRY	KYKGRKTQKI	YNDVEMSRLG	NK			512

SEQ ID NO: 84 moltype = AA length = 512
FEATURE Location/Qualifiers
VARIANT 24

-continued

note = Xaa can be any naturally occurring amino acid except
 Lys or Arg
 VARIANT 63
 note = Xaa can be any naturally occurring amino acid except
 Lys or Arg
 VARIANT 225
 note = Xaa can be any naturally occurring amino acid except
 Lys or Arg
 VARIANT 370
 note = Xaa can be any naturally occurring amino acid except
 Lys or Arg
 source 1..512
 mol_type = protein
 organism = Vesicular stomatitis virus
 SEQUENCE: 84
 MLRLFLFCFL ALGAHSKFTI VFPXHQKGNW KNVPSTYHYC PSSSDQNWHN DLTGVS LHVK 60
 IPXSHKAIQA DGWMCHAAKW VTTCDFRWYG PKYITHSIHS MSPBLEQCKT SIEQTKQGVW 120
 INPGFPQSC GYATVTDAEV VVQATPHHV LVDEYTGWEI DSQLVGGKCS KEVCQTVHNS 180
 TVWHADYKIT GLCESNLASV DITFFSEDGQ KTSLGKPN TG FRSNXFAYES GEKACRMQYC 240
 TQWGIRLPSG VWFELVDKDL FQAAKLPECP RGSSISAPSQ TSVDSVLIQD VERILDYSLC 300
 QETWSKIRAK LPVSPVDLSY LAPKNPGSGP AFTIINGTLK YFETRYIRVD ISNPIIPHMV 360
 GTMSGTTTEX ELWNDWYPYE DVEIGPNGVL KTPTGFKFPL YMIGHGMLDS DLHKSSQAQV 420
 FEHPHAKDAA SQLPDDETLF FGD TGLSKNP VELVEGWFS WKSTLASFFL IIGLGVALIF 480
 IIRIIVAIRY KYKGRKTQKI YNDVEMSRLG NK 512
 SEQ ID NO: 85 moltype = AA length = 501
 FEATURE Location/Qualifiers
 VARIANT 9
 note = Xaa can be any naturally occurring amino acid except
 Lys or Arg
 VARIANT 47
 note = Xaa can be any naturally occurring amino acid except
 Lys or Arg
 source 1..501
 mol_type = protein
 organism = Vesicular stomatitis virus
 SEQUENCE: 85
 KIEIVFPQXT TGDWKRVPHE YNYCPTSADK NSHGTQTGIP VELTMPXGLT THQVDGFMCH 60
 SALWMTTCDF RWYGPKYITH SIHNEEPTDY QCLEAIKAYK DGVSFNPGFP PQSCGYGTVT 120
 DAEAHIVTVT PHSVKVDEYT GEWIDPHFIG GRCKGQICET VHNSTKWFTS SDGESVCSQL 180
 FTLVGGTFPS DSEEITSMGL PETGIRSNYF PYVSTEGICK MPFCRKPGYK LKNDLWFQIT 240
 DPDLDKTVRD LPHIKDCDLS SSI VTPGEHA TDISLISDVE RILDYALCQN TWSKIEAGEP 300
 ITPVDLSYLG PKNPGAGPVF TIINGSLHYF MSKYLRVELE SPVIPRMEGK VAGTRIVRQL 360
 WDQWFPFGEV EIGPNGVLKT KQGYKFPLHI IGTGEVDNDI KMERIVKHWE HPHIEAAQTF 420
 LKKDDTEEV L YYDGTGVSKN PVELVEGWFS GWRSSIMGVL AVIIGFVILI FLIRLIGVLS 480
 SLFRQKRRPI YKSDVEMAHF R 501
 SEQ ID NO: 86 moltype = AA length = 501
 FEATURE Location/Qualifiers
 VARIANT 9
 note = Xaa can be any naturally occurring amino acid except
 Lys or Arg
 VARIANT 358
 note = Xaa can be any naturally occurring amino acid except
 Lys or Arg
 source 1..501
 mol_type = protein
 organism = Vesicular stomatitis virus
 SEQUENCE: 86
 KIEIVFPQXT TGDWKRVPHE YNYCPTSADK NSHGTQTGIP VELTMPKGLT THQVDGFMCH 60
 SALWMTTCDF RWYGPKYITH SIHNEEPTDY QCLEAIKAYK DGVSFNPGFP PQSCGYGTVT 120
 DAEAHIVTVT PHSVKVDEYT GEWIDPHFIG GRCKGQICET VHNSTKWFTS SDGESVCSQL 180
 FTLVGGTFPS DSEEITSMGL PETGIRSNYF PYVSTEGICK MPFCRKPGYK LKNDLWFQIT 240
 DPDLDKTVRD LPHIKDCDLS SSI VTPGEHA TDISLISDVE RILDYALCQN TWSKIEAGEP 300
 ITPVDLSYLG PKNPGAGPVF TIINGSLHYF MSKYLRVELE SPVIPRMEGK VAGTRIVXQL 360
 WDQWFPFGEV EIGPNGVLKT KQGYKFPLHI IGTGEVDNDI KMERIVKHWE HPHIEAAQTF 420
 LKKDDTEEV L YYDGTGVSKN PVELVEGWFS GWRSSIMGVL AVIIGFVILI FLIRLIGVLS 480
 SLFRQKRRPI YKSDVEMAHF R 501
 SEQ ID NO: 87 moltype = AA length = 501
 FEATURE Location/Qualifiers
 VARIANT 9
 note = Xaa can be any naturally occurring amino acid except
 Lys or Arg
 VARIANT 47
 note = Xaa can be any naturally occurring amino acid except
 Lys or Arg
 VARIANT 358
 note = Xaa can be any naturally occurring amino acid except

-continued

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source          Lys or Arg
                1..501
                mol_type = protein
                organism = Vesicular stomatitis virus

SEQUENCE: 87
KIEIVFPQXT  TGDWKRVPHE  YNYCPTSADK  NSHGTQTGIP  VELTMPXGLT  THQVDGFMCH  60
SALWMTTCDF  RWYGPKYITH  SIHNEEPTDY  QCLEAIKAYK  DGVSFNPGEF  PQSCGYGTVT  120
DAEAHIVTVT  PHSVKVDEYT  GEWIDPHFIG  GRCKGQICET  VHNSTKWFTS  SDGESVCSQL  180
FTLVGGTFFS  DSEETSMGL  PETGIRSNYF  PYVSTEGICK  MPFCRKPGYK  LKNDLWFQIT  240
DPDLDKTVRD  LPHIKDCDLS  SSIIVTPGEHA  TDISLISDVE  RILDYALCQN  TWSKIEAGEP  300
ITPVDSL SYLG  PKNPGAGPVF  TIINGSLHYF  MSKYL RVELE  SPVIPRMEGK  VAGTRIVXQL  360
WDQWFPFGEV  EIGPNGVLKT  KQGYKFPLHI  IGTGEVDNDI  KMERIVKHWE  HPHIEAAQTF  420
LKKDDTEEV L  YYGDTGVSKN  PVELVEGWFS  GWRSSIMGVL  AVIIGFVILI  FLIRLIGVLS  480
SLFRQKRRPI  YKSDVEMAHF  R  501

SEQ ID NO: 88      moltype = AA  length = 501
FEATURE           Location/Qualifiers
VARIANT           47
                  note = Xaa can be any naturally occurring amino acid except
                  Lys or Arg
source            1..501
                  mol_type = protein
                  organism = Vesicular stomatitis virus

SEQUENCE: 88
KIEIVFPQHT  TGDWKRVPHE  YNYCPTSADK  NSHGTQTGIP  VELTMPXGLT  THQVDGFMCH  60
SALWMTTCDF  RWYGPKYITH  SIHNEEPTDY  QCLEAIKAYK  DGVSFNPGEF  PQSCGYGTVT  120
DAEAHIVTVT  PHSVKVDEYT  GEWIDPHFIG  GRCKGQICET  VHNSTKWFTS  SDGESVCSQL  180
FTLVGGTFFS  DSEETSMGL  PETGIRSNYF  PYVSTEGICK  MPFCRKPGYK  LKNDLWFQIT  240
DPDLDKTVRD  LPHIKDCDLS  SSIIVTPGEHA  TDISLISDVE  RILDYALCQN  TWSKIEAGEP  300
ITPVDSL SYLG  PKNPGAGPVF  TIINGSLHYF  MSKYL RVELE  SPVIPRMEGK  VAGTRIVRQL  360
WDQWFPFGEV  EIGPNGVLKT  KQGYKFPLHI  IGTGEVDNDI  KMERIVKHWE  HPHIEAAQTF  420
LKKDDTEEV L  YYGDTGVSKN  PVELVEGWFS  GWRSSIMGVL  AVIIGFVILI  FLIRLIGVLS  480
SLFRQKRRPI  YKSDVEMAHF  R  501

SEQ ID NO: 89      moltype = AA  length = 501
FEATURE           Location/Qualifiers
VARIANT           209
                  note = Xaa can be any naturally occurring amino acid except
                  Lys or Arg
VARIANT           358
                  note = Xaa can be any naturally occurring amino acid except
                  Lys or Arg
source            1..501
                  mol_type = protein
                  organism = Vesicular stomatitis virus

SEQUENCE: 89
KIEIVFPQHT  TGDWKRVPHE  YNYCPTSADK  NSHGTQTGIP  VELTMPKGLT  THQVDGFMCH  60
SALWMTTCDF  RWYGPKYITH  SIHNEEPTDY  QCLEAIKAYK  DGVSFNPGEF  PQSCGYGTVT  120
DAEAHIVTVT  PHSVKVDEYT  GEWIDPHFIG  GRCKGQICET  VHNSTKWFTS  SDGESVCSQL  180
FTLVGGTFFS  DSEETSMGL  PETGIRSNXF  PYVSTEGICK  MPFCRKPGYK  LKNDLWFQIT  240
DPDLDKTVRD  LPHIKDCDLS  SSIIVTPGEHA  TDISLISDVE  RILDYALCQN  TWSKIEAGEP  300
ITPVDSL SYLG  PKNPGAGPVF  TIINGSLHYF  MSKYL RVELE  SPVIPRMEGK  VAGTRIVXQL  360
WDQWFPFGEV  EIGPNGVLKT  KQGYKFPLHI  IGTGEVDNDI  KMERIVKHWE  HPHIEAAQTF  420
LKKDDTEEV L  YYGDTGVSKN  PVELVEGWFS  GWRSSIMGVL  AVIIGFVILI  FLIRLIGVLS  480
SLFRQKRRPI  YKSDVEMAHF  R  501

SEQ ID NO: 90      moltype = AA  length = 501
FEATURE           Location/Qualifiers
VARIANT           47
                  note = Xaa can be any naturally occurring amino acid except
                  Lys or Arg
VARIANT           209
                  note = Xaa can be any naturally occurring amino acid except
                  Lys or Arg
VARIANT           358
                  note = Xaa can be any naturally occurring amino acid except
                  Lys or Arg
source            1..501
                  mol_type = protein
                  organism = Vesicular stomatitis virus

SEQUENCE: 90
KIEIVFPQHT  TGDWKRVPHE  YNYCPTSADK  NSHGTQTGIP  VELTMPXGLT  THQVDGFMCH  60
SALWMTTCDF  RWYGPKYITH  SIHNEEPTDY  QCLEAIKAYK  DGVSFNPGEF  PQSCGYGTVT  120
DAEAHIVTVT  PHSVKVDEYT  GEWIDPHFIG  GRCKGQICET  VHNSTKWFTS  SDGESVCSQL  180
FTLVGGTFFS  DSEETSMGL  PETGIRSNXF  PYVSTEGICK  MPFCRKPGYK  LKNDLWFQIT  240
DPDLDKTVRD  LPHIKDCDLS  SSIIVTPGEHA  TDISLISDVE  RILDYALCQN  TWSKIEAGEP  300
ITPVDSL SYLG  PKNPGAGPVF  TIINGSLHYF  MSKYL RVELE  SPVIPRMEGK  VAGTRIVXQL  360
WDQWFPFGEV  EIGPNGVLKT  KQGYKFPLHI  IGTGEVDNDI  KMERIVKHWE  HPHIEAAQTF  420
LKKDDTEEV L  YYGDTGVSKN  PVELVEGWFS  GWRSSIMGVL  AVIIGFVILI  FLIRLIGVLS  480

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SLFRQKRRPI YKSDVEMAHF R 501

SEQ ID NO: 91 moltype = AA length = 501
 FEATURE Location/Qualifiers
 VARIANT 9
 note = Xaa can be any naturally occurring amino acid except
 Lys or Arg
 VARIANT 47
 note = Xaa can be any naturally occurring amino acid except
 Lys or Arg
 VARIANT 209
 note = Xaa can be any naturally occurring amino acid except
 Lys or Arg
 VARIANT 358
 note = Xaa can be any naturally occurring amino acid except
 Lys or Arg
 source 1..501
 mol_type = protein
 organism = Vesicular stomatitis virus

SEQUENCE: 91
 KIEIVFPQXT TGDWKRVPHE YNYCPTSADK NSHGTQTGIP VELTMPXGLT THQVDGFMCH 60
 SALWMTTCDF RWYGPKYITH SIHNEPTDY QCLEAIKAYK DGVSFNPGFP PQSCGYGTVT 120
 DAEAHIVTVT PHSVKVDEYT GEWIDPHFIG GRCKGQICET VHNSTKWFTS SDGESVCSQL 180
 FTLVGGTFFS DSEEITSMGL PETGIRSNXF PYVSTEGICK MPFCRKPGYK LKNDLWFQIT 240
 DPDLDKTVRD LPHIKDCDLS SSIPTGGEHA TDISLISDVE RILDYALCQN TWSKIEAGEP 300
 ITPVDLSYLG PKNPGAGPVF TIINGSLHYF MSKYLRLVELE SPVIPRMEGK VAGTRIVXQL 360
 WDQWFPFGEV EIGPNGVLKT KQGYKFPLHI IGTGEVDNDI KMERIVKHWE HPHIEAAQTF 420
 LKKDDTEEVLY YGDTGVSKN PVELVEGWFS GWRSSIMGVL AVIIGFVILI FLIRLIGVLS 480
 SLFRQKRRPI YKSDVEMAHF R 501

SEQ ID NO: 92 moltype = AA length = 517
 FEATURE Location/Qualifiers
 VARIANT 25
 note = Xaa can be any naturally occurring amino acid except
 Lys or Arg
 VARIANT 63
 note = Xaa can be any naturally occurring amino acid except
 Lys or Arg
 source 1..517
 mol_type = protein
 organism = Vesicular stomatitis virus

SEQUENCE: 92
 MLSYLIFALV VSPILGKIEI VFPQXTTGDW KRPVPEYNYC PTSADKNSHG TQTGIPVELT 60
 MPXGLTTHQV DGFPMCHSALW MTTCDPRWYG PKYITHSIHN EEPTDYQCLE AIKAYKDGV 120
 FNPGFPPQSC GYGTVTDAEA HIVTVTPHSV KVDEYTGEWI DPHFIGGRCK GQICETVHNS 180
 TKWFTSSDGE SVCSQLFTLV GGTFFSDSEE ITSMGLPETG IRSNYFFPVVS TEGICKMPFC 240
 RKPQGYKLKND LWFQITDPLD DKTVRDLPHI KCDLSSSIV TPGEHATDIS LISDVERILD 300
 YALCQNTWSK IEAGEPITPV DLSYLGPKNP GAGPVFTIIN GSLHYFMSKY LRVELESPVI 360
 PRMEGKVAGT RIVRQLWDQW FPFGEVEIGP NGVLKTKQGY KFPLHIIGTG EVDNDIKMER 420
 IVKHWEHPHI EAAQTFLKKD DTEEVLYYGD TGVSKNPVEL VEGWFSGWRS SIMGVLAVII 480
 GFVILIFLIR LIGVLSLFR QKRRPIYKSD VEMAHF 517

SEQ ID NO: 93 moltype = AA length = 517
 FEATURE Location/Qualifiers
 VARIANT 25
 note = Xaa can be any naturally occurring amino acid except
 Lys or Arg
 VARIANT 374
 note = Xaa can be any naturally occurring amino acid except
 Lys or Arg
 source 1..517
 mol_type = protein
 organism = Vesicular stomatitis virus

SEQUENCE: 93
 MLSYLIFALV VSPILGKIEI VFPQXTTGDW KRPVPEYNYC PTSADKNSHG TQTGIPVELT 60
 MPKGLTTHQV DGFPMCHSALW MTTCDPRWYG PKYITHSIHN EEPTDYQCLE AIKAYKDGV 120
 FNPGFPPQSC GYGTVTDAEA HIVTVTPHSV KVDEYTGEWI DPHFIGGRCK GQICETVHNS 180
 TKWFTSSDGE SVCSQLFTLV GGTFFSDSEE ITSMGLPETG IRSNYFFPVVS TEGICKMPFC 240
 RKPQGYKLKND LWFQITDPLD DKTVRDLPHI KCDLSSSIV TPGEHATDIS LISDVERILD 300
 YALCQNTWSK IEAGEPITPV DLSYLGPKNP GAGPVFTIIN GSLHYFMSKY LRVELESPVI 360
 PRMEGKVAGT RIVRQLWDQW FPFGEVEIGP NGVLKTKQGY KFPLHIIGTG EVDNDIKMER 420
 IVKHWEHPHI EAAQTFLKKD DTEEVLYYGD TGVSKNPVEL VEGWFSGWRS SIMGVLAVII 480
 GFVILIFLIR LIGVLSLFR QKRRPIYKSD VEMAHF 517

SEQ ID NO: 94 moltype = AA length = 517
 FEATURE Location/Qualifiers
 VARIANT 25
 note = Xaa can be any naturally occurring amino acid except
 Lys or Arg

-continued

VARIANT 63
note = Xaa can be any naturally occurring amino acid except
Lys or Arg

VARIANT 374
note = Xaa can be any naturally occurring amino acid except
Lys or Arg

source 1..517
mol_type = protein
organism = Vesicular stomatitis virus

SEQUENCE: 94

MLSYLIFALV	VSPILGKIEI	VFPQXTTGDW	KRVPHEYNYC	PTSADKNSHG	TQTGIPVELT	60
MPXGLTTHQV	DGFMCHSALW	MTTCDFRWYG	PKYITHSIHN	EEPTDYQCLE	AIKAYKDGVS	120
FNPGFPPQSC	GYGTVTDAEA	HIVTVTPHSV	KVDEYTGWEI	DPHFIGGRCK	GQICETVHNS	180
TKWFTSSDGE	SVCSQLFTLV	GGTFFSDSEE	ITSMGLPETG	IRSNYFPYVS	TEGICKMPFC	240
RKPGYKLNKD	LWFQITDPLD	DKTVRDLPHI	KCDLSSSIV	TPGEHATDIS	LISDVERILD	300
YALCQNTWSK	IEAGEPITPV	DLSYLGPKNP	GAGPVFTIIN	GSLHYFMSKY	LRVELESPVI	360
PRMEGKVAGT	RIVXQLWDQW	FPFGEVEIGP	NGVLKTKQGY	KFPLHIIGTG	EVDNDIKMER	420
IVKHWEHPHI	EAAQTFLKDK	DTEEVLYYGD	TGVSKNPVEL	VEGWFSGWRS	SIMGVLAVII	480
GFVILIFLIR	LIGVLSLFR	QKRRPIYKSD	VEMAHFR			517

SEQ ID NO: 95 moltype = AA length = 517
FEATURE Location/Qualifiers
VARIANT 63
note = Xaa can be any naturally occurring amino acid except
Lys or Arg

VARIANT 225
note = Xaa can be any naturally occurring amino acid except
Lys or Arg

source 1..517
mol_type = protein
organism = Vesicular stomatitis virus

SEQUENCE: 95

MLSYLIFALV	VSPILGKIEI	VFPQHTTGDW	KRVPHEYNYC	PTSADKNSHG	TQTGIPVELT	60
MPXGLTTHQV	DGFMCHSALW	MTTCDFRWYG	PKYITHSIHN	EEPTDYQCLE	AIKAYKDGVS	120
FNPGFPPQSC	GYGTVTDAEA	HIVTVTPHSV	KVDEYTGWEI	DPHFIGGRCK	GQICETVHNS	180
TKWFTSSDGE	SVCSQLFTLV	GGTFFSDSEE	ITSMGLPETG	IRSNXFPYVS	TEGICKMPFC	240
RKPGYKLNKD	LWFQITDPLD	DKTVRDLPHI	KCDLSSSIV	TPGEHATDIS	LISDVERILD	300
YALCQNTWSK	IEAGEPITPV	DLSYLGPKNP	GAGPVFTIIN	GSLHYFMSKY	LRVELESPVI	360
PRMEGKVAGT	RIVRQLWDQW	FPFGEVEIGP	NGVLKTKQGY	KFPLHIIGTG	EVDNDIKMER	420
IVKHWEHPHI	EAAQTFLKDK	DTEEVLYYGD	TGVSKNPVEL	VEGWFSGWRS	SIMGVLAVII	480
GFVILIFLIR	LIGVLSLFR	QKRRPIYKSD	VEMAHFR			517

SEQ ID NO: 96 moltype = AA length = 517
FEATURE Location/Qualifiers
VARIANT 225
note = Xaa can be any naturally occurring amino acid except
Lys or Arg

VARIANT 374
note = Xaa can be any naturally occurring amino acid except
Lys or Arg

source 1..517
mol_type = protein
organism = Vesicular stomatitis virus

SEQUENCE: 96

MLSYLIFALV	VSPILGKIEI	VFPQHTTGDW	KRVPHEYNYC	PTSADKNSHG	TQTGIPVELT	60
MPKGLTTHQV	DGFMCHSALW	MTTCDFRWYG	PKYITHSIHN	EEPTDYQCLE	AIKAYKDGVS	120
FNPGFPPQSC	GYGTVTDAEA	HIVTVTPHSV	KVDEYTGWEI	DPHFIGGRCK	GQICETVHNS	180
TKWFTSSDGE	SVCSQLFTLV	GGTFFSDSEE	ITSMGLPETG	IRSNXFPYVS	TEGICKMPFC	240
RKPGYKLNKD	LWFQITDPLD	DKTVRDLPHI	KCDLSSSIV	TPGEHATDIS	LISDVERILD	300
YALCQNTWSK	IEAGEPITPV	DLSYLGPKNP	GAGPVFTIIN	GSLHYFMSKY	LRVELESPVI	360
PRMEGKVAGT	RIVXQLWDQW	FPFGEVEIGP	NGVLKTKQGY	KFPLHIIGTG	EVDNDIKMER	420
IVKHWEHPHI	EAAQTFLKDK	DTEEVLYYGD	TGVSKNPVEL	VEGWFSGWRS	SIMGVLAVII	480
GFVILIFLIR	LIGVLSLFR	QKRRPIYKSD	VEMAHFR			517

SEQ ID NO: 97 moltype = AA length = 517
FEATURE Location/Qualifiers
VARIANT 63
note = Xaa can be any naturally occurring amino acid except
Lys or Arg

VARIANT 225
note = Xaa can be any naturally occurring amino acid except
Lys or Arg

VARIANT 374
note = Xaa can be any naturally occurring amino acid except
Lys or Arg

source 1..517
mol_type = protein
organism = Vesicular stomatitis virus

SEQUENCE: 97

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MLSYLIFALV VSPILGKIEI VFPQHTTGDW KRPVPEYNYC PTSADKNSHG TQTGIPVELT 60
MPXGLTTHQV DGFMCBSALW MTTCDFRWYG PKYITHSIHN EEPTDYQCLE AIKAYKDGVs 120
FNPGFPPQSC GYGTVTDAEA HIVTVTPHSV KVDEYTGWEI DPHFIGGRCK GQICETVHNS 180
TKWFTSSDGE SVCSQLFTLV GGTFFSDSEE ITSMGLPETG IRSNXFPYVS TEGICKMPFC 240
RKPGYKLKND LWFQITDPL DKTVRDLPHI KCDLSSSIV TPGEHATDIS LISDVERILD 300
YALCQNTWSK IEAGEPITPV DLSYLGPKNP GAGPVFTIIN GSLHYFMSKY LRVELESPVI 360
PRMEGKVAGT RIVXQLWDQW FPFGEVEIGP NGVLKTKQGY KFPLHIIGTG EVDNDIKMER 420
IVKHWEHPHI EAAQTFLKDK DTEEVLYYGD TGVSKNPVEL VEGWFSGWRS SIMGVLAVII 480
GFVILIFLIR LIGVLSSLFR QKRRPIYKSD VEMAHR 517

SEQ ID NO: 98      moltype = AA length = 517
FEATURE           Location/Qualifiers
VARIANT           25
                  note = Xaa can be any naturally occurring amino acid except
                  Lys or Arg
VARIANT           63
                  note = Xaa can be any naturally occurring amino acid except
                  Lys or Arg
VARIANT           225
                  note = Xaa can be any naturally occurring amino acid except
                  Lys or Arg
VARIANT           374
                  note = Xaa can be any naturally occurring amino acid except
                  Lys or Arg
source            1..517
                  mol_type = protein
                  organism = Vesicular stomatitis virus

SEQUENCE: 98
MLSYLIFALV VSPILGKIEI VFPQHTTGDW KRPVPEYNYC PTSADKNSHG TQTGIPVELT 60
MPXGLTTHQV DGFMCBSALW MTTCDFRWYG PKYITHSIHN EEPTDYQCLE AIKAYKDGVs 120
FNPGFPPQSC GYGTVTDAEA HIVTVTPHSV KVDEYTGWEI DPHFIGGRCK GQICETVHNS 180
TKWFTSSDGE SVCSQLFTLV GGTFFSDSEE ITSMGLPETG IRSNXFPYVS TEGICKMPFC 240
RKPGYKLKND LWFQITDPL DKTVRDLPHI KCDLSSSIV TPGEHATDIS LISDVERILD 300
YALCQNTWSK IEAGEPITPV DLSYLGPKNP GAGPVFTIIN GSLHYFMSKY LRVELESPVI 360
PRMEGKVAGT RIVXQLWDQW FPFGEVEIGP NGVLKTKQGY KFPLHIIGTG EVDNDIKMER 420
IVKHWEHPHI EAAQTFLKDK DTEEVLYYGD TGVSKNPVEL VEGWFSGWRS SIMGVLAVII 480
GFVILIFLIR LIGVLSSLFR QKRRPIYKSD VEMAHR 517

SEQ ID NO: 99      moltype = AA length = 502
FEATURE           Location/Qualifiers
VARIANT           8
                  note = Xaa can be any naturally occurring amino acid except
                  Lys or Arg
VARIANT           47
                  note = Xaa can be any naturally occurring amino acid except
                  Lys or Arg
source            1..502
                  mol_type = protein
                  organism = Vesicular stomatitis virus

SEQUENCE: 99
KITISFPXSL KGDWRPVPKG YNYCPTSADK NLHGDLDIG LRLRAPXSKF GISADGWMCH 60
AARWITTCDF RYWGPKYITH SIHSFRPSND QCKEAIRLTN EGNWINPGFP PQSCGYASVT 120
DSESVVTVTV KHQVLVDEYS GSWIDSQFPG GSCTSPICDT VHNSTLWHAD HTLDSICDQE 180
FVAMDAVLFT ESGKFEEFGK PNSGIRSNYF PYESLKDVCQ MDFCKRKGFK LPSGVWFEIE 240
DAEKSHKAQV ELKIKRCPHG AVISAPNQNA ADINLIMDVE RILDYSLCQA TWSKIQNKEA 300
LTPIDISYLG PKNPGPGPAF TIINGTLHYF NTRYIRVDIA GPVTKEITGF VSGTSTSRVL 360
WDQWFPYGEN SIGPNGLLKT ASGYKYPLFM VGTGVLADAI HKLGATVIE HPHAKEAQKV 420
VDDSEVIFPG DTGVSKNPVE VVEGWFSGWR SSLMSIFGII LLIVCLVLIV RILIALKYCC 480
VRHKRTIYK EDLEMGRIPR RA 502

SEQ ID NO: 100     moltype = AA length = 502
FEATURE           Location/Qualifiers
VARIANT           8
                  note = Xaa can be any naturally occurring amino acid except
                  Lys or Arg
VARIANT           358
                  note = Xaa can be any naturally occurring amino acid except
                  Lys or Arg
source            1..502
                  mol_type = protein
                  organism = Vesicular stomatitis virus

SEQUENCE: 100
KITISFPXSL KGDWRPVPKG YNYCPTSADK NLHGDLDIG LRLRAPXSKF GISADGWMCH 60
AARWITTCDF RYWGPKYITH SIHSFRPSND QCKEAIRLTN EGNWINPGFP PQSCGYASVT 120
DSESVVTVTV KHQVLVDEYS GSWIDSQFPG GSCTSPICDT VHNSTLWHAD HTLDSICDQE 180
FVAMDAVLFT ESGKFEEFGK PNSGIRSNYF PYESLKDVCQ MDFCKRKGFK LPSGVWFEIE 240
DAEKSHKAQV ELKIKRCPHG AVISAPNQNA ADINLIMDVE RILDYSLCQA TWSKIQNKEA 300
LTPIDISYLG PKNPGPGPAF TIINGTLHYF NTRYIRVDIA GPVTKEITGF VSGTSTSRVL 360
WDQWFPYGEN SIGPNGLLKT ASGYKYPLFM VGTGVLADAI HKLGATVIE HPHAKEAQKV 420

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VDDSEVIFFG DTGVSKNPVE VVEGWFSGWR SSLMSIFGII LLIVCLVLIV RILIALKYCC 480
VRHKKRTIYK EDLEMGRIPR RA 502

SEQ ID NO: 101      moltype = AA length = 502
FEATURE            Location/Qualifiers
VARIANT            8
                    note = Xaa can be any naturally occurring amino acid except
                    Lys or Arg
VARIANT            47
                    note = Xaa can be any naturally occurring amino acid except
                    Lys or Arg
VARIANT            358
                    note = Xaa can be any naturally occurring amino acid except
                    Lys or Arg
source             1..502
                    mol_type = protein
                    organism = Vesicular stomatitis virus

SEQUENCE: 101
KITISFPXSL KGDWRPVPKG YNYCPTSADK NLHGLDIDIG LRLRAPXSEK GISADGWMCH 60
AARWITTCDF RWYGPKYITH SIHSFRPSND QCKEAIRLTN EGNWINPGFP PQSCGYASVT 120
DSESVVTVT KHQVLVDEYS GSWIDSQFPG GSCTSPICDT VHNSTLWHAD HTLDSICDQE 180
FVAMDAVLFT ESGKFEFEGK PNSGIRSNYF PYESLKDVCQ MDFCKRKGFK LPSGVWFEIE 240
DAEKSHKAQV ELKIKRCPHG AVISAPNQNA ADINLIMDVE RILDYSLCQA TWSKIQNKEA 300
LTPIDISYLG PKNPGPGPAF TIINGTLHYF NTRYIRVDIA GPVTKEITGF VSGTSTSXLV 360
WDQWFPYGEN SIGPNGLLKT ASGYKYPLFM VGTGVLDAI HKLGEATVIE HPHAKEAQKV 420
VDDSEVIFFG DTGVSKNPVE VVEGWFSGWR SSLMSIFGII LLIVCLVLIV RILIALKYCC 480
VRHKKRTIYK EDLEMGRIPR RA 502

SEQ ID NO: 102      moltype = AA length = 502
FEATURE            Location/Qualifiers
VARIANT            47
                    note = Xaa can be any naturally occurring amino acid except
                    Lys or Arg
VARIANT            209
                    note = Xaa can be any naturally occurring amino acid except
                    Lys or Arg
source             1..502
                    mol_type = protein
                    organism = Vesicular stomatitis virus

SEQUENCE: 102
KITISFPQSL KGDWRPVPKG YNYCPTSADK NLHGLDIDIG LRLRAPXSEK GISADGWMCH 60
AARWITTCDF RWYGPKYITH SIHSFRPSND QCKEAIRLTN EGNWINPGFP PQSCGYASVT 120
DSESVVTVT KHQVLVDEYS GSWIDSQFPG GSCTSPICDT VHNSTLWHAD HTLDSICDQE 180
FVAMDAVLFT ESGKFEFEGK PNSGIRSNXF PYESLKDVCQ MDFCKRKGFK LPSGVWFEIE 240
DAEKSHKAQV ELKIKRCPHG AVISAPNQNA ADINLIMDVE RILDYSLCQA TWSKIQNKEA 300
LTPIDISYLG PKNPGPGPAF TIINGTLHYF NTRYIRVDIA GPVTKEITGF VSGTSTSRLV 360
WDQWFPYGEN SIGPNGLLKT ASGYKYPLFM VGTGVLDAI HKLGEATVIE HPHAKEAQKV 420
VDDSEVIFFG DTGVSKNPVE VVEGWFSGWR SSLMSIFGII LLIVCLVLIV RILIALKYCC 480
VRHKKRTIYK EDLEMGRIPR RA 502

SEQ ID NO: 103      moltype = AA length = 502
FEATURE            Location/Qualifiers
VARIANT            209
                    note = Xaa can be any naturally occurring amino acid except
                    Lys or Arg
VARIANT            358
                    note = Xaa can be any naturally occurring amino acid except
                    Lys or Arg
source             1..502
                    mol_type = protein
                    organism = Vesicular stomatitis virus

SEQUENCE: 103
KITISFPQSL KGDWRPVPKG YNYCPTSADK NLHGLDIDIG LRLRAPXSEK GISADGWMCH 60
AARWITTCDF RWYGPKYITH SIHSFRPSND QCKEAIRLTN EGNWINPGFP PQSCGYASVT 120
DSESVVTVT KHQVLVDEYS GSWIDSQFPG GSCTSPICDT VHNSTLWHAD HTLDSICDQE 180
FVAMDAVLFT ESGKFEFEGK PNSGIRSNXF PYESLKDVCQ MDFCKRKGFK LPSGVWFEIE 240
DAEKSHKAQV ELKIKRCPHG AVISAPNQNA ADINLIMDVE RILDYSLCQA TWSKIQNKEA 300
LTPIDISYLG PKNPGPGPAF TIINGTLHYF NTRYIRVDIA GPVTKEITGF VSGTSTSXLV 360
WDQWFPYGEN SIGPNGLLKT ASGYKYPLFM VGTGVLDAI HKLGEATVIE HPHAKEAQKV 420
VDDSEVIFFG DTGVSKNPVE VVEGWFSGWR SSLMSIFGII LLIVCLVLIV RILIALKYCC 480
VRHKKRTIYK EDLEMGRIPR RA 502

SEQ ID NO: 104      moltype = AA length = 502
FEATURE            Location/Qualifiers
VARIANT            47
                    note = Xaa can be any naturally occurring amino acid except
                    Lys or Arg
VARIANT            209
                    note = Xaa can be any naturally occurring amino acid except

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VARIANT Lys or Arg
358
note = Xaa can be any naturally occurring amino acid except
Lys or Arg

source 1..502
mol_type = protein
organism = Vesicular stomatitis virus

SEQUENCE: 104

KITISFPQSL	KGDRPVPKG	YNYCPTSADK	NLHGDLIDIG	LRLRAPXSFK	GISADGWMCH	60
AARWITTCDF	RWYGPKYITH	SIHSFRPSND	QCKEAIRLTN	EGNWINPGFP	PQSCGYASVT	120
DSESVVTVT	KHQVLVDEYS	GSWIDSQFPG	GSCTSPICDT	VHNSTLWHAD	HTLDSICDQE	180
FVAMDAVLFT	ESGKFEEFGK	PNSGIRSNXF	PYESLKDVQC	MDFCRKRGFK	LPSGVWFEIE	240
DAEKSHKAQV	ELKIKRCPHG	AVISAPNQNA	ADINLIMDVE	RILDYSLCQA	TWSKIQNKEA	300
LTPIDISYLG	PKNPGPGPAF	TIINGTLHYF	NTRYIRVDIA	GPVTKEITGF	VSGTSTSXLV	360
WDQWFPYGEN	SIGPNGLLKT	ASGYKYPLFM	VTGVLADADI	HKLGEATVIE	HPHAKEAQKV	420
VDDSEVIFPG	DTGVSKNPVE	VVEGWFSGWR	SSLMSIFGII	LLIVCLVLIV	RILIALKYCC	480
VRHKKRTIYK	EDLEMGRIPR	RA				502

SEQ ID NO: 105 moltype = AA length = 502
FEATURE Location/Qualifiers
VARIANT 8
note = Xaa can be any naturally occurring amino acid except
Lys or Arg

VARIANT 47
note = Xaa can be any naturally occurring amino acid except
Lys or Arg

VARIANT 209
note = Xaa can be any naturally occurring amino acid except
Lys or Arg

VARIANT 358
note = Xaa can be any naturally occurring amino acid except
Lys or Arg

source 1..502
mol_type = protein
organism = Vesicular stomatitis virus

SEQUENCE: 105

KITISFPXSL	KGDRPVPKG	YNYCPTSADK	NLHGDLIDIG	LRLRAPXSFK	GISADGWMCH	60
AARWITTCDF	RWYGPKYITH	SIHSFRPSND	QCKEAIRLTN	EGNWINPGFP	PQSCGYASVT	120
DSESVVTVT	KHQVLVDEYS	GSWIDSQFPG	GSCTSPICDT	VHNSTLWHAD	HTLDSICDQE	180
FVAMDAVLFT	ESGKFEEFGK	PNSGIRSNXF	PYESLKDVQC	MDFCRKRGFK	LPSGVWFEIE	240
DAEKSHKAQV	ELKIKRCPHG	AVISAPNQNA	ADINLIMDVE	RILDYSLCQA	TWSKIQNKEA	300
LTPIDISYLG	PKNPGPGPAF	TIINGTLHYF	NTRYIRVDIA	GPVTKEITGF	VSGTSTSXLV	360
WDQWFPYGEN	SIGPNGLLKT	ASGYKYPLFM	VTGVLADADI	HKLGEATVIE	HPHAKEAQKV	420
VDDSEVIFPG	DTGVSKNPVE	VVEGWFSGWR	SSLMSIFGII	LLIVCLVLIV	RILIALKYCC	480
VRHKKRTIYK	EDLEMGRIPR	RA				502

SEQ ID NO: 106 moltype = AA length = 523
FEATURE Location/Qualifiers
VARIANT 29
note = Xaa can be any naturally occurring amino acid except
Lys or Arg

VARIANT 68
note = Xaa can be any naturally occurring amino acid except
Lys or Arg

source 1..523
mol_type = protein
organism = Vesicular stomatitis virus

SEQUENCE: 106

MKMKMVIAGL	ILCIGILPAI	GKITISFPXS	LKGDRPVPK	GYNYCPTSAD	KNLHGDLIDI	60
GLRLRAPXS	KGISADGWM	HAARWITTC	FRWYGPKYIT	HSIHSFRPSN	DQCKEAIRLT	120
NEGNWINPGF	PPQSCGYASV	TDSESVVTV	TKHQVLVDEY	SGSWIDSQFP	GGCTSPICD	180
TVHNSTLWHA	DHTLDSICDQ	EFVAMDAVL	TESGKFEEFG	KPNSGIRSNY	FPYESLKDV	240
QMDFCRKRGF	KLPSGVWFEI	EDAEKSHKAQ	VELKIKRCPH	GAVISAPNQ	AADINLIMDV	300
ERILDYSLCQ	ATWSKIQNKE	ALTPIDISYL	GPKNPGPGPA	FTIINGTLHY	FNTRYIRVDI	360
AGPVTKEITG	FVSGTSTSRV	LWDQWFPYGE	NSIGPNGLLK	TASGYKYPLF	MVGTGVLAD	420
IHLKGEATVI	EHPHAKEAQK	VDDSEVIF	GDTGVSKNPV	EVVEGWFSGW	RSSLMSIFGI	480
ILLIVCLVLI	VRILIALKYC	CVRHKKRTIY	KEDLEMGRIP	RRA		523

SEQ ID NO: 107 moltype = AA length = 523
FEATURE Location/Qualifiers
VARIANT 29
note = Xaa can be any naturally occurring amino acid except
Lys or Arg

VARIANT 379
note = Xaa can be any naturally occurring amino acid except
Lys or Arg

source 1..523
mol_type = protein
organism = Vesicular stomatitis virus

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SEQUENCE: 107

MKMKMVIAGL	ILCIGILPAI	GKITISFPXS	LKGDWRPVPK	GYNYCPTSAD	KNLHGDLDI	60
GLRLRAPKSF	KGISADGWMC	HAARWITTC	FRWYGPKYIT	HSIHSFRPSN	DQCKEAIRLT	120
NEGNWINPGF	PPQSCGYASV	TDSESVVTV	TKHQVLVDEY	SGSWIDSQFP	GGCTSPICD	180
TVHNSTLWHA	DHTLDSICDQ	EFVAMDAVL	TESGKFEEFG	KPNSGIRSNY	FPYESLKDVC	240
QMDPCKRKGF	KLPSGVWFIE	EDAEKSHKAQ	VELKIKRCPH	GAVISAPNQ	AADINLIMDV	300
ERILDYSLCQ	ATWSKIQNKE	ALTPIDISYL	GPKNPGPGPA	FTIINGTLHY	FNTRYIRVDI	360
AGPVTKEITG	FVSGTSTSV	LWDQWFPYGE	NSIGPNGLLK	TASGYKYPLF	MVGTGVLDAD	420
IHKLGATVI	EHPHAKAQK	VVDDSEVIF	GDTGVSKNPV	EVVEGWFSGW	RSSLMSIFGI	480
ILLIVCLVLI	VRILIALKYC	CVRHKKRTIY	KEDLEMGRIP	RAA		523

SEQ ID NO: 108 moltype = AA length = 523

FEATURE Location/Qualifiers

VARIANT 29

 note = Xaa can be any naturally occurring amino acid except

 Lys or Arg

VARIANT 68

 note = Xaa can be any naturally occurring amino acid except

 Lys or Arg

VARIANT 379

 note = Xaa can be any naturally occurring amino acid except

 Lys or Arg

source 1..523

 mol_type = protein

 organism = Vesicular stomatitis virus

SEQUENCE: 108

MKMKMVIAGL	ILCIGILPAI	GKITISFPXS	LKGDWRPVPK	GYNYCPTSAD	KNLHGDLDI	60
GLRLRAPKSF	KGISADGWMC	HAARWITTC	FRWYGPKYIT	HSIHSFRPSN	DQCKEAIRLT	120
NEGNWINPGF	PPQSCGYASV	TDSESVVTV	TKHQVLVDEY	SGSWIDSQFP	GGCTSPICD	180
TVHNSTLWHA	DHTLDSICDQ	EFVAMDAVL	TESGKFEEFG	KPNSGIRSNY	FPYESLKDVC	240
QMDPCKRKGF	KLPSGVWFIE	EDAEKSHKAQ	VELKIKRCPH	GAVISAPNQ	AADINLIMDV	300
ERILDYSLCQ	ATWSKIQNKE	ALTPIDISYL	GPKNPGPGPA	FTIINGTLHY	FNTRYIRVDI	360
AGPVTKEITG	FVSGTSTSV	LWDQWFPYGE	NSIGPNGLLK	TASGYKYPLF	MVGTGVLDAD	420
IHKLGATVI	EHPHAKAQK	VVDDSEVIF	GDTGVSKNPV	EVVEGWFSGW	RSSLMSIFGI	480
ILLIVCLVLI	VRILIALKYC	CVRHKKRTIY	KEDLEMGRIP	RAA		523

SEQ ID NO: 109 moltype = AA length = 523

FEATURE Location/Qualifiers

VARIANT 68

 note = Xaa can be any naturally occurring amino acid except

 Lys or Arg

VARIANT 230

 note = Xaa can be any naturally occurring amino acid except

 Lys or Arg

source 1..523

 mol_type = protein

 organism = Vesicular stomatitis virus

SEQUENCE: 109

MKMKMVIAGL	ILCIGILPAI	GKITISFPQS	LKGDWRPVPK	GYNYCPTSAD	KNLHGDLDI	60
GLRLRAPKSF	KGISADGWMC	HAARWITTC	FRWYGPKYIT	HSIHSFRPSN	DQCKEAIRLT	120
NEGNWINPGF	PPQSCGYASV	TDSESVVTV	TKHQVLVDEY	SGSWIDSQFP	GGCTSPICD	180
TVHNSTLWHA	DHTLDSICDQ	EFVAMDAVL	TESGKFEEFG	KPNSGIRSNX	FPYESLKDVC	240
QMDPCKRKGF	KLPSGVWFIE	EDAEKSHKAQ	VELKIKRCPH	GAVISAPNQ	AADINLIMDV	300
ERILDYSLCQ	ATWSKIQNKE	ALTPIDISYL	GPKNPGPGPA	FTIINGTLHY	FNTRYIRVDI	360
AGPVTKEITG	FVSGTSTSV	LWDQWFPYGE	NSIGPNGLLK	TASGYKYPLF	MVGTGVLDAD	420
IHKLGATVI	EHPHAKAQK	VVDDSEVIF	GDTGVSKNPV	EVVEGWFSGW	RSSLMSIFGI	480
ILLIVCLVLI	VRILIALKYC	CVRHKKRTIY	KEDLEMGRIP	RAA		523

SEQ ID NO: 110 moltype = AA length = 523

FEATURE Location/Qualifiers

VARIANT 230

 note = Xaa can be any naturally occurring amino acid except

 Lys or Arg

VARIANT 379

 note = Xaa can be any naturally occurring amino acid except

 Lys or Arg

source 1..523

 mol_type = protein

 organism = Vesicular stomatitis virus

SEQUENCE: 110

MKMKMVIAGL	ILCIGILPAI	GKITISFPQS	LKGDWRPVPK	GYNYCPTSAD	KNLHGDLDI	60
GLRLRAPKSF	KGISADGWMC	HAARWITTC	FRWYGPKYIT	HSIHSFRPSN	DQCKEAIRLT	120
NEGNWINPGF	PPQSCGYASV	TDSESVVTV	TKHQVLVDEY	SGSWIDSQFP	GGCTSPICD	180
TVHNSTLWHA	DHTLDSICDQ	EFVAMDAVL	TESGKFEEFG	KPNSGIRSNX	FPYESLKDVC	240
QMDPCKRKGF	KLPSGVWFIE	EDAEKSHKAQ	VELKIKRCPH	GAVISAPNQ	AADINLIMDV	300
ERILDYSLCQ	ATWSKIQNKE	ALTPIDISYL	GPKNPGPGPA	FTIINGTLHY	FNTRYIRVDI	360
AGPVTKEITG	FVSGTSTSV	LWDQWFPYGE	NSIGPNGLLK	TASGYKYPLF	MVGTGVLDAD	420
IHKLGATVI	EHPHAKAQK	VVDDSEVIF	GDTGVSKNPV	EVVEGWFSGW	RSSLMSIFGI	480
ILLIVCLVLI	VRILIALKYC	CVRHKKRTIY	KEDLEMGRIP	RAA		523

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SEQ ID NO: 111      moltype = AA  length = 523
FEATURE            Location/Qualifiers
VARIANT            68
                    note = Xaa can be any naturally occurring amino acid except
                    Lys or Arg
VARIANT            230
                    note = Xaa can be any naturally occurring amino acid except
                    Lys or Arg
VARIANT            379
                    note = Xaa can be any naturally occurring amino acid except
                    Lys or Arg
source             1..523
                    mol_type = protein
                    organism = Vesicular stomatitis virus

SEQUENCE: 111
MKMKMVIAGL ILCIGILPAI GKITISFPQS LKGDWRPVPK GYNYCPTSAD KNLHGDLDI 60
GLRLRAPXSF KGISADGWM C HAARWITTC FRWYGPKYIT HSIHSFRPSN DQCKEAIRLT 120
NEGNWINPGF PPQSCGYASV TDSESVVVTV TKHQVLVDEY SGSWIDSQFP GGSCTSPICD 180
TVHNSTLWHA DHTLDSICDQ EFVAMDAVL F TESGKFEEFG KPN SGIRSNX FPYESLKDVC 240
QMDFCRRKGF KLPSGVWF EI EDAEKSHKAQ VELKIKRCPH GAVISAPNQ N AADINLIMDV 300
ERILDYSLCQ ATWSKIQNKE ALTPIDISYL GPKNPGPGPA FTIINGTLHY FNTRYIRVDI 360
AGPVTKEITG FVSGTSTS XV LWDQWFPYGE NSIGPNGLLK TASGYKYPLF MVGTGVL DAD 420
IHKLG EATVI EHPHAK EAQK VVDDSEVIF F GDTGVSKNPV EVVEGW FSGW RSSLMSIFGI 480
ILLIVCLVLI VRILIALKYC CVRHKKRTIY KEDLEMGRIP RRA 523

SEQ ID NO: 112      moltype = AA  length = 523
FEATURE            Location/Qualifiers
VARIANT            29
                    note = Xaa can be any naturally occurring amino acid except
                    Lys or Arg
VARIANT            68
                    note = Xaa can be any naturally occurring amino acid except
                    Lys or Arg
VARIANT            230
                    note = Xaa can be any naturally occurring amino acid except
                    Lys or Arg
VARIANT            379
                    note = Xaa can be any naturally occurring amino acid except
                    Lys or Arg
source             1..523
                    mol_type = protein
                    organism = Vesicular stomatitis virus

SEQUENCE: 112
MKMKMVIAGL ILCIGILPAI GKITISFPXS LKGDWRPVPK GYNYCPTSAD KNLHGDLDI 60
GLRLRAPXSF KGISADGWM C HAARWITTC FRWYGPKYIT HSIHSFRPSN DQCKEAIRLT 120
NEGNWINPGF PPQSCGYASV TDSESVVVTV TKHQVLVDEY SGSWIDSQFP GGSCTSPICD 180
TVHNSTLWHA DHTLDSICDQ EFVAMDAVL F TESGKFEEFG KPN SGIRSNX FPYESLKDVC 240
QMDFCRRKGF KLPSGVWF EI EDAEKSHKAQ VELKIKRCPH GAVISAPNQ N AADINLIMDV 300
ERILDYSLCQ ATWSKIQNKE ALTPIDISYL GPKNPGPGPA FTIINGTLHY FNTRYIRVDI 360
AGPVTKEITG FVSGTSTS XV LWDQWFPYGE NSIGPNGLLK TASGYKYPLF MVGTGVL DAD 420
IHKLG EATVI EHPHAK EAQK VVDDSEVIF F GDTGVSKNPV EVVEGW FSGW RSSLMSIFGI 480
ILLIVCLVLI VRILIALKYC CVRHKKRTIY KEDLEMGRIP RRA 523

SEQ ID NO: 113      moltype = AA  length = 494
FEATURE            Location/Qualifiers
VARIANT            8
                    note = Xaa can be any naturally occurring amino acid except
                    Lys or Arg
VARIANT            47
                    note = Xaa can be any naturally occurring amino acid except
                    Lys or Arg
source             1..494
                    mol_type = protein
                    organism = Vesicular stomatitis virus

SEQUENCE: 113
KPTIVFPXSQ KGDWKDVPPN YRYCPSSADQ NWHGDL LGVN IRAKMPXVHK AIKADGWMCH 60
AAKWVTTCDY RWYGPQYITH SIHSFIPTKA QCEESIKQTK EGVWINPGFP PKNCGYASVS 120
DAESIIVQAT AHSVMI DEYS GDWLDSQFPT GRCTGSTCET IHNSTLWYAD YQVTGLCDSA 180
LVSTEVTFYS EDGLMTS IGR QNTGYRSNYF PYEKGAAACR MKYCTHEGIR LPSGVWFEMV 240
DKELLESVQM PECPAGLTIS APTQTSVDVS LILDVERMLD YSLCQETWSK VHSGLPISPV 300
DLGYIAPKNP GAGPAFTIVN GTLKYPDTRY LRIDIEGPVL KKMTGKVS GT PTKRELWTEW 360
FPYDDVEIGP NGVLKTP EGY KFPLYMIGHG LLDSDLQKTS QAEVFHHFQI AEA VQKLPPD 420
ETLFFGDTGI SKNPEVIEG VVDSNRSSVM AIVFAILLV ITVLMVRLCV AFRHFCCQKR 480
HKIYNDELMN QLRR 494

SEQ ID NO: 114      moltype = AA  length = 494
FEATURE            Location/Qualifiers
VARIANT            8

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note = Xaa can be any naturally occurring amino acid except
  Lys or Arg
VARIANT 354
note = Xaa can be any naturally occurring amino acid except
  Lys or Arg
source 1..494
mol_type = protein
organism = Vesicular stomatitis virus

SEQUENCE: 114
KFTIVFPXSQ KGDWKDVPPN YRYCPSSADQ NWHGDL LGVN IRAKMPKVHK AIKADGWMCH 60
AAKWVTTCDY RWYGPQYITH SIHSFIPTKA QCEESIKQTK EGVWINPGFP PKNCGYASVS 120
DAESIIVQAT AHSVMIDEYS GDWLDSQFPT GRCTGSTCET IHNSTLWYAD YQVTGLCDSA 180
LVSTEVTFYS EDGLMTSigr QNTGYRSNYF PYEKGAACR MKYCTHEGIR LPSGVWFEMV 240
DKELLESVQM PECPAGLTIS APTQTSVDVS LILDVERMLD YSLCQETWSK VHSGLPISPV 300
DLGYIAPKNP GAGPAFTIVN GTLKYPDTRY LRIDIEGPVL KKMTGKVSgt PTKXELWTEW 360
FPYDDVEIGP NGVLKTPEGY KFPLYMIGHG LLDSDLQKTS QAEVFHHPQI AEAQVQLPDD 420
ETLFFGDTGI SKNPVEVIEG WFSNWRSSVM AIVFAILLV ITVLMVRLCV AFRHFCCQKR 480
HKIYNDLEMN QLRR 494

SEQ ID NO: 115 moltype = AA length = 494
FEATURE Location/Qualifiers
VARIANT 8
note = Xaa can be any naturally occurring amino acid except
  Lys or Arg
VARIANT 47
note = Xaa can be any naturally occurring amino acid except
  Lys or Arg
VARIANT 354
note = Xaa can be any naturally occurring amino acid except
  Lys or Arg
source 1..494
mol_type = protein
organism = Vesicular stomatitis virus

SEQUENCE: 115
KFTIVFPXSQ KGDWKDVPPN YRYCPSSADQ NWHGDL LGVN IRAKMPKVHK AIKADGWMCH 60
AAKWVTTCDY RWYGPQYITH SIHSFIPTKA QCEESIKQTK EGVWINPGFP PKNCGYASVS 120
DAESIIVQAT AHSVMIDEYS GDWLDSQFPT GRCTGSTCET IHNSTLWYAD YQVTGLCDSA 180
LVSTEVTFYS EDGLMTSigr QNTGYRSNYF PYEKGAACR MKYCTHEGIR LPSGVWFEMV 240
DKELLESVQM PECPAGLTIS APTQTSVDVS LILDVERMLD YSLCQETWSK VHSGLPISPV 300
DLGYIAPKNP GAGPAFTIVN GTLKYPDTRY LRIDIEGPVL KKMTGKVSgt PTKXELWTEW 360
FPYDDVEIGP NGVLKTPEGY KFPLYMIGHG LLDSDLQKTS QAEVFHHPQI AEAQVQLPDD 420
ETLFFGDTGI SKNPVEVIEG WFSNWRSSVM AIVFAILLV ITVLMVRLCV AFRHFCCQKR 480
HKIYNDLEMN QLRR 494

SEQ ID NO: 116 moltype = AA length = 494
FEATURE Location/Qualifiers
VARIANT 47
note = Xaa can be any naturally occurring amino acid except
  Lys or Arg
VARIANT 209
note = Xaa can be any naturally occurring amino acid except
  Lys or Arg
source 1..494
mol_type = protein
organism = Vesicular stomatitis virus

SEQUENCE: 116
KFTIVFPXSQ KGDWKDVPPN YRYCPSSADQ NWHGDL LGVN IRAKMPKVHK AIKADGWMCH 60
AAKWVTTCDY RWYGPQYITH SIHSFIPTKA QCEESIKQTK EGVWINPGFP PKNCGYASVS 120
DAESIIVQAT AHSVMIDEYS GDWLDSQFPT GRCTGSTCET IHNSTLWYAD YQVTGLCDSA 180
LVSTEVTFYS EDGLMTSigr QNTGYRSNXF PYEKGAACR MKYCTHEGIR LPSGVWFEMV 240
DKELLESVQM PECPAGLTIS APTQTSVDVS LILDVERMLD YSLCQETWSK VHSGLPISPV 300
DLGYIAPKNP GAGPAFTIVN GTLKYPDTRY LRIDIEGPVL KKMTGKVSgt PTKRELWTEW 360
FPYDDVEIGP NGVLKTPEGY KFPLYMIGHG LLDSDLQKTS QAEVFHHPQI AEAQVQLPDD 420
ETLFFGDTGI SKNPVEVIEG WFSNWRSSVM AIVFAILLV ITVLMVRLCV AFRHFCCQKR 480
HKIYNDLEMN QLRR 494

SEQ ID NO: 117 moltype = AA length = 494
FEATURE Location/Qualifiers
VARIANT 209
note = Xaa can be any naturally occurring amino acid except
  Lys or Arg
VARIANT 354
note = Xaa can be any naturally occurring amino acid except
  Lys or Arg
source 1..494
mol_type = protein
organism = Vesicular stomatitis virus

SEQUENCE: 117
KFTIVFPXSQ KGDWKDVPPN YRYCPSSADQ NWHGDL LGVN IRAKMPKVHK AIKADGWMCH 60

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AAKWVTTCDY RWYGPQYITH SIHSFIPTKA QCEESIKQTK EGVWINPGFP PKNCGYASVS 120
DAESIIVQAT AHSVIMIDEYS GDWLDSQFPT GRCTGSTCET IHNSTLWYAD YQVTGLCDSA 180
LVSTEVTFYS EDGLMTSIGR QNTGYRSNXF PYEKGAACR MKYCTHEGIR LPSGVWFEMV 240
DKELLESVQM PECPAGLTIS APTQTSVDVS LILDVERMLD YSLCQETWSK VHSGLPISPV 300
DLGYIAPKNP GAGPAFTIVN GTLKYFDTRY LRIDIEGPVL KKMTGKVSQT PTKXELWTEW 360
FPYDDVEIGP NGVLKTPEGY KFPLYMIGHG LLDSDLQKTS QAEVFHHPQI AEAQKLPDD 420
ETLFFGDTGI SKNPVEVIEG WFSNWRSSVM AIVFAILLLV ITVLMVRLCV AFRHFCCQKR 480
HKIYNDLEMN QLRR 494

SEQ ID NO: 118      moltype = AA length = 494
FEATURE            Location/Qualifiers
VARIANT            47
                    note = Xaa can be any naturally occurring amino acid except
                    Lys or Arg
VARIANT            209
                    note = Xaa can be any naturally occurring amino acid except
                    Lys or Arg
VARIANT            354
                    note = Xaa can be any naturally occurring amino acid except
                    Lys or Arg
source              1..494
                    mol_type = protein
                    organism = Vesicular stomatitis virus

SEQUENCE: 118
KFTIVFPQSQ KGDWKDVPPN YRYCPSSADQ NWHGDLLGVN IRAKMPXVHK AIKADGWMCH 60
AAKWVTTCDY RWYGPQYITH SIHSFIPTKA QCEESIKQTK EGVWINPGFP PKNCGYASVS 120
DAESIIVQAT AHSVIMIDEYS GDWLDSQFPT GRCTGSTCET IHNSTLWYAD YQVTGLCDSA 180
LVSTEVTFYS EDGLMTSIGR QNTGYRSNXF PYEKGAACR MKYCTHEGIR LPSGVWFEMV 240
DKELLESVQM PECPAGLTIS APTQTSVDVS LILDVERMLD YSLCQETWSK VHSGLPISPV 300
DLGYIAPKNP GAGPAFTIVN GTLKYFDTRY LRIDIEGPVL KKMTGKVSQT PTKXELWTEW 360
FPYDDVEIGP NGVLKTPEGY KFPLYMIGHG LLDSDLQKTS QAEVFHHPQI AEAQKLPDD 420
ETLFFGDTGI SKNPVEVIEG WFSNWRSSVM AIVFAILLLV ITVLMVRLCV AFRHFCCQKR 480
HKIYNDLEMN QLRR 494

SEQ ID NO: 119      moltype = AA length = 494
FEATURE            Location/Qualifiers
VARIANT            8
                    note = Xaa can be any naturally occurring amino acid except
                    Lys or Arg
VARIANT            47
                    note = Xaa can be any naturally occurring amino acid except
                    Lys or Arg
VARIANT            209
                    note = Xaa can be any naturally occurring amino acid except
                    Lys or Arg
VARIANT            354
                    note = Xaa can be any naturally occurring amino acid except
                    Lys or Arg
source              1..494
                    mol_type = protein
                    organism = Vesicular stomatitis virus

SEQUENCE: 119
KFTIVFPXSQ KGDWKDVPPN YRYCPSSADQ NWHGDLLGVN IRAKMPXVHK AIKADGWMCH 60
AAKWVTTCDY RWYGPQYITH SIHSFIPTKA QCEESIKQTK EGVWINPGFP PKNCGYASVS 120
DAESIIVQAT AHSVIMIDEYS GDWLDSQFPT GRCTGSTCET IHNSTLWYAD YQVTGLCDSA 180
LVSTEVTFYS EDGLMTSIGR QNTGYRSNXF PYEKGAACR MKYCTHEGIR LPSGVWFEMV 240
DKELLESVQM PECPAGLTIS APTQTSVDVS LILDVERMLD YSLCQETWSK VHSGLPISPV 300
DLGYIAPKNP GAGPAFTIVN GTLKYFDTRY LRIDIEGPVL KKMTGKVSQT PTKXELWTEW 360
FPYDDVEIGP NGVLKTPEGY KFPLYMIGHG LLDSDLQKTS QAEVFHHPQI AEAQKLPDD 420
ETLFFGDTGI SKNPVEVIEG WFSNWRSSVM AIVFAILLLV ITVLMVRLCV AFRHFCCQKR 480
HKIYNDLEMN QLRR 494

SEQ ID NO: 120      moltype = AA length = 511
FEATURE            Location/Qualifiers
VARIANT            25
                    note = Xaa can be any naturally occurring amino acid except
                    Lys or Arg
VARIANT            64
                    note = Xaa can be any naturally occurring amino acid except
                    Lys or Arg
source              1..511
                    mol_type = protein
                    organism = Vesicular stomatitis virus

SEQUENCE: 120
MTPAFILCML LAGSSWAKFT IVFPXSQKGD WKDVPPNYRY CPSSADQNWHD GDLGVNIRA 60
KMPXVHKAIA ADGWMCHAAK WVTTCYRWY GPQYITHSIH SFIPTKAQCE ESIKQTKQEV 120
WINPGFPPKN CGYASVSDAE SIIVQATAHS VMIDEYSGDW LDSQFPTGRC TGSTCETIHN 180
STLWYADYQV TGLCDSALVS TEVTFYSEDG LMTSIGRQNT GYRSNYFPYE KGAAACRMKY 240
CTHEGIRLPS GVVWFEMV DKE LLESVQMEC PAGLTISAPT QTSVDVSLIL DVERMLDYSL 300

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CQETWSKVHS GLPISPVDLG YIAPKNPGAG PAFITVNGTL KYFDTRYLRI DIEGPVLKMK 360
 TGVSGTPTK RELWTEWFPY DDVEIGPNGV LKTPGYKFP LYMIGHGLLD SDLQKTSQAE 420
 VFHHPQIAEA VQKLPPDETLL FFGDTGISK N PVEVIEGWFS NWRSSVMAIV FAILLLVITV 480
 LMVRLCVAFR HFCCQKRHKI YNDLEMNQLR R 511

SEQ ID NO: 121 moltype = AA length = 511
 FEATURE Location/Qualifiers
 VARIANT 25
 note = Xaa can be any naturally occurring amino acid except
 Lys or Arg
 VARIANT 371
 note = Xaa can be any naturally occurring amino acid except
 Lys or Arg
 source 1..511
 mol_type = protein
 organism = Vesicular stomatitis virus

SEQUENCE: 121
 MTPAFILCML LAGSSWAKFT IVFPXSQKGD WKDVPPNYRY CPSSADQNH GDLGVLNIRA 60
 KMPKVHKAIAK ADGWMCHAAK WVTTCDYRWY GPQYITHSIH SFIPKAQCE ESIKQTKGV 120
 WINPGFPPKN CGYASVSDAE SIIVQATAHS VMIDEYSGDW LDSQFPTGRC TGSTCETIHN 180
 STLWYADYQV TGLCDSALVS TEVTFYSEDG LMTSIGRQNT GYRSNYFPYE KGAAACRMKY 240
 CTHEGIRLPS GVVWFEMVDE LLESVQMPEC PAGLTISAPT QTSVDVSLIL DVERMLDYSL 300
 CQETWSKVHS GLPISPVDLG YIAPKNPGAG PAFITVNGTL KYFDTRYLRI DIEGPVLKMK 360
 TGVSGTPTK RELWTEWFPY DDVEIGPNGV LKTPGYKFP LYMIGHGLLD SDLQKTSQAE 420
 VFHHPQIAEA VQKLPPDETLL FFGDTGISK N PVEVIEGWFS NWRSSVMAIV FAILLLVITV 480
 LMVRLCVAFR HFCCQKRHKI YNDLEMNQLR R 511

SEQ ID NO: 122 moltype = AA length = 511
 FEATURE Location/Qualifiers
 VARIANT 25
 note = Xaa can be any naturally occurring amino acid except
 Lys or Arg
 VARIANT 64
 note = Xaa can be any naturally occurring amino acid except
 Lys or Arg
 VARIANT 371
 note = Xaa can be any naturally occurring amino acid except
 Lys or Arg
 source 1..511
 mol_type = protein
 organism = Vesicular stomatitis virus

SEQUENCE: 122
 MTPAFILCML LAGSSWAKFT IVFPXSQKGD WKDVPPNYRY CPSSADQNH GDLGVLNIRA 60
 KMPKVHKAIAK ADGWMCHAAK WVTTCDYRWY GPQYITHSIH SFIPKAQCE ESIKQTKGV 120
 WINPGFPPKN CGYASVSDAE SIIVQATAHS VMIDEYSGDW LDSQFPTGRC TGSTCETIHN 180
 STLWYADYQV TGLCDSALVS TEVTFYSEDG LMTSIGRQNT GYRSNYFPYE KGAAACRMKY 240
 CTHEGIRLPS GVVWFEMVDE LLESVQMPEC PAGLTISAPT QTSVDVSLIL DVERMLDYSL 300
 CQETWSKVHS GLPISPVDLG YIAPKNPGAG PAFITVNGTL KYFDTRYLRI DIEGPVLKMK 360
 TGVSGTPTK RELWTEWFPY DDVEIGPNGV LKTPGYKFP LYMIGHGLLD SDLQKTSQAE 420
 VFHHPQIAEA VQKLPPDETLL FFGDTGISK N PVEVIEGWFS NWRSSVMAIV FAILLLVITV 480
 LMVRLCVAFR HFCCQKRHKI YNDLEMNQLR R 511

SEQ ID NO: 123 moltype = AA length = 511
 FEATURE Location/Qualifiers
 VARIANT 64
 note = Xaa can be any naturally occurring amino acid except
 Lys or Arg
 VARIANT 226
 note = Xaa can be any naturally occurring amino acid except
 Lys or Arg
 source 1..511
 mol_type = protein
 organism = Vesicular stomatitis virus

SEQUENCE: 123
 MTPAFILCML LAGSSWAKFT IVFPQSQKGD WKDVPPNYRY CPSSADQNH GDLGVLNIRA 60
 KMPKVHKAIAK ADGWMCHAAK WVTTCDYRWY GPQYITHSIH SFIPKAQCE ESIKQTKGV 120
 WINPGFPPKN CGYASVSDAE SIIVQATAHS VMIDEYSGDW LDSQFPTGRC TGSTCETIHN 180
 STLWYADYQV TGLCDSALVS TEVTFYSEDG LMTSIGRQNT GYRSNXPPEY KGAAACRMKY 240
 CTHEGIRLPS GVVWFEMVDE LLESVQMPEC PAGLTISAPT QTSVDVSLIL DVERMLDYSL 300
 CQETWSKVHS GLPISPVDLG YIAPKNPGAG PAFITVNGTL KYFDTRYLRI DIEGPVLKMK 360
 TGVSGTPTK RELWTEWFPY DDVEIGPNGV LKTPGYKFP LYMIGHGLLD SDLQKTSQAE 420
 VFHHPQIAEA VQKLPPDETLL FFGDTGISK N PVEVIEGWFS NWRSSVMAIV FAILLLVITV 480
 LMVRLCVAFR HFCCQKRHKI YNDLEMNQLR R 511

SEQ ID NO: 124 moltype = AA length = 511
 FEATURE Location/Qualifiers
 VARIANT 226
 note = Xaa can be any naturally occurring amino acid except
 Lys or Arg

-continued

VARIANT 371
note = Xaa can be any naturally occurring amino acid except
Lys or Arg

source 1..511
mol_type = protein
organism = Vesicular stomatitis virus

SEQUENCE: 124

MTPAFILCML	LAGSSWAKFT	IVFPQSQKGD	WKDVPPNYRY	CPSSADQNW	GDLLGVNIRA	60
KMPKVHKAIAK	ADGWMCHAAK	WVTTCDYRWY	GPQYITHSIH	SFIPTKAQCE	ESIKQTKEGV	120
WINPGFPPKN	CGYASVSDAE	SIIVQATAHS	VMIDEYSGDW	LDSQFPTGRC	TGSTCETIHN	180
STLWYADYQV	TGLCDSALVS	TEVTFYSEDG	LMTSIGRQNT	GYRSNXFPYE	KGAAACRMKY	240
CTHEGIRLPS	GVWFEMVDKE	LLESVQMPEC	PAGLTISAPT	QTSVDVSLIL	DVERMLDYSL	300
CQETWSKVHS	GLPISPVDLG	YIAPKNPGAG	PAFTIVNGTL	KYFDTRYLRI	DIEGPVLKMK	360
TGKVSQTPK	XELWTEWFPY	DDVEIGPNGV	LKTPEGYKFP	LYMIGHGLLD	SDLQKTSQAE	420
VFHHPQIAEA	VQKLPPDETL	FFGDTGISK	PVEVIEGWFS	NWRSSVMAIV	FAILLLVITV	480
LMVRLCVAFR	HFCCQKRHKI	YNDLEMNQLR	R			511

SEQ ID NO: 125 moltype = AA length = 511
FEATURE Location/Qualifiers
VARIANT 64
note = Xaa can be any naturally occurring amino acid except
Lys or Arg

VARIANT 226
note = Xaa can be any naturally occurring amino acid except
Lys or Arg

VARIANT 371
note = Xaa can be any naturally occurring amino acid except
Lys or Arg

source 1..511
mol_type = protein
organism = Vesicular stomatitis virus

SEQUENCE: 125

MTPAFILCML	LAGSSWAKFT	IVFPQSQKGD	WKDVPPNYRY	CPSSADQNW	GDLLGVNIRA	60
KMPKVHKAIAK	ADGWMCHAAK	WVTTCDYRWY	GPQYITHSIH	SFIPTKAQCE	ESIKQTKEGV	120
WINPGFPPKN	CGYASVSDAE	SIIVQATAHS	VMIDEYSGDW	LDSQFPTGRC	TGSTCETIHN	180
STLWYADYQV	TGLCDSALVS	TEVTFYSEDG	LMTSIGRQNT	GYRSNXFPYE	KGAAACRMKY	240
CTHEGIRLPS	GVWFEMVDKE	LLESVQMPEC	PAGLTISAPT	QTSVDVSLIL	DVERMLDYSL	300
CQETWSKVHS	GLPISPVDLG	YIAPKNPGAG	PAFTIVNGTL	KYFDTRYLRI	DIEGPVLKMK	360
TGKVSQTPK	XELWTEWFPY	DDVEIGPNGV	LKTPEGYKFP	LYMIGHGLLD	SDLQKTSQAE	420
VFHHPQIAEA	VQKLPPDETL	FFGDTGISK	PVEVIEGWFS	NWRSSVMAIV	FAILLLVITV	480
LMVRLCVAFR	HFCCQKRHKI	YNDLEMNQLR	R			511

SEQ ID NO: 126 moltype = AA length = 511
FEATURE Location/Qualifiers
VARIANT 25
note = Xaa can be any naturally occurring amino acid except
Lys or Arg

VARIANT 64
note = Xaa can be any naturally occurring amino acid except
Lys or Arg

VARIANT 226
note = Xaa can be any naturally occurring amino acid except
Lys or Arg

VARIANT 371
note = Xaa can be any naturally occurring amino acid except
Lys or Arg

source 1..511
mol_type = protein
organism = Vesicular stomatitis virus

SEQUENCE: 126

MTPAFILCML	LAGSSWAKFT	IVFPXSQKGD	WKDVPPNYRY	CPSSADQNW	GDLLGVNIRA	60
KMPKVHKAIAK	ADGWMCHAAK	WVTTCDYRWY	GPQYITHSIH	SFIPTKAQCE	ESIKQTKEGV	120
WINPGFPPKN	CGYASVSDAE	SIIVQATAHS	VMIDEYSGDW	LDSQFPTGRC	TGSTCETIHN	180
STLWYADYQV	TGLCDSALVS	TEVTFYSEDG	LMTSIGRQNT	GYRSNXFPYE	KGAAACRMKY	240
CTHEGIRLPS	GVWFEMVDKE	LLESVQMPEC	PAGLTISAPT	QTSVDVSLIL	DVERMLDYSL	300
CQETWSKVHS	GLPISPVDLG	YIAPKNPGAG	PAFTIVNGTL	KYFDTRYLRI	DIEGPVLKMK	360
TGKVSQTPK	XELWTEWFPY	DDVEIGPNGV	LKTPEGYKFP	LYMIGHGLLD	SDLQKTSQAE	420
VFHHPQIAEA	VQKLPPDETL	FFGDTGISK	PVEVIEGWFS	NWRSSVMAIV	FAILLLVITV	480
LMVRLCVAFR	HFCCQKRHKI	YNDLEMNQLR	R			511

SEQ ID NO: 127 moltype = AA length = 495
FEATURE Location/Qualifiers
VARIANT 8
note = Xaa can be any naturally occurring amino acid except
Lys or Arg

VARIANT 47
note = Xaa can be any naturally occurring amino acid except
Lys or Arg

source 1..495

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mol_type = protein
organism = Vesicular stomatitis virus

SEQUENCE: 127
KFSIVFPXSQ KGNWKNVPSS YHYCPSSSDQ NWHNDLLGIT MKVKMPXTHK AIQADGWMCH 60
AAKWITTCDF RYWGPKYITH SIHSIQPTSE QCKESIKQTK QGTWMSPGFP PQNCGYATVT 120
DSVAVVVQAT PHHVLVDEYT GEWIDSQFPN GKCETEECET VHNSTVWYSD YKVTGLCDAT 180
LVDTEITFFS EDGKKESIGK PNTGYRSNYF AYEKGDKVCK MNYCKHAGVR LPSGVWFEFV 240
DQDVYAAAKL PECPVGATIS APTQTSVDVS LILDVERILD YSLCQETWSK IRSKQPVSPV 300
DLSYLAPKNP GTGPAFTIIN GTLKYPETRY IRIDIDNPII SKMVGKISGS QTERELWTEW 360
FPYEGVEIGP NGILKTPTGY KFPLFMIGHG MLDSDLHKTS QAEVFEHPL AEAPKQLPEE 420
ETLFFGDTGI SKNPVELIEG WFSSWKSTVV TFFFAIGVFI LLYVVARIVI AVRYRYQGSN 480
NKRIYNDIEM SRFRK 495

SEQ ID NO: 128      moltype = AA length = 495
FEATURE            Location/Qualifiers
VARIANT            8
                    note = Xaa can be any naturally occurring amino acid except
                    Lys or Arg
VARIANT            354
                    note = Xaa can be any naturally occurring amino acid except
                    Lys or Arg
source              1..495
                    mol_type = protein
                    organism = Vesicular stomatitis virus

SEQUENCE: 128
KFSIVFPXSQ KGNWKNVPSS YHYCPSSSDQ NWHNDLLGIT MKVKMPXTHK AIQADGWMCH 60
AAKWITTCDF RYWGPKYITH SIHSIQPTSE QCKESIKQTK QGTWMSPGFP PQNCGYATVT 120
DSVAVVVQAT PHHVLVDEYT GEWIDSQFPN GKCETEECET VHNSTVWYSD YKVTGLCDAT 180
LVDTEITFFS EDGKKESIGK PNTGYRSNYF AYEKGDKVCK MNYCKHAGVR LPSGVWFEFV 240
DQDVYAAAKL PECPVGATIS APTQTSVDVS LILDVERILD YSLCQETWSK IRSKQPVSPV 300
DLSYLAPKNP GTGPAFTIIN GTLKYPETRY IRIDIDNPII SKMVGKISGS QTEXELWTEW 360
FPYEGVEIGP NGILKTPTGY KFPLFMIGHG MLDSDLHKTS QAEVFEHPL AEAPKQLPEE 420
ETLFFGDTGI SKNPVELIEG WFSSWKSTVV TFFFAIGVFI LLYVVARIVI AVRYRYQGSN 480
NKRIYNDIEM SRFRK 495

SEQ ID NO: 129      moltype = AA length = 495
FEATURE            Location/Qualifiers
VARIANT            8
                    note = Xaa can be any naturally occurring amino acid except
                    Lys or Arg
VARIANT            47
                    note = Xaa can be any naturally occurring amino acid except
                    Lys or Arg
VARIANT            354
                    note = Xaa can be any naturally occurring amino acid except
                    Lys or Arg
source              1..495
                    mol_type = protein
                    organism = Vesicular stomatitis virus

SEQUENCE: 129
KFSIVFPXSQ KGNWKNVPSS YHYCPSSSDQ NWHNDLLGIT MKVKMPXTHK AIQADGWMCH 60
AAKWITTCDF RYWGPKYITH SIHSIQPTSE QCKESIKQTK QGTWMSPGFP PQNCGYATVT 120
DSVAVVVQAT PHHVLVDEYT GEWIDSQFPN GKCETEECET VHNSTVWYSD YKVTGLCDAT 180
LVDTEITFFS EDGKKESIGK PNTGYRSNYF AYEKGDKVCK MNYCKHAGVR LPSGVWFEFV 240
DQDVYAAAKL PECPVGATIS APTQTSVDVS LILDVERILD YSLCQETWSK IRSKQPVSPV 300
DLSYLAPKNP GTGPAFTIIN GTLKYPETRY IRIDIDNPII SKMVGKISGS QTEXELWTEW 360
FPYEGVEIGP NGILKTPTGY KFPLFMIGHG MLDSDLHKTS QAEVFEHPL AEAPKQLPEE 420
ETLFFGDTGI SKNPVELIEG WFSSWKSTVV TFFFAIGVFI LLYVVARIVI AVRYRYQGSN 480
NKRIYNDIEM SRFRK 495

SEQ ID NO: 130      moltype = AA length = 495
FEATURE            Location/Qualifiers
VARIANT            47
                    note = Xaa can be any naturally occurring amino acid except
                    Lys or Arg
VARIANT            209
                    note = Xaa can be any naturally occurring amino acid except
                    Lys or Arg
source              1..495
                    mol_type = protein
                    organism = Vesicular stomatitis virus

SEQUENCE: 130
KFSIVFPXSQ KGNWKNVPSS YHYCPSSSDQ NWHNDLLGIT MKVKMPXTHK AIQADGWMCH 60
AAKWITTCDF RYWGPKYITH SIHSIQPTSE QCKESIKQTK QGTWMSPGFP PQNCGYATVT 120
DSVAVVVQAT PHHVLVDEYT GEWIDSQFPN GKCETEECET VHNSTVWYSD YKVTGLCDAT 180
LVDTEITFFS EDGKKESIGK PNTGYRSNXP AYEKGDKVCK MNYCKHAGVR LPSGVWFEFV 240
DQDVYAAAKL PECPVGATIS APTQTSVDVS LILDVERILD YSLCQETWSK IRSKQPVSPV 300
DLSYLAPKNP GTGPAFTIIN GTLKYPETRY IRIDIDNPII SKMVGKISGS QTERELWTEW 360
FPYEGVEIGP NGILKTPTGY KFPLFMIGHG MLDSDLHKTS QAEVFEHPL AEAPKQLPEE 420

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ETLFFGDTGI SKNPVELIEG WFSSWKSTVV TFFFAIGVFI LLYVVARIVI AVRYRYQGSN 480
NKRIYNDIEM SRFRK 495

SEQ ID NO: 131      moltype = AA length = 495
FEATURE            Location/Qualifiers
VARIANT            209
                    note = Xaa can be any naturally occurring amino acid except
                    Lys or Arg
VARIANT            354
                    note = Xaa can be any naturally occurring amino acid except
                    Lys or Arg
source              1..495
                    mol_type = protein
                    organism = Vesicular stomatitis virus

SEQUENCE: 131
KFSIVFPQSQ KGNWKNVPSS YHYCPSSSDQ NWHNDLLGIT MKVKMPKTHK AIQADGWMCH 60
AAKWITTCDF RWYGPKYITH SIHSIQPTSE QCKESIKQTK QGTWMSPGFP PQNCGYATVT 120
DSVAVVVQAT PHHVLVDEYT GEWIDSQFPN GKCETEECET VHNSTVWYSD YKVTGLCDAT 180
LVDTEITFFS EDGKKESIGK PNTGYRSNXF AYEKGDVKCK MNYCKHAGVR LPSGVWFEFV 240
DQDVYAAAKL PECPGVATIS APTQTSVDVS LILDVERILD YSLCQETWSK IRSKQPVSPV 300
DLSYLAPKNP GTGPAFTIIN GTLKYPETRY IRIDIDNPPI SKMVGKISGS QTEXELWTEW 360
FPYEGVEIGP NGILKTPTGY KFPLFMIGHG MLDSDLHKTS QAEVFEHPLH AEAPKQLPEE 420
ETLFFGDTGI SKNPVELIEG WFSSWKSTVV TFFFAIGVFI LLYVVARIVI AVRYRYQGSN 480
NKRIYNDIEM SRFRK 495

SEQ ID NO: 132      moltype = AA length = 495
FEATURE            Location/Qualifiers
VARIANT            47
                    note = Xaa can be any naturally occurring amino acid except
                    Lys or Arg
VARIANT            209
                    note = Xaa can be any naturally occurring amino acid except
                    Lys or Arg
VARIANT            354
                    note = Xaa can be any naturally occurring amino acid except
                    Lys or Arg
source              1..495
                    mol_type = protein
                    organism = Vesicular stomatitis virus

SEQUENCE: 132
KFSIVFPQSQ KGNWKNVPSS YHYCPSSSDQ NWHNDLLGIT MKVKMPXTHK AIQADGWMCH 60
AAKWITTCDF RWYGPKYITH SIHSIQPTSE QCKESIKQTK QGTWMSPGFP PQNCGYATVT 120
DSVAVVVQAT PHHVLVDEYT GEWIDSQFPN GKCETEECET VHNSTVWYSD YKVTGLCDAT 180
LVDTEITFFS EDGKKESIGK PNTGYRSNXF AYEKGDVKCK MNYCKHAGVR LPSGVWFEFV 240
DQDVYAAAKL PECPGVATIS APTQTSVDVS LILDVERILD YSLCQETWSK IRSKQPVSPV 300
DLSYLAPKNP GTGPAFTIIN GTLKYPETRY IRIDIDNPPI SKMVGKISGS QTEXELWTEW 360
FPYEGVEIGP NGILKTPTGY KFPLFMIGHG MLDSDLHKTS QAEVFEHPLH AEAPKQLPEE 420
ETLFFGDTGI SKNPVELIEG WFSSWKSTVV TFFFAIGVFI LLYVVARIVI AVRYRYQGSN 480
NKRIYNDIEM SRFRK 495

SEQ ID NO: 133      moltype = AA length = 495
FEATURE            Location/Qualifiers
VARIANT            8
                    note = Xaa can be any naturally occurring amino acid except
                    Lys or Arg
VARIANT            47
                    note = Xaa can be any naturally occurring amino acid except
                    Lys or Arg
VARIANT            209
                    note = Xaa can be any naturally occurring amino acid except
                    Lys or Arg
VARIANT            354
                    note = Xaa can be any naturally occurring amino acid except
                    Lys or Arg
source              1..495
                    mol_type = protein
                    organism = Vesicular stomatitis virus

SEQUENCE: 133
KFSIVFPXSQ KGNWKNVPSS YHYCPSSSDQ NWHNDLLGIT MKVKMPXTHK AIQADGWMCH 60
AAKWITTCDF RWYGPKYITH SIHSIQPTSE QCKESIKQTK QGTWMSPGFP PQNCGYATVT 120
DSVAVVVQAT PHHVLVDEYT GEWIDSQFPN GKCETEECET VHNSTVWYSD YKVTGLCDAT 180
LVDTEITFFS EDGKKESIGK PNTGYRSNXF AYEKGDVKCK MNYCKHAGVR LPSGVWFEFV 240
DQDVYAAAKL PECPGVATIS APTQTSVDVS LILDVERILD YSLCQETWSK IRSKQPVSPV 300
DLSYLAPKNP GTGPAFTIIN GTLKYPETRY IRIDIDNPPI SKMVGKISGS QTEXELWTEW 360
FPYEGVEIGP NGILKTPTGY KFPLFMIGHG MLDSDLHKTS QAEVFEHPLH AEAPKQLPEE 420
ETLFFGDTGI SKNPVELIEG WFSSWKSTVV TFFFAIGVFI LLYVVARIVI AVRYRYQGSN 480
NKRIYNDIEM SRFRK 495

SEQ ID NO: 134      moltype = AA length = 512

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FEATURE      Location/Qualifiers
VARIANT      25
              note = Xaa can be any naturally occurring amino acid except
              Lys or Arg
VARIANT      64
              note = Xaa can be any naturally occurring amino acid except
              Lys or Arg
source      1..512
            mol_type = protein
            organism = Vesicular stomatitis virus

SEQUENCE: 134
MNFLLLTFIV LPLCSHAKFS IVFPXSQKGN WKNVPSSYHY CPSSSDQNW NHDLGIMTKV 60
KMPXTHKAIQ ADGWMCHAAK WITTCDFRWY GPKYITHSIH SIQPTSEQCK ESIKQTKQGT 120
WMSPGFPPQN CGYATVTDV AVVVQATPHH VLVDEYTG EW IDSQFPNGKC ETEECETVHN 180
STVWYSYDKV TGLCDATLVD TEITFFSEDG KKE SIGKPNT GYRSNYFAYE KGDVKCKMNY 240
CKHAGVRLPS GVVWFEFVDQD VYAAAKLPEC PVGATISAPT QTSVDVSLIL DVERILDYSL 300
CQETWSKIRS KQPVSPVDLS YLAPKNPGTG PAF TIINGTL KYFETRYIRI DIDNPIISKM 360
VGKISGSQTE RELWTEWFPY EGVEIGPNGI LKTP TGYKFP LFMIGHGMLD SDLHKTSQAE 420
VFEHPLAE A PKQLPEEETL FFGDTGISK N PVELIEGWFS SWKSTVVTF FFAIGVFILLY 480
VVARIVIAVR YRYQGSNNKR IYNDIEMSRF RK 512

SEQ ID NO: 135      moltype = AA length = 512
FEATURE      Location/Qualifiers
VARIANT      25
              note = Xaa can be any naturally occurring amino acid except
              Lys or Arg
VARIANT      371
              note = Xaa can be any naturally occurring amino acid except
              Lys or Arg
source      1..512
            mol_type = protein
            organism = Vesicular stomatitis virus

SEQUENCE: 135
MNFLLLTFIV LPLCSHAKFS IVFPXSQKGN WKNVPSSYHY CPSSSDQNW NHDLGIMTKV 60
KMPKTHKAIQ ADGWMCHAAK WITTCDFRWY GPKYITHSIH SIQPTSEQCK ESIKQTKQGT 120
WMSPGFPPQN CGYATVTDV AVVVQATPHH VLVDEYTG EW IDSQFPNGKC ETEECETVHN 180
STVWYSYDKV TGLCDATLVD TEITFFSEDG KKE SIGKPNT GYRSNYFAYE KGDVKCKMNY 240
CKHAGVRLPS GVVWFEFVDQD VYAAAKLPEC PVGATISAPT QTSVDVSLIL DVERILDYSL 300
CQETWSKIRS KQPVSPVDLS YLAPKNPGTG PAF TIINGTL KYFETRYIRI DIDNPIISKM 360
VGKISGSQTE XELWTEWFPY EGVEIGPNGI LKTP TGYKFP LFMIGHGMLD SDLHKTSQAE 420
VFEHPLAE A PKQLPEEETL FFGDTGISK N PVELIEGWFS SWKSTVVTF FFAIGVFILLY 480
VVARIVIAVR YRYQGSNNKR IYNDIEMSRF RK 512

SEQ ID NO: 136      moltype = AA length = 512
FEATURE      Location/Qualifiers
VARIANT      25
              note = Xaa can be any naturally occurring amino acid except
              Lys or Arg
VARIANT      64
              note = Xaa can be any naturally occurring amino acid except
              Lys or Arg
VARIANT      371
              note = Xaa can be any naturally occurring amino acid except
              Lys or Arg
source      1..512
            mol_type = protein
            organism = Vesicular stomatitis virus

SEQUENCE: 136
MNFLLLTFIV LPLCSHAKFS IVFPXSQKGN WKNVPSSYHY CPSSSDQNW NHDLGIMTKV 60
KMPXTHKAIQ ADGWMCHAAK WITTCDFRWY GPKYITHSIH SIQPTSEQCK ESIKQTKQGT 120
WMSPGFPPQN CGYATVTDV AVVVQATPHH VLVDEYTG EW IDSQFPNGKC ETEECETVHN 180
STVWYSYDKV TGLCDATLVD TEITFFSEDG KKE SIGKPNT GYRSNYFAYE KGDVKCKMNY 240
CKHAGVRLPS GVVWFEFVDQD VYAAAKLPEC PVGATISAPT QTSVDVSLIL DVERILDYSL 300
CQETWSKIRS KQPVSPVDLS YLAPKNPGTG PAF TIINGTL KYFETRYIRI DIDNPIISKM 360
VGKISGSQTE XELWTEWFPY EGVEIGPNGI LKTP TGYKFP LFMIGHGMLD SDLHKTSQAE 420
VFEHPLAE A PKQLPEEETL FFGDTGISK N PVELIEGWFS SWKSTVVTF FFAIGVFILLY 480
VVARIVIAVR YRYQGSNNKR IYNDIEMSRF RK 512

SEQ ID NO: 137      moltype = AA length = 512
FEATURE      Location/Qualifiers
VARIANT      64
              note = Xaa can be any naturally occurring amino acid except
              Lys or Arg
VARIANT      226
              note = Xaa can be any naturally occurring amino acid except
              Lys or Arg
source      1..512
            mol_type = protein
            organism = Vesicular stomatitis virus

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-continued

SEQUENCE: 137

MNFLLLTFIV	LPLCSHAKFS	IVFPQSQKGN	WKNVPSSYHY	CPSSSDQNW	NDLLGITMKV	60
KMPXTHKAIQ	ADGWMCHAAK	WITTCDFRWY	GPKYITHSIH	SIQPTSEQCK	ESIKQTKQGT	120
WMSPGFPPQN	CGYATVTDV	AVVVQATPHH	VLVDEYTG	IDSQFPNGKC	ETEECETVHN	180
STVWYSYK	TGLCDATLVD	TEITFFSE	EDG	KKESIGKPNT	GYRSNXFAYE	240
CKHAGVRLPS	GVWFEFVDQD	VYAAAKLPEC	PVGATISAPT	QTSVDVSLIL	DVERILDYSL	300
CQETWSKIRS	KQPVSPVDLS	YLAPKNPGTG	PAFTIINGTL	KYFETRYIRI	DIDNPIISKM	360
VGKISGSQTE	RELWTEWFPY	EGVEIGPNGI	LKTPTGYKFP	LFMIGHGMLD	SDLHKTSQAE	420
VFEHPLHAEA	PKQLPEEETL	FFGDTGISK	N	PVELIEGWFS	SWKSTVVTF	480
VVARIVIAVR	YRYQGSNNKR	IYNDIEMSRF	RK			512

SEQ ID NO: 138 moltype = AA length = 512

FEATURE Location/Qualifiers

VARIANT 226

 note = Xaa can be any naturally occurring amino acid except

 Lys or Arg

VARIANT 371

 note = Xaa can be any naturally occurring amino acid except

 Lys or Arg

source 1..512

 mol_type = protein

 organism = Vesicular stomatitis virus

SEQUENCE: 138

MNFLLLTFIV	LPLCSHAKFS	IVFPQSQKGN	WKNVPSSYHY	CPSSSDQNW	NDLLGITMKV	60
KMPKTHKAIQ	ADGWMCHAAK	WITTCDFRWY	GPKYITHSIH	SIQPTSEQCK	ESIKQTKQGT	120
WMSPGFPPQN	CGYATVTDV	AVVVQATPHH	VLVDEYTG	IDSQFPNGKC	ETEECETVHN	180
STVWYSYK	TGLCDATLVD	TEITFFSE	EDG	KKESIGKPNT	GYRSNXFAYE	240
CKHAGVRLPS	GVWFEFVDQD	VYAAAKLPEC	PVGATISAPT	QTSVDVSLIL	DVERILDYSL	300
CQETWSKIRS	KQPVSPVDLS	YLAPKNPGTG	PAFTIINGTL	KYFETRYIRI	DIDNPIISKM	360
VGKISGSQTE	XELWTEWFPY	EGVEIGPNGI	LKTPTGYKFP	LFMIGHGMLD	SDLHKTSQAE	420
VFEHPLHAEA	PKQLPEEETL	FFGDTGISK	N	PVELIEGWFS	SWKSTVVTF	480
VVARIVIAVR	YRYQGSNNKR	IYNDIEMSRF	RK			512

SEQ ID NO: 139 moltype = AA length = 512

FEATURE Location/Qualifiers

VARIANT 64

 note = Xaa can be any naturally occurring amino acid except

 Lys or Arg

VARIANT 226

 note = Xaa can be any naturally occurring amino acid except

 Lys or Arg

VARIANT 371

 note = Xaa can be any naturally occurring amino acid except

 Lys or Arg

source 1..512

 mol_type = protein

 organism = Vesicular stomatitis virus

SEQUENCE: 139

MNFLLLTFIV	LPLCSHAKFS	IVFPQSQKGN	WKNVPSSYHY	CPSSSDQNW	NDLLGITMKV	60
KMPXTHKAIQ	ADGWMCHAAK	WITTCDFRWY	GPKYITHSIH	SIQPTSEQCK	ESIKQTKQGT	120
WMSPGFPPQN	CGYATVTDV	AVVVQATPHH	VLVDEYTG	IDSQFPNGKC	ETEECETVHN	180
STVWYSYK	TGLCDATLVD	TEITFFSE	EDG	KKESIGKPNT	GYRSNXFAYE	240
CKHAGVRLPS	GVWFEFVDQD	VYAAAKLPEC	PVGATISAPT	QTSVDVSLIL	DVERILDYSL	300
CQETWSKIRS	KQPVSPVDLS	YLAPKNPGTG	PAFTIINGTL	KYFETRYIRI	DIDNPIISKM	360
VGKISGSQTE	XELWTEWFPY	EGVEIGPNGI	LKTPTGYKFP	LFMIGHGMLD	SDLHKTSQAE	420
VFEHPLHAEA	PKQLPEEETL	FFGDTGISK	N	PVELIEGWFS	SWKSTVVTF	480
VVARIVIAVR	YRYQGSNNKR	IYNDIEMSRF	RK			512

SEQ ID NO: 140 moltype = AA length = 512

FEATURE Location/Qualifiers

VARIANT 25

 note = Xaa can be any naturally occurring amino acid except

 Lys or Arg

VARIANT 64

 note = Xaa can be any naturally occurring amino acid except

 Lys or Arg

VARIANT 226

 note = Xaa can be any naturally occurring amino acid except

 Lys or Arg

VARIANT 371

 note = Xaa can be any naturally occurring amino acid except

 Lys or Arg

source 1..512

 mol_type = protein

 organism = Vesicular stomatitis virus

SEQUENCE: 140

MNFLLLTFIV	LPLCSHAKFS	IVFPXSQKGN	WKNVPSSYHY	CPSSSDQNW	NDLLGITMKV	60
KMPXTHKAIQ	ADGWMCHAAK	WITTCDFRWY	GPKYITHSIH	SIQPTSEQCK	ESIKQTKQGT	120
WMSPGFPPQN	CGYATVTDV	AVVVQATPHH	VLVDEYTG	IDSQFPNGKC	ETEECETVHN	180

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STVWYSYDKV TGLCDATLVD TEITFFSEDG KKEISIGKPT GYRSNXFAYE KGDVKCKMNY 240
CKHAGVRLPS GVWFEFVDQD VYAAAKLPEC PVGATISAPT QTSVDVSLIL DVERILDYSL 300
CQETWSKIRS KQPVSPVDLS YLAPKNPGTG PARTIINGTL KYFETRYIRI DIDNPIISKM 360
VGKISGSQTE XELWTEWFPY EGVEIGPNGI LKTPPTGYKFP LFMIGHGMLD SDLHKTSQAE 420
VFEHPLAEAE PKQLPEEETL FFGDTGISKV PVELIEGWFS SWKSTVVTFE FAIGVFILLY 480
VVARIVIAVR YRYQGSNNKR IYNDIEMSRF RK 512

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SEQ ID NO: 141      moltype = AA  length = 496
FEATURE            Location/Qualifiers
VARIANT            8
                    note = Xaa can be any naturally occurring amino acid except
                    Lys or Arg
VARIANT            47
                    note = Xaa can be any naturally occurring amino acid except
                    Lys or Arg
source              1..496
                    mol_type = protein
                    organism = Vesicular stomatitis virus

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SEQUENCE: 141
KFTIVFPXNQ KGNWKNVPAN YQYCPSSSDL NWHNGLIGTS LQVKMPXSHK AIQADGWMCH 60
AAKWVTTCDF RWYGPKYVTH SIKSMIPTVD QCKESIAQTK QGTWLNPGFP PQSCGYASVT 120
DAEAVIVKAT PHQVLVDEYT GEWVDSQFPT GKCNKDICTP VHNSTTWHSY YKVTGLCDAN 180
LISMDITFFS EDGKLTSGLK EGTGFRSNYF AYENGDKACR MQYCKHWGVR LPSGVWFEMA 240
DKDIYNDAKF PDCPEGSSIA APSQTSVDVS LIQDVERILD YSLCQETWSK IRAHLPISPV 300
DLSYLSPKNP GTGPAFTIIN GTLKYFETRY IRVDIAGPII PQMRGVISGT TTERELWTDW 360
YPYEDVEIGP NGVLKTATGY KPPLYMIGHG MLDSDLHISS KAQVFEHPHI QDAASQLPDD 420
ETLFFGDTGL SKNPIELVEG WFSGWKSTIA SFFFIIGLVI GLYLVLRIGI ALCIKCRVQE 480
KRPKIYTDVE MNRLDR 496

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SEQ ID NO: 142      moltype = AA  length = 496
FEATURE            Location/Qualifiers
VARIANT            8
                    note = Xaa can be any naturally occurring amino acid except
                    Lys or Arg
VARIANT            354
                    note = Xaa can be any naturally occurring amino acid except
                    Lys or Arg
source              1..496
                    mol_type = protein
                    organism = Vesicular stomatitis virus

```

```

SEQUENCE: 142
KFTIVFPXNQ KGNWKNVPAN YQYCPSSSDL NWHNGLIGTS LQVKMPXSHK AIQADGWMCH 60
AAKWVTTCDF RWYGPKYVTH SIKSMIPTVD QCKESIAQTK QGTWLNPGFP PQSCGYASVT 120
DAEAVIVKAT PHQVLVDEYT GEWVDSQFPT GKCNKDICTP VHNSTTWHSY YKVTGLCDAN 180
LISMDITFFS EDGKLTSGLK EGTGFRSNYF AYENGDKACR MQYCKHWGVR LPSGVWFEMA 240
DKDIYNDAKF PDCPEGSSIA APSQTSVDVS LIQDVERILD YSLCQETWSK IRAHLPISPV 300
DLSYLSPKNP GTGPAFTIIN GTLKYFETRY IRVDIAGPII PQMRGVISGT TTEXELWTDW 360
YPYEDVEIGP NGVLKTATGY KPPLYMIGHG MLDSDLHISS KAQVFEHPHI QDAASQLPDD 420
ETLFFGDTGL SKNPIELVEG WFSGWKSTIA SFFFIIGLVI GLYLVLRIGI ALCIKCRVQE 480
KRPKIYTDVE MNRLDR 496

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SEQ ID NO: 143      moltype = AA  length = 496
FEATURE            Location/Qualifiers
VARIANT            8
                    note = Xaa can be any naturally occurring amino acid except
                    Lys or Arg
VARIANT            47
                    note = Xaa can be any naturally occurring amino acid except
                    Lys or Arg
VARIANT            354
                    note = Xaa can be any naturally occurring amino acid except
                    Lys or Arg
source              1..496
                    mol_type = protein
                    organism = Vesicular stomatitis virus

```

```

SEQUENCE: 143
KFTIVFPXNQ KGNWKNVPAN YQYCPSSSDL NWHNGLIGTS LQVKMPXSHK AIQADGWMCH 60
AAKWVTTCDF RWYGPKYVTH SIKSMIPTVD QCKESIAQTK QGTWLNPGFP PQSCGYASVT 120
DAEAVIVKAT PHQVLVDEYT GEWVDSQFPT GKCNKDICTP VHNSTTWHSY YKVTGLCDAN 180
LISMDITFFS EDGKLTSGLK EGTGFRSNYF AYENGDKACR MQYCKHWGVR LPSGVWFEMA 240
DKDIYNDAKF PDCPEGSSIA APSQTSVDVS LIQDVERILD YSLCQETWSK IRAHLPISPV 300
DLSYLSPKNP GTGPAFTIIN GTLKYFETRY IRVDIAGPII PQMRGVISGT TTEXELWTDW 360
YPYEDVEIGP NGVLKTATGY KPPLYMIGHG MLDSDLHISS KAQVFEHPHI QDAASQLPDD 420
ETLFFGDTGL SKNPIELVEG WFSGWKSTIA SFFFIIGLVI GLYLVLRIGI ALCIKCRVQE 480
KRPKIYTDVE MNRLDR 496

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SEQ ID NO: 144      moltype = AA  length = 496
FEATURE            Location/Qualifiers
VARIANT            47

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VARIANT note = Xaa can be any naturally occurring amino acid except
 Lys or Arg
 209
 note = Xaa can be any naturally occurring amino acid except
 Lys or Arg
 source 1..496
 mol_type = protein
 organism = Vesicular stomatitis virus

SEQUENCE: 144
 KFTIVFPHNQ KGNWKNVPAN YQYCPSSSDL NWHNGLIGTS LQVKMPKSHK AIQADGWMCH 60
 AAKWVTTCDF RWYGPKYVTH SIKSMIPTVD QCKESIAQTK QGTWLNPGFP PQSCGYASVT 120
 DAEAVIVKAT PHQVLVDEYT GEWVDSQFPT GKCNDICPT VHNSTTWHS YKVTGLCDAN 180
 LISMDITFFS EDGKLTSLGK EGTGFRSNXF AYENGDKACR MQYCKHWGVR LPSGVWFEMA 240
 DKDIYNDAKF PDCPEGSSIA APSQTSVDVS LIQDVERILD YSLCQETWSK IRAHLPISPV 300
 DLSYLSPKNP GTGPAFTIIN GTLKYFETRY IRVDIAGPII PQMRGVISGT TTERELWTDW 360
 YPYEDVEIGP NGVLKTATGY KPPLYMIGHG MLDSDLHISS KAQVFEHPHI QDAASQLPDD 420
 ETLFFGDTGL SKNPIELVEG WFSGWKSTIA SFFFIIGLVI GLYLVLRIGI ALCIKCRVQE 480
 KRPKIYTDVE MNRLDR 496

SEQ ID NO: 145 moltype = AA length = 496
 FEATURE Location/Qualifiers
 VARIANT 209
 note = Xaa can be any naturally occurring amino acid except
 Lys or Arg
 354
 note = Xaa can be any naturally occurring amino acid except
 Lys or Arg
 source 1..496
 mol_type = protein
 organism = Vesicular stomatitis virus

SEQUENCE: 145
 KFTIVFPHNQ KGNWKNVPAN YQYCPSSSDL NWHNGLIGTS LQVKMPKSHK AIQADGWMCH 60
 AAKWVTTCDF RWYGPKYVTH SIKSMIPTVD QCKESIAQTK QGTWLNPGFP PQSCGYASVT 120
 DAEAVIVKAT PHQVLVDEYT GEWVDSQFPT GKCNDICPT VHNSTTWHS YKVTGLCDAN 180
 LISMDITFFS EDGKLTSLGK EGTGFRSNXF AYENGDKACR MQYCKHWGVR LPSGVWFEMA 240
 DKDIYNDAKF PDCPEGSSIA APSQTSVDVS LIQDVERILD YSLCQETWSK IRAHLPISPV 300
 DLSYLSPKNP GTGPAFTIIN GTLKYFETRY IRVDIAGPII PQMRGVISGT TTEXELWTDW 360
 YPYEDVEIGP NGVLKTATGY KPPLYMIGHG MLDSDLHISS KAQVFEHPHI QDAASQLPDD 420
 ETLFFGDTGL SKNPIELVEG WFSGWKSTIA SFFFIIGLVI GLYLVLRIGI ALCIKCRVQE 480
 KRPKIYTDVE MNRLDR 496

SEQ ID NO: 146 moltype = AA length = 496
 FEATURE Location/Qualifiers
 VARIANT 47
 note = Xaa can be any naturally occurring amino acid except
 Lys or Arg
 209
 note = Xaa can be any naturally occurring amino acid except
 Lys or Arg
 354
 note = Xaa can be any naturally occurring amino acid except
 Lys or Arg
 source 1..496
 mol_type = protein
 organism = Vesicular stomatitis virus

SEQUENCE: 146
 KFTIVFPHNQ KGNWKNVPAN YQYCPSSSDL NWHNGLIGTS LQVKMPKSHK AIQADGWMCH 60
 AAKWVTTCDF RWYGPKYVTH SIKSMIPTVD QCKESIAQTK QGTWLNPGFP PQSCGYASVT 120
 DAEAVIVKAT PHQVLVDEYT GEWVDSQFPT GKCNDICPT VHNSTTWHS YKVTGLCDAN 180
 LISMDITFFS EDGKLTSLGK EGTGFRSNXF AYENGDKACR MQYCKHWGVR LPSGVWFEMA 240
 DKDIYNDAKF PDCPEGSSIA APSQTSVDVS LIQDVERILD YSLCQETWSK IRAHLPISPV 300
 DLSYLSPKNP GTGPAFTIIN GTLKYFETRY IRVDIAGPII PQMRGVISGT TTEXELWTDW 360
 YPYEDVEIGP NGVLKTATGY KPPLYMIGHG MLDSDLHISS KAQVFEHPHI QDAASQLPDD 420
 ETLFFGDTGL SKNPIELVEG WFSGWKSTIA SFFFIIGLVI GLYLVLRIGI ALCIKCRVQE 480
 KRPKIYTDVE MNRLDR 496

SEQ ID NO: 147 moltype = AA length = 496
 FEATURE Location/Qualifiers
 VARIANT 8
 note = Xaa can be any naturally occurring amino acid except
 Lys or Arg
 47
 note = Xaa can be any naturally occurring amino acid except
 Lys or Arg
 209
 note = Xaa can be any naturally occurring amino acid except
 Lys or Arg
 354
 note = Xaa can be any naturally occurring amino acid except

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source          Lys or Arg
                1..496
                mol_type = protein
                organism = Vesicular stomatitis virus

SEQUENCE: 147
KFTIVFPXNQ KGNWKNVPAN YQYCPSSSDL NWHNGLIGTS LQVKMPXSHK AIQADGWMCH 60
AAKWVTTCDF RWYGPKYVTH SIKSMIPTVD QCKESIAQTK QGTWLNPGFP PQSCGYASVT 120
DAEAVIVKAT PHQVLVDEYT GEWVDSQFPT GKCNDICPT VHNSTTWHSD YKVTGLCDAN 180
LISMDITFFS EDGKLTSLGK EGTGFRSNXF AYENGDKACR MQYCKHWGVR LPSGVWFEMA 240
DKDIYNDAKF PDCPEGSSIA APSQTSVDVS LIQDVERILD YSLCQETWSK IRAHLPISPV 300
DLSYLSPKNP GTGPAFTIIN GTLKYPETRY IRVDIAGPII PQMRGVISGT TTEXELWTDW 360
YPYEDVEIGP NGVLKTATGY KPPLYMIGHG MLDSDLHISS KAQVFEHPHI QDAASQLPDD 420
ETLFFGDTGL SKNPIELVEG WFSGWKSTIA SFFFIIGLVI GLYLVLIRIGI ALCKICRVQE 480
KRPKIYTDVE MNRLDR 496

SEQ ID NO: 148      moltype = AA length = 513
FEATURE            Location/Qualifiers
VARIANT            25
                    note = Xaa can be any naturally occurring amino acid except
                    Lys or Arg
VARIANT            64
                    note = Xaa can be any naturally occurring amino acid except
                    Lys or Arg
source             1..513
                    mol_type = protein
                    organism = Vesicular stomatitis virus

SEQUENCE: 148
MLVLYLLLSL LALGAQCKFT IVFPXNQKGN WKNVPANYQY CPSSSDLNWH NGLIGTSLQV 60
KMPXSHKAIQ ADGWMCHAAK WVTTCDFRWY GPKYVTHSIK SMIPTVDQCK ESIAQTKQGT 120
WLNPGFPPQS CGYASVTDAA AVIVKATPHQ VLVDEYTGGEW VDSQFPTGKC NKDICTPTVHN 180
STTWHSDYKV TGLCDANLIS MDITFFSEDG KLTSLGKEGT GFRSNYFAYE NGDKACRMQY 240
CKHWGVRLPS GVVWFEMADK IYNDAKFPDC PEGSSIAAPS QTSVDVSLIQ DVERILDYSL 300
CQETWSKIRA HLPISPDVLS YLSPKNPGTG PFTIINGTL KYFETRYIRV DIAGPIIPQM 360
RGVISGTTTE RELWTDWYPY EDVEIGPNGV LKTATGYKFP LYMIGHGMLD SDLHISSKAQ 420
VFEHPHIQDA ASQLPDDET L FFGDTGLSKN PIELVEGWFS GWKSTIASFF FIIGLVIGLY 480
LVLIRIGIALC IKCRVQEKRP KIYTDVEMNR LDR 513

SEQ ID NO: 149      moltype = AA length = 513
FEATURE            Location/Qualifiers
VARIANT            25
                    note = Xaa can be any naturally occurring amino acid except
                    Lys or Arg
VARIANT            371
                    note = Xaa can be any naturally occurring amino acid except
                    Lys or Arg
source             1..513
                    mol_type = protein
                    organism = Vesicular stomatitis virus

SEQUENCE: 149
MLVLYLLLSL LALGAQCKFT IVFPXNQKGN WKNVPANYQY CPSSSDLNWH NGLIGTSLQV 60
KMPXSHKAIQ ADGWMCHAAK WVTTCDFRWY GPKYVTHSIK SMIPTVDQCK ESIAQTKQGT 120
WLNPGFPPQS CGYASVTDAA AVIVKATPHQ VLVDEYTGGEW VDSQFPTGKC NKDICTPTVHN 180
STTWHSDYKV TGLCDANLIS MDITFFSEDG KLTSLGKEGT GFRSNYFAYE NGDKACRMQY 240
CKHWGVRLPS GVVWFEMADK IYNDAKFPDC PEGSSIAAPS QTSVDVSLIQ DVERILDYSL 300
CQETWSKIRA HLPISPDVLS YLSPKNPGTG PFTIINGTL KYFETRYIRV DIAGPIIPQM 360
RGVISGTTTE RELWTDWYPY EDVEIGPNGV LKTATGYKFP LYMIGHGMLD SDLHISSKAQ 420
VFEHPHIQDA ASQLPDDET L FFGDTGLSKN PIELVEGWFS GWKSTIASFF FIIGLVIGLY 480
LVLIRIGIALC IKCRVQEKRP KIYTDVEMNR LDR 513

SEQ ID NO: 150      moltype = AA length = 513
FEATURE            Location/Qualifiers
VARIANT            25
                    note = Xaa can be any naturally occurring amino acid except
                    Lys or Arg
VARIANT            64
                    note = Xaa can be any naturally occurring amino acid except
                    Lys or Arg
VARIANT            371
                    note = Xaa can be any naturally occurring amino acid except
                    Lys or Arg
source             1..513
                    mol_type = protein
                    organism = Vesicular stomatitis virus

SEQUENCE: 150
MLVLYLLLSL LALGAQCKFT IVFPXNQKGN WKNVPANYQY CPSSSDLNWH NGLIGTSLQV 60
KMPXSHKAIQ ADGWMCHAAK WVTTCDFRWY GPKYVTHSIK SMIPTVDQCK ESIAQTKQGT 120
WLNPGFPPQS CGYASVTDAA AVIVKATPHQ VLVDEYTGGEW VDSQFPTGKC NKDICTPTVHN 180
STTWHSDYKV TGLCDANLIS MDITFFSEDG KLTSLGKEGT GFRSNYFAYE NGDKACRMQY 240
CKHWGVRLPS GVVWFEMADK IYNDAKFPDC PEGSSIAAPS QTSVDVSLIQ DVERILDYSL 300

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CQETWSKIRA HLPISPVDLS YLSPKNPGTG PAFITTINGTL KYFETRYIRV DIAGPIIPQM 360
 RGVISGTTTE XELWTDWYPY EDVEIGPNGV LKTATGYKFP LYMIGHGMLD SDLHISSKAQ 420
 VFEHPHIQDA ASQLPDDETl FFGDTGLSKN PIELVEGWFS GWKSTIASFF FIIGLVIGLY 480
 LVLRIIGIALC IKCRVQEKRP KIYTDVEMNR LDR 513

SEQ ID NO: 151 moltype = AA length = 513
 FEATURE Location/Qualifiers
 VARIANT 64
 note = Xaa can be any naturally occurring amino acid except
 Lys or Arg
 VARIANT 226
 note = Xaa can be any naturally occurring amino acid except
 Lys or Arg
 source 1..513
 mol_type = protein
 organism = Vesicular stomatitis virus

SEQUENCE: 151
 MLVLYLLLSL LALGAQCKFT IVFPHNQKGN WKNVPANYQY CPSSSDLNWH NGLIGTSLQV 60
 KMPXSHKAIQ ADGWMCHAAC WVTTCDFRWY GPKYVTHSIK SMIPTVDQCK ESIAQTKQGT 120
 WLNPGFPPQS CGYASVTDAA AVIVKATPHQ VLVDEYTG EW VDSQFPTGKC NKDICPTVHN 180
 STTWHSYDYK TGLCDANLIS MDITFFSEDG KLTSLGKEGT GFRSNXPAYE NGDKACRMQY 240
 CKHWGVRLLPS GVVWFEMADKD IYNDAKFPDC PEGSSIAAPS QTSVDVSLIQ DVERILDYSL 300
 CQETWSKIRA HLPISPVDLS YLSPKNPGTG PAFITTINGTL KYFETRYIRV DIAGPIIPQM 360
 RGVISGTTTE RELWTDWYPY EDVEIGPNGV LKTATGYKFP LYMIGHGMLD SDLHISSKAQ 420
 VFEHPHIQDA ASQLPDDETl FFGDTGLSKN PIELVEGWFS GWKSTIASFF FIIGLVIGLY 480
 LVLRIIGIALC IKCRVQEKRP KIYTDVEMNR LDR 513

SEQ ID NO: 152 moltype = AA length = 513
 FEATURE Location/Qualifiers
 VARIANT 226
 note = Xaa can be any naturally occurring amino acid except
 Lys or Arg
 VARIANT 371
 note = Xaa can be any naturally occurring amino acid except
 Lys or Arg
 source 1..513
 mol_type = protein
 organism = Vesicular stomatitis virus

SEQUENCE: 152
 MLVLYLLLSL LALGAQCKFT IVFPHNQKGN WKNVPANYQY CPSSSDLNWH NGLIGTSLQV 60
 KMPXSHKAIQ ADGWMCHAAC WVTTCDFRWY GPKYVTHSIK SMIPTVDQCK ESIAQTKQGT 120
 WLNPGFPPQS CGYASVTDAA AVIVKATPHQ VLVDEYTG EW VDSQFPTGKC NKDICPTVHN 180
 STTWHSYDYK TGLCDANLIS MDITFFSEDG KLTSLGKEGT GFRSNXPAYE NGDKACRMQY 240
 CKHWGVRLLPS GVVWFEMADKD IYNDAKFPDC PEGSSIAAPS QTSVDVSLIQ DVERILDYSL 300
 CQETWSKIRA HLPISPVDLS YLSPKNPGTG PAFITTINGTL KYFETRYIRV DIAGPIIPQM 360
 RGVISGTTTE XELWTDWYPY EDVEIGPNGV LKTATGYKFP LYMIGHGMLD SDLHISSKAQ 420
 VFEHPHIQDA ASQLPDDETl FFGDTGLSKN PIELVEGWFS GWKSTIASFF FIIGLVIGLY 480
 LVLRIIGIALC IKCRVQEKRP KIYTDVEMNR LDR 513

SEQ ID NO: 153 moltype = AA length = 513
 FEATURE Location/Qualifiers
 VARIANT 64
 note = Xaa can be any naturally occurring amino acid except
 Lys or Arg
 VARIANT 226
 note = Xaa can be any naturally occurring amino acid except
 Lys or Arg
 VARIANT 371
 note = Xaa can be any naturally occurring amino acid except
 Lys or Arg
 source 1..513
 mol_type = protein
 organism = Vesicular stomatitis virus

SEQUENCE: 153
 MLVLYLLLSL LALGAQCKFT IVFPHNQKGN WKNVPANYQY CPSSSDLNWH NGLIGTSLQV 60
 KMPXSHKAIQ ADGWMCHAAC WVTTCDFRWY GPKYVTHSIK SMIPTVDQCK ESIAQTKQGT 120
 WLNPGFPPQS CGYASVTDAA AVIVKATPHQ VLVDEYTG EW VDSQFPTGKC NKDICPTVHN 180
 STTWHSYDYK TGLCDANLIS MDITFFSEDG KLTSLGKEGT GFRSNXPAYE NGDKACRMQY 240
 CKHWGVRLLPS GVVWFEMADKD IYNDAKFPDC PEGSSIAAPS QTSVDVSLIQ DVERILDYSL 300
 CQETWSKIRA HLPISPVDLS YLSPKNPGTG PAFITTINGTL KYFETRYIRV DIAGPIIPQM 360
 RGVISGTTTE XELWTDWYPY EDVEIGPNGV LKTATGYKFP LYMIGHGMLD SDLHISSKAQ 420
 VFEHPHIQDA ASQLPDDETl FFGDTGLSKN PIELVEGWFS GWKSTIASFF FIIGLVIGLY 480
 LVLRIIGIALC IKCRVQEKRP KIYTDVEMNR LDR 513

SEQ ID NO: 154 moltype = AA length = 513
 FEATURE Location/Qualifiers
 VARIANT 25
 note = Xaa can be any naturally occurring amino acid except
 Lys or Arg

-continued

VARIANT 64
note = Xaa can be any naturally occurring amino acid except
Lys or Arg

VARIANT 226
note = Xaa can be any naturally occurring amino acid except
Lys or Arg

VARIANT 371
note = Xaa can be any naturally occurring amino acid except
Lys or Arg

source 1..513
mol_type = protein
organism = Vesicular stomatitis virus

SEQUENCE: 154

MLVLYLLLSL	LALGAQCKFT	IVFPXNQKGN	WKNVPANYQY	CPSSSDLNWH	NGLIGTSLQV	60
KMPXSHKAIQ	ADGWMCHAAK	WVTTCDFRWY	GPKYVTHSIK	SMIPTVDQCK	ESIAQTKQGT	120
WLNPGFPPQS	CGYASVTDAE	AVIVKATPHQ	VLVDEYTGWE	VDSQFPTGKC	NKDICTVHN	180
STTWSDYK	TGLCDANLIS	MDITFFSEDG	KLTSLGKEGT	GFRSNXPAYE	NGDKACRMQY	240
CKHWGVRLLP	GVWFEMADKD	IYNDAKFPPD	PEGSSIAAPS	QTSVDVSLIQ	DVERILDYSL	300
CQETWSKIRA	HLPISPVDLS	YLSPKNPGTG	PAFTIINGTL	KYFETRYIRV	DIAGPIIPQM	360
RGVISGTTTE	XELWTDWYPY	EDVEIGPNGV	LKTATGYKFP	LYMIGHGMLD	SDLHISSKAQ	420
VFEHPHIQDA	ASQLPDDDEL	FFGDTGLSKN	PIELVEGWFS	GWKSTIASFF	FIIGLVIGLY	480
LVLRIIGIALC	IKCRVQEKRP	KIYTDVEMNR	LDR			513

SEQ ID NO: 155 moltype = AA length = 495
FEATURE Location/Qualifiers
source 1..495
mol_type = protein
organism = Vesicular stomatitis virus

SEQUENCE: 155

KFTIVFPHNQ	KGNWKNVPSN	YHYCPSSSDL	NWHNDLIGTA	IQVKMPASHK	AIQADGWMCH	60
ASKWVTTCDF	RWYGPKYITQ	SIRSFTPSVE	QCKESIEQTK	QGTWLNPGFP	PQSCGYATVT	120
DAEAVIVQVT	PHHVLVDEYT	GEWVDSQFIN	GKCSNYICPT	VHNSTTWHS	YKVKGLCDSN	180
LISMDITFFS	EDGELSSLGK	EGTGFRSNYF	AYETGGKACK	MQYCKHWGVR	LPSGVWFEMA	240
DKDLFAAARF	PECPEGSSIS	APSQTSVDVS	LIQDVERILD	YSLCQETWSK	IRAGLPISPV	300
DLSYLAPKNP	GTGPAFTIIN	GTLKYFETRY	IRVDIAAPIL	SRMVGMI	SGT TTERELWDDW	360
APYEDVEIGP	NGVLTSSGY	KFPLYMIGHG	MLDSDLHLSS	KAQVFEHPHI	QDAASQLPDD	420
ESLFFGDTGL	SKNPIELVEG	WFSSWKSSIA	SFFFIIGLII	GLFLVLRVGI	HLCIKLKHTK	480
KRQIYTDIEM	NRLGK					495

SEQ ID NO: 156 moltype = AA length = 495
FEATURE Location/Qualifiers
source 1..495
mol_type = protein
organism = Vesicular stomatitis virus

SEQUENCE: 156

KFTIVFPHNQ	KGNWKNVPSN	YHYCPSSSDL	NWHNDLIGTA	IQVKMPGSHK	AIQADGWMCH	60
ASKWVTTCDF	RWYGPKYITQ	SIRSFTPSVE	QCKESIEQTK	QGTWLNPGFP	PQSCGYATVT	120
DAEAVIVQVT	PHHVLVDEYT	GEWVDSQFIN	GKCSNYICPT	VHNSTTWHS	YKVKGLCDSN	180
LISMDITFFS	EDGELSSLGK	EGTGFRSNYF	AYETGGKACK	MQYCKHWGVR	LPSGVWFEMA	240
DKDLFAAARF	PECPEGSSIS	APSQTSVDVS	LIQDVERILD	YSLCQETWSK	IRAGLPISPV	300
DLSYLAPKNP	GTGPAFTIIN	GTLKYFETRY	IRVDIAAPIL	SRMVGMI	SGT TTERELWDDW	360
APYEDVEIGP	NGVLTSSGY	KFPLYMIGHG	MLDSDLHLSS	KAQVFEHPHI	QDAASQLPDD	420
ESLFFGDTGL	SKNPIELVEG	WFSSWKSSIA	SFFFIIGLII	GLFLVLRVGI	HLCIKLKHTK	480
KRQIYTDIEM	NRLGK					495

SEQ ID NO: 157 moltype = AA length = 495
FEATURE Location/Qualifiers
source 1..495
mol_type = protein
organism = Vesicular stomatitis virus

SEQUENCE: 157

KFTIVFPHNQ	KGNWKNVPSN	YHYCPSSSDL	NWHNDLIGTA	IQVKMPFSHK	AIQADGWMCH	60
ASKWVTTCDF	RWYGPKYITQ	SIRSFTPSVE	QCKESIEQTK	QGTWLNPGFP	PQSCGYATVT	120
DAEAVIVQVT	PHHVLVDEYT	GEWVDSQFIN	GKCSNYICPT	VHNSTTWHS	YKVKGLCDSN	180
LISMDITFFS	EDGELSSLGK	EGTGFRSNYF	AYETGGKACK	MQYCKHWGVR	LPSGVWFEMA	240
DKDLFAAARF	PECPEGSSIS	APSQTSVDVS	LIQDVERILD	YSLCQETWSK	IRAGLPISPV	300
DLSYLAPKNP	GTGPAFTIIN	GTLKYFETRY	IRVDIAAPIL	SRMVGMI	SGT TTERELWDDW	360
APYEDVEIGP	NGVLTSSGY	KFPLYMIGHG	MLDSDLHLSS	KAQVFEHPHI	QDAASQLPDD	420
ESLFFGDTGL	SKNPIELVEG	WFSSWKSSIA	SFFFIIGLII	GLFLVLRVGI	HLCIKLKHTK	480
KRQIYTDIEM	NRLGK					495

SEQ ID NO: 158 moltype = AA length = 495
FEATURE Location/Qualifiers
source 1..495
mol_type = protein
organism = Vesicular stomatitis virus

SEQUENCE: 158

KFTIVFPHNQ	KGNWKNVPSN	YHYCPSSSDL	NWHNDLIGTA	IQVKMPQSHK	AIQADGWMCH	60
ASKWVTTCDF	RWYGPKYITQ	SIRSFTPSVE	QCKESIEQTK	QGTWLNPGFP	PQSCGYATVT	120

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DAAEIVQVT	PHHVLVDEYT	GEWVDSQFIN	GKCSNYICPT	VHNSTTWHSD	YKVKGLCDSN	180
LISMDITFFS	EDGELSSLGK	EGTGFRSNYF	AYETGGKACK	MQYCKHWGVR	LPSGVWFEMA	240
DKDLFAAARF	PECPEGSSIS	APSQTSVDVS	LIQDVERILD	YSLCQETWSK	IRAGLPISPV	300
DLSYLAPKNP	GTGPAFTIIN	GTLKYFETRY	IRVDIAAPIL	SRMVGMIISGT	TTERELWDDW	360
APYEDVEIGP	NGVLTSSGY	KFPLYMIGHG	MLDSDLHLSS	KAQVFEHPHI	QDAASQLPDD	420
ESLFFGDTGL	SKNPIELVEG	WFSSWKSSIA	SFFFIIGLII	GLFLVLRVGI	HLCIKLKHTK	480
KRQIYTDIEM	NRLGK					495

SEQ ID NO: 159 moltype = AA length = 495
 FEATURE Location/Qualifiers
 source 1..495
 mol_type = protein
 organism = Vesicular stomatitis virus

SEQUENCE: 159

KFTIVFPNHQ	KGNWKNVPSN	YHYCPSSSDL	NWHNDLIGTA	IQVKMPKSHK	AIQADGWMCH	60
ASKWVTTCDF	RWYGPKYITQ	SIRSFTPSVE	QCKESIEQTK	QGTWLNPGFP	PQSCGYATVT	120
DAAEIVQVT	PHHVLVDEYT	GEWVDSQFIN	GKCSNYICPT	VHNSTTWHSD	YKVKGLCDSN	180
LISMDITFFS	EDGELSSLGK	EGTGFRSNYF	AYETGGKACK	MQYCKHWGVR	LPSGVWFEMA	240
DKDLFAAARF	PECPEGSSIS	APSQTSVDVS	LIQDVERILD	YSLCQETWSK	IRAGLPISPV	300
DLSYLAPKNP	GTGPAFTIIN	GTLKYFETRY	IRVDIAAPIL	SRMVGMIISGT	TTEAELWDDW	360
APYEDVEIGP	NGVLTSSGY	KFPLYMIGHG	MLDSDLHLSS	KAQVFEHPHI	QDAASQLPDD	420
ESLFFGDTGL	SKNPIELVEG	WFSSWKSSIA	SFFFIIGLII	GLFLVLRVGI	HLCIKLKHTK	480
KRQIYTDIEM	NRLGK					495

SEQ ID NO: 160 moltype = AA length = 495
 FEATURE Location/Qualifiers
 source 1..495
 mol_type = protein
 organism = Vesicular stomatitis virus

SEQUENCE: 160

KFTIVFPNHQ	KGNWKNVPSN	YHYCPSSSDL	NWHNDLIGTA	IQVKMPKSHK	AIQADGWMCH	60
ASKWVTTCDF	RWYGPKYITQ	SIRSFTPSVE	QCKESIEQTK	QGTWLNPGFP	PQSCGYATVT	120
DAAEIVQVT	PHHVLVDEYT	GEWVDSQFIN	GKCSNYICPT	VHNSTTWHSD	YKVKGLCDSN	180
LISMDITFFS	EDGELSSLGK	EGTGFRSNYF	AYETGGKACK	MQYCKHWGVR	LPSGVWFEMA	240
DKDLFAAARF	PECPEGSSIS	APSQTSVDVS	LIQDVERILD	YSLCQETWSK	IRAGLPISPV	300
DLSYLAPKNP	GTGPAFTIIN	GTLKYFETRY	IRVDIAAPIL	SRMVGMIISGT	TTEGELWDDW	360
APYEDVEIGP	NGVLTSSGY	KFPLYMIGHG	MLDSDLHLSS	KAQVFEHPHI	QDAASQLPDD	420
ESLFFGDTGL	SKNPIELVEG	WFSSWKSSIA	SFFFIIGLII	GLFLVLRVGI	HLCIKLKHTK	480
KRQIYTDIEM	NRLGK					495

SEQ ID NO: 161 moltype = AA length = 495
 FEATURE Location/Qualifiers
 source 1..495
 mol_type = protein
 organism = Vesicular stomatitis virus

SEQUENCE: 161

KFTIVFPNHQ	KGNWKNVPSN	YHYCPSSSDL	NWHNDLIGTA	IQVKMPKSHK	AIQADGWMCH	60
ASKWVTTCDF	RWYGPKYITQ	SIRSFTPSVE	QCKESIEQTK	QGTWLNPGFP	PQSCGYATVT	120
DAAEIVQVT	PHHVLVDEYT	GEWVDSQFIN	GKCSNYICPT	VHNSTTWHSD	YKVKGLCDSN	180
LISMDITFFS	EDGELSSLGK	EGTGFRSNYF	AYETGGKACK	MQYCKHWGVR	LPSGVWFEMA	240
DKDLFAAARF	PECPEGSSIS	APSQTSVDVS	LIQDVERILD	YSLCQETWSK	IRAGLPISPV	300
DLSYLAPKNP	GTGPAFTIIN	GTLKYFETRY	IRVDIAAPIL	SRMVGMIISGT	TTEFELWDDW	360
APYEDVEIGP	NGVLTSSGY	KFPLYMIGHG	MLDSDLHLSS	KAQVFEHPHI	QDAASQLPDD	420
ESLFFGDTGL	SKNPIELVEG	WFSSWKSSIA	SFFFIIGLII	GLFLVLRVGI	HLCIKLKHTK	480
KRQIYTDIEM	NRLGK					495

SEQ ID NO: 162 moltype = AA length = 495
 FEATURE Location/Qualifiers
 source 1..495
 mol_type = protein
 organism = Vesicular stomatitis virus

SEQUENCE: 162

KFTIVFPNHQ	KGNWKNVPSN	YHYCPSSSDL	NWHNDLIGTA	IQVKMPKSHK	AIQADGWMCH	60
ASKWVTTCDF	RWYGPKYITQ	SIRSFTPSVE	QCKESIEQTK	QGTWLNPGFP	PQSCGYATVT	120
DAAEIVQVT	PHHVLVDEYT	GEWVDSQFIN	GKCSNYICPT	VHNSTTWHSD	YKVKGLCDSN	180
LISMDITFFS	EDGELSSLGK	EGTGFRSNYF	AYETGGKACK	MQYCKHWGVR	LPSGVWFEMA	240
DKDLFAAARF	PECPEGSSIS	APSQTSVDVS	LIQDVERILD	YSLCQETWSK	IRAGLPISPV	300
DLSYLAPKNP	GTGPAFTIIN	GTLKYFETRY	IRVDIAAPIL	SRMVGMIISGT	TTEQELWDDW	360
APYEDVEIGP	NGVLTSSGY	KFPLYMIGHG	MLDSDLHLSS	KAQVFEHPHI	QDAASQLPDD	420
ESLFFGDTGL	SKNPIELVEG	WFSSWKSSIA	SFFFIIGLII	GLFLVLRVGI	HLCIKLKHTK	480
KRQIYTDIEM	NRLGK					495

SEQ ID NO: 163 moltype = AA length = 495
 FEATURE Location/Qualifiers
 source 1..495
 mol_type = protein
 organism = Vesicular stomatitis virus

SEQUENCE: 163

KFTIVFPNHQ	KGNWKNVPSN	YHYCPSSSDL	NWHNDLIGTA	IQVKMPASHK	AIQADGWMCH	60
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ASKWVTTCDF RWYGPKEYITQ SIRSFTPSVE QCKESIEQTK QGTWLNPGFP PQSCGYATVT 120
DAEAVIVQVT PHHVLVDEYT GEWVDSQFIN GKCSNYICPT VHNSTTWHS YKVKGLCDSN 180
LISMDITFFS EDGELSSLGK EGTGFRSNYF AYETGGKACK MQYCKHWGVR LPSGVWFEMA 240
DKDLFAAARF PECPEGSSIS APSQTSVDVS LIQDVERILD YSLCQETWSK IRAGLPISPV 300
DLSYLAPKNP GTGPAFTIIN GTLKYFETRY IRVDIAAPIL SRMVGMI SGT TTEAELWDDW 360
APYEDVEIGP NGVLR TSSGY KFPLYMIGHG MLDSDLHLSS KAQVFEHPHI QDAASQLPDD 420
ESLFFGDTGL SKNPIELVEG WFSSWKSSIA SFFFIIGLII GLFLVLRVGI HLCIKLKHTK 480
KRQIYTDIEM NRLGK 495

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SEQ ID NO: 164      moltype = AA length = 495
FEATURE            Location/Qualifiers
source              1..495
                    mol_type = protein
                    organism = Vesicular stomatitis virus

```

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SEQUENCE: 164
KFTIVFPNHQ KGNWKNVPSN YHYCPSSSDL NWHNDLIGTA IQVKMPASHK AIQADGWMCH 60
ASKWVTTCDF RWYGPKEYITQ SIRSFTPSVE QCKESIEQTK QGTWLNPGFP PQSCGYATVT 120
DAEAVIVQVT PHHVLVDEYT GEWVDSQFIN GKCSNYICPT VHNSTTWHS YKVKGLCDSN 180
LISMDITFFS EDGELSSLGK EGTGFRSNYF AYETGGKACK MQYCKHWGVR LPSGVWFEMA 240
DKDLFAAARF PECPEGSSIS APSQTSVDVS LIQDVERILD YSLCQETWSK IRAGLPISPV 300
DLSYLAPKNP GTGPAFTIIN GTLKYFETRY IRVDIAAPIL SRMVGMI SGT TTEGELWDDW 360
APYEDVEIGP NGVLR TSSGY KFPLYMIGHG MLDSDLHLSS KAQVFEHPHI QDAASQLPDD 420
ESLFFGDTGL SKNPIELVEG WFSSWKSSIA SFFFIIGLII GLFLVLRVGI HLCIKLKHTK 480
KRQIYTDIEM NRLGK 495

```

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SEQ ID NO: 165      moltype = AA length = 495
FEATURE            Location/Qualifiers
source              1..495
                    mol_type = protein
                    organism = Vesicular stomatitis virus

```

```

SEQUENCE: 165
KFTIVFPNHQ KGNWKNVPSN YHYCPSSSDL NWHNDLIGTA IQVKMPASHK AIQADGWMCH 60
ASKWVTTCDF RWYGPKEYITQ SIRSFTPSVE QCKESIEQTK QGTWLNPGFP PQSCGYATVT 120
DAEAVIVQVT PHHVLVDEYT GEWVDSQFIN GKCSNYICPT VHNSTTWHS YKVKGLCDSN 180
LISMDITFFS EDGELSSLGK EGTGFRSNYF AYETGGKACK MQYCKHWGVR LPSGVWFEMA 240
DKDLFAAARF PECPEGSSIS APSQTSVDVS LIQDVERILD YSLCQETWSK IRAGLPISPV 300
DLSYLAPKNP GTGPAFTIIN GTLKYFETRY IRVDIAAPIL SRMVGMI SGT TTEFELWDDW 360
APYEDVEIGP NGVLR TSSGY KFPLYMIGHG MLDSDLHLSS KAQVFEHPHI QDAASQLPDD 420
ESLFFGDTGL SKNPIELVEG WFSSWKSSIA SFFFIIGLII GLFLVLRVGI HLCIKLKHTK 480
KRQIYTDIEM NRLGK 495

```

```

SEQ ID NO: 166      moltype = AA length = 495
FEATURE            Location/Qualifiers
source              1..495
                    mol_type = protein
                    organism = Vesicular stomatitis virus

```

```

SEQUENCE: 166
KFTIVFPNHQ KGNWKNVPSN YHYCPSSSDL NWHNDLIGTA IQVKMPASHK AIQADGWMCH 60
ASKWVTTCDF RWYGPKEYITQ SIRSFTPSVE QCKESIEQTK QGTWLNPGFP PQSCGYATVT 120
DAEAVIVQVT PHHVLVDEYT GEWVDSQFIN GKCSNYICPT VHNSTTWHS YKVKGLCDSN 180
LISMDITFFS EDGELSSLGK EGTGFRSNYF AYETGGKACK MQYCKHWGVR LPSGVWFEMA 240
DKDLFAAARF PECPEGSSIS APSQTSVDVS LIQDVERILD YSLCQETWSK IRAGLPISPV 300
DLSYLAPKNP GTGPAFTIIN GTLKYFETRY IRVDIAAPIL SRMVGMI SGT TTEQELWDDW 360
APYEDVEIGP NGVLR TSSGY KFPLYMIGHG MLDSDLHLSS KAQVFEHPHI QDAASQLPDD 420
ESLFFGDTGL SKNPIELVEG WFSSWKSSIA SFFFIIGLII GLFLVLRVGI HLCIKLKHTK 480
KRQIYTDIEM NRLGK 495

```

```

SEQ ID NO: 167      moltype = AA length = 495
FEATURE            Location/Qualifiers
source              1..495
                    mol_type = protein
                    organism = Vesicular stomatitis virus

```

```

SEQUENCE: 167
KFTIVFPNHQ KGNWKNVPSN YHYCPSSSDL NWHNDLIGTA IQVKMPGSHK AIQADGWMCH 60
ASKWVTTCDF RWYGPKEYITQ SIRSFTPSVE QCKESIEQTK QGTWLNPGFP PQSCGYATVT 120
DAEAVIVQVT PHHVLVDEYT GEWVDSQFIN GKCSNYICPT VHNSTTWHS YKVKGLCDSN 180
LISMDITFFS EDGELSSLGK EGTGFRSNYF AYETGGKACK MQYCKHWGVR LPSGVWFEMA 240
DKDLFAAARF PECPEGSSIS APSQTSVDVS LIQDVERILD YSLCQETWSK IRAGLPISPV 300
DLSYLAPKNP GTGPAFTIIN GTLKYFETRY IRVDIAAPIL SRMVGMI SGT TTEAELWDDW 360
APYEDVEIGP NGVLR TSSGY KFPLYMIGHG MLDSDLHLSS KAQVFEHPHI QDAASQLPDD 420
ESLFFGDTGL SKNPIELVEG WFSSWKSSIA SFFFIIGLII GLFLVLRVGI HLCIKLKHTK 480
KRQIYTDIEM NRLGK 495

```

```

SEQ ID NO: 168      moltype = AA length = 495
FEATURE            Location/Qualifiers
source              1..495
                    mol_type = protein
                    organism = Vesicular stomatitis virus

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SEQUENCE: 168

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KFTIVFPNHQ KGNWKNVPSN YHYCPSSSDL NWHNDLIGTA IQVKMPGSHK AIQADGWMCH 60
ASKWVTTCDF RWYGPKYITQ SIRSFTPSVE QCKESIEQTK QGTWLNPGFP PQSCGYATVT 120
DAEAVIVQVT PHHVLVDEYT GEWVDSQFIN GKCSNYICPT VHNSTTWHSY YKVKGLCDSN 180
LISMDITFFS EDGELSSLGK EGTGFRSNYF AYETGGKACK MQYCKHWGVR LPSGVWFEMA 240
DKDLFAAARF PECPEGSSIS APSQTSVDVS LIQDVERILD YSLCQETWSK IRAGLPISPV 300
DLSYLAPKNP GTGPAFTIIN GTLKYFETRY IRVDIAAPIL SRMVGMIISGT TTEGELWDDW 360
APYEDVEIGP NGVLRSSGY KFPLYMIGHG MLDSDLHLSS KAQVFEHPHI QDAASQLPDD 420
ESLFFGDTGL SKNPIELVEG WFSWKSSIA SFFFIIGLII GLFLVLRVGI HLCIKLKHTK 480
KRQIYTDIEM NRLGK 495

```

```

SEQ ID NO: 169      moltype = AA length = 495
FEATURE            Location/Qualifiers
source              1..495
                    mol_type = protein
                    organism = Vesicular stomatitis virus

```

```

SEQUENCE: 169
KFTIVFPNHQ KGNWKNVPSN YHYCPSSSDL NWHNDLIGTA IQVKMPGSHK AIQADGWMCH 60
ASKWVTTCDF RWYGPKYITQ SIRSFTPSVE QCKESIEQTK QGTWLNPGFP PQSCGYATVT 120
DAEAVIVQVT PHHVLVDEYT GEWVDSQFIN GKCSNYICPT VHNSTTWHSY YKVKGLCDSN 180
LISMDITFFS EDGELSSLGK EGTGFRSNYF AYETGGKACK MQYCKHWGVR LPSGVWFEMA 240
DKDLFAAARF PECPEGSSIS APSQTSVDVS LIQDVERILD YSLCQETWSK IRAGLPISPV 300
DLSYLAPKNP GTGPAFTIIN GTLKYFETRY IRVDIAAPIL SRMVGMIISGT TTEGELWDDW 360
APYEDVEIGP NGVLRSSGY KFPLYMIGHG MLDSDLHLSS KAQVFEHPHI QDAASQLPDD 420
ESLFFGDTGL SKNPIELVEG WFSWKSSIA SFFFIIGLII GLFLVLRVGI HLCIKLKHTK 480
KRQIYTDIEM NRLGK 495

```

```

SEQ ID NO: 170      moltype = AA length = 495
FEATURE            Location/Qualifiers
source              1..495
                    mol_type = protein
                    organism = Vesicular stomatitis virus

```

```

SEQUENCE: 170
KFTIVFPNHQ KGNWKNVPSN YHYCPSSSDL NWHNDLIGTA IQVKMPGSHK AIQADGWMCH 60
ASKWVTTCDF RWYGPKYITQ SIRSFTPSVE QCKESIEQTK QGTWLNPGFP PQSCGYATVT 120
DAEAVIVQVT PHHVLVDEYT GEWVDSQFIN GKCSNYICPT VHNSTTWHSY YKVKGLCDSN 180
LISMDITFFS EDGELSSLGK EGTGFRSNYF AYETGGKACK MQYCKHWGVR LPSGVWFEMA 240
DKDLFAAARF PECPEGSSIS APSQTSVDVS LIQDVERILD YSLCQETWSK IRAGLPISPV 300
DLSYLAPKNP GTGPAFTIIN GTLKYFETRY IRVDIAAPIL SRMVGMIISGT TTEGELWDDW 360
APYEDVEIGP NGVLRSSGY KFPLYMIGHG MLDSDLHLSS KAQVFEHPHI QDAASQLPDD 420
ESLFFGDTGL SKNPIELVEG WFSWKSSIA SFFFIIGLII GLFLVLRVGI HLCIKLKHTK 480
KRQIYTDIEM NRLGK 495

```

```

SEQ ID NO: 171      moltype = AA length = 495
FEATURE            Location/Qualifiers
source              1..495
                    mol_type = protein
                    organism = Vesicular stomatitis virus

```

```

SEQUENCE: 171
KFTIVFPNHQ KGNWKNVPSN YHYCPSSSDL NWHNDLIGTA IQVKMPGSHK AIQADGWMCH 60
ASKWVTTCDF RWYGPKYITQ SIRSFTPSVE QCKESIEQTK QGTWLNPGFP PQSCGYATVT 120
DAEAVIVQVT PHHVLVDEYT GEWVDSQFIN GKCSNYICPT VHNSTTWHSY YKVKGLCDSN 180
LISMDITFFS EDGELSSLGK EGTGFRSNYF AYETGGKACK MQYCKHWGVR LPSGVWFEMA 240
DKDLFAAARF PECPEGSSIS APSQTSVDVS LIQDVERILD YSLCQETWSK IRAGLPISPV 300
DLSYLAPKNP GTGPAFTIIN GTLKYFETRY IRVDIAAPIL SRMVGMIISGT TTEGELWDDW 360
APYEDVEIGP NGVLRSSGY KFPLYMIGHG MLDSDLHLSS KAQVFEHPHI QDAASQLPDD 420
ESLFFGDTGL SKNPIELVEG WFSWKSSIA SFFFIIGLII GLFLVLRVGI HLCIKLKHTK 480
KRQIYTDIEM NRLGK 495

```

```

SEQ ID NO: 172      moltype = AA length = 495
FEATURE            Location/Qualifiers
source              1..495
                    mol_type = protein
                    organism = Vesicular stomatitis virus

```

```

SEQUENCE: 172
KFTIVFPNHQ KGNWKNVPSN YHYCPSSSDL NWHNDLIGTA IQVKMPGSHK AIQADGWMCH 60
ASKWVTTCDF RWYGPKYITQ SIRSFTPSVE QCKESIEQTK QGTWLNPGFP PQSCGYATVT 120
DAEAVIVQVT PHHVLVDEYT GEWVDSQFIN GKCSNYICPT VHNSTTWHSY YKVKGLCDSN 180
LISMDITFFS EDGELSSLGK EGTGFRSNYF AYETGGKACK MQYCKHWGVR LPSGVWFEMA 240
DKDLFAAARF PECPEGSSIS APSQTSVDVS LIQDVERILD YSLCQETWSK IRAGLPISPV 300
DLSYLAPKNP GTGPAFTIIN GTLKYFETRY IRVDIAAPIL SRMVGMIISGT TTEGELWDDW 360
APYEDVEIGP NGVLRSSGY KFPLYMIGHG MLDSDLHLSS KAQVFEHPHI QDAASQLPDD 420
ESLFFGDTGL SKNPIELVEG WFSWKSSIA SFFFIIGLII GLFLVLRVGI HLCIKLKHTK 480
KRQIYTDIEM NRLGK 495

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SEQ ID NO: 173      moltype = AA length = 495
FEATURE            Location/Qualifiers
source              1..495
                    mol_type = protein
                    organism = Vesicular stomatitis virus

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SEQUENCE: 173
KFTIVFPNHQ KGNWKNVPSN YHYCPSSSDL NWHNDLIGTA IQVKMPFESHK AIQADGWMCH 60
ASKWVTTCDP RWYGPKYITQ SIRSFPTPSVE QCKESIEQTK QGTWLNPGFP PQSCGYATVT 120
DAEAVIVQVT PHHVLVDEYT GEWVDSQFIN GKCSNYICPT VHNSTTWHS YKVKGLCDSN 180
LISMDITFFS EDGELSSLGK EGTGFRSNYF AYETGGKACK MQYCKHWGVR LPSGVWFEMA 240
DKDLFAAARF PECPEGSSIS APSQTSVDVS LIQDVERILD YSLCQETWSK IRAGLPISPV 300
DLSYLAPKNP GTGPAFTIIN GTLKYFETRY IRVDIAAPIL SRMVGMISGT TTEFELWDDW 360
APYEDVEIGP NGVLTSSGY KFLPLYMIGHG MLDSDLHLSS KAQVFEHPHI QDAASQLPDD 420
ESLFFGDTGL SKNPIELVEG WFSSWKSSIA SFFFIIGLII GLFLVLRVGI HLCIKLKHTK 480
KRQIYTDIEM NRLGK 495

SEQ ID NO: 174 moltype = AA length = 495
FEATURE Location/Qualifiers
source 1..495
 mol_type = protein
 organism = Vesicular stomatitis virus

SEQUENCE: 174
KFTIVFPNHQ KGNWKNVPSN YHYCPSSSDL NWHNDLIGTA IQVKMPFESHK AIQADGWMCH 60
ASKWVTTCDP RWYGPKYITQ SIRSFPTPSVE QCKESIEQTK QGTWLNPGFP PQSCGYATVT 120
DAEAVIVQVT PHHVLVDEYT GEWVDSQFIN GKCSNYICPT VHNSTTWHS YKVKGLCDSN 180
LISMDITFFS EDGELSSLGK EGTGFRSNYF AYETGGKACK MQYCKHWGVR LPSGVWFEMA 240
DKDLFAAARF PECPEGSSIS APSQTSVDVS LIQDVERILD YSLCQETWSK IRAGLPISPV 300
DLSYLAPKNP GTGPAFTIIN GTLKYFETRY IRVDIAAPIL SRMVGMISGT TTEQELWDDW 360
APYEDVEIGP NGVLTSSGY KFLPLYMIGHG MLDSDLHLSS KAQVFEHPHI QDAASQLPDD 420
ESLFFGDTGL SKNPIELVEG WFSSWKSSIA SFFFIIGLII GLFLVLRVGI HLCIKLKHTK 480
KRQIYTDIEM NRLGK 495

SEQ ID NO: 175 moltype = AA length = 495
FEATURE Location/Qualifiers
source 1..495
 mol_type = protein
 organism = Vesicular stomatitis virus

SEQUENCE: 175
KFTIVFPNHQ KGNWKNVPSN YHYCPSSSDL NWHNDLIGTA IQVKMPQSHK AIQADGWMCH 60
ASKWVTTCDP RWYGPKYITQ SIRSFPTPSVE QCKESIEQTK QGTWLNPGFP PQSCGYATVT 120
DAEAVIVQVT PHHVLVDEYT GEWVDSQFIN GKCSNYICPT VHNSTTWHS YKVKGLCDSN 180
LISMDITFFS EDGELSSLGK EGTGFRSNYF AYETGGKACK MQYCKHWGVR LPSGVWFEMA 240
DKDLFAAARF PECPEGSSIS APSQTSVDVS LIQDVERILD YSLCQETWSK IRAGLPISPV 300
DLSYLAPKNP GTGPAFTIIN GTLKYFETRY IRVDIAAPIL SRMVGMISGT TTEAELWDDW 360
APYEDVEIGP NGVLTSSGY KFLPLYMIGHG MLDSDLHLSS KAQVFEHPHI QDAASQLPDD 420
ESLFFGDTGL SKNPIELVEG WFSSWKSSIA SFFFIIGLII GLFLVLRVGI HLCIKLKHTK 480
KRQIYTDIEM NRLGK 495

SEQ ID NO: 176 moltype = AA length = 495
FEATURE Location/Qualifiers
source 1..495
 mol_type = protein
 organism = Vesicular stomatitis virus

SEQUENCE: 176
KFTIVFPNHQ KGNWKNVPSN YHYCPSSSDL NWHNDLIGTA IQVKMPQSHK AIQADGWMCH 60
ASKWVTTCDP RWYGPKYITQ SIRSFPTPSVE QCKESIEQTK QGTWLNPGFP PQSCGYATVT 120
DAEAVIVQVT PHHVLVDEYT GEWVDSQFIN GKCSNYICPT VHNSTTWHS YKVKGLCDSN 180
LISMDITFFS EDGELSSLGK EGTGFRSNYF AYETGGKACK MQYCKHWGVR LPSGVWFEMA 240
DKDLFAAARF PECPEGSSIS APSQTSVDVS LIQDVERILD YSLCQETWSK IRAGLPISPV 300
DLSYLAPKNP GTGPAFTIIN GTLKYFETRY IRVDIAAPIL SRMVGMISGT TTEGELWDDW 360
APYEDVEIGP NGVLTSSGY KFLPLYMIGHG MLDSDLHLSS KAQVFEHPHI QDAASQLPDD 420
ESLFFGDTGL SKNPIELVEG WFSSWKSSIA SFFFIIGLII GLFLVLRVGI HLCIKLKHTK 480
KRQIYTDIEM NRLGK 495

SEQ ID NO: 177 moltype = AA length = 495
FEATURE Location/Qualifiers
source 1..495
 mol_type = protein
 organism = Vesicular stomatitis virus

SEQUENCE: 177
KFTIVFPNHQ KGNWKNVPSN YHYCPSSSDL NWHNDLIGTA IQVKMPQSHK AIQADGWMCH 60
ASKWVTTCDP RWYGPKYITQ SIRSFPTPSVE QCKESIEQTK QGTWLNPGFP PQSCGYATVT 120
DAEAVIVQVT PHHVLVDEYT GEWVDSQFIN GKCSNYICPT VHNSTTWHS YKVKGLCDSN 180
LISMDITFFS EDGELSSLGK EGTGFRSNYF AYETGGKACK MQYCKHWGVR LPSGVWFEMA 240
DKDLFAAARF PECPEGSSIS APSQTSVDVS LIQDVERILD YSLCQETWSK IRAGLPISPV 300
DLSYLAPKNP GTGPAFTIIN GTLKYFETRY IRVDIAAPIL SRMVGMISGT TTEFELWDDW 360
APYEDVEIGP NGVLTSSGY KFLPLYMIGHG MLDSDLHLSS KAQVFEHPHI QDAASQLPDD 420
ESLFFGDTGL SKNPIELVEG WFSSWKSSIA SFFFIIGLII GLFLVLRVGI HLCIKLKHTK 480
KRQIYTDIEM NRLGK 495

SEQ ID NO: 178 moltype = AA length = 495
FEATURE Location/Qualifiers
source 1..495
 mol_type = protein

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organism = Vesicular stomatitis virus

SEQUENCE: 178
KFTIVFPNHQ KGNWKNVPSN YHYCPSSSDL NWHNDLIGTA IQVKMPQSHK AIQADGWMCH 60
ASKWVTTCDF RWYGPKYITQ SIRSFTPSVE QCKESIEQTK QGTWLNPGFP PQSCGYATVT 120
DAEAVIVQVT PHHVLVDEYT GEWVDSQFIN GKCSNYICPT VHNSTTWHSY YKVKGLCDSN 180
LISMDITFFS EDGELSSLGK EGTGFRSNYF AYETGGKACK MQYCKHWGVR LPSGVWFEMA 240
DKDLFAAARF PECPEGSSIS APSQTSVDVS LIQDVERILD YSLCQETWSK IRAGLPISPV 300
DLSYLAPKNP GTGPAFTIIN GTLKYFETRY IRVDIAAPIL SRMVGMIISGT TTEQELWDDW 360
APYEDVEIGP NGVLTSSGY KFPLYMIGHG MLDSDLHLSS KAQVFEHPHI QDAASQLPDD 420
ESLFFGDTGL SKNPVELVEG WFSSWKSSIA SFFFIIGLII GLFLVLRVGI HLCIKLKHTK 480
KRQIYTDIEM NRLGK 495

SEQ ID NO: 179      moltype = AA length = 496
FEATURE            Location/Qualifiers
source              1..496
                    mol_type = protein
                    organism = Vesicular stomatitis virus

SEQUENCE: 179
KFTIVFPHHQ KGNWKNVPST YHYCPSSSDQ NWHNDLTGVS LHVKIPASHK AIQADGWMCH 60
AAKWVTTCDF RWYGPKYITH SIHMSPTLE QCKTSIEQTK QGVWINPGFP PQSCGYATVT 120
DAEVVVVQAT PHHVLVDEYT GEWIDSQLVG GKCSKEVCQT VHNSTVWHAD YKITGLCESN 180
LASVDITFFS EDGQKTSLGK PNTGFRSNHF AYESGEKACR MQYCTQWGIR LPSGVWFELV 240
DKDLFQAARKL PECPRGSSIS APSQTSVDVS LIQDVERILD YSLCQETWSK IRAKLPVSPV 300
DLSYLAPKNP GSGPAFTIIN GTLKYFETRY IRVDISNPII PHMVGTMISGT TTERELWNDW 360
YPYEDVEIGP NGVLKPTGTF KFPLYMIGHG MLDSDLHKSS QAQVFEHPHA KDAASQLPDD 420
ETLFFGDTGL SKNPVELVEG WFSSWKSTLA SFFLIIGLGV ALIFIIRIIV AIRYKYKGRK 480
TQKIYNDVEM SRLGNK 496

SEQ ID NO: 180      moltype = AA length = 496
FEATURE            Location/Qualifiers
source              1..496
                    mol_type = protein
                    organism = Vesicular stomatitis virus

SEQUENCE: 180
KFTIVFPHHQ KGNWKNVPST YHYCPSSSDQ NWHNDLTGVS LHVKIPGSHK AIQADGWMCH 60
AAKWVTTCDF RWYGPKYITH SIHMSPTLE QCKTSIEQTK QGVWINPGFP PQSCGYATVT 120
DAEVVVVQAT PHHVLVDEYT GEWIDSQLVG GKCSKEVCQT VHNSTVWHAD YKITGLCESN 180
LASVDITFFS EDGQKTSLGK PNTGFRSNHF AYESGEKACR MQYCTQWGIR LPSGVWFELV 240
DKDLFQAARKL PECPRGSSIS APSQTSVDVS LIQDVERILD YSLCQETWSK IRAKLPVSPV 300
DLSYLAPKNP GSGPAFTIIN GTLKYFETRY IRVDISNPII PHMVGTMISGT TTERELWNDW 360
YPYEDVEIGP NGVLKPTGTF KFPLYMIGHG MLDSDLHKSS QAQVFEHPHA KDAASQLPDD 420
ETLFFGDTGL SKNPVELVEG WFSSWKSTLA SFFLIIGLGV ALIFIIRIIV AIRYKYKGRK 480
TQKIYNDVEM SRLGNK 496

SEQ ID NO: 181      moltype = AA length = 496
FEATURE            Location/Qualifiers
source              1..496
                    mol_type = protein
                    organism = Vesicular stomatitis virus

SEQUENCE: 181
KFTIVFPHHQ KGNWKNVPST YHYCPSSSDQ NWHNDLTGVS LHVKIPFSHK AIQADGWMCH 60
AAKWVTTCDF RWYGPKYITH SIHMSPTLE QCKTSIEQTK QGVWINPGFP PQSCGYATVT 120
DAEVVVVQAT PHHVLVDEYT GEWIDSQLVG GKCSKEVCQT VHNSTVWHAD YKITGLCESN 180
LASVDITFFS EDGQKTSLGK PNTGFRSNHF AYESGEKACR MQYCTQWGIR LPSGVWFELV 240
DKDLFQAARKL PECPRGSSIS APSQTSVDVS LIQDVERILD YSLCQETWSK IRAKLPVSPV 300
DLSYLAPKNP GSGPAFTIIN GTLKYFETRY IRVDISNPII PHMVGTMISGT TTERELWNDW 360
YPYEDVEIGP NGVLKPTGTF KFPLYMIGHG MLDSDLHKSS QAQVFEHPHA KDAASQLPDD 420
ETLFFGDTGL SKNPVELVEG WFSSWKSTLA SFFLIIGLGV ALIFIIRIIV AIRYKYKGRK 480
TQKIYNDVEM SRLGNK 496

SEQ ID NO: 182      moltype = AA length = 496
FEATURE            Location/Qualifiers
source              1..496
                    mol_type = protein
                    organism = Vesicular stomatitis virus

SEQUENCE: 182
KFTIVFPHHQ KGNWKNVPST YHYCPSSSDQ NWHNDLTGVS LHVKIPQSHK AIQADGWMCH 60
AAKWVTTCDF RWYGPKYITH SIHMSPTLE QCKTSIEQTK QGVWINPGFP PQSCGYATVT 120
DAEVVVVQAT PHHVLVDEYT GEWIDSQLVG GKCSKEVCQT VHNSTVWHAD YKITGLCESN 180
LASVDITFFS EDGQKTSLGK PNTGFRSNHF AYESGEKACR MQYCTQWGIR LPSGVWFELV 240
DKDLFQAARKL PECPRGSSIS APSQTSVDVS LIQDVERILD YSLCQETWSK IRAKLPVSPV 300
DLSYLAPKNP GSGPAFTIIN GTLKYFETRY IRVDISNPII PHMVGTMISGT TTERELWNDW 360
YPYEDVEIGP NGVLKPTGTF KFPLYMIGHG MLDSDLHKSS QAQVFEHPHA KDAASQLPDD 420
ETLFFGDTGL SKNPVELVEG WFSSWKSTLA SFFLIIGLGV ALIFIIRIIV AIRYKYKGRK 480
TQKIYNDVEM SRLGNK 496

SEQ ID NO: 183      moltype = AA length = 496
FEATURE            Location/Qualifiers
source              1..496

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mol_type = protein
organism = Vesicular stomatitis virus

SEQUENCE: 183
KFTIVFPHHQ KGNWKNVPST YHYCPSSSDQ NWHNDLTGVS LHVKIPKSHK AIQADGWMCH 60
AAKWVTTCDF RwygPKYITH SIHMSPTLE QCKTSIEQTK QGVWINPGFP PQSCGYATVT 120
DAEVVVVQAT PHHVLVDEYT GEWIDSQLVG GKCSKEVCQT VHNSTVWHAD YKITGLCESN 180
LASVDITFFS EDGQKTSLGK PNTGFRSNHF AYESGEKACR MQYCTQWGIR LPSGVWFELV 240
DKDLFQAACL PECPRGSSIS APSQTSVDVS LIQDVERILD YSLCQETWSK IRAKLPVSPV 300
DLSYLAPKNP GSGPAFTIIN GTLKYPETRY IRVDISNPII PHMVGTMSTG TTEAELWNDW 360
YPYEDVEIGP NGVLKTPTFG KFPLYMIGHG MLDSDLHKSS QAQVFEHPHA KDAASQLPDD 420
ETLFFGDTGL SKNPVELVEG WFSSWKSTLA SFFLIIGLGV ALIFIIRIIV AIRYKYKGRK 480
TQKIYNDVEM SRLGNK 496

SEQ ID NO: 184      moltype = AA length = 496
FEATURE            Location/Qualifiers
source              1..496
                    mol_type = protein
                    organism = Vesicular stomatitis virus

SEQUENCE: 184
KFTIVFPHHQ KGNWKNVPST YHYCPSSSDQ NWHNDLTGVS LHVKIPKSHK AIQADGWMCH 60
AAKWVTTCDF RwygPKYITH SIHMSPTLE QCKTSIEQTK QGVWINPGFP PQSCGYATVT 120
DAEVVVVQAT PHHVLVDEYT GEWIDSQLVG GKCSKEVCQT VHNSTVWHAD YKITGLCESN 180
LASVDITFFS EDGQKTSLGK PNTGFRSNHF AYESGEKACR MQYCTQWGIR LPSGVWFELV 240
DKDLFQAACL PECPRGSSIS APSQTSVDVS LIQDVERILD YSLCQETWSK IRAKLPVSPV 300
DLSYLAPKNP GSGPAFTIIN GTLKYPETRY IRVDISNPII PHMVGTMSTG TTEGELWNDW 360
YPYEDVEIGP NGVLKTPTFG KFPLYMIGHG MLDSDLHKSS QAQVFEHPHA KDAASQLPDD 420
ETLFFGDTGL SKNPVELVEG WFSSWKSTLA SFFLIIGLGV ALIFIIRIIV AIRYKYKGRK 480
TQKIYNDVEM SRLGNK 496

SEQ ID NO: 185      moltype = AA length = 496
FEATURE            Location/Qualifiers
source              1..496
                    mol_type = protein
                    organism = Vesicular stomatitis virus

SEQUENCE: 185
KFTIVFPHHQ KGNWKNVPST YHYCPSSSDQ NWHNDLTGVS LHVKIPKSHK AIQADGWMCH 60
AAKWVTTCDF RwygPKYITH SIHMSPTLE QCKTSIEQTK QGVWINPGFP PQSCGYATVT 120
DAEVVVVQAT PHHVLVDEYT GEWIDSQLVG GKCSKEVCQT VHNSTVWHAD YKITGLCESN 180
LASVDITFFS EDGQKTSLGK PNTGFRSNHF AYESGEKACR MQYCTQWGIR LPSGVWFELV 240
DKDLFQAACL PECPRGSSIS APSQTSVDVS LIQDVERILD YSLCQETWSK IRAKLPVSPV 300
DLSYLAPKNP GSGPAFTIIN GTLKYPETRY IRVDISNPII PHMVGTMSTG TTEFELWNDW 360
YPYEDVEIGP NGVLKTPTFG KFPLYMIGHG MLDSDLHKSS QAQVFEHPHA KDAASQLPDD 420
ETLFFGDTGL SKNPVELVEG WFSSWKSTLA SFFLIIGLGV ALIFIIRIIV AIRYKYKGRK 480
TQKIYNDVEM SRLGNK 496

SEQ ID NO: 186      moltype = AA length = 496
FEATURE            Location/Qualifiers
source              1..496
                    mol_type = protein
                    organism = Vesicular stomatitis virus

SEQUENCE: 186
KFTIVFPHHQ KGNWKNVPST YHYCPSSSDQ NWHNDLTGVS LHVKIPKSHK AIQADGWMCH 60
AAKWVTTCDF RwygPKYITH SIHMSPTLE QCKTSIEQTK QGVWINPGFP PQSCGYATVT 120
DAEVVVVQAT PHHVLVDEYT GEWIDSQLVG GKCSKEVCQT VHNSTVWHAD YKITGLCESN 180
LASVDITFFS EDGQKTSLGK PNTGFRSNHF AYESGEKACR MQYCTQWGIR LPSGVWFELV 240
DKDLFQAACL PECPRGSSIS APSQTSVDVS LIQDVERILD YSLCQETWSK IRAKLPVSPV 300
DLSYLAPKNP GSGPAFTIIN GTLKYPETRY IRVDISNPII PHMVGTMSTG TTEQELWNDW 360
YPYEDVEIGP NGVLKTPTFG KFPLYMIGHG MLDSDLHKSS QAQVFEHPHA KDAASQLPDD 420
ETLFFGDTGL SKNPVELVEG WFSSWKSTLA SFFLIIGLGV ALIFIIRIIV AIRYKYKGRK 480
TQKIYNDVEM SRLGNK 496

SEQ ID NO: 187      moltype = AA length = 496
FEATURE            Location/Qualifiers
source              1..496
                    mol_type = protein
                    organism = Vesicular stomatitis virus

SEQUENCE: 187
KFTIVFPHHQ KGNWKNVPST YHYCPSSSDQ NWHNDLTGVS LHVKIPASHK AIQADGWMCH 60
AAKWVTTCDF RwygPKYITH SIHMSPTLE QCKTSIEQTK QGVWINPGFP PQSCGYATVT 120
DAEVVVVQAT PHHVLVDEYT GEWIDSQLVG GKCSKEVCQT VHNSTVWHAD YKITGLCESN 180
LASVDITFFS EDGQKTSLGK PNTGFRSNHF AYESGEKACR MQYCTQWGIR LPSGVWFELV 240
DKDLFQAACL PECPRGSSIS APSQTSVDVS LIQDVERILD YSLCQETWSK IRAKLPVSPV 300
DLSYLAPKNP GSGPAFTIIN GTLKYPETRY IRVDISNPII PHMVGTMSTG TTEAELWNDW 360
YPYEDVEIGP NGVLKTPTFG KFPLYMIGHG MLDSDLHKSS QAQVFEHPHA KDAASQLPDD 420
ETLFFGDTGL SKNPVELVEG WFSSWKSTLA SFFLIIGLGV ALIFIIRIIV AIRYKYKGRK 480
TQKIYNDVEM SRLGNK 496

SEQ ID NO: 188      moltype = AA length = 496
FEATURE            Location/Qualifiers

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source                1..496
                      mol_type = protein
                      organism = Vesicular stomatitis virus

SEQUENCE: 188
KFTIVFPHHQ KGNWKNVPST YHYCPSSSDQ NWHNDLTGVS LHVKIPASHK AIQADGWMCH 60
AAKWVTTCDF RWYGPKYITH SIHMSPTLE QCKTSIEQTK QGVWINPGFP PQSCGYATVT 120
DAEVVVVQAT PHHVLVDEYT GEWIDSQLVG GKCSKEVCQT VHNSTVWHAD YKITGLCESN 180
LASVDITFFS EDGQKTSLGK PNTGFRSNHF AYESGEKACR MQYCTQWGIR LPSGVWFELV 240
DKDLFQAACL PECPRGSSIS APSQTSVDVS LIQDVERILD YSLCQETWSK IRAKLPVSPV 300
DLSYLAPKNP GSGPAFTIIN GTLKYFETRY IRVDISNPPI PHMVGTMSTG TTEGELWNDW 360
YPYEDVEIGP NGVLKTPTFG KFPLYMIGHG MLDSDLHKSS QAQVFEHPHA KDAASQLPDD 420
ETLFFGDTGL SKNPVELVEG WFSSWKSTLA SFFLIIGLGV ALIFIIRIIV AIRYKYKGRK 480
TQKIYNDVEM SRLGNK                                     496

SEQ ID NO: 189        moltype = AA length = 496
FEATURE              Location/Qualifiers
source                1..496
                      mol_type = protein
                      organism = Vesicular stomatitis virus

SEQUENCE: 189
KFTIVFPHHQ KGNWKNVPST YHYCPSSSDQ NWHNDLTGVS LHVKIPASHK AIQADGWMCH 60
AAKWVTTCDF RWYGPKYITH SIHMSPTLE QCKTSIEQTK QGVWINPGFP PQSCGYATVT 120
DAEVVVVQAT PHHVLVDEYT GEWIDSQLVG GKCSKEVCQT VHNSTVWHAD YKITGLCESN 180
LASVDITFFS EDGQKTSLGK PNTGFRSNHF AYESGEKACR MQYCTQWGIR LPSGVWFELV 240
DKDLFQAACL PECPRGSSIS APSQTSVDVS LIQDVERILD YSLCQETWSK IRAKLPVSPV 300
DLSYLAPKNP GSGPAFTIIN GTLKYFETRY IRVDISNPPI PHMVGTMSTG TTEGELWNDW 360
YPYEDVEIGP NGVLKTPTFG KFPLYMIGHG MLDSDLHKSS QAQVFEHPHA KDAASQLPDD 420
ETLFFGDTGL SKNPVELVEG WFSSWKSTLA SFFLIIGLGV ALIFIIRIIV AIRYKYKGRK 480
TQKIYNDVEM SRLGNK                                     496

SEQ ID NO: 190        moltype = AA length = 496
FEATURE              Location/Qualifiers
source                1..496
                      mol_type = protein
                      organism = Vesicular stomatitis virus

SEQUENCE: 190
KFTIVFPHHQ KGNWKNVPST YHYCPSSSDQ NWHNDLTGVS LHVKIPASHK AIQADGWMCH 60
AAKWVTTCDF RWYGPKYITH SIHMSPTLE QCKTSIEQTK QGVWINPGFP PQSCGYATVT 120
DAEVVVVQAT PHHVLVDEYT GEWIDSQLVG GKCSKEVCQT VHNSTVWHAD YKITGLCESN 180
LASVDITFFS EDGQKTSLGK PNTGFRSNHF AYESGEKACR MQYCTQWGIR LPSGVWFELV 240
DKDLFQAACL PECPRGSSIS APSQTSVDVS LIQDVERILD YSLCQETWSK IRAKLPVSPV 300
DLSYLAPKNP GSGPAFTIIN GTLKYFETRY IRVDISNPPI PHMVGTMSTG TTEGELWNDW 360
YPYEDVEIGP NGVLKTPTFG KFPLYMIGHG MLDSDLHKSS QAQVFEHPHA KDAASQLPDD 420
ETLFFGDTGL SKNPVELVEG WFSSWKSTLA SFFLIIGLGV ALIFIIRIIV AIRYKYKGRK 480
TQKIYNDVEM SRLGNK                                     496

SEQ ID NO: 191        moltype = AA length = 496
FEATURE              Location/Qualifiers
source                1..496
                      mol_type = protein
                      organism = Vesicular stomatitis virus

SEQUENCE: 191
KFTIVFPHHQ KGNWKNVPST YHYCPSSSDQ NWHNDLTGVS LHVKIPGSHK AIQADGWMCH 60
AAKWVTTCDF RWYGPKYITH SIHMSPTLE QCKTSIEQTK QGVWINPGFP PQSCGYATVT 120
DAEVVVVQAT PHHVLVDEYT GEWIDSQLVG GKCSKEVCQT VHNSTVWHAD YKITGLCESN 180
LASVDITFFS EDGQKTSLGK PNTGFRSNHF AYESGEKACR MQYCTQWGIR LPSGVWFELV 240
DKDLFQAACL PECPRGSSIS APSQTSVDVS LIQDVERILD YSLCQETWSK IRAKLPVSPV 300
DLSYLAPKNP GSGPAFTIIN GTLKYFETRY IRVDISNPPI PHMVGTMSTG TTEAELWNDW 360
YPYEDVEIGP NGVLKTPTFG KFPLYMIGHG MLDSDLHKSS QAQVFEHPHA KDAASQLPDD 420
ETLFFGDTGL SKNPVELVEG WFSSWKSTLA SFFLIIGLGV ALIFIIRIIV AIRYKYKGRK 480
TQKIYNDVEM SRLGNK                                     496

SEQ ID NO: 192        moltype = AA length = 496
FEATURE              Location/Qualifiers
source                1..496
                      mol_type = protein
                      organism = Vesicular stomatitis virus

SEQUENCE: 192
KFTIVFPHHQ KGNWKNVPST YHYCPSSSDQ NWHNDLTGVS LHVKIPGSHK AIQADGWMCH 60
AAKWVTTCDF RWYGPKYITH SIHMSPTLE QCKTSIEQTK QGVWINPGFP PQSCGYATVT 120
DAEVVVVQAT PHHVLVDEYT GEWIDSQLVG GKCSKEVCQT VHNSTVWHAD YKITGLCESN 180
LASVDITFFS EDGQKTSLGK PNTGFRSNHF AYESGEKACR MQYCTQWGIR LPSGVWFELV 240
DKDLFQAACL PECPRGSSIS APSQTSVDVS LIQDVERILD YSLCQETWSK IRAKLPVSPV 300
DLSYLAPKNP GSGPAFTIIN GTLKYFETRY IRVDISNPPI PHMVGTMSTG TTEGELWNDW 360
YPYEDVEIGP NGVLKTPTFG KFPLYMIGHG MLDSDLHKSS QAQVFEHPHA KDAASQLPDD 420
ETLFFGDTGL SKNPVELVEG WFSSWKSTLA SFFLIIGLGV ALIFIIRIIV AIRYKYKGRK 480
TQKIYNDVEM SRLGNK                                     496

SEQ ID NO: 193        moltype = AA length = 496

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-continued

FEATURE
source Location/Qualifiers
1..496
mol_type = protein
organism = Vesicular stomatitis virus

SEQUENCE: 193

KFTIVFPHHQ	KGNWKNVPST	YHYCPSSSDQ	NWHNDLTGVS	LHVKIPGSHK	AIQADGWMCH	60
AAKWVTTCDF	RWYGPKYITH	SIHMSPTLE	QCKTSIEQTK	QGVWINPGFP	PQSCGYATVT	120
DAEVVVVQAT	PHHVLVDEYT	GEWIDSQLVG	GKCSKEVCQT	VHNSTVWHAD	YKITGLCESN	180
LASVDITFFS	EDGQKTSLGK	PNTGFRSNHF	AYESGEKACR	MQYCTQWGIR	LPSGVWFELV	240
DKDLFQAACL	PECPRGSSIS	APSQTSVDVS	LIQDVERILD	YSLCQETWSK	IRAKLPVSPV	300
DLSYLAPKNP	GSGPAFTIIN	GTLKYFETRY	IRVDISNPPII	PHMVGTMSTGT	TTEFELWNDW	360
YPYEDVEIGP	NGVLKTPTFG	KFPLYMIGHG	MLDSDLHKSS	QAQVFEHPHA	KDAASQLPDD	420
ETLFFGDTGL	SKNPVELVEG	WFSSWKSTLA	SFFLIIGLGV	ALIFIIRIIV	AIRYKYKGRK	480
TQKIYNDVEM	SRLGNK					496

SEQ ID NO: 194 moltype = AA length = 496

FEATURE
source Location/Qualifiers
1..496
mol_type = protein
organism = Vesicular stomatitis virus

SEQUENCE: 194

KFTIVFPHHQ	KGNWKNVPST	YHYCPSSSDQ	NWHNDLTGVS	LHVKIPGSHK	AIQADGWMCH	60
AAKWVTTCDF	RWYGPKYITH	SIHMSPTLE	QCKTSIEQTK	QGVWINPGFP	PQSCGYATVT	120
DAEVVVVQAT	PHHVLVDEYT	GEWIDSQLVG	GKCSKEVCQT	VHNSTVWHAD	YKITGLCESN	180
LASVDITFFS	EDGQKTSLGK	PNTGFRSNHF	AYESGEKACR	MQYCTQWGIR	LPSGVWFELV	240
DKDLFQAACL	PECPRGSSIS	APSQTSVDVS	LIQDVERILD	YSLCQETWSK	IRAKLPVSPV	300
DLSYLAPKNP	GSGPAFTIIN	GTLKYFETRY	IRVDISNPPII	PHMVGTMSTGT	TTEQELWNDW	360
YPYEDVEIGP	NGVLKTPTFG	KFPLYMIGHG	MLDSDLHKSS	QAQVFEHPHA	KDAASQLPDD	420
ETLFFGDTGL	SKNPVELVEG	WFSSWKSTLA	SFFLIIGLGV	ALIFIIRIIV	AIRYKYKGRK	480
TQKIYNDVEM	SRLGNK					496

SEQ ID NO: 195 moltype = AA length = 496

FEATURE
source Location/Qualifiers
1..496
mol_type = protein
organism = Vesicular stomatitis virus

SEQUENCE: 195

KFTIVFPHHQ	KGNWKNVPST	YHYCPSSSDQ	NWHNDLTGVS	LHVKIPFSHK	AIQADGWMCH	60
AAKWVTTCDF	RWYGPKYITH	SIHMSPTLE	QCKTSIEQTK	QGVWINPGFP	PQSCGYATVT	120
DAEVVVVQAT	PHHVLVDEYT	GEWIDSQLVG	GKCSKEVCQT	VHNSTVWHAD	YKITGLCESN	180
LASVDITFFS	EDGQKTSLGK	PNTGFRSNHF	AYESGEKACR	MQYCTQWGIR	LPSGVWFELV	240
DKDLFQAACL	PECPRGSSIS	APSQTSVDVS	LIQDVERILD	YSLCQETWSK	IRAKLPVSPV	300
DLSYLAPKNP	GSGPAFTIIN	GTLKYFETRY	IRVDISNPPII	PHMVGTMSTGT	TTEAELWNDW	360
YPYEDVEIGP	NGVLKTPTFG	KFPLYMIGHG	MLDSDLHKSS	QAQVFEHPHA	KDAASQLPDD	420
ETLFFGDTGL	SKNPVELVEG	WFSSWKSTLA	SFFLIIGLGV	ALIFIIRIIV	AIRYKYKGRK	480
TQKIYNDVEM	SRLGNK					496

SEQ ID NO: 196 moltype = AA length = 496

FEATURE
source Location/Qualifiers
1..496
mol_type = protein
organism = Vesicular stomatitis virus

SEQUENCE: 196

KFTIVFPHHQ	KGNWKNVPST	YHYCPSSSDQ	NWHNDLTGVS	LHVKIPFSHK	AIQADGWMCH	60
AAKWVTTCDF	RWYGPKYITH	SIHMSPTLE	QCKTSIEQTK	QGVWINPGFP	PQSCGYATVT	120
DAEVVVVQAT	PHHVLVDEYT	GEWIDSQLVG	GKCSKEVCQT	VHNSTVWHAD	YKITGLCESN	180
LASVDITFFS	EDGQKTSLGK	PNTGFRSNHF	AYESGEKACR	MQYCTQWGIR	LPSGVWFELV	240
DKDLFQAACL	PECPRGSSIS	APSQTSVDVS	LIQDVERILD	YSLCQETWSK	IRAKLPVSPV	300
DLSYLAPKNP	GSGPAFTIIN	GTLKYFETRY	IRVDISNPPII	PHMVGTMSTGT	TTEGELWNDW	360
YPYEDVEIGP	NGVLKTPTFG	KFPLYMIGHG	MLDSDLHKSS	QAQVFEHPHA	KDAASQLPDD	420
ETLFFGDTGL	SKNPVELVEG	WFSSWKSTLA	SFFLIIGLGV	ALIFIIRIIV	AIRYKYKGRK	480
TQKIYNDVEM	SRLGNK					496

SEQ ID NO: 197 moltype = AA length = 496

FEATURE
source Location/Qualifiers
1..496
mol_type = protein
organism = Vesicular stomatitis virus

SEQUENCE: 197

KFTIVFPHHQ	KGNWKNVPST	YHYCPSSSDQ	NWHNDLTGVS	LHVKIPFSHK	AIQADGWMCH	60
AAKWVTTCDF	RWYGPKYITH	SIHMSPTLE	QCKTSIEQTK	QGVWINPGFP	PQSCGYATVT	120
DAEVVVVQAT	PHHVLVDEYT	GEWIDSQLVG	GKCSKEVCQT	VHNSTVWHAD	YKITGLCESN	180
LASVDITFFS	EDGQKTSLGK	PNTGFRSNHF	AYESGEKACR	MQYCTQWGIR	LPSGVWFELV	240
DKDLFQAACL	PECPRGSSIS	APSQTSVDVS	LIQDVERILD	YSLCQETWSK	IRAKLPVSPV	300
DLSYLAPKNP	GSGPAFTIIN	GTLKYFETRY	IRVDISNPPII	PHMVGTMSTGT	TTEFELWNDW	360
YPYEDVEIGP	NGVLKTPTFG	KFPLYMIGHG	MLDSDLHKSS	QAQVFEHPHA	KDAASQLPDD	420
ETLFFGDTGL	SKNPVELVEG	WFSSWKSTLA	SFFLIIGLGV	ALIFIIRIIV	AIRYKYKGRK	480
TQKIYNDVEM	SRLGNK					496

-continued

SEQ ID NO: 198 moltype = AA length = 496
 FEATURE Location/Qualifiers
 source 1..496
 mol_type = protein
 organism = Vesicular stomatitis virus

SEQUENCE: 198

KFTIVFPHHQ	KGNWKNVPST	YHYCPSSSDQ	NWHNDLTGVS	LHVKIPFSHK	AIQADGWMCH	60
AAKWVTTCDF	RWYGPKYITH	SIHMSPTLE	QCKTSIEQTK	QGVWINPGFP	PQSCGYATVT	120
DAEVVVVQAT	PHHVLVDEYT	GEWIDSQLVG	GKCSKEVCQT	VHNSTVWHAD	YKITGLCESN	180
LASVDITFFS	EDGQKTSLGK	PNTGFRSNHF	AYESGEKACR	MQYCTQWGIR	LPSGVWFELV	240
DKDLFQAACL	PECPRGSSIS	APSQTSVDVS	LIQDVERILD	YSLCQETWSK	IRAKLPVSPV	300
DLSYLAPKNP	GS GPAFTIIN	GTLKYFETRY	IRVDISNPII	PHMVGTMSTG	TTEQELWNDW	360
YPYEDVEIGP	NGVLKTPTGF	KFPLYMIGHG	MLDSDLHKSS	QAQVFEHPHA	KDAASQLPDD	420
ETLFFGDTGL	SKNPVELVEG	WFSSWKSTLA	SFFLIIGLGV	ALIFIIRIIV	AIRYKYKGRK	480
TQKIYNDVEM	SRLGNK					496

SEQ ID NO: 199 moltype = AA length = 496
 FEATURE Location/Qualifiers
 source 1..496
 mol_type = protein
 organism = Vesicular stomatitis virus

SEQUENCE: 199

KFTIVFPHHQ	KGNWKNVPST	YHYCPSSSDQ	NWHNDLTGVS	LHVKIPQSHK	AIQADGWMCH	60
AAKWVTTCDF	RWYGPKYITH	SIHMSPTLE	QCKTSIEQTK	QGVWINPGFP	PQSCGYATVT	120
DAEVVVVQAT	PHHVLVDEYT	GEWIDSQLVG	GKCSKEVCQT	VHNSTVWHAD	YKITGLCESN	180
LASVDITFFS	EDGQKTSLGK	PNTGFRSNHF	AYESGEKACR	MQYCTQWGIR	LPSGVWFELV	240
DKDLFQAACL	PECPRGSSIS	APSQTSVDVS	LIQDVERILD	YSLCQETWSK	IRAKLPVSPV	300
DLSYLAPKNP	GS GPAFTIIN	GTLKYFETRY	IRVDISNPII	PHMVGTMSTG	TTEAELWNDW	360
YPYEDVEIGP	NGVLKTPTGF	KFPLYMIGHG	MLDSDLHKSS	QAQVFEHPHA	KDAASQLPDD	420
ETLFFGDTGL	SKNPVELVEG	WFSSWKSTLA	SFFLIIGLGV	ALIFIIRIIV	AIRYKYKGRK	480
TQKIYNDVEM	SRLGNK					496

SEQ ID NO: 200 moltype = AA length = 496
 FEATURE Location/Qualifiers
 source 1..496
 mol_type = protein
 organism = Vesicular stomatitis virus

SEQUENCE: 200

KFTIVFPHHQ	KGNWKNVPST	YHYCPSSSDQ	NWHNDLTGVS	LHVKIPQSHK	AIQADGWMCH	60
AAKWVTTCDF	RWYGPKYITH	SIHMSPTLE	QCKTSIEQTK	QGVWINPGFP	PQSCGYATVT	120
DAEVVVVQAT	PHHVLVDEYT	GEWIDSQLVG	GKCSKEVCQT	VHNSTVWHAD	YKITGLCESN	180
LASVDITFFS	EDGQKTSLGK	PNTGFRSNHF	AYESGEKACR	MQYCTQWGIR	LPSGVWFELV	240
DKDLFQAACL	PECPRGSSIS	APSQTSVDVS	LIQDVERILD	YSLCQETWSK	IRAKLPVSPV	300
DLSYLAPKNP	GS GPAFTIIN	GTLKYFETRY	IRVDISNPII	PHMVGTMSTG	TTEGELWNDW	360
YPYEDVEIGP	NGVLKTPTGF	KFPLYMIGHG	MLDSDLHKSS	QAQVFEHPHA	KDAASQLPDD	420
ETLFFGDTGL	SKNPVELVEG	WFSSWKSTLA	SFFLIIGLGV	ALIFIIRIIV	AIRYKYKGRK	480
TQKIYNDVEM	SRLGNK					496

SEQ ID NO: 201 moltype = AA length = 496
 FEATURE Location/Qualifiers
 source 1..496
 mol_type = protein
 organism = Vesicular stomatitis virus

SEQUENCE: 201

KFTIVFPHHQ	KGNWKNVPST	YHYCPSSSDQ	NWHNDLTGVS	LHVKIPQSHK	AIQADGWMCH	60
AAKWVTTCDF	RWYGPKYITH	SIHMSPTLE	QCKTSIEQTK	QGVWINPGFP	PQSCGYATVT	120
DAEVVVVQAT	PHHVLVDEYT	GEWIDSQLVG	GKCSKEVCQT	VHNSTVWHAD	YKITGLCESN	180
LASVDITFFS	EDGQKTSLGK	PNTGFRSNHF	AYESGEKACR	MQYCTQWGIR	LPSGVWFELV	240
DKDLFQAACL	PECPRGSSIS	APSQTSVDVS	LIQDVERILD	YSLCQETWSK	IRAKLPVSPV	300
DLSYLAPKNP	GS GPAFTIIN	GTLKYFETRY	IRVDISNPII	PHMVGTMSTG	TTEFELWNDW	360
YPYEDVEIGP	NGVLKTPTGF	KFPLYMIGHG	MLDSDLHKSS	QAQVFEHPHA	KDAASQLPDD	420
ETLFFGDTGL	SKNPVELVEG	WFSSWKSTLA	SFFLIIGLGV	ALIFIIRIIV	AIRYKYKGRK	480
TQKIYNDVEM	SRLGNK					496

SEQ ID NO: 202 moltype = AA length = 496
 FEATURE Location/Qualifiers
 source 1..496
 mol_type = protein
 organism = Vesicular stomatitis virus

SEQUENCE: 202

KFTIVFPHHQ	KGNWKNVPST	YHYCPSSSDQ	NWHNDLTGVS	LHVKIPQSHK	AIQADGWMCH	60
AAKWVTTCDF	RWYGPKYITH	SIHMSPTLE	QCKTSIEQTK	QGVWINPGFP	PQSCGYATVT	120
DAEVVVVQAT	PHHVLVDEYT	GEWIDSQLVG	GKCSKEVCQT	VHNSTVWHAD	YKITGLCESN	180
LASVDITFFS	EDGQKTSLGK	PNTGFRSNHF	AYESGEKACR	MQYCTQWGIR	LPSGVWFELV	240
DKDLFQAACL	PECPRGSSIS	APSQTSVDVS	LIQDVERILD	YSLCQETWSK	IRAKLPVSPV	300
DLSYLAPKNP	GS GPAFTIIN	GTLKYFETRY	IRVDISNPII	PHMVGTMSTG	TTEQELWNDW	360
YPYEDVEIGP	NGVLKTPTGF	KFPLYMIGHG	MLDSDLHKSS	QAQVFEHPHA	KDAASQLPDD	420
ETLFFGDTGL	SKNPVELVEG	WFSSWKSTLA	SFFLIIGLGV	ALIFIIRIIV	AIRYKYKGRK	480
TQKIYNDVEM	SRLGNK					496

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SEQ ID NO: 203 moltype = AA length = 501
FEATURE Location/Qualifiers
source 1..501
 mol_type = protein
 organism = Vesicular stomatitis virus

SEQUENCE: 203

KIEIVFPQHT	TGDWKRVPHE	YNYCPTSADK	NSHGTQTGIP	VELTMPAGLT	THQVDGFMCH	60
SALWMTTCDF	RWYGPKYITH	SIHNEEPTDY	QCLEAIKAYK	DGVSFNPGFP	PQSCGYGTVT	120
DAEAHIVTVT	PHSVKVDEYT	GEWIDPHFIG	GRCKGQICET	VHNSTKWFTS	SDGESVCSQL	180
FTLVGGTFFS	DSEETSMGL	PETGIRSNYF	PYVSTEGICK	MPFCRKPGYK	LKNDLWFQIT	240
DPDLDKTVRD	LPHIKDCDLS	SSIIVTPGEHA	TDISLISDVE	RILDYALCQN	TWSKIEAGEP	300
ITPVDLSYLG	PKNPGAGPVF	TIINGSLHYF	MSKYLRVELE	SPVIPRMEGK	VAGTRIVRQL	360
WDQWFFPGEV	EIGPNGVLKT	KQGYKFPLHI	IGTGEVDNDI	KMERIVKHWE	HPHIEAAQTF	420
LKKDDTEEV	YYGDTGVSKN	PVELVEGWFS	GWRSSIMGVL	AVIIGFVILI	FLIRLIGVLS	480
SLFRQKRRPI	YKSDVEMAHF	R				501

SEQ ID NO: 204 moltype = AA length = 501
FEATURE Location/Qualifiers
source 1..501
 mol_type = protein
 organism = Vesicular stomatitis virus

SEQUENCE: 204

KIEIVFPQHT	TGDWKRVPHE	YNYCPTSADK	NSHGTQTGIP	VELTMPGGLT	THQVDGFMCH	60
SALWMTTCDF	RWYGPKYITH	SIHNEEPTDY	QCLEAIKAYK	DGVSFNPGFP	PQSCGYGTVT	120
DAEAHIVTVT	PHSVKVDEYT	GEWIDPHFIG	GRCKGQICET	VHNSTKWFTS	SDGESVCSQL	180
FTLVGGTFFS	DSEETSMGL	PETGIRSNYF	PYVSTEGICK	MPFCRKPGYK	LKNDLWFQIT	240
DPDLDKTVRD	LPHIKDCDLS	SSIIVTPGEHA	TDISLISDVE	RILDYALCQN	TWSKIEAGEP	300
ITPVDLSYLG	PKNPGAGPVF	TIINGSLHYF	MSKYLRVELE	SPVIPRMEGK	VAGTRIVRQL	360
WDQWFFPGEV	EIGPNGVLKT	KQGYKFPLHI	IGTGEVDNDI	KMERIVKHWE	HPHIEAAQTF	420
LKKDDTEEV	YYGDTGVSKN	PVELVEGWFS	GWRSSIMGVL	AVIIGFVILI	FLIRLIGVLS	480
SLFRQKRRPI	YKSDVEMAHF	R				501

SEQ ID NO: 205 moltype = AA length = 501
FEATURE Location/Qualifiers
source 1..501
 mol_type = protein
 organism = Vesicular stomatitis virus

SEQUENCE: 205

KIEIVFPQHT	TGDWKRVPHE	YNYCPTSADK	NSHGTQTGIP	VELTMPGGLT	THQVDGFMCH	60
SALWMTTCDF	RWYGPKYITH	SIHNEEPTDY	QCLEAIKAYK	DGVSFNPGFP	PQSCGYGTVT	120
DAEAHIVTVT	PHSVKVDEYT	GEWIDPHFIG	GRCKGQICET	VHNSTKWFTS	SDGESVCSQL	180
FTLVGGTFFS	DSEETSMGL	PETGIRSNYF	PYVSTEGICK	MPFCRKPGYK	LKNDLWFQIT	240
DPDLDKTVRD	LPHIKDCDLS	SSIIVTPGEHA	TDISLISDVE	RILDYALCQN	TWSKIEAGEP	300
ITPVDLSYLG	PKNPGAGPVF	TIINGSLHYF	MSKYLRVELE	SPVIPRMEGK	VAGTRIVRQL	360
WDQWFFPGEV	EIGPNGVLKT	KQGYKFPLHI	IGTGEVDNDI	KMERIVKHWE	HPHIEAAQTF	420
LKKDDTEEV	YYGDTGVSKN	PVELVEGWFS	GWRSSIMGVL	AVIIGFVILI	FLIRLIGVLS	480
SLFRQKRRPI	YKSDVEMAHF	R				501

SEQ ID NO: 206 moltype = AA length = 501
FEATURE Location/Qualifiers
source 1..501
 mol_type = protein
 organism = Vesicular stomatitis virus

SEQUENCE: 206

KIEIVFPQHT	TGDWKRVPHE	YNYCPTSADK	NSHGTQTGIP	VELTMPQGLT	THQVDGFMCH	60
SALWMTTCDF	RWYGPKYITH	SIHNEEPTDY	QCLEAIKAYK	DGVSFNPGFP	PQSCGYGTVT	120
DAEAHIVTVT	PHSVKVDEYT	GEWIDPHFIG	GRCKGQICET	VHNSTKWFTS	SDGESVCSQL	180
FTLVGGTFFS	DSEETSMGL	PETGIRSNYF	PYVSTEGICK	MPFCRKPGYK	LKNDLWFQIT	240
DPDLDKTVRD	LPHIKDCDLS	SSIIVTPGEHA	TDISLISDVE	RILDYALCQN	TWSKIEAGEP	300
ITPVDLSYLG	PKNPGAGPVF	TIINGSLHYF	MSKYLRVELE	SPVIPRMEGK	VAGTRIVRQL	360
WDQWFFPGEV	EIGPNGVLKT	KQGYKFPLHI	IGTGEVDNDI	KMERIVKHWE	HPHIEAAQTF	420
LKKDDTEEV	YYGDTGVSKN	PVELVEGWFS	GWRSSIMGVL	AVIIGFVILI	FLIRLIGVLS	480
SLFRQKRRPI	YKSDVEMAHF	R				501

SEQ ID NO: 207 moltype = AA length = 501
FEATURE Location/Qualifiers
source 1..501
 mol_type = protein
 organism = Vesicular stomatitis virus

SEQUENCE: 207

KIEIVFPQHT	TGDWKRVPHE	YNYCPTSADK	NSHGTQTGIP	VELTMPKGLT	THQVDGFMCH	60
SALWMTTCDF	RWYGPKYITH	SIHNEEPTDY	QCLEAIKAYK	DGVSFNPGFP	PQSCGYGTVT	120
DAEAHIVTVT	PHSVKVDEYT	GEWIDPHFIG	GRCKGQICET	VHNSTKWFTS	SDGESVCSQL	180
FTLVGGTFFS	DSEETSMGL	PETGIRSNYF	PYVSTEGICK	MPFCRKPGYK	LKNDLWFQIT	240
DPDLDKTVRD	LPHIKDCDLS	SSIIVTPGEHA	TDISLISDVE	RILDYALCQN	TWSKIEAGEP	300
ITPVDLSYLG	PKNPGAGPVF	TIINGSLHYF	MSKYLRVELE	SPVIPRMEGK	VAGTRIVAQL	360
WDQWFFPGEV	EIGPNGVLKT	KQGYKFPLHI	IGTGEVDNDI	KMERIVKHWE	HPHIEAAQTF	420
LKKDDTEEV	YYGDTGVSKN	PVELVEGWFS	GWRSSIMGVL	AVIIGFVILI	FLIRLIGVLS	480

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SLFRQKRRPI YKSDVEMAHF R 501

SEQ ID NO: 208 moltype = AA length = 501
 FEATURE Location/Qualifiers
 source 1..501
 mol_type = protein
 organism = Vesicular stomatitis virus

SEQUENCE: 208
 KIEIVFPQHT TGDWKRVPHE YNYCPTSADK NSHGTQTGIP VELTMPKGLT THQVDGFMCH 60
 SALWMTTCDF RWYGPKYITH SIHNEEPTDY QCLEAIKAYK DGVSFNPGEF PQSCGYGTVT 120
 DAEAHIVTVT PHSVKVDEYT GEWIDPHFIG GRCKGQICET VHNSTKWFTS SDGESVCSQL 180
 FTLVGGTFFS DSEETSMGL PETGIRSNYF PYVSTEGICK MPFCRKPGYK LKNDLWFQIT 240
 DPDLDKTVRD LPHIKDCDLS SSIIVTPGEHA TDISLISDVE RILDYALCQN TWSKIEAGEP 300
 ITPVDLSYLG PKNPGAGPVF TIINGSLHYF MSKYLRVELE SPVIPRMEGK VAGTRIVFQL 360
 WDQWFFPGEV EIGPNGVLKT KQGYKFPLHI IGTGEVDNDI KMERIVKHWE HPHIEAAQTF 420
 LKKDDTEEV L YYGDTGVSKN PVELVEGWFS GWRSSIMGVL AVIIGFVILI FLIRLIGVLS 480
 SLFRQKRRPI YKSDVEMAHF R 501

SEQ ID NO: 209 moltype = AA length = 501
 FEATURE Location/Qualifiers
 source 1..501
 mol_type = protein
 organism = Vesicular stomatitis virus

SEQUENCE: 209
 KIEIVFPQHT TGDWKRVPHE YNYCPTSADK NSHGTQTGIP VELTMPKGLT THQVDGFMCH 60
 SALWMTTCDF RWYGPKYITH SIHNEEPTDY QCLEAIKAYK DGVSFNPGEF PQSCGYGTVT 120
 DAEAHIVTVT PHSVKVDEYT GEWIDPHFIG GRCKGQICET VHNSTKWFTS SDGESVCSQL 180
 FTLVGGTFFS DSEETSMGL PETGIRSNYF PYVSTEGICK MPFCRKPGYK LKNDLWFQIT 240
 DPDLDKTVRD LPHIKDCDLS SSIIVTPGEHA TDISLISDVE RILDYALCQN TWSKIEAGEP 300
 ITPVDLSYLG PKNPGAGPVF TIINGSLHYF MSKYLRVELE SPVIPRMEGK VAGTRIVGQL 360
 WDQWFFPGEV EIGPNGVLKT KQGYKFPLHI IGTGEVDNDI KMERIVKHWE HPHIEAAQTF 420
 LKKDDTEEV L YYGDTGVSKN PVELVEGWFS GWRSSIMGVL AVIIGFVILI FLIRLIGVLS 480
 SLFRQKRRPI YKSDVEMAHF R 501

SEQ ID NO: 210 moltype = AA length = 501
 FEATURE Location/Qualifiers
 source 1..501
 mol_type = protein
 organism = Vesicular stomatitis virus

SEQUENCE: 210
 KIEIVFPQHT TGDWKRVPHE YNYCPTSADK NSHGTQTGIP VELTMPKGLT THQVDGFMCH 60
 SALWMTTCDF RWYGPKYITH SIHNEEPTDY QCLEAIKAYK DGVSFNPGEF PQSCGYGTVT 120
 DAEAHIVTVT PHSVKVDEYT GEWIDPHFIG GRCKGQICET VHNSTKWFTS SDGESVCSQL 180
 FTLVGGTFFS DSEETSMGL PETGIRSNYF PYVSTEGICK MPFCRKPGYK LKNDLWFQIT 240
 DPDLDKTVRD LPHIKDCDLS SSIIVTPGEHA TDISLISDVE RILDYALCQN TWSKIEAGEP 300
 ITPVDLSYLG PKNPGAGPVF TIINGSLHYF MSKYLRVELE SPVIPRMEGK VAGTRIVQQL 360
 WDQWFFPGEV EIGPNGVLKT KQGYKFPLHI IGTGEVDNDI KMERIVKHWE HPHIEAAQTF 420
 LKKDDTEEV L YYGDTGVSKN PVELVEGWFS GWRSSIMGVL AVIIGFVILI FLIRLIGVLS 480
 SLFRQKRRPI YKSDVEMAHF R 501

SEQ ID NO: 211 moltype = AA length = 501
 FEATURE Location/Qualifiers
 source 1..501
 mol_type = protein
 organism = Vesicular stomatitis virus

SEQUENCE: 211
 KIEIVFPQHT TGDWKRVPHE YNYCPTSADK NSHGTQTGIP VELTMPAGLT THQVDGFMCH 60
 SALWMTTCDF RWYGPKYITH SIHNEEPTDY QCLEAIKAYK DGVSFNPGEF PQSCGYGTVT 120
 DAEAHIVTVT PHSVKVDEYT GEWIDPHFIG GRCKGQICET VHNSTKWFTS SDGESVCSQL 180
 FTLVGGTFFS DSEETSMGL PETGIRSNYF PYVSTEGICK MPFCRKPGYK LKNDLWFQIT 240
 DPDLDKTVRD LPHIKDCDLS SSIIVTPGEHA TDISLISDVE RILDYALCQN TWSKIEAGEP 300
 ITPVDLSYLG PKNPGAGPVF TIINGSLHYF MSKYLRVELE SPVIPRMEGK VAGTRIVAQL 360
 WDQWFFPGEV EIGPNGVLKT KQGYKFPLHI IGTGEVDNDI KMERIVKHWE HPHIEAAQTF 420
 LKKDDTEEV L YYGDTGVSKN PVELVEGWFS GWRSSIMGVL AVIIGFVILI FLIRLIGVLS 480
 SLFRQKRRPI YKSDVEMAHF R 501

SEQ ID NO: 212 moltype = AA length = 501
 FEATURE Location/Qualifiers
 source 1..501
 mol_type = protein
 organism = Vesicular stomatitis virus

SEQUENCE: 212
 KIEIVFPQHT TGDWKRVPHE YNYCPTSADK NSHGTQTGIP VELTMPAGLT THQVDGFMCH 60
 SALWMTTCDF RWYGPKYITH SIHNEEPTDY QCLEAIKAYK DGVSFNPGEF PQSCGYGTVT 120
 DAEAHIVTVT PHSVKVDEYT GEWIDPHFIG GRCKGQICET VHNSTKWFTS SDGESVCSQL 180
 FTLVGGTFFS DSEETSMGL PETGIRSNYF PYVSTEGICK MPFCRKPGYK LKNDLWFQIT 240
 DPDLDKTVRD LPHIKDCDLS SSIIVTPGEHA TDISLISDVE RILDYALCQN TWSKIEAGEP 300
 ITPVDLSYLG PKNPGAGPVF TIINGSLHYF MSKYLRVELE SPVIPRMEGK VAGTRIVFQL 360
 WDQWFFPGEV EIGPNGVLKT KQGYKFPLHI IGTGEVDNDI KMERIVKHWE HPHIEAAQTF 420

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LKKDDTEEVLY YGDTGVSKN PVELVEGWFS GWRSSIMGVL AVIIGFVILI FLIRLIGVLS 480
SLFRQKRRPI YKSDVEMAHF R 501

SEQ ID NO: 213 moltype = AA length = 501
FEATURE Location/Qualifiers
source 1..501
 mol_type = protein
 organism = Vesicular stomatitis virus

SEQUENCE: 213
KIEIVFPQHT TGDWKRVPHE YNYCPTSADK NSHGTQTGIP VELTMPAGLT THQVDGFMCH 60
SALWMTTCDF RWYGPKYITH SIHNEEPTDY QCLEAIKAYK DGVSFNPGFP PQSCGYGTVT 120
DAEAHIVTVT PHSVKVDEYT GEWIDPHFIG GRCKGQICET VHNSTKWFTS SDGESVCSQL 180
FTLVGGTFPS DSEETSMGL PETGIRSNYF PYVSTEGICK MPFCRKPGYK LKNDLWFQIT 240
DPDLDKTVRD LPHIKDCDLS SSIIVTPGEHA TDISLISDVE RILDYALCQN TWSKIEAGEP 300
ITPVDLSYLG PKNPGAGPVF TIINGSLHYF MSKYLRVELE SPVIPRMEGK VAGTRIVGQL 360
WDQWFFPGEV EIGPNGVLKT KQGYKFPLHI IGTGEVDNDI KMERIVKHWE HPHIEAAQTF 420
LKKDDTEEVLY YGDTGVSKN PVELVEGWFS GWRSSIMGVL AVIIGFVILI FLIRLIGVLS 480
SLFRQKRRPI YKSDVEMAHF R 501

SEQ ID NO: 214 moltype = AA length = 501
FEATURE Location/Qualifiers
source 1..501
 mol_type = protein
 organism = Vesicular stomatitis virus

SEQUENCE: 214
KIEIVFPQHT TGDWKRVPHE YNYCPTSADK NSHGTQTGIP VELTMPAGLT THQVDGFMCH 60
SALWMTTCDF RWYGPKYITH SIHNEEPTDY QCLEAIKAYK DGVSFNPGFP PQSCGYGTVT 120
DAEAHIVTVT PHSVKVDEYT GEWIDPHFIG GRCKGQICET VHNSTKWFTS SDGESVCSQL 180
FTLVGGTFPS DSEETSMGL PETGIRSNYF PYVSTEGICK MPFCRKPGYK LKNDLWFQIT 240
DPDLDKTVRD LPHIKDCDLS SSIIVTPGEHA TDISLISDVE RILDYALCQN TWSKIEAGEP 300
ITPVDLSYLG PKNPGAGPVF TIINGSLHYF MSKYLRVELE SPVIPRMEGK VAGTRIVGQL 360
WDQWFFPGEV EIGPNGVLKT KQGYKFPLHI IGTGEVDNDI KMERIVKHWE HPHIEAAQTF 420
LKKDDTEEVLY YGDTGVSKN PVELVEGWFS GWRSSIMGVL AVIIGFVILI FLIRLIGVLS 480
SLFRQKRRPI YKSDVEMAHF R 501

SEQ ID NO: 215 moltype = AA length = 501
FEATURE Location/Qualifiers
source 1..501
 mol_type = protein
 organism = Vesicular stomatitis virus

SEQUENCE: 215
KIEIVFPQHT TGDWKRVPHE YNYCPTSADK NSHGTQTGIP VELTMPFGLT THQVDGFMCH 60
SALWMTTCDF RWYGPKYITH SIHNEEPTDY QCLEAIKAYK DGVSFNPGFP PQSCGYGTVT 120
DAEAHIVTVT PHSVKVDEYT GEWIDPHFIG GRCKGQICET VHNSTKWFTS SDGESVCSQL 180
FTLVGGTFPS DSEETSMGL PETGIRSNYF PYVSTEGICK MPFCRKPGYK LKNDLWFQIT 240
DPDLDKTVRD LPHIKDCDLS SSIIVTPGEHA TDISLISDVE RILDYALCQN TWSKIEAGEP 300
ITPVDLSYLG PKNPGAGPVF TIINGSLHYF MSKYLRVELE SPVIPRMEGK VAGTRIVGQL 360
WDQWFFPGEV EIGPNGVLKT KQGYKFPLHI IGTGEVDNDI KMERIVKHWE HPHIEAAQTF 420
LKKDDTEEVLY YGDTGVSKN PVELVEGWFS GWRSSIMGVL AVIIGFVILI FLIRLIGVLS 480
SLFRQKRRPI YKSDVEMAHF R 501

SEQ ID NO: 216 moltype = AA length = 501
FEATURE Location/Qualifiers
source 1..501
 mol_type = protein
 organism = Vesicular stomatitis virus

SEQUENCE: 216
KIEIVFPQHT TGDWKRVPHE YNYCPTSADK NSHGTQTGIP VELTMPFGLT THQVDGFMCH 60
SALWMTTCDF RWYGPKYITH SIHNEEPTDY QCLEAIKAYK DGVSFNPGFP PQSCGYGTVT 120
DAEAHIVTVT PHSVKVDEYT GEWIDPHFIG GRCKGQICET VHNSTKWFTS SDGESVCSQL 180
FTLVGGTFPS DSEETSMGL PETGIRSNYF PYVSTEGICK MPFCRKPGYK LKNDLWFQIT 240
DPDLDKTVRD LPHIKDCDLS SSIIVTPGEHA TDISLISDVE RILDYALCQN TWSKIEAGEP 300
ITPVDLSYLG PKNPGAGPVF TIINGSLHYF MSKYLRVELE SPVIPRMEGK VAGTRIVGQL 360
WDQWFFPGEV EIGPNGVLKT KQGYKFPLHI IGTGEVDNDI KMERIVKHWE HPHIEAAQTF 420
LKKDDTEEVLY YGDTGVSKN PVELVEGWFS GWRSSIMGVL AVIIGFVILI FLIRLIGVLS 480
SLFRQKRRPI YKSDVEMAHF R 501

SEQ ID NO: 217 moltype = AA length = 501
FEATURE Location/Qualifiers
source 1..501
 mol_type = protein
 organism = Vesicular stomatitis virus

SEQUENCE: 217
KIEIVFPQHT TGDWKRVPHE YNYCPTSADK NSHGTQTGIP VELTMPFGLT THQVDGFMCH 60
SALWMTTCDF RWYGPKYITH SIHNEEPTDY QCLEAIKAYK DGVSFNPGFP PQSCGYGTVT 120
DAEAHIVTVT PHSVKVDEYT GEWIDPHFIG GRCKGQICET VHNSTKWFTS SDGESVCSQL 180
FTLVGGTFPS DSEETSMGL PETGIRSNYF PYVSTEGICK MPFCRKPGYK LKNDLWFQIT 240
DPDLDKTVRD LPHIKDCDLS SSIIVTPGEHA TDISLISDVE RILDYALCQN TWSKIEAGEP 300
ITPVDLSYLG PKNPGAGPVF TIINGSLHYF MSKYLRVELE SPVIPRMEGK VAGTRIVGQL 360

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WDQWFFPGEV EIGPNGVLKT KQGYKFPLHI IGTGEVDNDI KMERIVKHWE HPHIEAAQTF 420
LKDDTEEV L YYGDTGVSKN PVELVEGWFS GWRSSIMGVL AVIIGFVILI FLIRLIGVLS 480
SLFRQKRRPI YKSDVEMAHF R 501

```

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SEQ ID NO: 218      moltype = AA  length = 501
FEATURE            Location/Qualifiers
source              1..501
                    mol_type = protein
                    organism = Vesicular stomatitis virus

```

```

SEQUENCE: 218
KIEIVFPQHT TGDWKRVPHE YNYCPTSADK NSHGTQTGIP VELTMPFGLT THQVDGFMCH 60
SALWMTTCDF RYWGPKYITH SIHNEEPTDY QCLEAIKAYK DGVSFNPGFP PQSCGYGTVT 120
DAEAHIVTVT PHSVKVDEYT GEWIDPHFIG GRCKGQICET VHNSTKWFTS SDGESVCSQL 180
FTLVGGTFFS DSEETSMGL PETGIRSNYF PYVSTEGICK MPFCRKPGYK LKNDLWFQIT 240
DPDLDKTVRD LPHIKDCDLS SSIIVTPGEHA TDISLISDVE RILDYALCQN TWSKIEAGEP 300
ITPVDLSYLG PKNPGAGPVF TIINGSLHYF MSKYLRVELE SPVIPRMEGK VAGTRIVQQL 360
WDQWFFPGEV EIGPNGVLKT KQGYKFPLHI IGTGEVDNDI KMERIVKHWE HPHIEAAQTF 420
LKDDTEEV L YYGDTGVSKN PVELVEGWFS GWRSSIMGVL AVIIGFVILI FLIRLIGVLS 480
SLFRQKRRPI YKSDVEMAHF R 501

```

```

SEQ ID NO: 219      moltype = AA  length = 501
FEATURE            Location/Qualifiers
source              1..501
                    mol_type = protein
                    organism = Vesicular stomatitis virus

```

```

SEQUENCE: 219
KIEIVFPQHT TGDWKRVPHE YNYCPTSADK NSHGTQTGIP VELTMPGGLT THQVDGFMCH 60
SALWMTTCDF RYWGPKYITH SIHNEEPTDY QCLEAIKAYK DGVSFNPGFP PQSCGYGTVT 120
DAEAHIVTVT PHSVKVDEYT GEWIDPHFIG GRCKGQICET VHNSTKWFTS SDGESVCSQL 180
FTLVGGTFFS DSEETSMGL PETGIRSNYF PYVSTEGICK MPFCRKPGYK LKNDLWFQIT 240
DPDLDKTVRD LPHIKDCDLS SSIIVTPGEHA TDISLISDVE RILDYALCQN TWSKIEAGEP 300
ITPVDLSYLG PKNPGAGPVF TIINGSLHYF MSKYLRVELE SPVIPRMEGK VAGTRIVQQL 360
WDQWFFPGEV EIGPNGVLKT KQGYKFPLHI IGTGEVDNDI KMERIVKHWE HPHIEAAQTF 420
LKDDTEEV L YYGDTGVSKN PVELVEGWFS GWRSSIMGVL AVIIGFVILI FLIRLIGVLS 480
SLFRQKRRPI YKSDVEMAHF R 501

```

```

SEQ ID NO: 220      moltype = AA  length = 501
FEATURE            Location/Qualifiers
source              1..501
                    mol_type = protein
                    organism = Vesicular stomatitis virus

```

```

SEQUENCE: 220
KIEIVFPQHT TGDWKRVPHE YNYCPTSADK NSHGTQTGIP VELTMPGGLT THQVDGFMCH 60
SALWMTTCDF RYWGPKYITH SIHNEEPTDY QCLEAIKAYK DGVSFNPGFP PQSCGYGTVT 120
DAEAHIVTVT PHSVKVDEYT GEWIDPHFIG GRCKGQICET VHNSTKWFTS SDGESVCSQL 180
FTLVGGTFFS DSEETSMGL PETGIRSNYF PYVSTEGICK MPFCRKPGYK LKNDLWFQIT 240
DPDLDKTVRD LPHIKDCDLS SSIIVTPGEHA TDISLISDVE RILDYALCQN TWSKIEAGEP 300
ITPVDLSYLG PKNPGAGPVF TIINGSLHYF MSKYLRVELE SPVIPRMEGK VAGTRIVQQL 360
WDQWFFPGEV EIGPNGVLKT KQGYKFPLHI IGTGEVDNDI KMERIVKHWE HPHIEAAQTF 420
LKDDTEEV L YYGDTGVSKN PVELVEGWFS GWRSSIMGVL AVIIGFVILI FLIRLIGVLS 480
SLFRQKRRPI YKSDVEMAHF R 501

```

```

SEQ ID NO: 221      moltype = AA  length = 501
FEATURE            Location/Qualifiers
source              1..501
                    mol_type = protein
                    organism = Vesicular stomatitis virus

```

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SEQUENCE: 221
KIEIVFPQHT TGDWKRVPHE YNYCPTSADK NSHGTQTGIP VELTMPGGLT THQVDGFMCH 60
SALWMTTCDF RYWGPKYITH SIHNEEPTDY QCLEAIKAYK DGVSFNPGFP PQSCGYGTVT 120
DAEAHIVTVT PHSVKVDEYT GEWIDPHFIG GRCKGQICET VHNSTKWFTS SDGESVCSQL 180
FTLVGGTFFS DSEETSMGL PETGIRSNYF PYVSTEGICK MPFCRKPGYK LKNDLWFQIT 240
DPDLDKTVRD LPHIKDCDLS SSIIVTPGEHA TDISLISDVE RILDYALCQN TWSKIEAGEP 300
ITPVDLSYLG PKNPGAGPVF TIINGSLHYF MSKYLRVELE SPVIPRMEGK VAGTRIVQQL 360
WDQWFFPGEV EIGPNGVLKT KQGYKFPLHI IGTGEVDNDI KMERIVKHWE HPHIEAAQTF 420
LKDDTEEV L YYGDTGVSKN PVELVEGWFS GWRSSIMGVL AVIIGFVILI FLIRLIGVLS 480
SLFRQKRRPI YKSDVEMAHF R 501

```

```

SEQ ID NO: 222      moltype = AA  length = 501
FEATURE            Location/Qualifiers
source              1..501
                    mol_type = protein
                    organism = Vesicular stomatitis virus

```

```

SEQUENCE: 222
KIEIVFPQHT TGDWKRVPHE YNYCPTSADK NSHGTQTGIP VELTMPGGLT THQVDGFMCH 60
SALWMTTCDF RYWGPKYITH SIHNEEPTDY QCLEAIKAYK DGVSFNPGFP PQSCGYGTVT 120
DAEAHIVTVT PHSVKVDEYT GEWIDPHFIG GRCKGQICET VHNSTKWFTS SDGESVCSQL 180
FTLVGGTFFS DSEETSMGL PETGIRSNYF PYVSTEGICK MPFCRKPGYK LKNDLWFQIT 240
DPDLDKTVRD LPHIKDCDLS SSIIVTPGEHA TDISLISDVE RILDYALCQN TWSKIEAGEP 300

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ITPVDLSYLG PKNPGAGPVF TIINGSLHYF MSKYLRLVELE SPVIPRMEGK VAGTRIVQQL 360
WDQWFFPGEV EIGPNGVLKT KQGYKFPLHI IGTGEVDNDI KMERIVKHWE HPHIEAAQTF 420
LKDDTDEEVL YYGDTGVSKN PVELVEGWFS GWRSSIMGVL AVIIGFVILI FLIRLIGVLS 480
SLFRQKRRPI YKSDVEMAHF R 501
```

```
SEQ ID NO: 223      moltype = AA length = 501
FEATURE            Location/Qualifiers
source              1..501
                    mol_type = protein
                    organism = Vesicular stomatitis virus
```

```
SEQUENCE: 223
KIEIVFPQHT TGDWKRVPHE YNYCPTSADK NSHGTQTGIP VELTMPQGLT THQVDGFMCH 60
SALWMTTCDF RWYGPKYITH SIHNEEPTDY QCLEAIKAYK DGVSFNPGEF PQSCGYGTVT 120
DAEAHIVTVT PHSVKVDEYT GEWIDPHFIG GRCKGQICET VHNSTKWFTS SDGESVCSQL 180
FTLVGGTFFS DSEETSMGL PETGIRSNYF PYVSTEGICK MPFCRKPGYK LKNDLWFQIT 240
DPDLDKTVRD LPHIKDCDLS SSVTPGEHA TDISLISDVE RILDYALCQN TWSKIEAGEP 300
ITPVDLSYLG PKNPGAGPVF TIINGSLHYF MSKYLRLVELE SPVIPRMEGK VAGTRIVQQL 360
WDQWFFPGEV EIGPNGVLKT KQGYKFPLHI IGTGEVDNDI KMERIVKHWE HPHIEAAQTF 420
LKDDTDEEVL YYGDTGVSKN PVELVEGWFS GWRSSIMGVL AVIIGFVILI FLIRLIGVLS 480
SLFRQKRRPI YKSDVEMAHF R 501
```

```
SEQ ID NO: 224      moltype = AA length = 501
FEATURE            Location/Qualifiers
source              1..501
                    mol_type = protein
                    organism = Vesicular stomatitis virus
```

```
SEQUENCE: 224
KIEIVFPQHT TGDWKRVPHE YNYCPTSADK NSHGTQTGIP VELTMPQGLT THQVDGFMCH 60
SALWMTTCDF RWYGPKYITH SIHNEEPTDY QCLEAIKAYK DGVSFNPGEF PQSCGYGTVT 120
DAEAHIVTVT PHSVKVDEYT GEWIDPHFIG GRCKGQICET VHNSTKWFTS SDGESVCSQL 180
FTLVGGTFFS DSEETSMGL PETGIRSNYF PYVSTEGICK MPFCRKPGYK LKNDLWFQIT 240
DPDLDKTVRD LPHIKDCDLS SSVTPGEHA TDISLISDVE RILDYALCQN TWSKIEAGEP 300
ITPVDLSYLG PKNPGAGPVF TIINGSLHYF MSKYLRLVELE SPVIPRMEGK VAGTRIVQQL 360
WDQWFFPGEV EIGPNGVLKT KQGYKFPLHI IGTGEVDNDI KMERIVKHWE HPHIEAAQTF 420
LKDDTDEEVL YYGDTGVSKN PVELVEGWFS GWRSSIMGVL AVIIGFVILI FLIRLIGVLS 480
SLFRQKRRPI YKSDVEMAHF R 501
```

```
SEQ ID NO: 225      moltype = AA length = 501
FEATURE            Location/Qualifiers
source              1..501
                    mol_type = protein
                    organism = Vesicular stomatitis virus
```

```
SEQUENCE: 225
KIEIVFPQHT TGDWKRVPHE YNYCPTSADK NSHGTQTGIP VELTMPQGLT THQVDGFMCH 60
SALWMTTCDF RWYGPKYITH SIHNEEPTDY QCLEAIKAYK DGVSFNPGEF PQSCGYGTVT 120
DAEAHIVTVT PHSVKVDEYT GEWIDPHFIG GRCKGQICET VHNSTKWFTS SDGESVCSQL 180
FTLVGGTFFS DSEETSMGL PETGIRSNYF PYVSTEGICK MPFCRKPGYK LKNDLWFQIT 240
DPDLDKTVRD LPHIKDCDLS SSVTPGEHA TDISLISDVE RILDYALCQN TWSKIEAGEP 300
ITPVDLSYLG PKNPGAGPVF TIINGSLHYF MSKYLRLVELE SPVIPRMEGK VAGTRIVQQL 360
WDQWFFPGEV EIGPNGVLKT KQGYKFPLHI IGTGEVDNDI KMERIVKHWE HPHIEAAQTF 420
LKDDTDEEVL YYGDTGVSKN PVELVEGWFS GWRSSIMGVL AVIIGFVILI FLIRLIGVLS 480
SLFRQKRRPI YKSDVEMAHF R 501
```

```
SEQ ID NO: 226      moltype = AA length = 501
FEATURE            Location/Qualifiers
source              1..501
                    mol_type = protein
                    organism = Vesicular stomatitis virus
```

```
SEQUENCE: 226
KIEIVFPQHT TGDWKRVPHE YNYCPTSADK NSHGTQTGIP VELTMPQGLT THQVDGFMCH 60
SALWMTTCDF RWYGPKYITH SIHNEEPTDY QCLEAIKAYK DGVSFNPGEF PQSCGYGTVT 120
DAEAHIVTVT PHSVKVDEYT GEWIDPHFIG GRCKGQICET VHNSTKWFTS SDGESVCSQL 180
FTLVGGTFFS DSEETSMGL PETGIRSNYF PYVSTEGICK MPFCRKPGYK LKNDLWFQIT 240
DPDLDKTVRD LPHIKDCDLS SSVTPGEHA TDISLISDVE RILDYALCQN TWSKIEAGEP 300
ITPVDLSYLG PKNPGAGPVF TIINGSLHYF MSKYLRLVELE SPVIPRMEGK VAGTRIVQQL 360
WDQWFFPGEV EIGPNGVLKT KQGYKFPLHI IGTGEVDNDI KMERIVKHWE HPHIEAAQTF 420
LKDDTDEEVL YYGDTGVSKN PVELVEGWFS GWRSSIMGVL AVIIGFVILI FLIRLIGVLS 480
SLFRQKRRPI YKSDVEMAHF R 501
```

```
SEQ ID NO: 227      moltype = AA length = 502
FEATURE            Location/Qualifiers
source              1..502
                    mol_type = protein
                    organism = Vesicular stomatitis virus
```

```
SEQUENCE: 227
KITISFPQSL KGDWRPVPKG YNYCPTSADK NLHGDLDIDG LRLRAPASFK GISADGWMCH 60
AARWITTCDF RWYGPKYITH SIHSFRPSND QCKEAIRLTN EGNWINPGFP PQSCGYASVT 120
DSESVVTVT KHVVLVDEYS GSWIDSQFPG GSCTSPICDT VHNSTLWHAD HTLDSICDQE 180
FVAMDAVLFT ESGKFEFEGK PNSGIRSNYF PYESLKDVQC MDFCKRKGFK LPSGVWFEIE 240
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DAEKSHKAQV	ELKIKRCPHG	AVISAPNQNA	ADINLIMDVE	RILDYSLCQA	TWSKIQNKEA	300
LTPIDISYLG	PKNPGPGPAF	TIINGTLHYF	NTRYIRVDIA	GPVTKEITGF	VSGTSTSRVL	360
WDQWFPYGEN	SIGPNGLLKT	ASGYKYPLFM	VTGVLDADI	HKLGEATVIE	HPHAKEAQKV	420
VDDSEVIFPG	DTGVSKNPVE	VVEGWFSGWR	SSLMSIFGII	LLIVCLVLIV	RILIALKYCC	480
VRHKKRTIYK	EDLEMGRIPR	RA				502

SEQ ID NO: 228 moltype = AA length = 502
 FEATURE Location/Qualifiers
 source 1..502
 mol_type = protein
 organism = Vesicular stomatitis virus

SEQUENCE: 228

KITISFPQSL	KGDWRPVPKG	YNYCPTSADK	NLHGDLDIDG	LRLRAPFSFK	GISADGWMCH	60
AARWITTCDF	RWYGPKYITH	SIHSFRPSND	QCKEAIRLTN	EGNWINPGFP	PQSCGYASVT	120
DSESVVTVT	KHQVLVDEYS	GSWIDSQFPG	GSCTSPICDT	VHNSTLWHAD	HTLDSICDQE	180
FVAMDAVLFT	ESGKFEFEGK	PNSGIRSNYF	PYESLKDVQC	MDPCKRKGFK	LPSGVWFEIE	240
DAEKSHKAQV	ELKIKRCPHG	AVISAPNQNA	ADINLIMDVE	RILDYSLCQA	TWSKIQNKEA	300
LTPIDISYLG	PKNPGPGPAF	TIINGTLHYF	NTRYIRVDIA	GPVTKEITGF	VSGTSTSRVL	360
WDQWFPYGEN	SIGPNGLLKT	ASGYKYPLFM	VTGVLDADI	HKLGEATVIE	HPHAKEAQKV	420
VDDSEVIFPG	DTGVSKNPVE	VVEGWFSGWR	SSLMSIFGII	LLIVCLVLIV	RILIALKYCC	480
VRHKKRTIYK	EDLEMGRIPR	RA				502

SEQ ID NO: 229 moltype = AA length = 502
 FEATURE Location/Qualifiers
 source 1..502
 mol_type = protein
 organism = Vesicular stomatitis virus

SEQUENCE: 229

KITISFPQSL	KGDWRPVPKG	YNYCPTSADK	NLHGDLDIDG	LRLRAPGSFK	GISADGWMCH	60
AARWITTCDF	RWYGPKYITH	SIHSFRPSND	QCKEAIRLTN	EGNWINPGFP	PQSCGYASVT	120
DSESVVTVT	KHQVLVDEYS	GSWIDSQFPG	GSCTSPICDT	VHNSTLWHAD	HTLDSICDQE	180
FVAMDAVLFT	ESGKFEFEGK	PNSGIRSNYF	PYESLKDVQC	MDPCKRKGFK	LPSGVWFEIE	240
DAEKSHKAQV	ELKIKRCPHG	AVISAPNQNA	ADINLIMDVE	RILDYSLCQA	TWSKIQNKEA	300
LTPIDISYLG	PKNPGPGPAF	TIINGTLHYF	NTRYIRVDIA	GPVTKEITGF	VSGTSTSRVL	360
WDQWFPYGEN	SIGPNGLLKT	ASGYKYPLFM	VTGVLDADI	HKLGEATVIE	HPHAKEAQKV	420
VDDSEVIFPG	DTGVSKNPVE	VVEGWFSGWR	SSLMSIFGII	LLIVCLVLIV	RILIALKYCC	480
VRHKKRTIYK	EDLEMGRIPR	RA				502

SEQ ID NO: 230 moltype = AA length = 502
 FEATURE Location/Qualifiers
 source 1..502
 mol_type = protein
 organism = Vesicular stomatitis virus

SEQUENCE: 230

KITISFPQSL	KGDWRPVPKG	YNYCPTSADK	NLHGDLDIDG	LRLRAPQSFK	GISADGWMCH	60
AARWITTCDF	RWYGPKYITH	SIHSFRPSND	QCKEAIRLTN	EGNWINPGFP	PQSCGYASVT	120
DSESVVTVT	KHQVLVDEYS	GSWIDSQFPG	GSCTSPICDT	VHNSTLWHAD	HTLDSICDQE	180
FVAMDAVLFT	ESGKFEFEGK	PNSGIRSNYF	PYESLKDVQC	MDPCKRKGFK	LPSGVWFEIE	240
DAEKSHKAQV	ELKIKRCPHG	AVISAPNQNA	ADINLIMDVE	RILDYSLCQA	TWSKIQNKEA	300
LTPIDISYLG	PKNPGPGPAF	TIINGTLHYF	NTRYIRVDIA	GPVTKEITGF	VSGTSTSRVL	360
WDQWFPYGEN	SIGPNGLLKT	ASGYKYPLFM	VTGVLDADI	HKLGEATVIE	HPHAKEAQKV	420
VDDSEVIFPG	DTGVSKNPVE	VVEGWFSGWR	SSLMSIFGII	LLIVCLVLIV	RILIALKYCC	480
VRHKKRTIYK	EDLEMGRIPR	RA				502

SEQ ID NO: 231 moltype = AA length = 502
 FEATURE Location/Qualifiers
 source 1..502
 mol_type = protein
 organism = Vesicular stomatitis virus

SEQUENCE: 231

KITISFPQSL	KGDWRPVPKG	YNYCPTSADK	NLHGDLDIDG	LRLRAPKSFK	GISADGWMCH	60
AARWITTCDF	RWYGPKYITH	SIHSFRPSND	QCKEAIRLTN	EGNWINPGFP	PQSCGYASVT	120
DSESVVTVT	KHQVLVDEYS	GSWIDSQFPG	GSCTSPICDT	VHNSTLWHAD	HTLDSICDQE	180
FVAMDAVLFT	ESGKFEFEGK	PNSGIRSNYF	PYESLKDVQC	MDPCKRKGFK	LPSGVWFEIE	240
DAEKSHKAQV	ELKIKRCPHG	AVISAPNQNA	ADINLIMDVE	RILDYSLCQA	TWSKIQNKEA	300
LTPIDISYLG	PKNPGPGPAF	TIINGTLHYF	NTRYIRVDIA	GPVTKEITGF	VSGTSTSAVL	360
WDQWFPYGEN	SIGPNGLLKT	ASGYKYPLFM	VTGVLDADI	HKLGEATVIE	HPHAKEAQKV	420
VDDSEVIFPG	DTGVSKNPVE	VVEGWFSGWR	SSLMSIFGII	LLIVCLVLIV	RILIALKYCC	480
VRHKKRTIYK	EDLEMGRIPR	RA				502

SEQ ID NO: 232 moltype = AA length = 502
 FEATURE Location/Qualifiers
 source 1..502
 mol_type = protein
 organism = Vesicular stomatitis virus

SEQUENCE: 232

KITISFPQSL	KGDWRPVPKG	YNYCPTSADK	NLHGDLDIDG	LRLRAPKSFK	GISADGWMCH	60
AARWITTCDF	RWYGPKYITH	SIHSFRPSND	QCKEAIRLTN	EGNWINPGFP	PQSCGYASVT	120
DSESVVTVT	KHQVLVDEYS	GSWIDSQFPG	GSCTSPICDT	VHNSTLWHAD	HTLDSICDQE	180

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FVAMDAVLFT ESGKFEEFGK PNSGIRSNYF PYESLKDVCQ MDFCKRKGFK LPSGVWFEIE 240
DAEKSHKAQV ELKIKRCPHG AVISAPNQNA ADINLIMDVE RILDYSLCQA TWSKIQNKEA 300
LTPIDISYLG PKNPGPGPAF TIINGTLHYF NTRYIRVDIA GPVTKEITGF VSGTSTSFLV 360
WDQWFPYGEN SIGPGLLKT ASGYKYPLFM VGTGVLDAI HKLGEATVIE HPHAKEAQKV 420
VDDSEVIFFG DTGVSKNPVE VVEGWFSGWR SSLMSIFGII LLIVCLVLIV RILIALKYCC 480
VRHKKRTIYK EDLEMGRIPR RA 502

```

```

SEQ ID NO: 233      moltype = AA length = 502
FEATURE            Location/Qualifiers
source             1..502
                   mol_type = protein
                   organism = Vesicular stomatitis virus

```

```

SEQUENCE: 233
KITISFPQSL KGDWRPVPKG YNYCPTSADK NLHGLDIDIG LRLRAPKSFK GISADGWMCH 60
AARWITTCDF RWYGPYIITH SIHSFRPSND QCKEAIRLTN EGNWINPGFP PQSCGYASVT 120
DSESVVTVT KHQVLVDEYS GSWIDSQFPG GSCTSPICDT VHNSTLWHAD HTLDSICDQE 180
FVAMDAVLFT ESGKFEEFGK PNSGIRSNYF PYESLKDVCQ MDFCKRKGFK LPSGVWFEIE 240
DAEKSHKAQV ELKIKRCPHG AVISAPNQNA ADINLIMDVE RILDYSLCQA TWSKIQNKEA 300
LTPIDISYLG PKNPGPGPAF TIINGTLHYF NTRYIRVDIA GPVTKEITGF VSGTSTSGVL 360
WDQWFPYGEN SIGPGLLKT ASGYKYPLFM VGTGVLDAI HKLGEATVIE HPHAKEAQKV 420
VDDSEVIFFG DTGVSKNPVE VVEGWFSGWR SSLMSIFGII LLIVCLVLIV RILIALKYCC 480
VRHKKRTIYK EDLEMGRIPR RA 502

```

```

SEQ ID NO: 234      moltype = AA length = 502
FEATURE            Location/Qualifiers
source             1..502
                   mol_type = protein
                   organism = Vesicular stomatitis virus

```

```

SEQUENCE: 234
KITISFPQSL KGDWRPVPKG YNYCPTSADK NLHGLDIDIG LRLRAPKSFK GISADGWMCH 60
AARWITTCDF RWYGPYIITH SIHSFRPSND QCKEAIRLTN EGNWINPGFP PQSCGYASVT 120
DSESVVTVT KHQVLVDEYS GSWIDSQFPG GSCTSPICDT VHNSTLWHAD HTLDSICDQE 180
FVAMDAVLFT ESGKFEEFGK PNSGIRSNYF PYESLKDVCQ MDFCKRKGFK LPSGVWFEIE 240
DAEKSHKAQV ELKIKRCPHG AVISAPNQNA ADINLIMDVE RILDYSLCQA TWSKIQNKEA 300
LTPIDISYLG PKNPGPGPAF TIINGTLHYF NTRYIRVDIA GPVTKEITGF VSGTSTSQVL 360
WDQWFPYGEN SIGPGLLKT ASGYKYPLFM VGTGVLDAI HKLGEATVIE HPHAKEAQKV 420
VDDSEVIFFG DTGVSKNPVE VVEGWFSGWR SSLMSIFGII LLIVCLVLIV RILIALKYCC 480
VRHKKRTIYK EDLEMGRIPR RA 502

```

```

SEQ ID NO: 235      moltype = AA length = 502
FEATURE            Location/Qualifiers
source             1..502
                   mol_type = protein
                   organism = Vesicular stomatitis virus

```

```

SEQUENCE: 235
KITISFPQSL KGDWRPVPKG YNYCPTSADK NLHGLDIDIG LRLRAPKSFK GISADGWMCH 60
AARWITTCDF RWYGPYIITH SIHSFRPSND QCKEAIRLTN EGNWINPGFP PQSCGYASVT 120
DSESVVTVT KHQVLVDEYS GSWIDSQFPG GSCTSPICDT VHNSTLWHAD HTLDSICDQE 180
FVAMDAVLFT ESGKFEEFGK PNSGIRSNYF PYESLKDVCQ MDFCKRKGFK LPSGVWFEIE 240
DAEKSHKAQV ELKIKRCPHG AVISAPNQNA ADINLIMDVE RILDYSLCQA TWSKIQNKEA 300
LTPIDISYLG PKNPGPGPAF TIINGTLHYF NTRYIRVDIA GPVTKEITGF VSGTSTSQVL 360
WDQWFPYGEN SIGPGLLKT ASGYKYPLFM VGTGVLDAI HKLGEATVIE HPHAKEAQKV 420
VDDSEVIFFG DTGVSKNPVE VVEGWFSGWR SSLMSIFGII LLIVCLVLIV RILIALKYCC 480
VRHKKRTIYK EDLEMGRIPR RA 502

```

```

SEQ ID NO: 236      moltype = AA length = 502
FEATURE            Location/Qualifiers
source             1..502
                   mol_type = protein
                   organism = Vesicular stomatitis virus

```

```

SEQUENCE: 236
KITISFPQSL KGDWRPVPKG YNYCPTSADK NLHGLDIDIG LRLRAPKSFK GISADGWMCH 60
AARWITTCDF RWYGPYIITH SIHSFRPSND QCKEAIRLTN EGNWINPGFP PQSCGYASVT 120
DSESVVTVT KHQVLVDEYS GSWIDSQFPG GSCTSPICDT VHNSTLWHAD HTLDSICDQE 180
FVAMDAVLFT ESGKFEEFGK PNSGIRSNYF PYESLKDVCQ MDFCKRKGFK LPSGVWFEIE 240
DAEKSHKAQV ELKIKRCPHG AVISAPNQNA ADINLIMDVE RILDYSLCQA TWSKIQNKEA 300
LTPIDISYLG PKNPGPGPAF TIINGTLHYF NTRYIRVDIA GPVTKEITGF VSGTSTSFLV 360
WDQWFPYGEN SIGPGLLKT ASGYKYPLFM VGTGVLDAI HKLGEATVIE HPHAKEAQKV 420
VDDSEVIFFG DTGVSKNPVE VVEGWFSGWR SSLMSIFGII LLIVCLVLIV RILIALKYCC 480
VRHKKRTIYK EDLEMGRIPR RA 502

```

```

SEQ ID NO: 237      moltype = AA length = 502
FEATURE            Location/Qualifiers
source             1..502
                   mol_type = protein
                   organism = Vesicular stomatitis virus

```

```

SEQUENCE: 237
KITISFPQSL KGDWRPVPKG YNYCPTSADK NLHGLDIDIG LRLRAPKSFK GISADGWMCH 60
AARWITTCDF RWYGPYIITH SIHSFRPSND QCKEAIRLTN EGNWINPGFP PQSCGYASVT 120

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DSESVVTVT	KHQVLVDEYS	GSWIDSQFP	GSCTSPICDT	VHNSTLWHAD	HTLDSICDQE	180
FVAMDAVLFT	ESGKFEEFGK	PNSGIRSNYF	PYESLKDVQ	MDPCKRKGF	LPSGVWFEIE	240
DAEKSHKAQV	ELKIKRCPHG	AVISAPNQNA	ADINLIMDVE	RILDYSLCQA	TWSKIQNKEA	300
LTPIDISYLG	PKNPGPGPAF	TIINGTLHYF	NTRYIRVDIA	GPVTKEITGF	VSGTSTSGVL	360
WDQWFPYGEN	SIGPNGLLKT	ASGYKYPLFM	VTGVLDADI	HKLGEATVIE	HPHAKEAQKV	420
VDDSEVIFPG	DTGVSKNPVE	VVEGWFSGWR	SSLMSIFGII	LLIVCLVLIV	RILIALKYCC	480
VRHKKRTIYK	EDLEMGRIPR	RA				502

SEQ ID NO: 238 moltype = AA length = 502
 FEATURE Location/Qualifiers
 source 1..502
 mol_type = protein
 organism = Vesicular stomatitis virus

SEQUENCE: 238

KITISFPQSL	KGDWRPVPKG	YNYCPTSADK	NLHGDLDIG	LRLRAPFSFK	GISADGWMCH	60
AARWITTCDF	RWYGPKYITH	SIHSFRPSND	QCKEAIRLTN	EGNWINPGFP	PQSCGYASVT	120
DSESVVTVT	KHQVLVDEYS	GSWIDSQFP	GSCTSPICDT	VHNSTLWHAD	HTLDSICDQE	180
FVAMDAVLFT	ESGKFEEFGK	PNSGIRSNYF	PYESLKDVQ	MDPCKRKGF	LPSGVWFEIE	240
DAEKSHKAQV	ELKIKRCPHG	AVISAPNQNA	ADINLIMDVE	RILDYSLCQA	TWSKIQNKEA	300
LTPIDISYLG	PKNPGPGPAF	TIINGTLHYF	NTRYIRVDIA	GPVTKEITGF	VSGTSTSGVL	360
WDQWFPYGEN	SIGPNGLLKT	ASGYKYPLFM	VTGVLDADI	HKLGEATVIE	HPHAKEAQKV	420
VDDSEVIFPG	DTGVSKNPVE	VVEGWFSGWR	SSLMSIFGII	LLIVCLVLIV	RILIALKYCC	480
VRHKKRTIYK	EDLEMGRIPR	RA				502

SEQ ID NO: 239 moltype = AA length = 502
 FEATURE Location/Qualifiers
 source 1..502
 mol_type = protein
 organism = Vesicular stomatitis virus

SEQUENCE: 239

KITISFPQSL	KGDWRPVPKG	YNYCPTSADK	NLHGDLDIG	LRLRAPFSFK	GISADGWMCH	60
AARWITTCDF	RWYGPKYITH	SIHSFRPSND	QCKEAIRLTN	EGNWINPGFP	PQSCGYASVT	120
DSESVVTVT	KHQVLVDEYS	GSWIDSQFP	GSCTSPICDT	VHNSTLWHAD	HTLDSICDQE	180
FVAMDAVLFT	ESGKFEEFGK	PNSGIRSNYF	PYESLKDVQ	MDPCKRKGF	LPSGVWFEIE	240
DAEKSHKAQV	ELKIKRCPHG	AVISAPNQNA	ADINLIMDVE	RILDYSLCQA	TWSKIQNKEA	300
LTPIDISYLG	PKNPGPGPAF	TIINGTLHYF	NTRYIRVDIA	GPVTKEITGF	VSGTSTSAVL	360
WDQWFPYGEN	SIGPNGLLKT	ASGYKYPLFM	VTGVLDADI	HKLGEATVIE	HPHAKEAQKV	420
VDDSEVIFPG	DTGVSKNPVE	VVEGWFSGWR	SSLMSIFGII	LLIVCLVLIV	RILIALKYCC	480
VRHKKRTIYK	EDLEMGRIPR	RA				502

SEQ ID NO: 240 moltype = AA length = 502
 FEATURE Location/Qualifiers
 source 1..502
 mol_type = protein
 organism = Vesicular stomatitis virus

SEQUENCE: 240

KITISFPQSL	KGDWRPVPKG	YNYCPTSADK	NLHGDLDIG	LRLRAPFSFK	GISADGWMCH	60
AARWITTCDF	RWYGPKYITH	SIHSFRPSND	QCKEAIRLTN	EGNWINPGFP	PQSCGYASVT	120
DSESVVTVT	KHQVLVDEYS	GSWIDSQFP	GSCTSPICDT	VHNSTLWHAD	HTLDSICDQE	180
FVAMDAVLFT	ESGKFEEFGK	PNSGIRSNYF	PYESLKDVQ	MDPCKRKGF	LPSGVWFEIE	240
DAEKSHKAQV	ELKIKRCPHG	AVISAPNQNA	ADINLIMDVE	RILDYSLCQA	TWSKIQNKEA	300
LTPIDISYLG	PKNPGPGPAF	TIINGTLHYF	NTRYIRVDIA	GPVTKEITGF	VSGTSTSFVL	360
WDQWFPYGEN	SIGPNGLLKT	ASGYKYPLFM	VTGVLDADI	HKLGEATVIE	HPHAKEAQKV	420
VDDSEVIFPG	DTGVSKNPVE	VVEGWFSGWR	SSLMSIFGII	LLIVCLVLIV	RILIALKYCC	480
VRHKKRTIYK	EDLEMGRIPR	RA				502

SEQ ID NO: 241 moltype = AA length = 502
 FEATURE Location/Qualifiers
 source 1..502
 mol_type = protein
 organism = Vesicular stomatitis virus

SEQUENCE: 241

KITISFPQSL	KGDWRPVPKG	YNYCPTSADK	NLHGDLDIG	LRLRAPFSFK	GISADGWMCH	60
AARWITTCDF	RWYGPKYITH	SIHSFRPSND	QCKEAIRLTN	EGNWINPGFP	PQSCGYASVT	120
DSESVVTVT	KHQVLVDEYS	GSWIDSQFP	GSCTSPICDT	VHNSTLWHAD	HTLDSICDQE	180
FVAMDAVLFT	ESGKFEEFGK	PNSGIRSNYF	PYESLKDVQ	MDPCKRKGF	LPSGVWFEIE	240
DAEKSHKAQV	ELKIKRCPHG	AVISAPNQNA	ADINLIMDVE	RILDYSLCQA	TWSKIQNKEA	300
LTPIDISYLG	PKNPGPGPAF	TIINGTLHYF	NTRYIRVDIA	GPVTKEITGF	VSGTSTSGVL	360
WDQWFPYGEN	SIGPNGLLKT	ASGYKYPLFM	VTGVLDADI	HKLGEATVIE	HPHAKEAQKV	420
VDDSEVIFPG	DTGVSKNPVE	VVEGWFSGWR	SSLMSIFGII	LLIVCLVLIV	RILIALKYCC	480
VRHKKRTIYK	EDLEMGRIPR	RA				502

SEQ ID NO: 242 moltype = AA length = 502
 FEATURE Location/Qualifiers
 source 1..502
 mol_type = protein
 organism = Vesicular stomatitis virus

SEQUENCE: 242

KITISFPQSL	KGDWRPVPKG	YNYCPTSADK	NLHGDLDIG	LRLRAPFSFK	GISADGWMCH	60
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AARWITTCDF RWYGPKYITH SIHSFRPSND QCKEAIRLTN EGNWINPGFP PQSCGYASVT 120
DSESVVTVT KHQVLVDEYS GSWIDSQFPF GSCTSPICDT VHNSTLWHAD HTLDSICDQE 180
FVAMDAVLFT ESGKFEEFGK PNSGIRSNYF PYESLKDVCQ MDFCKRKGFK LPSGVWFEIE 240
DAEKSHKAQV ELKIKRCPHG AVISAPNQNA ADINLIMDVE RILDYSLCQA TWSKIQNKEA 300
LTPIDISYLG PKNPGPGPAF TIINGTLHYF NTRYIRVDIA GPVTKEITGF VSGTSTSQVL 360
WDQWFPYGEN SIGPNGLLKT ASGYKYPLFM VGTGVLDAI HKLGEATVIE HPHAKEAQKV 420
VDDSEVIFFG DTGVSKNPVE VVEGWFSQWR SSLMSIFGII LLIVCLVLIV RILIALKYCC 480
VRHKKRTIYK EDLEMGRIPR RA 502

```

```

SEQ ID NO: 243      moltype = AA length = 502
FEATURE            Location/Qualifiers
source              1..502
                    mol_type = protein
                    organism = Vesicular stomatitis virus

```

```

SEQUENCE: 243
KITISFPQSL KGDWRPVPKG YNYCPTSADK NLHGLDIDIG LRLRAPGSFK GISADGWMCH 60
AARWITTCDF RWYGPKYITH SIHSFRPSND QCKEAIRLTN EGNWINPGFP PQSCGYASVT 120
DSESVVTVT KHQVLVDEYS GSWIDSQFPF GSCTSPICDT VHNSTLWHAD HTLDSICDQE 180
FVAMDAVLFT ESGKFEEFGK PNSGIRSNYF PYESLKDVCQ MDFCKRKGFK LPSGVWFEIE 240
DAEKSHKAQV ELKIKRCPHG AVISAPNQNA ADINLIMDVE RILDYSLCQA TWSKIQNKEA 300
LTPIDISYLG PKNPGPGPAF TIINGTLHYF NTRYIRVDIA GPVTKEITGF VSGTSTSQVL 360
WDQWFPYGEN SIGPNGLLKT ASGYKYPLFM VGTGVLDAI HKLGEATVIE HPHAKEAQKV 420
VDDSEVIFFG DTGVSKNPVE VVEGWFSQWR SSLMSIFGII LLIVCLVLIV RILIALKYCC 480
VRHKKRTIYK EDLEMGRIPR RA 502

```

```

SEQ ID NO: 244      moltype = AA length = 502
FEATURE            Location/Qualifiers
source              1..502
                    mol_type = protein
                    organism = Vesicular stomatitis virus

```

```

SEQUENCE: 244
KITISFPQSL KGDWRPVPKG YNYCPTSADK NLHGLDIDIG LRLRAPGSFK GISADGWMCH 60
AARWITTCDF RWYGPKYITH SIHSFRPSND QCKEAIRLTN EGNWINPGFP PQSCGYASVT 120
DSESVVTVT KHQVLVDEYS GSWIDSQFPF GSCTSPICDT VHNSTLWHAD HTLDSICDQE 180
FVAMDAVLFT ESGKFEEFGK PNSGIRSNYF PYESLKDVCQ MDFCKRKGFK LPSGVWFEIE 240
DAEKSHKAQV ELKIKRCPHG AVISAPNQNA ADINLIMDVE RILDYSLCQA TWSKIQNKEA 300
LTPIDISYLG PKNPGPGPAF TIINGTLHYF NTRYIRVDIA GPVTKEITGF VSGTSTSQVL 360
WDQWFPYGEN SIGPNGLLKT ASGYKYPLFM VGTGVLDAI HKLGEATVIE HPHAKEAQKV 420
VDDSEVIFFG DTGVSKNPVE VVEGWFSQWR SSLMSIFGII LLIVCLVLIV RILIALKYCC 480
VRHKKRTIYK EDLEMGRIPR RA 502

```

```

SEQ ID NO: 245      moltype = AA length = 502
FEATURE            Location/Qualifiers
source              1..502
                    mol_type = protein
                    organism = Vesicular stomatitis virus

```

```

SEQUENCE: 245
KITISFPQSL KGDWRPVPKG YNYCPTSADK NLHGLDIDIG LRLRAPGSFK GISADGWMCH 60
AARWITTCDF RWYGPKYITH SIHSFRPSND QCKEAIRLTN EGNWINPGFP PQSCGYASVT 120
DSESVVTVT KHQVLVDEYS GSWIDSQFPF GSCTSPICDT VHNSTLWHAD HTLDSICDQE 180
FVAMDAVLFT ESGKFEEFGK PNSGIRSNYF PYESLKDVCQ MDFCKRKGFK LPSGVWFEIE 240
DAEKSHKAQV ELKIKRCPHG AVISAPNQNA ADINLIMDVE RILDYSLCQA TWSKIQNKEA 300
LTPIDISYLG PKNPGPGPAF TIINGTLHYF NTRYIRVDIA GPVTKEITGF VSGTSTSQVL 360
WDQWFPYGEN SIGPNGLLKT ASGYKYPLFM VGTGVLDAI HKLGEATVIE HPHAKEAQKV 420
VDDSEVIFFG DTGVSKNPVE VVEGWFSQWR SSLMSIFGII LLIVCLVLIV RILIALKYCC 480
VRHKKRTIYK EDLEMGRIPR RA 502

```

```

SEQ ID NO: 246      moltype = AA length = 502
FEATURE            Location/Qualifiers
source              1..502
                    mol_type = protein
                    organism = Vesicular stomatitis virus

```

```

SEQUENCE: 246
KITISFPQSL KGDWRPVPKG YNYCPTSADK NLHGLDIDIG LRLRAPGSFK GISADGWMCH 60
AARWITTCDF RWYGPKYITH SIHSFRPSND QCKEAIRLTN EGNWINPGFP PQSCGYASVT 120
DSESVVTVT KHQVLVDEYS GSWIDSQFPF GSCTSPICDT VHNSTLWHAD HTLDSICDQE 180
FVAMDAVLFT ESGKFEEFGK PNSGIRSNYF PYESLKDVCQ MDFCKRKGFK LPSGVWFEIE 240
DAEKSHKAQV ELKIKRCPHG AVISAPNQNA ADINLIMDVE RILDYSLCQA TWSKIQNKEA 300
LTPIDISYLG PKNPGPGPAF TIINGTLHYF NTRYIRVDIA GPVTKEITGF VSGTSTSQVL 360
WDQWFPYGEN SIGPNGLLKT ASGYKYPLFM VGTGVLDAI HKLGEATVIE HPHAKEAQKV 420
VDDSEVIFFG DTGVSKNPVE VVEGWFSQWR SSLMSIFGII LLIVCLVLIV RILIALKYCC 480
VRHKKRTIYK EDLEMGRIPR RA 502

```

```

SEQ ID NO: 247      moltype = AA length = 502
FEATURE            Location/Qualifiers
source              1..502
                    mol_type = protein
                    organism = Vesicular stomatitis virus

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```

SEQUENCE: 247

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-continued

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KITISFPQSL KGDWRPVPKG YNYCPTSADK NLHGDLIDIG LRLRAPQSFK GISADGWMCH 60
AARWITTCDF RYWGPKYITH SIHSFRPSND QCKEAIRLTN EGNWINPGFP PQSCGYASVT 120
DSESVVTVT KHQVLVDEYS GSWIDSQFPG GSCTSPICDT VHNSTLWHAD HTLDSICDQE 180
FVAMDAVLFT ESGKFEFEGK PNSGIRSNYF PYESLKDVCQ MDFCKRKGFK LPSGVWFEIE 240
DAEKSHKAQV ELKIKRCPHG AVISAPNQNA ADINLIMDVE RILDYSLCQA TWSKIQNKEA 300
LTPIDISYLG PKNPGPGPAF TIINGTLHYF NTRYIRVDIA GPVTKEITGF VSGTSTSAVL 360
WDQWFPYGEN SIGPNGLLKT ASGYKYPLFM VGTGVLDAI HKLGEATVIE HPHAKEAQKV 420
VDDSEVIFPG DTGVSKNPVE VVEGWFSGWR SSLMSIFGII LLIVCLVLIV RILIALKYCC 480
VRHKKRTIYK EDLEMGRIPR RA 502

```

```

SEQ ID NO: 248      moltype = AA length = 502
FEATURE            Location/Qualifiers
source              1..502
                    mol_type = protein
                    organism = Vesicular stomatitis virus

```

```

SEQUENCE: 248
KITISFPQSL KGDWRPVPKG YNYCPTSADK NLHGDLIDIG LRLRAPQSFK GISADGWMCH 60
AARWITTCDF RYWGPKYITH SIHSFRPSND QCKEAIRLTN EGNWINPGFP PQSCGYASVT 120
DSESVVTVT KHQVLVDEYS GSWIDSQFPG GSCTSPICDT VHNSTLWHAD HTLDSICDQE 180
FVAMDAVLFT ESGKFEFEGK PNSGIRSNYF PYESLKDVCQ MDFCKRKGFK LPSGVWFEIE 240
DAEKSHKAQV ELKIKRCPHG AVISAPNQNA ADINLIMDVE RILDYSLCQA TWSKIQNKEA 300
LTPIDISYLG PKNPGPGPAF TIINGTLHYF NTRYIRVDIA GPVTKEITGF VSGTSTSAVL 360
WDQWFPYGEN SIGPNGLLKT ASGYKYPLFM VGTGVLDAI HKLGEATVIE HPHAKEAQKV 420
VDDSEVIFPG DTGVSKNPVE VVEGWFSGWR SSLMSIFGII LLIVCLVLIV RILIALKYCC 480
VRHKKRTIYK EDLEMGRIPR RA 502

```

```

SEQ ID NO: 249      moltype = AA length = 502
FEATURE            Location/Qualifiers
source              1..502
                    mol_type = protein
                    organism = Vesicular stomatitis virus

```

```

SEQUENCE: 249
KITISFPQSL KGDWRPVPKG YNYCPTSADK NLHGDLIDIG LRLRAPQSFK GISADGWMCH 60
AARWITTCDF RYWGPKYITH SIHSFRPSND QCKEAIRLTN EGNWINPGFP PQSCGYASVT 120
DSESVVTVT KHQVLVDEYS GSWIDSQFPG GSCTSPICDT VHNSTLWHAD HTLDSICDQE 180
FVAMDAVLFT ESGKFEFEGK PNSGIRSNYF PYESLKDVCQ MDFCKRKGFK LPSGVWFEIE 240
DAEKSHKAQV ELKIKRCPHG AVISAPNQNA ADINLIMDVE RILDYSLCQA TWSKIQNKEA 300
LTPIDISYLG PKNPGPGPAF TIINGTLHYF NTRYIRVDIA GPVTKEITGF VSGTSTSGVL 360
WDQWFPYGEN SIGPNGLLKT ASGYKYPLFM VGTGVLDAI HKLGEATVIE HPHAKEAQKV 420
VDDSEVIFPG DTGVSKNPVE VVEGWFSGWR SSLMSIFGII LLIVCLVLIV RILIALKYCC 480
VRHKKRTIYK EDLEMGRIPR RA 502

```

```

SEQ ID NO: 250      moltype = AA length = 502
FEATURE            Location/Qualifiers
source              1..502
                    mol_type = protein
                    organism = Vesicular stomatitis virus

```

```

SEQUENCE: 250
KITISFPQSL KGDWRPVPKG YNYCPTSADK NLHGDLIDIG LRLRAPQSFK GISADGWMCH 60
AARWITTCDF RYWGPKYITH SIHSFRPSND QCKEAIRLTN EGNWINPGFP PQSCGYASVT 120
DSESVVTVT KHQVLVDEYS GSWIDSQFPG GSCTSPICDT VHNSTLWHAD HTLDSICDQE 180
FVAMDAVLFT ESGKFEFEGK PNSGIRSNYF PYESLKDVCQ MDFCKRKGFK LPSGVWFEIE 240
DAEKSHKAQV ELKIKRCPHG AVISAPNQNA ADINLIMDVE RILDYSLCQA TWSKIQNKEA 300
LTPIDISYLG PKNPGPGPAF TIINGTLHYF NTRYIRVDIA GPVTKEITGF VSGTSTSQVL 360
WDQWFPYGEN SIGPNGLLKT ASGYKYPLFM VGTGVLDAI HKLGEATVIE HPHAKEAQKV 420
VDDSEVIFPG DTGVSKNPVE VVEGWFSGWR SSLMSIFGII LLIVCLVLIV RILIALKYCC 480
VRHKKRTIYK EDLEMGRIPR RA 502

```

```

SEQ ID NO: 251      moltype = AA length = 494
FEATURE            Location/Qualifiers
source              1..494
                    mol_type = protein
                    organism = Vesicular stomatitis virus

```

```

SEQUENCE: 251
KFTIVFPQSQ KGDWKDVPPN YRYCPSSADQ NWHGDLGVLN IRAKMPAVHK AIKADGWMCH 60
AAKWVTTCDY RYWGPKYITH SIHSFIPTKA QCEESIKQTK EGVWINPGFP PKNCGYASVS 120
DAESIIVQAT AHSVMTSIGR GDWLDQFPT GRCTGSTCET IHNSTLWYAD YQVTGLCDSA 180
LVSTEVTFYS EDGVMISIGR QNTGYRSNYF PYEKGAACR MKYCTHEGIR LPSGVWFEMV 240
DKELLESVQM PECPAGLTIS APTQTSVDVS LILDVERMLD YSLCQETWSK VHSGLPISPV 300
DLGYIAPKNP GAGPAFTIVN GTLKYPDTRY LRIDIEGPVL KKMTGKVSQT PTKRELWTEW 360
FPYDDVEIGP NGVLKTPREGY KPPLYMIGHG LLDSDLQKTS QAEVFHHFQI AEAVQKLPPD 420
ETLFFGDTGI SKNPVEVIEG WFSNWRSSVM AIVFAILLLV ITVLMVRLCV AFRHFCCQKR 480
HKIYNDLEMN QLRR 494

```

```

SEQ ID NO: 252      moltype = AA length = 494
FEATURE            Location/Qualifiers
source              1..494
                    mol_type = protein
                    organism = Vesicular stomatitis virus

```

-continued

SEQUENCE: 252
 KFTIVFPQSQ KGDWKDVPPN YRYCPSSADQ NWHGDLLGVN IRAKMPFVHK AIKADGWMCH 60
 AAKWVTTCDY RWYGPQYITH SIHSFIPTKA QCEESIKQTK EGVWINPGFP PKNCGYASVS 120
 DAESIIVQAT AHSVIMIDEYS GDWLDSQFPT GRCTGSTCET IHNSTLWYAD YQVTGLCDSA 180
 LVSTEVTFYS EDGLMTSigr QNTGYRSNYF PYEKGAACR MKYCTHEGIR LPSGVWFEMV 240
 DKELLESVQM PECPAGLTIS APTQTSVDVS LILDVERMLD YSLCQETWSK VHSGLPISPV 300
 DLGYIAPKNP GAGPAFTIVN GTLKYPDTRY LRIDIEGPVL KKMTGKVSgt PTKRELWTEW 360
 FPYDDVEIGP NGVLKTPEGY KFLPYMIGHG LLDSDLQKTS QAEVFHHPQI AEAQVQLPDD 420
 ETLFFGDTGI SKNPVEVIEG WFSNWRSSVM AIVFAILLLV ITVLMVRLCV AFRHFCCQKR 480
 HKIYNDLEMN QLRR 494

SEQ ID NO: 253 moltype = AA length = 494
 FEATURE Location/Qualifiers
 source 1..494
 mol_type = protein
 organism = Vesicular stomatitis virus

SEQUENCE: 253
 KFTIVFPQSQ KGDWKDVPPN YRYCPSSADQ NWHGDLLGVN IRAKMPGVHK AIKADGWMCH 60
 AAKWVTTCDY RWYGPQYITH SIHSFIPTKA QCEESIKQTK EGVWINPGFP PKNCGYASVS 120
 DAESIIVQAT AHSVIMIDEYS GDWLDSQFPT GRCTGSTCET IHNSTLWYAD YQVTGLCDSA 180
 LVSTEVTFYS EDGLMTSigr QNTGYRSNYF PYEKGAACR MKYCTHEGIR LPSGVWFEMV 240
 DKELLESVQM PECPAGLTIS APTQTSVDVS LILDVERMLD YSLCQETWSK VHSGLPISPV 300
 DLGYIAPKNP GAGPAFTIVN GTLKYPDTRY LRIDIEGPVL KKMTGKVSgt PTKRELWTEW 360
 FPYDDVEIGP NGVLKTPEGY KFLPYMIGHG LLDSDLQKTS QAEVFHHPQI AEAQVQLPDD 420
 ETLFFGDTGI SKNPVEVIEG WFSNWRSSVM AIVFAILLLV ITVLMVRLCV AFRHFCCQKR 480
 HKIYNDLEMN QLRR 494

SEQ ID NO: 254 moltype = AA length = 494
 FEATURE Location/Qualifiers
 source 1..494
 mol_type = protein
 organism = Vesicular stomatitis virus

SEQUENCE: 254
 KFTIVFPQSQ KGDWKDVPPN YRYCPSSADQ NWHGDLLGVN IRAKMPQVHK AIKADGWMCH 60
 AAKWVTTCDY RWYGPQYITH SIHSFIPTKA QCEESIKQTK EGVWINPGFP PKNCGYASVS 120
 DAESIIVQAT AHSVIMIDEYS GDWLDSQFPT GRCTGSTCET IHNSTLWYAD YQVTGLCDSA 180
 LVSTEVTFYS EDGLMTSigr QNTGYRSNYF PYEKGAACR MKYCTHEGIR LPSGVWFEMV 240
 DKELLESVQM PECPAGLTIS APTQTSVDVS LILDVERMLD YSLCQETWSK VHSGLPISPV 300
 DLGYIAPKNP GAGPAFTIVN GTLKYPDTRY LRIDIEGPVL KKMTGKVSgt PTKRELWTEW 360
 FPYDDVEIGP NGVLKTPEGY KFLPYMIGHG LLDSDLQKTS QAEVFHHPQI AEAQVQLPDD 420
 ETLFFGDTGI SKNPVEVIEG WFSNWRSSVM AIVFAILLLV ITVLMVRLCV AFRHFCCQKR 480
 HKIYNDLEMN QLRR 494

SEQ ID NO: 255 moltype = AA length = 494
 FEATURE Location/Qualifiers
 source 1..494
 mol_type = protein
 organism = Vesicular stomatitis virus

SEQUENCE: 255
 KFTIVFPQSQ KGDWKDVPPN YRYCPSSADQ NWHGDLLGVN IRAKMPKVHK AIKADGWMCH 60
 AAKWVTTCDY RWYGPQYITH SIHSFIPTKA QCEESIKQTK EGVWINPGFP PKNCGYASVS 120
 DAESIIVQAT AHSVIMIDEYS GDWLDSQFPT GRCTGSTCET IHNSTLWYAD YQVTGLCDSA 180
 LVSTEVTFYS EDGLMTSigr QNTGYRSNYF PYEKGAACR MKYCTHEGIR LPSGVWFEMV 240
 DKELLESVQM PECPAGLTIS APTQTSVDVS LILDVERMLD YSLCQETWSK VHSGLPISPV 300
 DLGYIAPKNP GAGPAFTIVN GTLKYPDTRY LRIDIEGPVL KKMTGKVSgt PTKAELWTEW 360
 FPYDDVEIGP NGVLKTPEGY KFLPYMIGHG LLDSDLQKTS QAEVFHHPQI AEAQVQLPDD 420
 ETLFFGDTGI SKNPVEVIEG WFSNWRSSVM AIVFAILLLV ITVLMVRLCV AFRHFCCQKR 480
 HKIYNDLEMN QLRR 494

SEQ ID NO: 256 moltype = AA length = 494
 FEATURE Location/Qualifiers
 source 1..494
 mol_type = protein
 organism = Vesicular stomatitis virus

SEQUENCE: 256
 KFTIVFPQSQ KGDWKDVPPN YRYCPSSADQ NWHGDLLGVN IRAKMPKVHK AIKADGWMCH 60
 AAKWVTTCDY RWYGPQYITH SIHSFIPTKA QCEESIKQTK EGVWINPGFP PKNCGYASVS 120
 DAESIIVQAT AHSVIMIDEYS GDWLDSQFPT GRCTGSTCET IHNSTLWYAD YQVTGLCDSA 180
 LVSTEVTFYS EDGLMTSigr QNTGYRSNYF PYEKGAACR MKYCTHEGIR LPSGVWFEMV 240
 DKELLESVQM PECPAGLTIS APTQTSVDVS LILDVERMLD YSLCQETWSK VHSGLPISPV 300
 DLGYIAPKNP GAGPAFTIVN GTLKYPDTRY LRIDIEGPVL KKMTGKVSgt PTKFELWTEW 360
 FPYDDVEIGP NGVLKTPEGY KFLPYMIGHG LLDSDLQKTS QAEVFHHPQI AEAQVQLPDD 420
 ETLFFGDTGI SKNPVEVIEG WFSNWRSSVM AIVFAILLLV ITVLMVRLCV AFRHFCCQKR 480
 HKIYNDLEMN QLRR 494

SEQ ID NO: 257 moltype = AA length = 494
 FEATURE Location/Qualifiers
 source 1..494
 mol_type = protein

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                                organism = Vesicular stomatitis virus
SEQUENCE: 257
KFTIVFPQSQ KGDWKDVPPN YRYCPSSADQ NWHGDLLGVN IRAKMPKVHK AIKADGWMCH 60
AAKWVTTCDY RWYGPQYITH SIHSFIPTKA QCEESIKQTK EGVWINPGFP PKNCGYASVS 120
DAESIIVQAT AHSVMIIDEYS GDWLDSQFPT GRCTGSTCET IHNSTLWYAD YQVTGLCDSA 180
LVSTEVTFYS EDGLMTSigr QNTGYRSNYF PYEKGAACR MKYCTHEGIR LPSGVWFEMV 240
DKELLESVQM PECPAGLTIS APTQTSVDVS LILDVERMLD YSLCQETWSK VHSGLPISPV 300
DLGYIAPKNP GAGPAFTIVN GTLKYFDTRY LRIDIEGPVL KKMTGKVSgt PTKGELWTEW 360
FPYDDVEIGP NGVLKTPEGY KPPLYMIGHG LLDSDLQKTS QAEVFHHPQI AEAQVQLPDD 420
ETLFFGDTGI SKNPVEVIEG WFSNWRSSVM AIVFAILLLV ITVLMVRLCV AFRHFCCQKR 480
HKIYNDELMN QLRR 494

SEQ ID NO: 258          moltype = AA length = 494
FEATURE                Location/Qualifiers
source                 1..494
                        mol_type = protein
                        organism = Vesicular stomatitis virus

SEQUENCE: 258
KFTIVFPQSQ KGDWKDVPPN YRYCPSSADQ NWHGDLLGVN IRAKMPKVHK AIKADGWMCH 60
AAKWVTTCDY RWYGPQYITH SIHSFIPTKA QCEESIKQTK EGVWINPGFP PKNCGYASVS 120
DAESIIVQAT AHSVMIIDEYS GDWLDSQFPT GRCTGSTCET IHNSTLWYAD YQVTGLCDSA 180
LVSTEVTFYS EDGLMTSigr QNTGYRSNYF PYEKGAACR MKYCTHEGIR LPSGVWFEMV 240
DKELLESVQM PECPAGLTIS APTQTSVDVS LILDVERMLD YSLCQETWSK VHSGLPISPV 300
DLGYIAPKNP GAGPAFTIVN GTLKYFDTRY LRIDIEGPVL KKMTGKVSgt PTKQELWTEW 360
FPYDDVEIGP NGVLKTPEGY KPPLYMIGHG LLDSDLQKTS QAEVFHHPQI AEAQVQLPDD 420
ETLFFGDTGI SKNPVEVIEG WFSNWRSSVM AIVFAILLLV ITVLMVRLCV AFRHFCCQKR 480
HKIYNDELMN QLRR 494

SEQ ID NO: 259          moltype = AA length = 494
FEATURE                Location/Qualifiers
source                 1..494
                        mol_type = protein
                        organism = Vesicular stomatitis virus

SEQUENCE: 259
KFTIVFPQSQ KGDWKDVPPN YRYCPSSADQ NWHGDLLGVN IRAKMPAVHK AIKADGWMCH 60
AAKWVTTCDY RWYGPQYITH SIHSFIPTKA QCEESIKQTK EGVWINPGFP PKNCGYASVS 120
DAESIIVQAT AHSVMIIDEYS GDWLDSQFPT GRCTGSTCET IHNSTLWYAD YQVTGLCDSA 180
LVSTEVTFYS EDGLMTSigr QNTGYRSNYF PYEKGAACR MKYCTHEGIR LPSGVWFEMV 240
DKELLESVQM PECPAGLTIS APTQTSVDVS LILDVERMLD YSLCQETWSK VHSGLPISPV 300
DLGYIAPKNP GAGPAFTIVN GTLKYFDTRY LRIDIEGPVL KKMTGKVSgt PTKAELWTEW 360
FPYDDVEIGP NGVLKTPEGY KPPLYMIGHG LLDSDLQKTS QAEVFHHPQI AEAQVQLPDD 420
ETLFFGDTGI SKNPVEVIEG WFSNWRSSVM AIVFAILLLV ITVLMVRLCV AFRHFCCQKR 480
HKIYNDELMN QLRR 494

SEQ ID NO: 260          moltype = AA length = 494
FEATURE                Location/Qualifiers
source                 1..494
                        mol_type = protein
                        organism = Vesicular stomatitis virus

SEQUENCE: 260
KFTIVFPQSQ KGDWKDVPPN YRYCPSSADQ NWHGDLLGVN IRAKMPAVHK AIKADGWMCH 60
AAKWVTTCDY RWYGPQYITH SIHSFIPTKA QCEESIKQTK EGVWINPGFP PKNCGYASVS 120
DAESIIVQAT AHSVMIIDEYS GDWLDSQFPT GRCTGSTCET IHNSTLWYAD YQVTGLCDSA 180
LVSTEVTFYS EDGLMTSigr QNTGYRSNYF PYEKGAACR MKYCTHEGIR LPSGVWFEMV 240
DKELLESVQM PECPAGLTIS APTQTSVDVS LILDVERMLD YSLCQETWSK VHSGLPISPV 300
DLGYIAPKNP GAGPAFTIVN GTLKYFDTRY LRIDIEGPVL KKMTGKVSgt PTKFELWTEW 360
FPYDDVEIGP NGVLKTPEGY KPPLYMIGHG LLDSDLQKTS QAEVFHHPQI AEAQVQLPDD 420
ETLFFGDTGI SKNPVEVIEG WFSNWRSSVM AIVFAILLLV ITVLMVRLCV AFRHFCCQKR 480
HKIYNDELMN QLRR 494

SEQ ID NO: 261          moltype = AA length = 494
FEATURE                Location/Qualifiers
source                 1..494
                        mol_type = protein
                        organism = Vesicular stomatitis virus

SEQUENCE: 261
KFTIVFPQSQ KGDWKDVPPN YRYCPSSADQ NWHGDLLGVN IRAKMPAVHK AIKADGWMCH 60
AAKWVTTCDY RWYGPQYITH SIHSFIPTKA QCEESIKQTK EGVWINPGFP PKNCGYASVS 120
DAESIIVQAT AHSVMIIDEYS GDWLDSQFPT GRCTGSTCET IHNSTLWYAD YQVTGLCDSA 180
LVSTEVTFYS EDGLMTSigr QNTGYRSNYF PYEKGAACR MKYCTHEGIR LPSGVWFEMV 240
DKELLESVQM PECPAGLTIS APTQTSVDVS LILDVERMLD YSLCQETWSK VHSGLPISPV 300
DLGYIAPKNP GAGPAFTIVN GTLKYFDTRY LRIDIEGPVL KKMTGKVSgt PTKGELWTEW 360
FPYDDVEIGP NGVLKTPEGY KPPLYMIGHG LLDSDLQKTS QAEVFHHPQI AEAQVQLPDD 420
ETLFFGDTGI SKNPVEVIEG WFSNWRSSVM AIVFAILLLV ITVLMVRLCV AFRHFCCQKR 480
HKIYNDELMN QLRR 494

SEQ ID NO: 262          moltype = AA length = 494
FEATURE                Location/Qualifiers
source                 1..494

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mol_type = protein
organism = Vesicular stomatitis virus

SEQUENCE: 262
KFTIVFPQSQ KGDWKDVPPN YRYCPSSADQ NWHGDL LGVN IRAKMPAVHK AIKADGWMCH 60
AAKWVTTCDY RWYGPQYITH SIHSFIPTKA QCEESIKQTK EGVWINPGFP PKNCGYASVS 120
DAESIIVQAT AHSVMI DEYS GDWLDSQFPT GRCTGSTCET IHNSTLWYAD YQVTGLCDSA 180
LVSTEVTFYS EDGLMTS IGR QNTGYRSNYF PYEKGAACR MKYCTHEGIR LPSGVWFEMV 240
DKELLESVQM PECPAGLTIS APTQTSVDVS LILDVERMLD YSLCQETWSK VHSGLPISPV 300
DLGYIAPKNP GAGPAFTIVN GTLKYFDTRY LRIDIEGPVL KKMTGKVS GT PTKQELWTEW 360
FPYDDVEIGP NGVLKTPEGY KFPLYMIGHG LLSDSLQKTS QAEVFHHPQI AEA VQKL PDD 420
ETLFFGDTGI SKNPVEVIEG WFSNWRSSVM AIVFAILLV ITVLMVRLCV AFRHFCCQKR 480
HKIYNDLEMN QLRR 494

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SEQ ID NO: 263      moltype = AA length = 494
FEATURE            Location/Qualifiers
source              1..494
                    mol_type = protein
                    organism = Vesicular stomatitis virus

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SEQUENCE: 263
KFTIVFPQSQ KGDWKDVPPN YRYCPSSADQ NWHGDL LGVN IRAKMPFVHK AIKADGWMCH 60
AAKWVTTCDY RWYGPQYITH SIHSFIPTKA QCEESIKQTK EGVWINPGFP PKNCGYASVS 120
DAESIIVQAT AHSVMI DEYS GDWLDSQFPT GRCTGSTCET IHNSTLWYAD YQVTGLCDSA 180
LVSTEVTFYS EDGLMTS IGR QNTGYRSNYF PYEKGAACR MKYCTHEGIR LPSGVWFEMV 240
DKELLESVQM PECPAGLTIS APTQTSVDVS LILDVERMLD YSLCQETWSK VHSGLPISPV 300
DLGYIAPKNP GAGPAFTIVN GTLKYFDTRY LRIDIEGPVL KKMTGKVS GT PTKAELWTEW 360
FPYDDVEIGP NGVLKTPEGY KFPLYMIGHG LLSDSLQKTS QAEVFHHPQI AEA VQKL PDD 420
ETLFFGDTGI SKNPVEVIEG WFSNWRSSVM AIVFAILLV ITVLMVRLCV AFRHFCCQKR 480
HKIYNDLEMN QLRR 494

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SEQ ID NO: 264      moltype = AA length = 494
FEATURE            Location/Qualifiers
source              1..494
                    mol_type = protein
                    organism = Vesicular stomatitis virus

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SEQUENCE: 264
KFTIVFPQSQ KGDWKDVPPN YRYCPSSADQ NWHGDL LGVN IRAKMPFVHK AIKADGWMCH 60
AAKWVTTCDY RWYGPQYITH SIHSFIPTKA QCEESIKQTK EGVWINPGFP PKNCGYASVS 120
DAESIIVQAT AHSVMI DEYS GDWLDSQFPT GRCTGSTCET IHNSTLWYAD YQVTGLCDSA 180
LVSTEVTFYS EDGLMTS IGR QNTGYRSNYF PYEKGAACR MKYCTHEGIR LPSGVWFEMV 240
DKELLESVQM PECPAGLTIS APTQTSVDVS LILDVERMLD YSLCQETWSK VHSGLPISPV 300
DLGYIAPKNP GAGPAFTIVN GTLKYFDTRY LRIDIEGPVL KKMTGKVS GT PTKFELWTEW 360
FPYDDVEIGP NGVLKTPEGY KFPLYMIGHG LLSDSLQKTS QAEVFHHPQI AEA VQKL PDD 420
ETLFFGDTGI SKNPVEVIEG WFSNWRSSVM AIVFAILLV ITVLMVRLCV AFRHFCCQKR 480
HKIYNDLEMN QLRR 494

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SEQ ID NO: 265      moltype = AA length = 494
FEATURE            Location/Qualifiers
source              1..494
                    mol_type = protein
                    organism = Vesicular stomatitis virus

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SEQUENCE: 265
KFTIVFPQSQ KGDWKDVPPN YRYCPSSADQ NWHGDL LGVN IRAKMPFVHK AIKADGWMCH 60
AAKWVTTCDY RWYGPQYITH SIHSFIPTKA QCEESIKQTK EGVWINPGFP PKNCGYASVS 120
DAESIIVQAT AHSVMI DEYS GDWLDSQFPT GRCTGSTCET IHNSTLWYAD YQVTGLCDSA 180
LVSTEVTFYS EDGLMTS IGR QNTGYRSNYF PYEKGAACR MKYCTHEGIR LPSGVWFEMV 240
DKELLESVQM PECPAGLTIS APTQTSVDVS LILDVERMLD YSLCQETWSK VHSGLPISPV 300
DLGYIAPKNP GAGPAFTIVN GTLKYFDTRY LRIDIEGPVL KKMTGKVS GT PTKGELWTEW 360
FPYDDVEIGP NGVLKTPEGY KFPLYMIGHG LLSDSLQKTS QAEVFHHPQI AEA VQKL PDD 420
ETLFFGDTGI SKNPVEVIEG WFSNWRSSVM AIVFAILLV ITVLMVRLCV AFRHFCCQKR 480
HKIYNDLEMN QLRR 494

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SEQ ID NO: 266      moltype = AA length = 494
FEATURE            Location/Qualifiers
source              1..494
                    mol_type = protein
                    organism = Vesicular stomatitis virus

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SEQUENCE: 266
KFTIVFPQSQ KGDWKDVPPN YRYCPSSADQ NWHGDL LGVN IRAKMPFVHK AIKADGWMCH 60
AAKWVTTCDY RWYGPQYITH SIHSFIPTKA QCEESIKQTK EGVWINPGFP PKNCGYASVS 120
DAESIIVQAT AHSVMI DEYS GDWLDSQFPT GRCTGSTCET IHNSTLWYAD YQVTGLCDSA 180
LVSTEVTFYS EDGLMTS IGR QNTGYRSNYF PYEKGAACR MKYCTHEGIR LPSGVWFEMV 240
DKELLESVQM PECPAGLTIS APTQTSVDVS LILDVERMLD YSLCQETWSK VHSGLPISPV 300
DLGYIAPKNP GAGPAFTIVN GTLKYFDTRY LRIDIEGPVL KKMTGKVS GT PTKQELWTEW 360
FPYDDVEIGP NGVLKTPEGY KFPLYMIGHG LLSDSLQKTS QAEVFHHPQI AEA VQKL PDD 420
ETLFFGDTGI SKNPVEVIEG WFSNWRSSVM AIVFAILLV ITVLMVRLCV AFRHFCCQKR 480
HKIYNDLEMN QLRR 494

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SEQ ID NO: 267      moltype = AA length = 494
FEATURE            Location/Qualifiers

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source 1..494
mol_type = protein
organism = Vesicular stomatitis virus

SEQUENCE: 267
KFTIVFPQSQ KGDWKDVPPN YRYCPSSADQ NWHGDLLGVN IRAKMPGVHK AIKADGWMCH 60
AAKWVTTCDY RWYGPQYITH SIHSFIPTKA QCEESIKQTK EGVWINPGFP PKNCGYASVS 120
DAESIIVQAT AHSVIMIDEYS GDWLDSQFPT GRCTGSTCET IHNSTLWYAD YQVTGLCDSA 180
LVSTEVTFYS EDGLMTSigr QNTGYRSNYF PYEKGAAACR MKYCTHEGIR LPSGVWFEMV 240
DKELLESVQM PECPAGLTIS APTQTSVDVS LILDVERMLD YSLCQETWSK VHSGLPISPV 300
DLGYIAPKNP GAGPAFTIVN GTLKYFDTRY LRIDIEGPVL KKMTGKVSgt PTKAELWTEW 360
FPYDDVEIGP NGVLKTPEGY KFPLYMIGHG LLDSDLQKTS QAEVFHHPQI AEAQVQLPDD 420
ETLFFGDTGI SKNPVEVIEG WFSNWRSSVM AIVFAILLLV ITVLMVRLCV AFRHFCCQKR 480
HKIYNDLEMN QLRR 494

SEQ ID NO: 268 moltype = AA length = 494
FEATURE Location/Qualifiers
source 1..494
mol_type = protein
organism = Vesicular stomatitis virus

SEQUENCE: 268
KFTIVFPQSQ KGDWKDVPPN YRYCPSSADQ NWHGDLLGVN IRAKMPGVHK AIKADGWMCH 60
AAKWVTTCDY RWYGPQYITH SIHSFIPTKA QCEESIKQTK EGVWINPGFP PKNCGYASVS 120
DAESIIVQAT AHSVIMIDEYS GDWLDSQFPT GRCTGSTCET IHNSTLWYAD YQVTGLCDSA 180
LVSTEVTFYS EDGLMTSigr QNTGYRSNYF PYEKGAAACR MKYCTHEGIR LPSGVWFEMV 240
DKELLESVQM PECPAGLTIS APTQTSVDVS LILDVERMLD YSLCQETWSK VHSGLPISPV 300
DLGYIAPKNP GAGPAFTIVN GTLKYFDTRY LRIDIEGPVL KKMTGKVSgt PTKFELWTEW 360
FPYDDVEIGP NGVLKTPEGY KFPLYMIGHG LLDSDLQKTS QAEVFHHPQI AEAQVQLPDD 420
ETLFFGDTGI SKNPVEVIEG WFSNWRSSVM AIVFAILLLV ITVLMVRLCV AFRHFCCQKR 480
HKIYNDLEMN QLRR 494

SEQ ID NO: 269 moltype = AA length = 494
FEATURE Location/Qualifiers
source 1..494
mol_type = protein
organism = Vesicular stomatitis virus

SEQUENCE: 269
KFTIVFPQSQ KGDWKDVPPN YRYCPSSADQ NWHGDLLGVN IRAKMPGVHK AIKADGWMCH 60
AAKWVTTCDY RWYGPQYITH SIHSFIPTKA QCEESIKQTK EGVWINPGFP PKNCGYASVS 120
DAESIIVQAT AHSVIMIDEYS GDWLDSQFPT GRCTGSTCET IHNSTLWYAD YQVTGLCDSA 180
LVSTEVTFYS EDGLMTSigr QNTGYRSNYF PYEKGAAACR MKYCTHEGIR LPSGVWFEMV 240
DKELLESVQM PECPAGLTIS APTQTSVDVS LILDVERMLD YSLCQETWSK VHSGLPISPV 300
DLGYIAPKNP GAGPAFTIVN GTLKYFDTRY LRIDIEGPVL KKMTGKVSgt PTKGELWTEW 360
FPYDDVEIGP NGVLKTPEGY KFPLYMIGHG LLDSDLQKTS QAEVFHHPQI AEAQVQLPDD 420
ETLFFGDTGI SKNPVEVIEG WFSNWRSSVM AIVFAILLLV ITVLMVRLCV AFRHFCCQKR 480
HKIYNDLEMN QLRR 494

SEQ ID NO: 270 moltype = AA length = 494
FEATURE Location/Qualifiers
source 1..494
mol_type = protein
organism = Vesicular stomatitis virus

SEQUENCE: 270
KFTIVFPQSQ KGDWKDVPPN YRYCPSSADQ NWHGDLLGVN IRAKMPGVHK AIKADGWMCH 60
AAKWVTTCDY RWYGPQYITH SIHSFIPTKA QCEESIKQTK EGVWINPGFP PKNCGYASVS 120
DAESIIVQAT AHSVIMIDEYS GDWLDSQFPT GRCTGSTCET IHNSTLWYAD YQVTGLCDSA 180
LVSTEVTFYS EDGLMTSigr QNTGYRSNYF PYEKGAAACR MKYCTHEGIR LPSGVWFEMV 240
DKELLESVQM PECPAGLTIS APTQTSVDVS LILDVERMLD YSLCQETWSK VHSGLPISPV 300
DLGYIAPKNP GAGPAFTIVN GTLKYFDTRY LRIDIEGPVL KKMTGKVSgt PTKQELWTEW 360
FPYDDVEIGP NGVLKTPEGY KFPLYMIGHG LLDSDLQKTS QAEVFHHPQI AEAQVQLPDD 420
ETLFFGDTGI SKNPVEVIEG WFSNWRSSVM AIVFAILLLV ITVLMVRLCV AFRHFCCQKR 480
HKIYNDLEMN QLRR 494

SEQ ID NO: 271 moltype = AA length = 494
FEATURE Location/Qualifiers
source 1..494
mol_type = protein
organism = Vesicular stomatitis virus

SEQUENCE: 271
KFTIVFPQSQ KGDWKDVPPN YRYCPSSADQ NWHGDLLGVN IRAKMPQVHK AIKADGWMCH 60
AAKWVTTCDY RWYGPQYITH SIHSFIPTKA QCEESIKQTK EGVWINPGFP PKNCGYASVS 120
DAESIIVQAT AHSVIMIDEYS GDWLDSQFPT GRCTGSTCET IHNSTLWYAD YQVTGLCDSA 180
LVSTEVTFYS EDGLMTSigr QNTGYRSNYF PYEKGAAACR MKYCTHEGIR LPSGVWFEMV 240
DKELLESVQM PECPAGLTIS APTQTSVDVS LILDVERMLD YSLCQETWSK VHSGLPISPV 300
DLGYIAPKNP GAGPAFTIVN GTLKYFDTRY LRIDIEGPVL KKMTGKVSgt PTKAELWTEW 360
FPYDDVEIGP NGVLKTPEGY KFPLYMIGHG LLDSDLQKTS QAEVFHHPQI AEAQVQLPDD 420
ETLFFGDTGI SKNPVEVIEG WFSNWRSSVM AIVFAILLLV ITVLMVRLCV AFRHFCCQKR 480
HKIYNDLEMN QLRR 494

SEQ ID NO: 272 moltype = AA length = 494

-continued

FEATURE
source Location/Qualifiers
1..494
mol_type = protein
organism = Vesicular stomatitis virus

SEQUENCE: 272

KFTIVFPQSQ	KGDWKDVPPN	YRYCPSSADQ	NWHGDLLGVN	IRAKMPQVHK	AIKADGWMCH	60
AAKWVTTCDY	RWYGPQYITH	SIHSFIPTKA	QCEESIKQTK	EGVWINPGFP	PKNCGYASVS	120
DAESIIVQAT	AHSMIDEYS	GDWLDSQFPT	GRCTGSTCET	IHNSTLWYAD	YQVTGLCDSA	180
LVSTEVTFYS	EDGLMTSigr	QNTGYRSNYF	PYEKGAAACR	MKYCTHEGIR	LPSGVWFEMV	240
DKELLESVQM	PECPAGLTIS	APTQTSVDVS	LILDVERMLD	YSLCQETWSK	VHSGLPISPV	300
DLGYIAPKNP	GAGPAFTIVN	GTLKYFDTRY	LRIDIEGPVL	KKMTGKVSgt	PTKFELWTEW	360
FPYDDVEIGP	NGVLKTPEGY	KFPLYMIGHG	LLDSDLQKTS	QAEVFHHPQI	AEAVQKLPPD	420
ETLFFGDTGI	SKNPVEVIEG	WFSNWRSSVM	AIVFAILLLV	ITVLMVRLCV	AFRHFCCQKR	480
HKIYNDLEMN	QLRR					494

SEQ ID NO: 273 moltype = AA length = 494

FEATURE
source Location/Qualifiers
1..494
mol_type = protein
organism = Vesicular stomatitis virus

SEQUENCE: 273

KFTIVFPQSQ	KGDWKDVPPN	YRYCPSSADQ	NWHGDLLGVN	IRAKMPQVHK	AIKADGWMCH	60
AAKWVTTCDY	RWYGPQYITH	SIHSFIPTKA	QCEESIKQTK	EGVWINPGFP	PKNCGYASVS	120
DAESIIVQAT	AHSMIDEYS	GDWLDSQFPT	GRCTGSTCET	IHNSTLWYAD	YQVTGLCDSA	180
LVSTEVTFYS	EDGLMTSigr	QNTGYRSNYF	PYEKGAAACR	MKYCTHEGIR	LPSGVWFEMV	240
DKELLESVQM	PECPAGLTIS	APTQTSVDVS	LILDVERMLD	YSLCQETWSK	VHSGLPISPV	300
DLGYIAPKNP	GAGPAFTIVN	GTLKYFDTRY	LRIDIEGPVL	KKMTGKVSgt	PTKGELWTEW	360
FPYDDVEIGP	NGVLKTPEGY	KFPLYMIGHG	LLDSDLQKTS	QAEVFHHPQI	AEAVQKLPPD	420
ETLFFGDTGI	SKNPVEVIEG	WFSNWRSSVM	AIVFAILLLV	ITVLMVRLCV	AFRHFCCQKR	480
HKIYNDLEMN	QLRR					494

SEQ ID NO: 274 moltype = AA length = 494

FEATURE
source Location/Qualifiers
1..494
mol_type = protein
organism = Vesicular stomatitis virus

SEQUENCE: 274

KFTIVFPQSQ	KGDWKDVPPN	YRYCPSSADQ	NWHGDLLGVN	IRAKMPQVHK	AIKADGWMCH	60
AAKWVTTCDY	RWYGPQYITH	SIHSFIPTKA	QCEESIKQTK	EGVWINPGFP	PKNCGYASVS	120
DAESIIVQAT	AHSMIDEYS	GDWLDSQFPT	GRCTGSTCET	IHNSTLWYAD	YQVTGLCDSA	180
LVSTEVTFYS	EDGLMTSigr	QNTGYRSNYF	PYEKGAAACR	MKYCTHEGIR	LPSGVWFEMV	240
DKELLESVQM	PECPAGLTIS	APTQTSVDVS	LILDVERMLD	YSLCQETWSK	VHSGLPISPV	300
DLGYIAPKNP	GAGPAFTIVN	GTLKYFDTRY	LRIDIEGPVL	KKMTGKVSgt	PTKQELWTEW	360
FPYDDVEIGP	NGVLKTPEGY	KFPLYMIGHG	LLDSDLQKTS	QAEVFHHPQI	AEAVQKLPPD	420
ETLFFGDTGI	SKNPVEVIEG	WFSNWRSSVM	AIVFAILLLV	ITVLMVRLCV	AFRHFCCQKR	480
HKIYNDLEMN	QLRR					494

SEQ ID NO: 275 moltype = AA length = 495

FEATURE
source Location/Qualifiers
1..495
mol_type = protein
organism = Vesicular stomatitis virus

SEQUENCE: 275

KFSIVFPQSQ	KGNWKNVPSS	YHYCPSSSDQ	NWHNDLLGIT	MKVKMPATHK	AIQADGWMCH	60
AAKWITTCDF	RWYGPQYITH	SIHSIQPTSE	QCKESIKQTK	QGTWMSPGFP	PQNCGYATVT	120
DSVAVVVQAT	PHHVLVDEYT	GEWIDSQFPN	GKCETEECET	VHNSTVWYSD	YKVTGLCDAT	180
LVDTEITFFS	EDGKKEISIGK	PNTGYRSNYF	AYEKGDKVCK	MNYCKHAGVR	LPSGVWFEPV	240
DQDVYAAAKL	PECPVGATIS	APTQTSVDVS	LILDVERILD	YSLCQETWSK	IRSKQPVSPV	300
DLSYLAPKNP	GTGPAFTIIN	GTLKYFETRY	IRIDIDNPII	SKMVGKISGS	QTERELWTEW	360
FPYEGVEIGP	NGILKTPTGY	KFPLFMIGHG	MLDSDLHKTS	QAEVFEHPHL	AEAPKQLPEE	420
ETLFFGDTGI	SKNPVELIEG	WFSSWKSTVV	TFFFAIGVFI	LLYVVVARIVI	AVRIRYQGSN	480
NKRIYNDIEM	SRFRK					495

SEQ ID NO: 276 moltype = AA length = 495

FEATURE
source Location/Qualifiers
1..495
mol_type = protein
organism = Vesicular stomatitis virus

SEQUENCE: 276

KFSIVFPQSQ	KGNWKNVPSS	YHYCPSSSDQ	NWHNDLLGIT	MKVKMPFTHK	AIQADGWMCH	60
AAKWITTCDF	RWYGPQYITH	SIHSIQPTSE	QCKESIKQTK	QGTWMSPGFP	PQNCGYATVT	120
DSVAVVVQAT	PHHVLVDEYT	GEWIDSQFPN	GKCETEECET	VHNSTVWYSD	YKVTGLCDAT	180
LVDTEITFFS	EDGKKEISIGK	PNTGYRSNYF	AYEKGDKVCK	MNYCKHAGVR	LPSGVWFEPV	240
DQDVYAAAKL	PECPVGATIS	APTQTSVDVS	LILDVERILD	YSLCQETWSK	IRSKQPVSPV	300
DLSYLAPKNP	GTGPAFTIIN	GTLKYFETRY	IRIDIDNPII	SKMVGKISGS	QTERELWTEW	360
FPYEGVEIGP	NGILKTPTGY	KFPLFMIGHG	MLDSDLHKTS	QAEVFEHPHL	AEAPKQLPEE	420
ETLFFGDTGI	SKNPVELIEG	WFSSWKSTVV	TFFFAIGVFI	LLYVVVARIVI	AVRIRYQGSN	480
NKRIYNDIEM	SRFRK					495

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SEQ ID NO: 277 moltype = AA length = 495
 FEATURE Location/Qualifiers
 source 1..495
 mol_type = protein
 organism = Vesicular stomatitis virus

SEQUENCE: 277

KFSIVFPQSQ	KGWKNVPSS	YHYCPSSSDQ	NWHDLLGIT	MKVKMPGTHK	AIQADGWMCH	60
AAKWITTCDF	RWYGPKYITH	SIHSIQPTSE	QCKESIKQTK	QGTWMSPGFP	PQNCGYATVT	120
DSVAVVVQAT	PHHVLVDEYT	GEWIDSQFPN	GKCETEECET	VHNSTVWYSD	YKVTGLCDAT	180
LVDTEITFFS	EDGKKEISIGK	PNTGYRSNYF	AYEKGDKVCK	MNYCKHAGVR	LPSGVWFEFV	240
DQDVYAAAKL	PECPVGATIS	APTQTSVDVS	LILDVERILD	YSLCQETWSK	IRSKQPVSPV	300
DLSYLAPKNP	GTGPAFTIIN	GTLKYFETRY	IRIDIDNPPII	SKMVGKISGS	QTERELWTEW	360
FPYEGVEIGP	NGILKTPTGY	KFPLFMIGHG	MLDSDLHKTS	QAEVFEHPL	AEAPKQLPEE	420
ETLFFGDTGI	SKNPVELIEG	WFSSWKSTVV	TFFFAIGVFI	LLYVVARIVI	AVRIRYQGSN	480
NKRIYNDIEM	SRFRK					495

SEQ ID NO: 278 moltype = AA length = 495
 FEATURE Location/Qualifiers
 source 1..495
 mol_type = protein
 organism = Vesicular stomatitis virus

SEQUENCE: 278

KFSIVFPQSQ	KGWKNVPSS	YHYCPSSSDQ	NWHDLLGIT	MKVKMPQTHK	AIQADGWMCH	60
AAKWITTCDF	RWYGPKYITH	SIHSIQPTSE	QCKESIKQTK	QGTWMSPGFP	PQNCGYATVT	120
DSVAVVVQAT	PHHVLVDEYT	GEWIDSQFPN	GKCETEECET	VHNSTVWYSD	YKVTGLCDAT	180
LVDTEITFFS	EDGKKEISIGK	PNTGYRSNYF	AYEKGDKVCK	MNYCKHAGVR	LPSGVWFEFV	240
DQDVYAAAKL	PECPVGATIS	APTQTSVDVS	LILDVERILD	YSLCQETWSK	IRSKQPVSPV	300
DLSYLAPKNP	GTGPAFTIIN	GTLKYFETRY	IRIDIDNPPII	SKMVGKISGS	QTERELWTEW	360
FPYEGVEIGP	NGILKTPTGY	KFPLFMIGHG	MLDSDLHKTS	QAEVFEHPL	AEAPKQLPEE	420
ETLFFGDTGI	SKNPVELIEG	WFSSWKSTVV	TFFFAIGVFI	LLYVVARIVI	AVRIRYQGSN	480
NKRIYNDIEM	SRFRK					495

SEQ ID NO: 279 moltype = AA length = 495
 FEATURE Location/Qualifiers
 source 1..495
 mol_type = protein
 organism = Vesicular stomatitis virus

SEQUENCE: 279

KFSIVFPQSQ	KGWKNVPSS	YHYCPSSSDQ	NWHDLLGIT	MKVKMPKTHK	AIQADGWMCH	60
AAKWITTCDF	RWYGPKYITH	SIHSIQPTSE	QCKESIKQTK	QGTWMSPGFP	PQNCGYATVT	120
DSVAVVVQAT	PHHVLVDEYT	GEWIDSQFPN	GKCETEECET	VHNSTVWYSD	YKVTGLCDAT	180
LVDTEITFFS	EDGKKEISIGK	PNTGYRSNYF	AYEKGDKVCK	MNYCKHAGVR	LPSGVWFEFV	240
DQDVYAAAKL	PECPVGATIS	APTQTSVDVS	LILDVERILD	YSLCQETWSK	IRSKQPVSPV	300
DLSYLAPKNP	GTGPAFTIIN	GTLKYFETRY	IRIDIDNPPII	SKMVGKISGS	QTEAELWTEW	360
FPYEGVEIGP	NGILKTPTGY	KFPLFMIGHG	MLDSDLHKTS	QAEVFEHPL	AEAPKQLPEE	420
ETLFFGDTGI	SKNPVELIEG	WFSSWKSTVV	TFFFAIGVFI	LLYVVARIVI	AVRIRYQGSN	480
NKRIYNDIEM	SRFRK					495

SEQ ID NO: 280 moltype = AA length = 495
 FEATURE Location/Qualifiers
 source 1..495
 mol_type = protein
 organism = Vesicular stomatitis virus

SEQUENCE: 280

KFSIVFPQSQ	KGWKNVPSS	YHYCPSSSDQ	NWHDLLGIT	MKVKMPKTHK	AIQADGWMCH	60
AAKWITTCDF	RWYGPKYITH	SIHSIQPTSE	QCKESIKQTK	QGTWMSPGFP	PQNCGYATVT	120
DSVAVVVQAT	PHHVLVDEYT	GEWIDSQFPN	GKCETEECET	VHNSTVWYSD	YKVTGLCDAT	180
LVDTEITFFS	EDGKKEISIGK	PNTGYRSNYF	AYEKGDKVCK	MNYCKHAGVR	LPSGVWFEFV	240
DQDVYAAAKL	PECPVGATIS	APTQTSVDVS	LILDVERILD	YSLCQETWSK	IRSKQPVSPV	300
DLSYLAPKNP	GTGPAFTIIN	GTLKYFETRY	IRIDIDNPPII	SKMVGKISGS	QTEFELWTEW	360
FPYEGVEIGP	NGILKTPTGY	KFPLFMIGHG	MLDSDLHKTS	QAEVFEHPL	AEAPKQLPEE	420
ETLFFGDTGI	SKNPVELIEG	WFSSWKSTVV	TFFFAIGVFI	LLYVVARIVI	AVRIRYQGSN	480
NKRIYNDIEM	SRFRK					495

SEQ ID NO: 281 moltype = AA length = 495
 FEATURE Location/Qualifiers
 source 1..495
 mol_type = protein
 organism = Vesicular stomatitis virus

SEQUENCE: 281

KFSIVFPQSQ	KGWKNVPSS	YHYCPSSSDQ	NWHDLLGIT	MKVKMPKTHK	AIQADGWMCH	60
AAKWITTCDF	RWYGPKYITH	SIHSIQPTSE	QCKESIKQTK	QGTWMSPGFP	PQNCGYATVT	120
DSVAVVVQAT	PHHVLVDEYT	GEWIDSQFPN	GKCETEECET	VHNSTVWYSD	YKVTGLCDAT	180
LVDTEITFFS	EDGKKEISIGK	PNTGYRSNYF	AYEKGDKVCK	MNYCKHAGVR	LPSGVWFEFV	240
DQDVYAAAKL	PECPVGATIS	APTQTSVDVS	LILDVERILD	YSLCQETWSK	IRSKQPVSPV	300
DLSYLAPKNP	GTGPAFTIIN	GTLKYFETRY	IRIDIDNPPII	SKMVGKISGS	QTEGELWTEW	360
FPYEGVEIGP	NGILKTPTGY	KFPLFMIGHG	MLDSDLHKTS	QAEVFEHPL	AEAPKQLPEE	420
ETLFFGDTGI	SKNPVELIEG	WFSSWKSTVV	TFFFAIGVFI	LLYVVARIVI	AVRIRYQGSN	480
NKRIYNDIEM	SRFRK					495

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SEQ ID NO: 282      moltype = AA length = 495
FEATURE            Location/Qualifiers
source             1..495
                   mol_type = protein
                   organism = Vesicular stomatitis virus

SEQUENCE: 282
KFSIVFPQSQ KGNWKNVPSS YHYCPSSSDQ NWHNDLLGIT MKVKMPKTHK AIQADGWMCH 60
AAKWITTCDF RWYGPKYITH SIHSIQPTSE QCKESIKQTK QGTWMSPGFP PQNCGYATVT 120
DSVAVVVQAT PHHVLVDEYT GEWIDSQFPN GKCETEECET VHNSTVWYSD YKVTGLCDAT 180
LVDTEITFFS EDGKKEISIGK PNTGYRSNYF AYEKGDKVCK MNYCKHAGVR LPSGVWFEFV 240
DQDVYAAAKL PECPVGATIS APTQTSVDVS LILDVERILD YSLCQETWSK IRSKQPVSPV 300
DLSYLAPKNP GTGPAFTIIN GTLKYPETRY IRIDIDNPPII SKMVGKISGS QTEQELWTEW 360
FPYEGVEIGP NGILKTPTGY KFPLFMIGHG MLDSDLHKTS QAEVFEHPL AEAPKQLPEE 420
ETLFFGDTGI SKNPVELIEG WFSSWKSTVV TFFFAIGVFI LLYVVARIVI AVRYRYQGSN 480
NKRIYNDIEM SRFRK 495

SEQ ID NO: 283      moltype = AA length = 495
FEATURE            Location/Qualifiers
source             1..495
                   mol_type = protein
                   organism = Vesicular stomatitis virus

SEQUENCE: 283
KFSIVFPQSQ KGNWKNVPSS YHYCPSSSDQ NWHNDLLGIT MKVKMPATHK AIQADGWMCH 60
AAKWITTCDF RWYGPKYITH SIHSIQPTSE QCKESIKQTK QGTWMSPGFP PQNCGYATVT 120
DSVAVVVQAT PHHVLVDEYT GEWIDSQFPN GKCETEECET VHNSTVWYSD YKVTGLCDAT 180
LVDTEITFFS EDGKKEISIGK PNTGYRSNYF AYEKGDKVCK MNYCKHAGVR LPSGVWFEFV 240
DQDVYAAAKL PECPVGATIS APTQTSVDVS LILDVERILD YSLCQETWSK IRSKQPVSPV 300
DLSYLAPKNP GTGPAFTIIN GTLKYPETRY IRIDIDNPPII SKMVGKISGS QTEAELWTEW 360
FPYEGVEIGP NGILKTPTGY KFPLFMIGHG MLDSDLHKTS QAEVFEHPL AEAPKQLPEE 420
ETLFFGDTGI SKNPVELIEG WFSSWKSTVV TFFFAIGVFI LLYVVARIVI AVRYRYQGSN 480
NKRIYNDIEM SRFRK 495

SEQ ID NO: 284      moltype = AA length = 495
FEATURE            Location/Qualifiers
source             1..495
                   mol_type = protein
                   organism = Vesicular stomatitis virus

SEQUENCE: 284
KFSIVFPQSQ KGNWKNVPSS YHYCPSSSDQ NWHNDLLGIT MKVKMPATHK AIQADGWMCH 60
AAKWITTCDF RWYGPKYITH SIHSIQPTSE QCKESIKQTK QGTWMSPGFP PQNCGYATVT 120
DSVAVVVQAT PHHVLVDEYT GEWIDSQFPN GKCETEECET VHNSTVWYSD YKVTGLCDAT 180
LVDTEITFFS EDGKKEISIGK PNTGYRSNYF AYEKGDKVCK MNYCKHAGVR LPSGVWFEFV 240
DQDVYAAAKL PECPVGATIS APTQTSVDVS LILDVERILD YSLCQETWSK IRSKQPVSPV 300
DLSYLAPKNP GTGPAFTIIN GTLKYPETRY IRIDIDNPPII SKMVGKISGS QTEFELWTEW 360
FPYEGVEIGP NGILKTPTGY KFPLFMIGHG MLDSDLHKTS QAEVFEHPL AEAPKQLPEE 420
ETLFFGDTGI SKNPVELIEG WFSSWKSTVV TFFFAIGVFI LLYVVARIVI AVRYRYQGSN 480
NKRIYNDIEM SRFRK 495

SEQ ID NO: 285      moltype = AA length = 495
FEATURE            Location/Qualifiers
source             1..495
                   mol_type = protein
                   organism = Vesicular stomatitis virus

SEQUENCE: 285
KFSIVFPQSQ KGNWKNVPSS YHYCPSSSDQ NWHNDLLGIT MKVKMPATHK AIQADGWMCH 60
AAKWITTCDF RWYGPKYITH SIHSIQPTSE QCKESIKQTK QGTWMSPGFP PQNCGYATVT 120
DSVAVVVQAT PHHVLVDEYT GEWIDSQFPN GKCETEECET VHNSTVWYSD YKVTGLCDAT 180
LVDTEITFFS EDGKKEISIGK PNTGYRSNYF AYEKGDKVCK MNYCKHAGVR LPSGVWFEFV 240
DQDVYAAAKL PECPVGATIS APTQTSVDVS LILDVERILD YSLCQETWSK IRSKQPVSPV 300
DLSYLAPKNP GTGPAFTIIN GTLKYPETRY IRIDIDNPPII SKMVGKISGS QTEGELWTEW 360
FPYEGVEIGP NGILKTPTGY KFPLFMIGHG MLDSDLHKTS QAEVFEHPL AEAPKQLPEE 420
ETLFFGDTGI SKNPVELIEG WFSSWKSTVV TFFFAIGVFI LLYVVARIVI AVRYRYQGSN 480
NKRIYNDIEM SRFRK 495

SEQ ID NO: 286      moltype = AA length = 495
FEATURE            Location/Qualifiers
source             1..495
                   mol_type = protein
                   organism = Vesicular stomatitis virus

SEQUENCE: 286
KFSIVFPQSQ KGNWKNVPSS YHYCPSSSDQ NWHNDLLGIT MKVKMPATHK AIQADGWMCH 60
AAKWITTCDF RWYGPKYITH SIHSIQPTSE QCKESIKQTK QGTWMSPGFP PQNCGYATVT 120
DSVAVVVQAT PHHVLVDEYT GEWIDSQFPN GKCETEECET VHNSTVWYSD YKVTGLCDAT 180
LVDTEITFFS EDGKKEISIGK PNTGYRSNYF AYEKGDKVCK MNYCKHAGVR LPSGVWFEFV 240
DQDVYAAAKL PECPVGATIS APTQTSVDVS LILDVERILD YSLCQETWSK IRSKQPVSPV 300
DLSYLAPKNP GTGPAFTIIN GTLKYPETRY IRIDIDNPPII SKMVGKISGS QTEQELWTEW 360
FPYEGVEIGP NGILKTPTGY KFPLFMIGHG MLDSDLHKTS QAEVFEHPL AEAPKQLPEE 420
ETLFFGDTGI SKNPVELIEG WFSSWKSTVV TFFFAIGVFI LLYVVARIVI AVRYRYQGSN 480

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NKRIYNDIEM SRFRK 495

SEQ ID NO: 287 moltype = AA length = 495
 FEATURE Location/Qualifiers
 source 1..495
 mol_type = protein
 organism = Vesicular stomatitis virus

SEQUENCE: 287

KFSIVFPQSQ	KGNWKNVPSS	YHYCPSSSDQ	NWHNDLLGIT	MKVKMPFTHK	AIQADGWMCH	60
AAKWITTCDF	RWYGPKYITH	SIHSIQPTSE	QCKESIKQTK	QGTWMSPGFP	PQNCGYATVT	120
DSVAVVVQAT	PHHVLVDEYT	GEWIDSQFPN	GKCETEECET	VHNSTVWYSD	YKVTGLCDAT	180
LVDTEITFFS	EDGKKEISIGK	PNTGYRSNYF	AYEKGDKVCK	MNYCKHAGVR	LPSGVWFEFV	240
DQDVYAAAKL	PECPVGATIS	APTQTSVDVS	LILDVERILD	YSLCQETWSK	IRSKQPVSPV	300
DLSYLAPKNP	GTGPAFTIIN	GTLKYFETRY	IRIDIDNPPII	SKMVGKISGS	QTEAELWTEW	360
FPYEGVEIGP	NGILKTPTGY	KFPLFMIGHG	MLDSDLHKTS	QAEVFEHPLH	AEAPKQLPEE	420
ETLFFGDTGI	SKNPVELIEG	WFSSWKSTVV	TFFFAIGVFI	LLYVVARIVI	AVRIRYQGSN	480
NKRIYNDIEM	SRFRK					495

SEQ ID NO: 288 moltype = AA length = 495
 FEATURE Location/Qualifiers
 source 1..495
 mol_type = protein
 organism = Vesicular stomatitis virus

SEQUENCE: 288

KFSIVFPQSQ	KGNWKNVPSS	YHYCPSSSDQ	NWHNDLLGIT	MKVKMPFTHK	AIQADGWMCH	60
AAKWITTCDF	RWYGPKYITH	SIHSIQPTSE	QCKESIKQTK	QGTWMSPGFP	PQNCGYATVT	120
DSVAVVVQAT	PHHVLVDEYT	GEWIDSQFPN	GKCETEECET	VHNSTVWYSD	YKVTGLCDAT	180
LVDTEITFFS	EDGKKEISIGK	PNTGYRSNYF	AYEKGDKVCK	MNYCKHAGVR	LPSGVWFEFV	240
DQDVYAAAKL	PECPVGATIS	APTQTSVDVS	LILDVERILD	YSLCQETWSK	IRSKQPVSPV	300
DLSYLAPKNP	GTGPAFTIIN	GTLKYFETRY	IRIDIDNPPII	SKMVGKISGS	QTEFELWTEW	360
FPYEGVEIGP	NGILKTPTGY	KFPLFMIGHG	MLDSDLHKTS	QAEVFEHPLH	AEAPKQLPEE	420
ETLFFGDTGI	SKNPVELIEG	WFSSWKSTVV	TFFFAIGVFI	LLYVVARIVI	AVRIRYQGSN	480
NKRIYNDIEM	SRFRK					495

SEQ ID NO: 289 moltype = AA length = 495
 FEATURE Location/Qualifiers
 source 1..495
 mol_type = protein
 organism = Vesicular stomatitis virus

SEQUENCE: 289

KFSIVFPQSQ	KGNWKNVPSS	YHYCPSSSDQ	NWHNDLLGIT	MKVKMPFTHK	AIQADGWMCH	60
AAKWITTCDF	RWYGPKYITH	SIHSIQPTSE	QCKESIKQTK	QGTWMSPGFP	PQNCGYATVT	120
DSVAVVVQAT	PHHVLVDEYT	GEWIDSQFPN	GKCETEECET	VHNSTVWYSD	YKVTGLCDAT	180
LVDTEITFFS	EDGKKEISIGK	PNTGYRSNYF	AYEKGDKVCK	MNYCKHAGVR	LPSGVWFEFV	240
DQDVYAAAKL	PECPVGATIS	APTQTSVDVS	LILDVERILD	YSLCQETWSK	IRSKQPVSPV	300
DLSYLAPKNP	GTGPAFTIIN	GTLKYFETRY	IRIDIDNPPII	SKMVGKISGS	QTEGELWTEW	360
FPYEGVEIGP	NGILKTPTGY	KFPLFMIGHG	MLDSDLHKTS	QAEVFEHPLH	AEAPKQLPEE	420
ETLFFGDTGI	SKNPVELIEG	WFSSWKSTVV	TFFFAIGVFI	LLYVVARIVI	AVRIRYQGSN	480
NKRIYNDIEM	SRFRK					495

SEQ ID NO: 290 moltype = AA length = 495
 FEATURE Location/Qualifiers
 source 1..495
 mol_type = protein
 organism = Vesicular stomatitis virus

SEQUENCE: 290

KFSIVFPQSQ	KGNWKNVPSS	YHYCPSSSDQ	NWHNDLLGIT	MKVKMPFTHK	AIQADGWMCH	60
AAKWITTCDF	RWYGPKYITH	SIHSIQPTSE	QCKESIKQTK	QGTWMSPGFP	PQNCGYATVT	120
DSVAVVVQAT	PHHVLVDEYT	GEWIDSQFPN	GKCETEECET	VHNSTVWYSD	YKVTGLCDAT	180
LVDTEITFFS	EDGKKEISIGK	PNTGYRSNYF	AYEKGDKVCK	MNYCKHAGVR	LPSGVWFEFV	240
DQDVYAAAKL	PECPVGATIS	APTQTSVDVS	LILDVERILD	YSLCQETWSK	IRSKQPVSPV	300
DLSYLAPKNP	GTGPAFTIIN	GTLKYFETRY	IRIDIDNPPII	SKMVGKISGS	QTEQELWTEW	360
FPYEGVEIGP	NGILKTPTGY	KFPLFMIGHG	MLDSDLHKTS	QAEVFEHPLH	AEAPKQLPEE	420
ETLFFGDTGI	SKNPVELIEG	WFSSWKSTVV	TFFFAIGVFI	LLYVVARIVI	AVRIRYQGSN	480
NKRIYNDIEM	SRFRK					495

SEQ ID NO: 291 moltype = AA length = 495
 FEATURE Location/Qualifiers
 source 1..495
 mol_type = protein
 organism = Vesicular stomatitis virus

SEQUENCE: 291

KFSIVFPQSQ	KGNWKNVPSS	YHYCPSSSDQ	NWHNDLLGIT	MKVKMPGTHK	AIQADGWMCH	60
AAKWITTCDF	RWYGPKYITH	SIHSIQPTSE	QCKESIKQTK	QGTWMSPGFP	PQNCGYATVT	120
DSVAVVVQAT	PHHVLVDEYT	GEWIDSQFPN	GKCETEECET	VHNSTVWYSD	YKVTGLCDAT	180
LVDTEITFFS	EDGKKEISIGK	PNTGYRSNYF	AYEKGDKVCK	MNYCKHAGVR	LPSGVWFEFV	240
DQDVYAAAKL	PECPVGATIS	APTQTSVDVS	LILDVERILD	YSLCQETWSK	IRSKQPVSPV	300
DLSYLAPKNP	GTGPAFTIIN	GTLKYFETRY	IRIDIDNPPII	SKMVGKISGS	QTEAELWTEW	360
FPYEGVEIGP	NGILKTPTGY	KFPLFMIGHG	MLDSDLHKTS	QAEVFEHPLH	AEAPKQLPEE	420

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ETLFFGDTGI SKNPVELIEG WFSSWKSTVV TFFFAIGVFI LLYVVARIVI AVRYRYQGSN 480
NKRIYNDIEM SRFRK 495

SEQ ID NO: 292 moltype = AA length = 495
FEATURE Location/Qualifiers
source 1..495
 mol_type = protein
 organism = Vesicular stomatitis virus

SEQUENCE: 292
KFSIVFPQSQ KGNWKNVPSS YHYCPSSSDQ NWHNDLLGIT MKVKMPGTHK AIQADGWMCH 60
AAKWITTCDF RWYGPKYITH SIHSIQPTSE QCKESIKQTK QGTWMSPGFP PQNCGYATVT 120
DSVAVVVQAT PHHVLVDEYT GEWIDSQFPN GKCETEECET VHNSTVWYSD YKVTGLCDAT 180
LVDTEITFFS EDGKKEISIGK PNTGYRSNYF AYEKGDKVCK MNYCKHAGVR LPSGVWFEFV 240
DQDVYAAAKL PECPVGATIS APTQTSVDVS LILDVERILD YSLCQETWSK IRSKQPVSPV 300
DLSYLAPKNP GTGPAFTIIN GTLKYFETRY IRIDIDNPII SKMVGKISGS QTEFELWTEW 360
FPYEGVEIGP NGILKTPGTG KPPLFMIGHG MLDSDLHKTS QAEVFEHPLH AEAPKQLPEE 420
ETLFFGDTGI SKNPVELIEG WFSSWKSTVV TFFFAIGVFI LLYVVARIVI AVRYRYQGSN 480
NKRIYNDIEM SRFRK 495

SEQ ID NO: 293 moltype = AA length = 495
FEATURE Location/Qualifiers
source 1..495
 mol_type = protein
 organism = Vesicular stomatitis virus

SEQUENCE: 293
KFSIVFPQSQ KGNWKNVPSS YHYCPSSSDQ NWHNDLLGIT MKVKMPGTHK AIQADGWMCH 60
AAKWITTCDF RWYGPKYITH SIHSIQPTSE QCKESIKQTK QGTWMSPGFP PQNCGYATVT 120
DSVAVVVQAT PHHVLVDEYT GEWIDSQFPN GKCETEECET VHNSTVWYSD YKVTGLCDAT 180
LVDTEITFFS EDGKKEISIGK PNTGYRSNYF AYEKGDKVCK MNYCKHAGVR LPSGVWFEFV 240
DQDVYAAAKL PECPVGATIS APTQTSVDVS LILDVERILD YSLCQETWSK IRSKQPVSPV 300
DLSYLAPKNP GTGPAFTIIN GTLKYFETRY IRIDIDNPII SKMVGKISGS QTEGELWTEW 360
FPYEGVEIGP NGILKTPGTG KPPLFMIGHG MLDSDLHKTS QAEVFEHPLH AEAPKQLPEE 420
ETLFFGDTGI SKNPVELIEG WFSSWKSTVV TFFFAIGVFI LLYVVARIVI AVRYRYQGSN 480
NKRIYNDIEM SRFRK 495

SEQ ID NO: 294 moltype = AA length = 495
FEATURE Location/Qualifiers
source 1..495
 mol_type = protein
 organism = Vesicular stomatitis virus

SEQUENCE: 294
KFSIVFPQSQ KGNWKNVPSS YHYCPSSSDQ NWHNDLLGIT MKVKMPGTHK AIQADGWMCH 60
AAKWITTCDF RWYGPKYITH SIHSIQPTSE QCKESIKQTK QGTWMSPGFP PQNCGYATVT 120
DSVAVVVQAT PHHVLVDEYT GEWIDSQFPN GKCETEECET VHNSTVWYSD YKVTGLCDAT 180
LVDTEITFFS EDGKKEISIGK PNTGYRSNYF AYEKGDKVCK MNYCKHAGVR LPSGVWFEFV 240
DQDVYAAAKL PECPVGATIS APTQTSVDVS LILDVERILD YSLCQETWSK IRSKQPVSPV 300
DLSYLAPKNP GTGPAFTIIN GTLKYFETRY IRIDIDNPII SKMVGKISGS QTEQELWTEW 360
FPYEGVEIGP NGILKTPGTG KPPLFMIGHG MLDSDLHKTS QAEVFEHPLH AEAPKQLPEE 420
ETLFFGDTGI SKNPVELIEG WFSSWKSTVV TFFFAIGVFI LLYVVARIVI AVRYRYQGSN 480
NKRIYNDIEM SRFRK 495

SEQ ID NO: 295 moltype = AA length = 495
FEATURE Location/Qualifiers
source 1..495
 mol_type = protein
 organism = Vesicular stomatitis virus

SEQUENCE: 295
KFSIVFPQSQ KGNWKNVPSS YHYCPSSSDQ NWHNDLLGIT MKVKMPQTHK AIQADGWMCH 60
AAKWITTCDF RWYGPKYITH SIHSIQPTSE QCKESIKQTK QGTWMSPGFP PQNCGYATVT 120
DSVAVVVQAT PHHVLVDEYT GEWIDSQFPN GKCETEECET VHNSTVWYSD YKVTGLCDAT 180
LVDTEITFFS EDGKKEISIGK PNTGYRSNYF AYEKGDKVCK MNYCKHAGVR LPSGVWFEFV 240
DQDVYAAAKL PECPVGATIS APTQTSVDVS LILDVERILD YSLCQETWSK IRSKQPVSPV 300
DLSYLAPKNP GTGPAFTIIN GTLKYFETRY IRIDIDNPII SKMVGKISGS QTEAELWTEW 360
FPYEGVEIGP NGILKTPGTG KPPLFMIGHG MLDSDLHKTS QAEVFEHPLH AEAPKQLPEE 420
ETLFFGDTGI SKNPVELIEG WFSSWKSTVV TFFFAIGVFI LLYVVARIVI AVRYRYQGSN 480
NKRIYNDIEM SRFRK 495

SEQ ID NO: 296 moltype = AA length = 495
FEATURE Location/Qualifiers
source 1..495
 mol_type = protein
 organism = Vesicular stomatitis virus

SEQUENCE: 296
KFSIVFPQSQ KGNWKNVPSS YHYCPSSSDQ NWHNDLLGIT MKVKMPQTHK AIQADGWMCH 60
AAKWITTCDF RWYGPKYITH SIHSIQPTSE QCKESIKQTK QGTWMSPGFP PQNCGYATVT 120
DSVAVVVQAT PHHVLVDEYT GEWIDSQFPN GKCETEECET VHNSTVWYSD YKVTGLCDAT 180
LVDTEITFFS EDGKKEISIGK PNTGYRSNYF AYEKGDKVCK MNYCKHAGVR LPSGVWFEFV 240
DQDVYAAAKL PECPVGATIS APTQTSVDVS LILDVERILD YSLCQETWSK IRSKQPVSPV 300
DLSYLAPKNP GTGPAFTIIN GTLKYFETRY IRIDIDNPII SKMVGKISGS QTEFELWTEW 360

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FPYEGVEIGP NGILKTPTGY KFPLFMIGHG MLDSDLHKTS QAEVFEHPL AEAPKQLPEE 420
ETLFFGDTGI SKNPVELIEG WFSSWKSTVV TFFFAIGVFI LLYVVARIVI AVRYRYQGSN 480
NKRIYNDIEM SRFRK 495

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SEQ ID NO: 297      moltype = AA length = 495
FEATURE            Location/Qualifiers
source             1..495
                   mol_type = protein
                   organism = Vesicular stomatitis virus

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SEQUENCE: 297
KFSIVFPQSQ KGNWKNVPSS YHYCPSSSDQ NWHNDLLGIT MKVKMPQTHK AIQADGWMCH 60
AAKWITTCDF RWYGPKYIITH SIHSIQPTSE QCKESIKQTK QGTWMSPGFP PQNCGYATVT 120
DSVAVVVQAT PHHVLVDEYT GEWIDSQFPN GKCETEECET VHNSTVWYSD YKVTGLCDAT 180
LVDTEITFFS EDGKKEISIGK PNTGYRSNYF AYEKGDKVCK MNYCKHAGVR LPSGVWFEFV 240
DQDVYAAAKL PECPVGATIS APTQTSVDVS LILDVERILD YSLCQETWSK IRSKQPVSPV 300
DLSYLAPKNP GTGPAFTIIN GTLKYPETRY IRIDIDNP II SKMVGKISGS QTEGELWTEW 360
FPYEGVEIGP NGILKTPTGY KFPLFMIGHG MLDSDLHKTS QAEVFEHPL AEAPKQLPEE 420
ETLFFGDTGI SKNPVELIEG WFSSWKSTVV TFFFAIGVFI LLYVVARIVI AVRYRYQGSN 480
NKRIYNDIEM SRFRK 495

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SEQ ID NO: 298      moltype = AA length = 495
FEATURE            Location/Qualifiers
source             1..495
                   mol_type = protein
                   organism = Vesicular stomatitis virus

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SEQUENCE: 298
KFSIVFPQSQ KGNWKNVPSS YHYCPSSSDQ NWHNDLLGIT MKVKMPQTHK AIQADGWMCH 60
AAKWITTCDF RWYGPKYIITH SIHSIQPTSE QCKESIKQTK QGTWMSPGFP PQNCGYATVT 120
DSVAVVVQAT PHHVLVDEYT GEWIDSQFPN GKCETEECET VHNSTVWYSD YKVTGLCDAT 180
LVDTEITFFS EDGKKEISIGK PNTGYRSNYF AYEKGDKVCK MNYCKHAGVR LPSGVWFEFV 240
DQDVYAAAKL PECPVGATIS APTQTSVDVS LILDVERILD YSLCQETWSK IRSKQPVSPV 300
DLSYLAPKNP GTGPAFTIIN GTLKYPETRY IRIDIDNP II SKMVGKISGS QTEGELWTEW 360
FPYEGVEIGP NGILKTPTGY KFPLFMIGHG MLDSDLHKTS QAEVFEHPL AEAPKQLPEE 420
ETLFFGDTGI SKNPVELIEG WFSSWKSTVV TFFFAIGVFI LLYVVARIVI AVRYRYQGSN 480
NKRIYNDIEM SRFRK 495

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SEQ ID NO: 299      moltype = AA length = 496
FEATURE            Location/Qualifiers
source             1..496
                   mol_type = protein
                   organism = Vesicular stomatitis virus

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SEQUENCE: 299
KFTIVFPHNQ KGNWKNVPAN YQYCPSSSDL NWHNGLIGTS LQVKMPASHK AIQADGWMCH 60
AAKWVTTCDF RWYGPKYVTH SIKSMIPTVD QCKESIAQTK QGTWLNPGFP PQSCGYASVT 120
DAEAVIVKAT PHQVLVDEYT GEWVDSQFPT GCKNKDICTP VHNSTTWHS D YKVTGLCDAN 180
LISMDITFFS EDGKLTSLGK EGTGFRSNYF AYENGDKACR MQYCKHWGVR LPSGVWFEMA 240
DKDIYNDAKF PDCPEGSSIA APSQTSVDVS LIQDVERILD YSLCQETWSK IRAHLPISPV 300
DLSYLSPKNP GTGPAFTIIN GTLKYPETRY IRVDIAGPII PQMRGVISGT TTERELWTDW 360
YPYEDVEIGP NGVLKTATGY KFPLYMIGHG MLDSDLHISS KAQVFEHPI QDAASQLPDD 420
ETLFFGDTGL SKNPIELVEG WFSGWKSTIA SFFFIIGLVI GLYLVLIRIGI ALCIKCRVQE 480
KRPKIYTDVE MNRLDR 496

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SEQ ID NO: 300      moltype = AA length = 496
FEATURE            Location/Qualifiers
source             1..496
                   mol_type = protein
                   organism = Vesicular stomatitis virus

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SEQUENCE: 300
KFTIVFPHNQ KGNWKNVPAN YQYCPSSSDL NWHNGLIGTS LQVKMPFSHK AIQADGWMCH 60
AAKWVTTCDF RWYGPKYVTH SIKSMIPTVD QCKESIAQTK QGTWLNPGFP PQSCGYASVT 120
DAEAVIVKAT PHQVLVDEYT GEWVDSQFPT GCKNKDICTP VHNSTTWHS D YKVTGLCDAN 180
LISMDITFFS EDGKLTSLGK EGTGFRSNYF AYENGDKACR MQYCKHWGVR LPSGVWFEMA 240
DKDIYNDAKF PDCPEGSSIA APSQTSVDVS LIQDVERILD YSLCQETWSK IRAHLPISPV 300
DLSYLSPKNP GTGPAFTIIN GTLKYPETRY IRVDIAGPII PQMRGVISGT TTERELWTDW 360
YPYEDVEIGP NGVLKTATGY KFPLYMIGHG MLDSDLHISS KAQVFEHPI QDAASQLPDD 420
ETLFFGDTGL SKNPIELVEG WFSGWKSTIA SFFFIIGLVI GLYLVLIRIGI ALCIKCRVQE 480
KRPKIYTDVE MNRLDR 496

```

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SEQ ID NO: 301      moltype = AA length = 496
FEATURE            Location/Qualifiers
source             1..496
                   mol_type = protein
                   organism = Vesicular stomatitis virus

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SEQUENCE: 301
KFTIVFPHNQ KGNWKNVPAN YQYCPSSSDL NWHNGLIGTS LQVKMPGSHK AIQADGWMCH 60
AAKWVTTCDF RWYGPKYVTH SIKSMIPTVD QCKESIAQTK QGTWLNPGFP PQSCGYASVT 120
DAEAVIVKAT PHQVLVDEYT GEWVDSQFPT GCKNKDICTP VHNSTTWHS D YKVTGLCDAN 180
LISMDITFFS EDGKLTSLGK EGTGFRSNYF AYENGDKACR MQYCKHWGVR LPSGVWFEMA 240
DKDIYNDAKF PDCPEGSSIA APSQTSVDVS LIQDVERILD YSLCQETWSK IRAHLPISPV 300

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DLSYLSPKNP	GTGPAFTIIN	GTLKYFETRY	IRVDIAGPII	PQMRGVISGT	TTERELWTDW	360
YPYEDVEIGP	NGVLKTATGY	KFPLYMIGHG	MLDSLHISS	KAQVFEHPHI	QDAASQLPDD	420
ETLFFGDTGL	SKNPIELVEG	WFSGWKSTIA	SFFFIIGLVI	GLYLVLRIGI	ALCIKCRVQE	480
KRPKIYTDVE	MNRLDR					496

SEQ ID NO: 302 moltype = AA length = 496
FEATURE Location/Qualifiers
source 1..496
 mol_type = protein
 organism = Vesicular stomatitis virus

SEQUENCE: 302

KFTIVFPHNQ	KGNWKNVPAN	YQYCPSSSDL	NWHNGLIGTS	LQVKMPQSHK	AIQADGWMCH	60
AAKWVTTCDF	RWYGPKYVTH	SIKSMIPTVD	QCKESIAQTK	QGTWLNPGFP	PQSCGYASVT	120
DAEAVIVKAT	PHQVLVDEYT	GEWVDSQFPT	GKCNKDICTP	VHNSTTWHS	YKVTGLCDAN	180
LISMDITFFS	EDGKLTSLGK	EGTGFRSNYF	AYENGDKACR	MQYCKHWGVR	LPSGVWFEMA	240
DKDIYNDAKF	PDCPEGSSIA	APSQTSVDVS	LIQDVERILD	YSLCQETWSK	IRAHLPISPV	300
DLSYLSPKNP	GTGPAFTIIN	GTLKYFETRY	IRVDIAGPII	PQMRGVISGT	TTERELWTDW	360
YPYEDVEIGP	NGVLKTATGY	KFPLYMIGHG	MLDSLHISS	KAQVFEHPHI	QDAASQLPDD	420
ETLFFGDTGL	SKNPIELVEG	WFSGWKSTIA	SFFFIIGLVI	GLYLVLRIGI	ALCIKCRVQE	480
KRPKIYTDVE	MNRLDR					496

SEQ ID NO: 303 moltype = AA length = 496
FEATURE Location/Qualifiers
source 1..496
 mol_type = protein
 organism = Vesicular stomatitis virus

SEQUENCE: 303

KFTIVFPHNQ	KGNWKNVPAN	YQYCPSSSDL	NWHNGLIGTS	LQVKMPKSHK	AIQADGWMCH	60
AAKWVTTCDF	RWYGPKYVTH	SIKSMIPTVD	QCKESIAQTK	QGTWLNPGFP	PQSCGYASVT	120
DAEAVIVKAT	PHQVLVDEYT	GEWVDSQFPT	GKCNKDICTP	VHNSTTWHS	YKVTGLCDAN	180
LISMDITFFS	EDGKLTSLGK	EGTGFRSNYF	AYENGDKACR	MQYCKHWGVR	LPSGVWFEMA	240
DKDIYNDAKF	PDCPEGSSIA	APSQTSVDVS	LIQDVERILD	YSLCQETWSK	IRAHLPISPV	300
DLSYLSPKNP	GTGPAFTIIN	GTLKYFETRY	IRVDIAGPII	PQMRGVISGT	TTERELWTDW	360
YPYEDVEIGP	NGVLKTATGY	KFPLYMIGHG	MLDSLHISS	KAQVFEHPHI	QDAASQLPDD	420
ETLFFGDTGL	SKNPIELVEG	WFSGWKSTIA	SFFFIIGLVI	GLYLVLRIGI	ALCIKCRVQE	480
KRPKIYTDVE	MNRLDR					496

SEQ ID NO: 304 moltype = AA length = 496
FEATURE Location/Qualifiers
source 1..496
 mol_type = protein
 organism = Vesicular stomatitis virus

SEQUENCE: 304

KFTIVFPHNQ	KGNWKNVPAN	YQYCPSSSDL	NWHNGLIGTS	LQVKMPKSHK	AIQADGWMCH	60
AAKWVTTCDF	RWYGPKYVTH	SIKSMIPTVD	QCKESIAQTK	QGTWLNPGFP	PQSCGYASVT	120
DAEAVIVKAT	PHQVLVDEYT	GEWVDSQFPT	GKCNKDICTP	VHNSTTWHS	YKVTGLCDAN	180
LISMDITFFS	EDGKLTSLGK	EGTGFRSNYF	AYENGDKACR	MQYCKHWGVR	LPSGVWFEMA	240
DKDIYNDAKF	PDCPEGSSIA	APSQTSVDVS	LIQDVERILD	YSLCQETWSK	IRAHLPISPV	300
DLSYLSPKNP	GTGPAFTIIN	GTLKYFETRY	IRVDIAGPII	PQMRGVISGT	TTEFELWTDW	360
YPYEDVEIGP	NGVLKTATGY	KFPLYMIGHG	MLDSLHISS	KAQVFEHPHI	QDAASQLPDD	420
ETLFFGDTGL	SKNPIELVEG	WFSGWKSTIA	SFFFIIGLVI	GLYLVLRIGI	ALCIKCRVQE	480
KRPKIYTDVE	MNRLDR					496

SEQ ID NO: 305 moltype = AA length = 496
FEATURE Location/Qualifiers
source 1..496
 mol_type = protein
 organism = Vesicular stomatitis virus

SEQUENCE: 305

KFTIVFPHNQ	KGNWKNVPAN	YQYCPSSSDL	NWHNGLIGTS	LQVKMPKSHK	AIQADGWMCH	60
AAKWVTTCDF	RWYGPKYVTH	SIKSMIPTVD	QCKESIAQTK	QGTWLNPGFP	PQSCGYASVT	120
DAEAVIVKAT	PHQVLVDEYT	GEWVDSQFPT	GKCNKDICTP	VHNSTTWHS	YKVTGLCDAN	180
LISMDITFFS	EDGKLTSLGK	EGTGFRSNYF	AYENGDKACR	MQYCKHWGVR	LPSGVWFEMA	240
DKDIYNDAKF	PDCPEGSSIA	APSQTSVDVS	LIQDVERILD	YSLCQETWSK	IRAHLPISPV	300
DLSYLSPKNP	GTGPAFTIIN	GTLKYFETRY	IRVDIAGPII	PQMRGVISGT	TTEGELWTDW	360
YPYEDVEIGP	NGVLKTATGY	KFPLYMIGHG	MLDSLHISS	KAQVFEHPHI	QDAASQLPDD	420
ETLFFGDTGL	SKNPIELVEG	WFSGWKSTIA	SFFFIIGLVI	GLYLVLRIGI	ALCIKCRVQE	480
KRPKIYTDVE	MNRLDR					496

SEQ ID NO: 306 moltype = AA length = 496
FEATURE Location/Qualifiers
source 1..496
 mol_type = protein
 organism = Vesicular stomatitis virus

SEQUENCE: 306

KFTIVFPHNQ	KGNWKNVPAN	YQYCPSSSDL	NWHNGLIGTS	LQVKMPKSHK	AIQADGWMCH	60
AAKWVTTCDF	RWYGPKYVTH	SIKSMIPTVD	QCKESIAQTK	QGTWLNPGFP	PQSCGYASVT	120
DAEAVIVKAT	PHQVLVDEYT	GEWVDSQFPT	GKCNKDICTP	VHNSTTWHS	YKVTGLCDAN	180
LISMDITFFS	EDGKLTSLGK	EGTGFRSNYF	AYENGDKACR	MQYCKHWGVR	LPSGVWFEMA	240

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DKDIYNDAKF	PDCPEGSSIA	APSQTSVDVS	LIQDVERILD	YSLCQETWSK	IRAHLPISPV	300
DLSYLSPKNP	GTGPAFTIIN	GTLKYFETRY	IRVDIAGPII	PQMRGVISGT	TTEQELWTDW	360
YPYEDVEIGP	NGVLKTATGY	KFPLYMIGHG	MLDSDLHISS	KAQVFEHPHI	QDAASQLPDD	420
ETLFFGDTGL	SKNPIELVEG	WFSGWKSTIA	SFFFIIGLVI	GLYLVLRIGI	ALCIKCRVQE	480
KRPKIYTDVE	MNRLDR					496

SEQ ID NO: 307 moltype = AA length = 496
 FEATURE Location/Qualifiers
 source 1..496
 mol_type = protein
 organism = Vesicular stomatitis virus

SEQUENCE: 307

KFTIVFPHNQ	KGNWKNVPAN	YQYCPSSSDL	NWHNGLIGTS	LQVKMPASHK	AIQADGWMCH	60
AAKWVTTCDF	RWYGPKYVTH	SIKSMIPTVD	QCKESIAQTK	QGTWLNPGFP	PQSCGYASVT	120
DAEAVIVKAT	PHQVLVDEYT	GEWVDSQFPT	GKCNKDICTP	VHNSTTWHS	YKVTGLCDAN	180
LISMDITFFS	EDGKLTSLGK	EGTGFRSNYP	AYENGDKACR	MQYCKHWGVR	LPSGVWFEMA	240
DKDIYNDAKF	PDCPEGSSIA	APSQTSVDVS	LIQDVERILD	YSLCQETWSK	IRAHLPISPV	300
DLSYLSPKNP	GTGPAFTIIN	GTLKYFETRY	IRVDIAGPII	PQMRGVISGT	TTEAELWTDW	360
YPYEDVEIGP	NGVLKTATGY	KFPLYMIGHG	MLDSDLHISS	KAQVFEHPHI	QDAASQLPDD	420
ETLFFGDTGL	SKNPIELVEG	WFSGWKSTIA	SFFFIIGLVI	GLYLVLRIGI	ALCIKCRVQE	480
KRPKIYTDVE	MNRLDR					496

SEQ ID NO: 308 moltype = AA length = 496
 FEATURE Location/Qualifiers
 source 1..496
 mol_type = protein
 organism = Vesicular stomatitis virus

SEQUENCE: 308

KFTIVFPHNQ	KGNWKNVPAN	YQYCPSSSDL	NWHNGLIGTS	LQVKMPASHK	AIQADGWMCH	60
AAKWVTTCDF	RWYGPKYVTH	SIKSMIPTVD	QCKESIAQTK	QGTWLNPGFP	PQSCGYASVT	120
DAEAVIVKAT	PHQVLVDEYT	GEWVDSQFPT	GKCNKDICTP	VHNSTTWHS	YKVTGLCDAN	180
LISMDITFFS	EDGKLTSLGK	EGTGFRSNYP	AYENGDKACR	MQYCKHWGVR	LPSGVWFEMA	240
DKDIYNDAKF	PDCPEGSSIA	APSQTSVDVS	LIQDVERILD	YSLCQETWSK	IRAHLPISPV	300
DLSYLSPKNP	GTGPAFTIIN	GTLKYFETRY	IRVDIAGPII	PQMRGVISGT	TTEFELWTDW	360
YPYEDVEIGP	NGVLKTATGY	KFPLYMIGHG	MLDSDLHISS	KAQVFEHPHI	QDAASQLPDD	420
ETLFFGDTGL	SKNPIELVEG	WFSGWKSTIA	SFFFIIGLVI	GLYLVLRIGI	ALCIKCRVQE	480
KRPKIYTDVE	MNRLDR					496

SEQ ID NO: 309 moltype = AA length = 496
 FEATURE Location/Qualifiers
 source 1..496
 mol_type = protein
 organism = Vesicular stomatitis virus

SEQUENCE: 309

KFTIVFPHNQ	KGNWKNVPAN	YQYCPSSSDL	NWHNGLIGTS	LQVKMPASHK	AIQADGWMCH	60
AAKWVTTCDF	RWYGPKYVTH	SIKSMIPTVD	QCKESIAQTK	QGTWLNPGFP	PQSCGYASVT	120
DAEAVIVKAT	PHQVLVDEYT	GEWVDSQFPT	GKCNKDICTP	VHNSTTWHS	YKVTGLCDAN	180
LISMDITFFS	EDGKLTSLGK	EGTGFRSNYP	AYENGDKACR	MQYCKHWGVR	LPSGVWFEMA	240
DKDIYNDAKF	PDCPEGSSIA	APSQTSVDVS	LIQDVERILD	YSLCQETWSK	IRAHLPISPV	300
DLSYLSPKNP	GTGPAFTIIN	GTLKYFETRY	IRVDIAGPII	PQMRGVISGT	TTEGELWTDW	360
YPYEDVEIGP	NGVLKTATGY	KFPLYMIGHG	MLDSDLHISS	KAQVFEHPHI	QDAASQLPDD	420
ETLFFGDTGL	SKNPIELVEG	WFSGWKSTIA	SFFFIIGLVI	GLYLVLRIGI	ALCIKCRVQE	480
KRPKIYTDVE	MNRLDR					496

SEQ ID NO: 310 moltype = AA length = 496
 FEATURE Location/Qualifiers
 source 1..496
 mol_type = protein
 organism = Vesicular stomatitis virus

SEQUENCE: 310

KFTIVFPHNQ	KGNWKNVPAN	YQYCPSSSDL	NWHNGLIGTS	LQVKMPASHK	AIQADGWMCH	60
AAKWVTTCDF	RWYGPKYVTH	SIKSMIPTVD	QCKESIAQTK	QGTWLNPGFP	PQSCGYASVT	120
DAEAVIVKAT	PHQVLVDEYT	GEWVDSQFPT	GKCNKDICTP	VHNSTTWHS	YKVTGLCDAN	180
LISMDITFFS	EDGKLTSLGK	EGTGFRSNYP	AYENGDKACR	MQYCKHWGVR	LPSGVWFEMA	240
DKDIYNDAKF	PDCPEGSSIA	APSQTSVDVS	LIQDVERILD	YSLCQETWSK	IRAHLPISPV	300
DLSYLSPKNP	GTGPAFTIIN	GTLKYFETRY	IRVDIAGPII	PQMRGVISGT	TTEQELWTDW	360
YPYEDVEIGP	NGVLKTATGY	KFPLYMIGHG	MLDSDLHISS	KAQVFEHPHI	QDAASQLPDD	420
ETLFFGDTGL	SKNPIELVEG	WFSGWKSTIA	SFFFIIGLVI	GLYLVLRIGI	ALCIKCRVQE	480
KRPKIYTDVE	MNRLDR					496

SEQ ID NO: 311 moltype = AA length = 496
 FEATURE Location/Qualifiers
 source 1..496
 mol_type = protein
 organism = Vesicular stomatitis virus

SEQUENCE: 311

KFTIVFPHNQ	KGNWKNVPAN	YQYCPSSSDL	NWHNGLIGTS	LQVKMPFSHK	AIQADGWMCH	60
AAKWVTTCDF	RWYGPKYVTH	SIKSMIPTVD	QCKESIAQTK	QGTWLNPGFP	PQSCGYASVT	120
DAEAVIVKAT	PHQVLVDEYT	GEWVDSQFPT	GKCNKDICTP	VHNSTTWHS	YKVTGLCDAN	180

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LISMDITFFS	EDGKLTSLGK	EGTGFRSNYP	AYENGDKACR	MQYCKHWGVR	LPSGVWFEMA	240
DKDIYNDAKF	PDCPEGSSIA	APSQTSVDVS	LIQDVERILD	YSLCQETWSK	IRAHLPISPV	300
DLSYLSPKNP	GTGPAFTIIN	GTLKYFETRY	IRVDIAGPII	PQMRGVISGT	TTEAELWTDW	360
YPYEDVEIGP	NGVLKTATGY	KFPLYMIGHG	MLDSDLHISS	KAQVFEHPHI	QDAASQLPDD	420
ETLFFGDTGL	SKNPIELVEG	WFSGWKSTIA	SFFFIIGLVI	GLYLVLRIGI	ALCIKCRVQE	480
KRPKIYTDVE	MNRLDR					496

SEQ ID NO: 312 moltype = AA length = 496
 FEATURE Location/Qualifiers
 source 1..496
 mol_type = protein
 organism = Vesicular stomatitis virus

SEQUENCE: 312

KFTIVFPHNQ	KGNWKNVPAN	YQYCPSSSDL	NWHNGLIGTS	LQVKMPFSHK	AIQADGWMCH	60
AAKWVTTCDF	RWYGPKYVTH	SIKSMIPTVD	QCKESIAQTK	QGTWLNPGFP	PQSCGYASVT	120
DAEAVIVKAT	PHQVLVDEYT	GEWVDSQFPT	GKCNKDICTP	VHNSTTWHS	YKVTGLCDAN	180
LISMDITFFS	EDGKLTSLGK	EGTGFRSNYP	AYENGDKACR	MQYCKHWGVR	LPSGVWFEMA	240
DKDIYNDAKF	PDCPEGSSIA	APSQTSVDVS	LIQDVERILD	YSLCQETWSK	IRAHLPISPV	300
DLSYLSPKNP	GTGPAFTIIN	GTLKYFETRY	IRVDIAGPII	PQMRGVISGT	TTEFELWTDW	360
YPYEDVEIGP	NGVLKTATGY	KFPLYMIGHG	MLDSDLHISS	KAQVFEHPHI	QDAASQLPDD	420
ETLFFGDTGL	SKNPIELVEG	WFSGWKSTIA	SFFFIIGLVI	GLYLVLRIGI	ALCIKCRVQE	480
KRPKIYTDVE	MNRLDR					496

SEQ ID NO: 313 moltype = AA length = 496
 FEATURE Location/Qualifiers
 source 1..496
 mol_type = protein
 organism = Vesicular stomatitis virus

SEQUENCE: 313

KFTIVFPHNQ	KGNWKNVPAN	YQYCPSSSDL	NWHNGLIGTS	LQVKMPFSHK	AIQADGWMCH	60
AAKWVTTCDF	RWYGPKYVTH	SIKSMIPTVD	QCKESIAQTK	QGTWLNPGFP	PQSCGYASVT	120
DAEAVIVKAT	PHQVLVDEYT	GEWVDSQFPT	GKCNKDICTP	VHNSTTWHS	YKVTGLCDAN	180
LISMDITFFS	EDGKLTSLGK	EGTGFRSNYP	AYENGDKACR	MQYCKHWGVR	LPSGVWFEMA	240
DKDIYNDAKF	PDCPEGSSIA	APSQTSVDVS	LIQDVERILD	YSLCQETWSK	IRAHLPISPV	300
DLSYLSPKNP	GTGPAFTIIN	GTLKYFETRY	IRVDIAGPII	PQMRGVISGT	TTEGELWTDW	360
YPYEDVEIGP	NGVLKTATGY	KFPLYMIGHG	MLDSDLHISS	KAQVFEHPHI	QDAASQLPDD	420
ETLFFGDTGL	SKNPIELVEG	WFSGWKSTIA	SFFFIIGLVI	GLYLVLRIGI	ALCIKCRVQE	480
KRPKIYTDVE	MNRLDR					496

SEQ ID NO: 314 moltype = AA length = 496
 FEATURE Location/Qualifiers
 source 1..496
 mol_type = protein
 organism = Vesicular stomatitis virus

SEQUENCE: 314

KFTIVFPHNQ	KGNWKNVPAN	YQYCPSSSDL	NWHNGLIGTS	LQVKMPFSHK	AIQADGWMCH	60
AAKWVTTCDF	RWYGPKYVTH	SIKSMIPTVD	QCKESIAQTK	QGTWLNPGFP	PQSCGYASVT	120
DAEAVIVKAT	PHQVLVDEYT	GEWVDSQFPT	GKCNKDICTP	VHNSTTWHS	YKVTGLCDAN	180
LISMDITFFS	EDGKLTSLGK	EGTGFRSNYP	AYENGDKACR	MQYCKHWGVR	LPSGVWFEMA	240
DKDIYNDAKF	PDCPEGSSIA	APSQTSVDVS	LIQDVERILD	YSLCQETWSK	IRAHLPISPV	300
DLSYLSPKNP	GTGPAFTIIN	GTLKYFETRY	IRVDIAGPII	PQMRGVISGT	TTEQELWTDW	360
YPYEDVEIGP	NGVLKTATGY	KFPLYMIGHG	MLDSDLHISS	KAQVFEHPHI	QDAASQLPDD	420
ETLFFGDTGL	SKNPIELVEG	WFSGWKSTIA	SFFFIIGLVI	GLYLVLRIGI	ALCIKCRVQE	480
KRPKIYTDVE	MNRLDR					496

SEQ ID NO: 315 moltype = AA length = 496
 FEATURE Location/Qualifiers
 source 1..496
 mol_type = protein
 organism = Vesicular stomatitis virus

SEQUENCE: 315

KFTIVFPHNQ	KGNWKNVPAN	YQYCPSSSDL	NWHNGLIGTS	LQVKMPGSHK	AIQADGWMCH	60
AAKWVTTCDF	RWYGPKYVTH	SIKSMIPTVD	QCKESIAQTK	QGTWLNPGFP	PQSCGYASVT	120
DAEAVIVKAT	PHQVLVDEYT	GEWVDSQFPT	GKCNKDICTP	VHNSTTWHS	YKVTGLCDAN	180
LISMDITFFS	EDGKLTSLGK	EGTGFRSNYP	AYENGDKACR	MQYCKHWGVR	LPSGVWFEMA	240
DKDIYNDAKF	PDCPEGSSIA	APSQTSVDVS	LIQDVERILD	YSLCQETWSK	IRAHLPISPV	300
DLSYLSPKNP	GTGPAFTIIN	GTLKYFETRY	IRVDIAGPII	PQMRGVISGT	TTEAELWTDW	360
YPYEDVEIGP	NGVLKTATGY	KFPLYMIGHG	MLDSDLHISS	KAQVFEHPHI	QDAASQLPDD	420
ETLFFGDTGL	SKNPIELVEG	WFSGWKSTIA	SFFFIIGLVI	GLYLVLRIGI	ALCIKCRVQE	480
KRPKIYTDVE	MNRLDR					496

SEQ ID NO: 316 moltype = AA length = 496
 FEATURE Location/Qualifiers
 source 1..496
 mol_type = protein
 organism = Vesicular stomatitis virus

SEQUENCE: 316

KFTIVFPHNQ	KGNWKNVPAN	YQYCPSSSDL	NWHNGLIGTS	LQVKMPGSHK	AIQADGWMCH	60
AAKWVTTCDF	RWYGPKYVTH	SIKSMIPTVD	QCKESIAQTK	QGTWLNPGFP	PQSCGYASVT	120

-continued

DAAEAVIVKAT	PHQVLVDEYT	GEWVDSQFPT	GKCNKDICTP	VHNSTTWHSD	YKVTGLCDAN	180
LISMDITFFS	EDGKLTSLGK	EGTGFRSNYF	AYENGDKACR	MQYCKHWGVR	LPSGVWFEMA	240
DKDIYNDAKF	PDCPEGSSIA	APSQTSVDVS	LIQDVERILD	YSLCQETWSK	IRAHLPISPV	300
DLSYLSPKNP	GTGPAFTIIN	GTLKYFETRY	IRVDIAGPII	PQMRGVISGT	TTEFELWTDW	360
YPYEDVEIGP	NGVLKTATGY	KFPLYMIGHG	MLDSLHISS	KAQVFEHPHI	QDAASQLPDD	420
ETLFFGDTGL	SKNPIELVEG	WFSGWKSTIA	SFFFIIGLVI	GLYLVLRIGI	ALCIKCRVQE	480
KRPKIYTDVE	MNRLLDR					496

SEQ ID NO: 317 moltype = AA length = 496
 FEATURE Location/Qualifiers
 source 1..496
 mol_type = protein
 organism = Vesicular stomatitis virus

SEQUENCE: 317

KFTIVFPHNQ	KGNWKNVPAN	YQYCPSSSDL	NWHNGLIGTS	LQVKMPGSHK	AIQADGWMCH	60
AAKWVTTCDF	RWYGPKYVTH	SIKSMIPTVD	QCKESIAQTK	QGTWLNPGFP	PQSCGYASVT	120
DAAEAVIVKAT	PHQVLVDEYT	GEWVDSQFPT	GKCNKDICTP	VHNSTTWHSD	YKVTGLCDAN	180
LISMDITFFS	EDGKLTSLGK	EGTGFRSNYF	AYENGDKACR	MQYCKHWGVR	LPSGVWFEMA	240
DKDIYNDAKF	PDCPEGSSIA	APSQTSVDVS	LIQDVERILD	YSLCQETWSK	IRAHLPISPV	300
DLSYLSPKNP	GTGPAFTIIN	GTLKYFETRY	IRVDIAGPII	PQMRGVISGT	TTEGELWTDW	360
YPYEDVEIGP	NGVLKTATGY	KFPLYMIGHG	MLDSLHISS	KAQVFEHPHI	QDAASQLPDD	420
ETLFFGDTGL	SKNPIELVEG	WFSGWKSTIA	SFFFIIGLVI	GLYLVLRIGI	ALCIKCRVQE	480
KRPKIYTDVE	MNRLLDR					496

SEQ ID NO: 318 moltype = AA length = 496
 FEATURE Location/Qualifiers
 source 1..496
 mol_type = protein
 organism = Vesicular stomatitis virus

SEQUENCE: 318

KFTIVFPHNQ	KGNWKNVPAN	YQYCPSSSDL	NWHNGLIGTS	LQVKMPGSHK	AIQADGWMCH	60
AAKWVTTCDF	RWYGPKYVTH	SIKSMIPTVD	QCKESIAQTK	QGTWLNPGFP	PQSCGYASVT	120
DAAEAVIVKAT	PHQVLVDEYT	GEWVDSQFPT	GKCNKDICTP	VHNSTTWHSD	YKVTGLCDAN	180
LISMDITFFS	EDGKLTSLGK	EGTGFRSNYF	AYENGDKACR	MQYCKHWGVR	LPSGVWFEMA	240
DKDIYNDAKF	PDCPEGSSIA	APSQTSVDVS	LIQDVERILD	YSLCQETWSK	IRAHLPISPV	300
DLSYLSPKNP	GTGPAFTIIN	GTLKYFETRY	IRVDIAGPII	PQMRGVISGT	TTEQELWTDW	360
YPYEDVEIGP	NGVLKTATGY	KFPLYMIGHG	MLDSLHISS	KAQVFEHPHI	QDAASQLPDD	420
ETLFFGDTGL	SKNPIELVEG	WFSGWKSTIA	SFFFIIGLVI	GLYLVLRIGI	ALCIKCRVQE	480
KRPKIYTDVE	MNRLLDR					496

SEQ ID NO: 319 moltype = AA length = 496
 FEATURE Location/Qualifiers
 source 1..496
 mol_type = protein
 organism = Vesicular stomatitis virus

SEQUENCE: 319

KFTIVFPHNQ	KGNWKNVPAN	YQYCPSSSDL	NWHNGLIGTS	LQVKMPQSHK	AIQADGWMCH	60
AAKWVTTCDF	RWYGPKYVTH	SIKSMIPTVD	QCKESIAQTK	QGTWLNPGFP	PQSCGYASVT	120
DAAEAVIVKAT	PHQVLVDEYT	GEWVDSQFPT	GKCNKDICTP	VHNSTTWHSD	YKVTGLCDAN	180
LISMDITFFS	EDGKLTSLGK	EGTGFRSNYF	AYENGDKACR	MQYCKHWGVR	LPSGVWFEMA	240
DKDIYNDAKF	PDCPEGSSIA	APSQTSVDVS	LIQDVERILD	YSLCQETWSK	IRAHLPISPV	300
DLSYLSPKNP	GTGPAFTIIN	GTLKYFETRY	IRVDIAGPII	PQMRGVISGT	TTEAELWTDW	360
YPYEDVEIGP	NGVLKTATGY	KFPLYMIGHG	MLDSLHISS	KAQVFEHPHI	QDAASQLPDD	420
ETLFFGDTGL	SKNPIELVEG	WFSGWKSTIA	SFFFIIGLVI	GLYLVLRIGI	ALCIKCRVQE	480
KRPKIYTDVE	MNRLLDR					496

SEQ ID NO: 320 moltype = AA length = 496
 FEATURE Location/Qualifiers
 source 1..496
 mol_type = protein
 organism = Vesicular stomatitis virus

SEQUENCE: 320

KFTIVFPHNQ	KGNWKNVPAN	YQYCPSSSDL	NWHNGLIGTS	LQVKMPQSHK	AIQADGWMCH	60
AAKWVTTCDF	RWYGPKYVTH	SIKSMIPTVD	QCKESIAQTK	QGTWLNPGFP	PQSCGYASVT	120
DAAEAVIVKAT	PHQVLVDEYT	GEWVDSQFPT	GKCNKDICTP	VHNSTTWHSD	YKVTGLCDAN	180
LISMDITFFS	EDGKLTSLGK	EGTGFRSNYF	AYENGDKACR	MQYCKHWGVR	LPSGVWFEMA	240
DKDIYNDAKF	PDCPEGSSIA	APSQTSVDVS	LIQDVERILD	YSLCQETWSK	IRAHLPISPV	300
DLSYLSPKNP	GTGPAFTIIN	GTLKYFETRY	IRVDIAGPII	PQMRGVISGT	TTEFELWTDW	360
YPYEDVEIGP	NGVLKTATGY	KFPLYMIGHG	MLDSLHISS	KAQVFEHPHI	QDAASQLPDD	420
ETLFFGDTGL	SKNPIELVEG	WFSGWKSTIA	SFFFIIGLVI	GLYLVLRIGI	ALCIKCRVQE	480
KRPKIYTDVE	MNRLLDR					496

SEQ ID NO: 321 moltype = AA length = 496
 FEATURE Location/Qualifiers
 source 1..496
 mol_type = protein
 organism = Vesicular stomatitis virus

SEQUENCE: 321

KFTIVFPHNQ	KGNWKNVPAN	YQYCPSSSDL	NWHNGLIGTS	LQVKMPQSHK	AIQADGWMCH	60
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AAKWVTTCDP RWYGPKYVTH SIKSMIPTVD QCKESIAQTK QGTWLNPGFP PQSCGYASVT 120
DAEAVIVKAT PHQVLVDEYT GEWVDSQFPT GKCNDICPT VHNSTTWHSY YKVTGLCDAN 180
LISMDITFFS EDGKLTSLGK EGTGFRSNYF AYENGDKACR MQYCKHWGVR LPSGVWFEMA 240
DKDIYNDAKF PDCPEGSSIA APSQTSVDVS LIQDVERILD YSLCQETWSK IRAHLPISPV 300
DLSYLSPKNP GTGPAFTIIN GTLKYPETRY IRVDIAGPII PQMRGVISGT TTEGELWTDW 360
YPYEDVEIGP NGVLKTATGY KFPLYMIGHG MLDSDLHISS KAQVFEHPHI QDAASQLPDD 420
ETLFFGDTGL SKNPIELVEG WFSGWKSTIA SFFFIIIGLVI GLYLVLIRIGI ALCIKCRVQE 480
KRPKIYTDVE MNRLDR 496

```

```

SEQ ID NO: 322      moltype = AA length = 496
FEATURE            Location/Qualifiers
source             1..496
                  mol_type = protein
                  organism = Vesicular stomatitis virus

```

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SEQUENCE: 322
KFTIVFPHNQ KGNWKNVPAN YQYCPSSSDL NWHNGLIGTS LQVKMPQSHK AIQADGWMCH 60
AAKWVTTCDP RWYGPKYVTH SIKSMIPTVD QCKESIAQTK QGTWLNPGFP PQSCGYASVT 120
DAEAVIVKAT PHQVLVDEYT GEWVDSQFPT GKCNDICPT VHNSTTWHSY YKVTGLCDAN 180
LISMDITFFS EDGKLTSLGK EGTGFRSNYF AYENGDKACR MQYCKHWGVR LPSGVWFEMA 240
DKDIYNDAKF PDCPEGSSIA APSQTSVDVS LIQDVERILD YSLCQETWSK IRAHLPISPV 300
DLSYLSPKNP GTGPAFTIIN GTLKYPETRY IRVDIAGPII PQMRGVISGT TTEGELWTDW 360
YPYEDVEIGP NGVLKTATGY KFPLYMIGHG MLDSDLHISS KAQVFEHPHI QDAASQLPDD 420
ETLFFGDTGL SKNPIELVEG WFSGWKSTIA SFFFIIIGLVI GLYLVLIRIGI ALCIKCRVQE 480
KRPKIYTDVE MNRLDR 496

```

```

SEQ ID NO: 323      moltype = DNA length = 32
FEATURE            Location/Qualifiers
misc_feature       1..32
                  note = CR2 of human LDL-R
misc_feature       15
                  note = n is a, c, g, or t
misc_feature       27
                  note = n is a, c, g, or t
source             1..32
                  mol_type = other DNA
                  organism = synthetic construct

```

```

SEQUENCE: 323
svtcksgdsc ggrvnrcwrc dgvdcdngsd gc 32

```

```

SEQ ID NO: 324      moltype = AA length = 39
FEATURE            Location/Qualifiers
REGION            1..39
                  note = CR3 of human LDL-R
source            1..39
                  mol_type = protein
                  organism = synthetic construct

```

```

SEQUENCE: 324
PKTCSQDEFR CHDGKCISRQ FVCDSDRDCL DGSDEASCP 39

```

```

SEQ ID NO: 325      moltype = AA length = 42
FEATURE            Location/Qualifiers
REGION            1..42
                  note = CR1 of human LDL-R
source            1..42
                  mol_type = protein
                  organism = synthetic construct

```

```

SEQUENCE: 325
VGDR CERNEF QCQDGKCISY KWVCDGSAEC QDGSDESQET CL 42

```

```

SEQ ID NO: 326      moltype = AA length = 41
FEATURE            Location/Qualifiers
REGION            1..41
                  note = CR2 of human LDL-R
source            1..41
                  mol_type = protein
                  organism = synthetic construct

```

```

SEQUENCE: 326
SVTCKSGDFS CGGRVNRICP QFWRCDGQVD CDNGSDEQGC P 41

```

```

SEQ ID NO: 327      moltype = AA length = 39
FEATURE            Location/Qualifiers
REGION            1..39
                  note = CR3 of human LDL-R
source            1..39
                  mol_type = protein
                  organism = synthetic construct

```

```

SEQUENCE: 327
PKTCSQDEFR CHDGKCISRQ FVCDSDRDCL DGSDEASCP 39

```

-continued

SEQ ID NO: 328	moltype = AA length = 41	
FEATURE	Location/Qualifiers	
REGION	1..41	
	note = CR4 of human LDL-R	
source	1..41	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 328		
VLTCGPASFQ CNSSTCIPKL WACDNDPDCE DGSDEWPQRC R		41
SEQ ID NO: 329	moltype = AA length = 40	
FEATURE	Location/Qualifiers	
REGION	1..40	
	note = CR5 of human LDL-R	
source	1..40	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 329		
DSSPCSAFEF HCLSGECIHS SWRCDGGPDC KDKSDEENCA		40
SEQ ID NO: 330	moltype = AA length = 39	
FEATURE	Location/Qualifiers	
REGION	1..39	
	note = CR6 of human LDL-R	
source	1..39	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 330		
VATCRPDEFQ CSDGNCIHGS RQCDREYDCK DMSDEVGCV		39
SEQ ID NO: 331	moltype = AA length = 41	
FEATURE	Location/Qualifiers	
REGION	1..41	
	note = CR7 of human LDL-R	
source	1..41	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 331		
VTLCGPNKF KCHSGECITL DKVCNMARDC RDWSDEPIKE C		41
SEQ ID NO: 332	moltype = DNA length = 41	
FEATURE	Location/Qualifiers	
misc_feature	1..41	
	note = primer sense	
source	1..41	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 332		
tctcatcatt ttggcaaaga tgaagtcct tttgtactta g		41
SEQ ID NO: 333	moltype = DNA length = 36	
FEATURE	Location/Qualifiers	
misc_feature	1..36	
	note = primer antisense1	
source	1..36	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 333		
ttgctcacca tgcaattcac cccaatgaat aaaaag		36
SEQ ID NO: 334	moltype = DNA length = 31	
FEATURE	Location/Qualifiers	
misc_feature	1..31	
	note = primer antisense351	
source	1..31	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 334		
gctcaccata gttccactga tcattccgac c		31
SEQ ID NO: 335	moltype = DNA length = 31	
FEATURE	Location/Qualifiers	
misc_feature	1..31	
	note = primer sense	
source	1..31	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 335		
cattggggtg aattgcatgg tgagcaaggg c		31

-continued

```

SEQ ID NO: 336      moltype = DNA  length = 31
FEATURE            Location/Qualifiers
misc_feature        1..31
                    note = primer sense
source              1..31
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 336
aatgatcagt ggaactatgg tgagcaaggg c                               31

SEQ ID NO: 337      moltype = DNA  length = 31
FEATURE            Location/Qualifiers
misc_feature        1..31
                    note = primer antisense1
source              1..31
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 337
cattggggtg aattgcatgg tgagcaaggg c                               31

SEQ ID NO: 338      moltype = DNA  length = 32
FEATURE            Location/Qualifiers
misc_feature        1..32
                    note = primer antisense351
source              1..32
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 338
gttccttttc tgtggtcttg tacagctcgt cc                             32

SEQ ID NO: 339      moltype = DNA  length = 37
FEATURE            Location/Qualifiers
misc_feature        1..37
                    note = primer sense1
source              1..37
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 339
gagctgtaca agaagttcac catagttttt ccacaca                       37

SEQ ID NO: 340      moltype = DNA  length = 28
FEATURE            Location/Qualifiers
misc_feature        1..28
                    note = primer sense351
source              1..28
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 340
ctgtacaaga ccacagaaag ggaactgt                                  28

SEQ ID NO: 341      moltype = DNA  length = 34
FEATURE            Location/Qualifiers
misc_feature        1..34
                    note = primer antisense
source              1..34
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 341
ccgccggga gctcggttact ttccaagtcg gttc                           34

```

The invention claimed is:

1. A polypeptide comprising an amino acid sequence at least 95% identical to the amino acid sequence set forth in SEQ ID NO: 1, wherein the polypeptide comprises an amino acid mutation at position 47 and an amino acid mutation at position 354.

2. The polypeptide of claim 1, wherein the polypeptide comprises an amino acid sequence at least 96% identical to the amino acid sequence set forth in SEQ ID NO: 1, wherein the polypeptide comprises an amino acid mutation at position 47 and/or an amino acid mutation at position 354.

3. The polypeptide of claim 1, wherein the polypeptide comprises an amino acid sequence at least 97% identical to the amino acid sequence set forth in SEQ ID NO: 1, wherein

the polypeptide comprises an amino acid mutation at position 47 and/or an amino acid mutation at position 354.

4. The polypeptide of claim 1, wherein the polypeptide comprises an amino acid sequence at least 98% identical to the amino acid sequence set forth in SEQ ID NO: 1, wherein the polypeptide comprises an amino acid mutation at position 47 and/or an amino acid mutation at position 354.

5. The polypeptide of claim 1, wherein the polypeptide comprises an amino acid sequence at least 99% identical to the amino acid sequence set forth in SEQ ID NO: 1, wherein the polypeptide comprises an amino acid mutation at position 47 and/or an amino acid mutation at position 354.

6. The polypeptide of claim 2, wherein the polypeptide comprises:

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- (a) an amino acid mutation at position 47 and position 354; or
- (b) an amino acid mutation at position 47 or position 354.
- 7. A nucleic acid molecule encoding the polypeptide of claim 1.
- 8. A nucleic acid molecule encoding the polypeptide of claim 2.
- 9. A nucleic acid molecule encoding the polypeptide of claim 3.
- 10. A nucleic acid molecule encoding the polypeptide of claim 4.
- 11. A nucleic acid molecule encoding the polypeptide of claim 5.
- 12. A nucleic acid molecule encoding the polypeptide of claim 6.
- 13. A recombinant virus expressing the polypeptide of claim 1.
- 14. A recombinant virus expressing the polypeptide of claim 2.
- 15. A recombinant virus expressing the polypeptide of claim 3.
- 16. A recombinant virus expressing the polypeptide of claim 4.
- 17. A recombinant virus expressing the polypeptide of claim 5.
- 18. A recombinant virus expressing the polypeptide of claim 6.
- 19. A composition comprising the polypeptide of claim 1.
- 20. A composition comprising the polypeptide of claim 2.

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- 21. A composition comprising the polypeptide of claim 3.
- 22. A composition comprising the polypeptide of claim 4.
- 23. A composition comprising the polypeptide of claim 5.
- 24. A composition comprising the polypeptide of claim 6.
- 25. A membrane comprising the polypeptide of claim 1, wherein the polypeptide is anchored in the membrane, and wherein the membrane is the membrane of a liposome, vesicle, exosome, or nanoparticle.
- 26. A membrane comprising the polypeptide of claim 2, wherein the polypeptide is anchored in the membrane, and wherein the membrane is the membrane of a liposome, vesicle, exosome, or nanoparticle.
- 27. A membrane comprising the polypeptide of claim 3, wherein the polypeptide is anchored in the membrane, and wherein the membrane is the membrane of a liposome, vesicle, exosome, or nanoparticle.
- 28. A membrane comprising the polypeptide of claim 4, wherein the polypeptide is anchored in the membrane, and wherein the membrane is the membrane of a liposome, vesicle, exosome, or nanoparticle.
- 29. A membrane comprising the polypeptide of claim 5, wherein the polypeptide is anchored in the membrane, and wherein the membrane is the membrane of a liposome, vesicle, exosome, or nanoparticle.
- 30. A membrane comprising the polypeptide of claim 6, wherein the polypeptide is anchored in the membrane, and wherein the membrane is the membrane of a liposome, vesicle, exosome, or nanoparticle.

* * * * *